SGA 7-I: Monodromy Groups in Algebraic Geometry

Directed by A. Grothendieck, with the collaboration of M. Raynaud and D. S. Rim.

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Seminar on Algebraic Geometry at Bois-Marie, 1967–1969

Monodromy Groups in Algebraic Geometry

(SGA 7 I)

Directed by A. Grothendieck

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Introduction

Let S be a scheme and let $f : X \to S$ be a morphism of schemes. If f is proper and smooth, and if the prime number ℓ is invertible on S, the ℓ-adic cohomology groups

$$H^i(X_{\bar{s}}, \mathbb{Z}_\ell)$$

of the geometric fibres of f form a local ℓ-adic system on S. For f assumed only proper, these groups are the fibres of an ℓ-adic sheaf $R^i f_* \mathbb{Z}_\ell$ on S. The theory of vanishing cycles relates the ramification of this sheaf on S to the singularities of f.

We shall consider only the case where S is a henselian trait, i.e. the spectrum of a henselian discrete valuation ring. This is in practice the essential case. I refer to (I 2.1) for a heuristic description of the theory. Let $S = \operatorname{Spec}(V)$, let $k(\bar{\eta})$ be an algebraic closure of the field of fractions $k(\eta)$ of V, and let $k(\bar{s})$ be the corresponding algebraic closure of the residue field $k(s)$; thus $k(\bar{s})$ is the residue field of the normalization $\bar{V}$ of V in $k(\bar{\eta})$. In the particular case where X is proper and flat over S, of relative dimension n, and where f has only one point $x \in X_s$ at which it is not smooth, one defines $\operatorname{Gal}(\bar{\eta}/\eta)$-modules Φ^i, purely local in nature in a neighbourhood of x, zero for $i \notin [0, n]$ (I 4.2), and one constructs a long exact sequence of $\operatorname{Gal}(\bar{\eta}/\eta)$-modules

$$\cdots \longrightarrow H^i(X_s, \mathbb{Z}_\ell) \xrightarrow{\mathrm{sp}} H^i(X_{\bar{\eta}}, \mathbb{Z}_\ell) \longrightarrow \Phi^i \longrightarrow H^{i+1}(X_s, \mathbb{Z}_\ell) \longrightarrow \cdots .$$

The $\operatorname{Gal}(\bar{s}/s)$-module $H^i(X_s, \mathbb{Z}_\ell)$ is regarded as a $\operatorname{Gal}(\bar{\eta}/\eta)$-module with trivial inertia action; sp is the specialization map. Criteria are also given, at present only in characteristic 0, for the Φ^i with i small to vanish; for example, if X is moreover locally a complete intersection, then $\Phi^i = 0$ for $i \neq n$ in the situation of I 4.5.

The monodromy theorem asserts that the action of the inertia subgroup I of $\operatorname{Gal}(\bar{\eta}/\eta)$ is quasi-unipotent: for $T \in I$, there exist integers $N > 0$, $M > 0$ such that the endomorphism

$$(T^M - I)^N$$

of $H^i(X_{\bar{\eta}}, \mathbb{Z}_\ell)$ is zero. Two proofs of this theorem are given. The first (I 1.2), arithmetic in nature, applies as soon as the cohomology groups considered have reasonable finiteness properties. The key point is that, when $k(s)$ is of finite type over its prime subfield, the action of I is quasi-unipotent for every ℓ-adic representation of $\operatorname{Gal}(\bar{\eta}/\eta)$. The second proof, more geometric, requires purity and resolution of singularities; it is at present valid only in characteristic 0 or for H^1. It gives useful information about the exponent of nilpotence N ($N \leq i + 1$ for H^i), and, as will be seen later, lends itself well, over $\mathbb{C}$, to comparison with transcendental theory.

Applied to H^1 of abelian varieties, i.e. to their Tate modules, these results make it possible to study reduction modulo p. Thus two proofs are given of the stable reduction theorem for abelian varieties: after ramification, the connected special fibre of the Neron model is an extension of an abelian variety by a torus. The first proof (I 6) takes up again the arithmetic method of the monodromy theorem. The second (IX 3.6) rests on a much finer analysis of the Neron model and of its polarization properties.

Grothendieck's Exposes I to V were not written up. They have been summarized in Expose I. The results stated there are proved succinctly, but essentially completely. Expose II applies the method of Lefschetz pencils and (I 5.3) to the study of the fundamental group. For S a surface over an algebraically closed field k, of characteristic exponent p, one shows that the profinite group $\pi_1^{(p')}(S, s)$, completed away from p, is of finite pro-(p') presentation. As explained above, Exposes III to V do not exist.

Expose VI contains, with some complements, Schlessinger's deformation theory. Care is taken there to account for infinitesimal automorphisms of the objects being classified, and not to pass too abruptly to the functor of isomorphism classes of objects. A new construction is also given of

$$\mathrm{Ext}^1(L_{X/S}, -),$$

where $L_{X/S}$ is the relative cotangent complex. This expose serves in the rest of the seminar especially through the study made there of deformations of ordinary quadratic singularities.

Exposes VII and VIII are devoted to the theory of biextensions. This theory plays an essential role in Expose IX, to express what happens to a polarization when one passes from an abelian variety to its Neron model. Expose IX contains the proof of the stable reduction theorem for abelian varieties and various applications. The sequel to this seminar, SGA 7 II, by P. Deligne and N. Katz, will appear later.

Bures-sur-Yvette, May 1972,
P. Deligne

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Expose I. Summary of the first exposes of A. Grothendieck

written up by P. Deligne

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0. Preliminaries

0.0. Terminology

We recall the following definitions.

(0.0.1) A *trait* is the spectrum of a discrete valuation ring. Its generic and closed points are often denoted by η and s.

(0.0.2) A ring is *strictly local* if it is henselian local with separably closed residue field. A scheme is strictly local if it is the spectrum of such a ring.

(0.0.3) A geometric point of S in this expose is a morphism $\bar{x} : \mathrm{Spec}(k) \to S$ with image $x \in S$, where $k(x) \to k$ and k is a separable closure of $k(x)$. For a henselian trait $S = \mathrm{Spec}(V)$, one often writes $\bar{\eta}$ for a generic geometric point of S. The residue field of the valuation ring $\bar{V}$, the integral closure of V in $k(\bar{\eta})$, is a separably closed algebraic extension of $k(s)$ and defines a geometric point $\bar{s}$ of S localized at s.

(0.0.4) The strict localization of X at the geometric point $\bar{x}$ is as in SGA 4 VIII 4.4.

(0.0.5) A local, henselian local, or strictly local S-scheme is said to be essentially of finite type over S if it is the spectrum of a local ring, respectively the henselization of such a local ring, respectively the strict localization, of a scheme of finite type over S.

(0.0.6) $F \mapsto F(n)$ denotes the Tate twist.

(0.0.7) An endomorphism T of a finite-dimensional vector space is quasi-unipotent if its eigenvalues are roots of unity, i.e. if there exist $N \geq 1$ and $M \geq 1$ such that $(T^N - 1)^M = 0$.

0.1. Reminders on ℓ-adic cohomology

Let X be a scheme of finite type over an algebraically closed field k of characteristic exponent p, and let ℓ be a prime prime to p.

The ℓ-adic cohomology groups of X were defined in SGA 4 and SGA 5:

a) $H^i(X, \mathbb{Z}/\ell^n)$ is the i-th cohomology group of the etale site X_{et} with values in $\mathbb{Z}/\ell^n$;

b) $H^i(X, \mathbb{Z}_\ell) = \varprojlim_n H^i(X, \mathbb{Z}/\ell^n)$;

c) $H^i(X, \mathbb{Q}_\ell) = H^i(X, \mathbb{Z}_\ell) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$.

The ℓ-adic cohomology groups with proper supports are defined similarly from the groups $H^i_c(X, \mathbb{Z}/\ell^n)$ of SGA 4 XVII.

In each of the following cases one can prove that these groups $H^i(X)$ have reasonable finiteness properties: cohomology with proper support; X proper over k; X smooth over k (using duality and Poincare duality); when Hironaka resolution of singularities is available for schemes of finite type over k of dimension at most $\dim X$; and for $i = 0, 1$.

In all these cases, the groups $H^i(X, \mathbb{Z}/\ell^n)$ are finite, the groups $H^i(X, \mathbb{Z}_\ell)$ are of finite type, the groups $H^i(X, \mathbb{Q}_\ell)$ are finite-dimensional, and, except perhaps in the last case, one has split short exact sequences

$$0 \to H^i(X, \mathbb{Z}_\ell) \otimes \mathbb{Z}/\ell^n \to H^i(X, \mathbb{Z}/\ell^n) \to \mathrm{Tor}_1(H^{i+1}(X, \mathbb{Z}_\ell), \mathbb{Z}/\ell^n) \to 0.$$

Moreover, the proof of the finiteness theorem gives each time a proof of generic constructibility: if k is the field of fractions of an integral noetherian scheme S and if X is the generic fibre of $f : X_1 \to S$ of finite type, there exists a nonempty open $U \subset S$ such that, over U, the sheaves $R^i f_* \mathbb{Z}/\ell^n$ are locally constant and have formation compatible with every base change.

0.2. Geometric cohomology

We no longer suppose k algebraically closed. Let $\bar{k}$ be an algebraic closure (or separable closure, which amounts to the same thing). We consider the geometric cohomology groups $H^i(X_{\bar{k}})$, where $X_{\bar{k}}$ is obtained from X by extending scalars from k to $\bar{k}$. By transport of structure, $\mathrm{Gal}(\bar{k}/k)$ acts on these groups. In $\mathbb{Q}_\ell$-cohomology, in the cases (0.1.1) to (0.1.5), one obtains continuous representations of $\mathrm{Gal}(\bar{k}/k)$ in a linear group $\mathrm{GL}(n, \mathbb{Q}_\ell)$.

0.3. Reminders on traits

Let S be a henselian trait, with notation $\eta, s, V, \bar{V}, \bar{\eta}, \bar{s}$ as in (0.0.1) and (0.0.3). Let p be the characteristic exponent of $k(s)$. The map $\mathrm{Gal}(\bar{\eta}/\eta) \to \mathrm{Gal}(\bar{s}/s)$ has kernel the inertia group I. The subfield of $k(\bar{\eta})$ defined by I is the maximal unramified extension of $k(\eta)$.

The profinite group I has a largest pro-p subgroup P. The quotient I/P is the tame inertia group, and canonically

$$I/P \simeq \varprojlim_{(n,p)=1} \mu_n(k(\bar{s})) \simeq \widehat{\mathbb{Z}}^{(p')}(1)(k(\bar{s})).$$

If $\chi_n : I/P \to \mu_n(k(\bar{s}))$ is the canonical map and if u is an element of valuation $1/n$, then for $\sigma \in I$,

$$\sigma(u) = \chi_n(\sigma)\, u \quad (\text{mod elements of valuation } > 1/n).$$

In summary,

$$\mathrm{Gal}(\bar{\eta}/\eta) \supset I \supset P, \quad \mathrm{Gal}(\bar{\eta}/\eta)/I \simeq \mathrm{Gal}(\bar{s}/s), \quad I/P \simeq \widehat{\mathbb{Z}}^{(p')}(1)(k(\bar{s})),$$

where P is a pro-p group (trivial if $p = 1$), and conjugation by $\mathrm{Gal}(\bar{\eta}/\eta)/I$ on I/P is the natural action of $\mathrm{Gal}(\bar{s}/s)$ on $\widehat{\mathbb{Z}}^{(p')}(1)(k(\bar{s}))$.

0.4. Abhyankar's lemma

More generally, let S be strictly local regular, let D be a regular divisor in S, and let p' be the characteristic exponent of the generic point of D. Let $\bar{\eta}$ be a geometric generic point of S and let $\bar{s}$ be the geometric point localized at the closed point s which is deduced from it. By Abhyankar's lemma the fundamental group $\pi_1(S - D, \bar{\eta})$ has an analogous devissage:

$$\pi_1(S - D, \bar{\eta}) \supset I \supset P, \quad \pi_1(S - D, \bar{\eta})/I \simeq \mathrm{Gal}(\bar{s}/s), \quad I/P \simeq \widehat{\mathbb{Z}}^{(p')}(1)(k(\bar{s})),$$

where P has no quotient of order prime to p'. The action by inner automorphisms of $\pi_1(S - D, \bar{\eta})/I$ on I/P is the natural action of $\mathrm{Gal}(\bar{s}/s)$ on $\widehat{\mathbb{Z}}^{(p')}(1)(k(\bar{s}))$.

0.5. Neron desingularization

The method of Neron desingularization gives the following result.

Lemma 0.5.1. Let $S \to S_1$ be a morphism of henselian traits, with $S = \mathrm{Spec}(V)$, $S_1 = \mathrm{Spec}(V_1)$, and let ϖ be a uniformizer of V_1. Suppose that ϖ is also a uniformizer of V, and that the extensions $k(s)/k(s_1)$ and $k(\eta)/k(\eta_1)$ are separable. Then V is the filtered inductive limit of henselian V_1-algebras, smooth and essentially of finite type over V_1.

To construct the desired V_1-algebras, one takes $V_1 \subset B \subset V$ with B of finite type, and applies Neron desingularization to $\mathrm{Spec}(B)$.

Let $S = \mathrm{Spec}(V)$ be a henselian trait. If S is of equal characteristic, with uniformizer ϖ, (0.5.1) applies to $S_1 = \mathbb{F}_p[\varpi]_{(\varpi)}$ after the usual separability check; hence S is a limit of spectra of henselian algebras, smooth and essentially of finite type over a trait of finite type. If S is complete of mixed characteristic, with perfect residue field k, then V is an Eisenstein extension of the Cohen-Witt ring $W(k)$; after a small perturbation of the coefficients one reduces to a trait whose residue field is a radical extension of a field of finite type over $\mathbb{F}_p$, and again (0.5.1) applies.

1. Arithmetic proof of the monodromy theorem

The reader will find in [5], Appendix, a proof of the following result of A. Grothendieck.

Proposition 1.1. Let S be a henselian trait, let ℓ be a prime prime to the characteristic exponent p of $k(s)$, and let ρ be a continuous representation of $\mathrm{Gal}(\bar\eta/\eta)$ in $\mathrm{GL}(n, \mathbb{Q}_\ell)$. Suppose that no finite extension of $k(s)$ contains all roots of unity of order a power of ℓ. Then there exists an open subgroup I_1 of the inertia group I such that $\rho(\sigma)$ is unipotent for $\sigma \in I_1$.

The preceding condition is automatically satisfied when $k(s)$ is of finite type over the prime field, or purely inseparable over such a field.

Monodromy theorem 1.2. With the preceding notation, let X be a scheme of finite type over η, and let H be a cohomology space of $X_{\bar\eta}$, with coefficients in $\mathbb{Q}_\ell$, of one of the types (0.1.1)–(0.1.5). If the condition of Proposition 1.1 is satisfied, the action of any element of I on H is quasi-unipotent.

The limit arguments of 0.5 allow one to remove this hypothesis.

Variant 1.3. The condition of Proposition 1.1 is superfluous.

Let H' be a $\mathbb{Z}_\ell$-cohomology space of $X_{\bar\eta}$ which gives rise to H, and put $H_1 = H'/\text{torsion}$. One has $H = H_1 \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$. After replacing the trait by a finite extension, one may suppose that $\mathrm{Gal}(\bar\eta/\eta)$ acts trivially on $H_1/\ell H_1$, and, in mixed characteristic, that the trait is complete with perfect residue field. Using the presentation of the trait as a limit from 0.5, the representation comes from a constructible constant-twisted sheaf on $S_i - D_i$, and the devissages (0.3.1) and (0.4.1) give commutative diagrams

$$\begin{array}{ccccc} \mathrm{Gal}(\bar\eta/\eta) & \twoheadrightarrow & I & \twoheadrightarrow & P \\ \downarrow & & \downarrow & & \downarrow \\ \pi_1(S_i - D_i, \bar\eta) & \twoheadrightarrow & I_i & \twoheadrightarrow & P_i \end{array} \qquad \begin{array}{ccc} I/P & \xrightarrow{\sim} & \widehat{\mathbb{Z}}^{(p')}(1) \\ \downarrow & & \| \\ I_i/P_i & \xrightarrow{\sim} & \widehat{\mathbb{Z}}^{(p')}(1). \end{array}$$

Since $\mathrm{Ker}(\mathrm{GL}(H_1) \to \mathrm{GL}(H_1/\ell H_1))$ is a pro-ℓ group, P_i acts trivially, and one applies the following variant.

Variant 1.4. With the notation of 0.4, let ρ be a continuous representation of $\mathrm{Gal}(\bar\eta/\eta)$ in

$\mathrm{GL}(n, \mathbb{Q}_\ell)$. Suppose that $k(s)$ satisfies the condition of 1.1 and that $\rho(P)$ is finite. Then the action of I is quasi-unipotent.

2. Vanishing cycles

Let S be a strictly local trait, and keep the notation of 0.3; here $s = \bar{s}$. Let $f : X \to S$ be a scheme of finite type over S. The formalism of vanishing cycles is a tool for studying the relation between the cohomology of X_s and that of $X_{\bar\eta}$, and the inertia action on the latter, from local information on X and on the morphism f.

In transcendental theory, if $f : X \to D$ is a proper morphism from an analytic space to a disk, the vanishing cycles appear as follows. After shrinking D, X is a topological fibration over the punctured disk D^*, so the fibres X_t for $t \in D^*$ are all isomorphic. One represents the special fibre X_0 as obtained from the general fibre by contraction of polyhedra:

$$\begin{tikzcd} X_t \arrow[rr, "\text{contraction}"] \arrow[dr] & & X_0 \arrow[dl] \\ & D & \end{tikzcd}$$

The method of vanishing cycles then studies the difference between the cohomology of X_t and that of X_0 by means of the Leray spectral sequence of this contraction.

The geometric theory below is very close to this transcendental theory: the general point $t \in D$ is replaced by $\bar\eta$, the geometric generic point of S. Conversely, the transcendental theory can be modelled on the geometric one by replacing t or $\bar\eta$ by the universal covering of D^*.

Proposition 2.3. Let $u : X' \to X$ be etale, x' a geometric point of X' over the geometric point x of X_s. The homomorphism

$$(R^i\Psi)_x \longrightarrow (R^i\Psi)_{x'}$$

induced by u is an isomorphism.

Corollary 2.4. The formation of the Ψ^i and Φ^i commutes with strict localization at a point x.

Let A be the coefficient ring. If I acts on the geometric generic fibre, one has the exact sequence of sheaves

$$0 \to A_{X_s} \to R\Psi A \to R\Phi A \to 0$$

in the derived sense; for f proper this gives the long exact sequence

$$\cdots \to H^i(X_s, A) \to H^i(X_{\bar\eta}, A) \to H^i(X_s, R\Phi A) \to H^{i+1}(X_s, A) \to \cdots .$$

3. Geometric proof of the monodromy theorem

Let S be strictly local, X a normal excellent scheme regular away from a divisor D, and suppose that, etale locally, the special fibre is a divisor with normal crossings

$$X_s = \sum_{i \in C} a_i D_i.$$

The proof uses purity to describe the prime-to-p coverings of the complement and then takes limits. If x is a point of the special fibre and if C is the set of components through x, set

$$K : \bigoplus_{i \in C} \mathbb{Z}(1) \xrightarrow{(a_i)} \mathbb{Z}(1),$$

concentrated in degrees $-1, 0$.

Theorem 3.3. Assuming purity, the groups of tame vanishing cycles are described by

$$R^*\Psi_x \simeq H^*(K \otimes A) \otimes A[t], \qquad \deg(t) = 2,$$

and the inertia action on $R^1\Psi_x$ is obtained from a transitive action on a finite set of cardinal prime to p, determined by the multiplicities a_i.

Corollary 3.4. Let N be the least common multiple of the multiplicities of the components of the special fibre. If at most q' components meet at a point, then for T in tame inertia

$$(T^N - 1)^{q'+1} = 0 \quad \text{on } R^q\Psi.$$

Theorem 3.5. Let C be a smooth projective curve over the field of fractions of a discrete valuation ring. The inertia group acts on

$$H^1(C_{\bar{\eta}}, \mathbb{Q}_\ell)$$

by quasi-unipotent automorphisms of level 2: for some N and all $T \in I$,

$$(T^N - 1)^2 = 0.$$

If A is an abelian variety, the same conclusion holds for $H^1(A_{\bar{\eta}}, \mathbb{Q}_\ell)$ and for the Tate module. More generally, for every scheme of finite type over η, inertia acts quasi-unipotently on $H^i(X_{\bar{\eta}}, \mathbb{Q}_\ell)$; in characteristic zero, for X proper and smooth one obtains $(T-1)^{i+1} = 0$ on H^i after passing to an open subgroup of inertia.

4. Vanishing criteria for sheaves of vanishing cycles

Let S be a strictly local trait and $f : X \to S$ of finite type. Put $n = \dim X_{\bar{\eta}}$. If X' is the schematic closure of X_η in X, then on $X_s - X'_s$ one has $\Phi^0 = A$ and the other Φ^i vanish; on X' the Φ^i and Ψ^i relative to X/S or X'/S agree, and $\Phi^0 = 0$. Since X' is flat over S, this often reduces the problem to the flat case.

Theorem 4.2. One has $\Phi^i = 0$ for $i > n$. More precisely, if x is a point of X_s whose closure has dimension k and $\bar{x}$ is a geometric point localized at x, then

$$\Phi^i_{\bar{x}} = 0 \qquad (i > n - k).$$

Corollary 4.3. If f is proper and flat and the non-smooth locus of f in X_s has dimension $< d$, the specialization morphism

$$H^i(X_s) \longrightarrow H^i(X_{\bar{\eta}})$$

is an isomorphism for $i > n + d + 1$ and an epimorphism for $i = n + d + 1$.

To prove in certain cases that the small Φ^i vanish, one uses local duality, directly or through the local Lefschetz theorems of SGA 2 XIV. This tool is available only in characteristic 0, or in equal characteristic p when resolution of singularities is known in the dimensions considered.

Theorem 4.5. Suppose that S has characteristic 0. Let $f : X \to S$ be of finite type and let x be a closed point of X_s. Suppose that x is an isolated non-smooth point in its fibre and that X has relative geometric depth $\geq n$ at x, i.e. near x, X is identified with a subscheme defined by k equations in a smooth scheme of relative dimension N at x, with $n < N - k$. Then

$$\Phi^i_{\bar{x}} = 0 \qquad (i < n).$$

Corollary 4.6. For S a henselian trait of characteristic 0, X/S a relative complete intersection of relative dimension n, and $x \in X_s$ an isolated non-smooth point in its fibre, one has

$$\Phi^i_{\bar{x}} = 0 \qquad (i \neq n).$$

Corollary 4.7. Under the hypotheses of 4.6, $\Phi^n_{\bar{x}}$ is a flat A-module.

Variant 4.8. If in 4.5 one suppresses the isolatedness condition and lets Z be the non-smooth locus of f, then

$$\Phi^i = 0 \qquad (i < n - \dim Z).$$

Proposition 4.10. Under the general hypotheses of 4.5, suppose that x is an isolated non-smooth point in its fibre; that for every point y of $X_{\bar{\eta}}$ specializing to x, with closure of dimension $d(y)$, one has $\mathrm{prof}_{\mathrm{et},y}(X_{\bar{\eta}}) \geq n - d(y)$; and that $\mathrm{prof}_{\mathrm{et},x}(X_s) \geq n$. Then

$$\Phi^i_{\bar{x}} = 0 \qquad (i < n - 1).$$

5. Action of monodromy on fundamental groups

In this section, which corresponds to Grothendieck's expose of 19 November 1968, we use the semistable reduction theorem for curves. Artin and Winters gave a direct proof; it can also be deduced from the analogous theorem for abelian varieties. We shall also use Schlessinger's theory as presented in Rim's Expose VI.

Let k be algebraically closed, let C be a projective curve over k, and let $x_0, \ldots, x_n$ be finitely many closed points of C. We say that $X = (C; x_0, \ldots, x_n)$ is stable if C is reduced and has only ordinary double points with distinct tangents, the x_i are distinct smooth points of C, and if C_j is an irreducible component of C and E is the set of points of its normalization lying over a singular point of C or one of the x_i, then E has at least 3 elements when C_j has genus 0 and at least 1 element when C_j has genus 1.

The last condition is equivalent to the absence of non-trivial infinitesimal automorphisms, and also to the ampleness of the invertible sheaf

$$\omega\Bigl(\sum x_i\Bigr),$$

where ω is the dualizing sheaf.

Proposition 5.1.1. Let $X = (C; x_0, \ldots, x_n)$ be stable over the generic point η of a trait S, with C_η smooth. There exists a finite separable extension η' of η and a stable curve X' over the normalization S' of S in η' such that

$$X' \otimes_{S'} \eta' = X \otimes_\eta \eta'.$$

Let X_η be a geometrically connected scheme of finite type over the field of fractions of a strictly local trait S. For a geometric point ξ of $X_{\bar{\eta}}$ one has an exact sequence

$$0 \to \pi_1(X_{\bar{\eta}}, \xi) \to \pi_1(X, \xi) \to \mathrm{Gal}(\bar{\eta}/\eta) \to 0, \qquad (5.2.1)$$

and hence a homomorphism from $\mathrm{Gal}(\bar{\eta}/\eta)$ to the group of outer automorphisms of $\pi_1(X_{\bar{\eta}})$. Passing to the maximal prime-to-p quotient gives

$$\mathrm{Gal}(\bar{\eta}/\eta) \to \mathrm{Aut}_{\mathrm{ext}}(\pi_1^{(p')}(X_{\bar{\eta}})). \qquad (5.2.2)$$

If X_η has a rational point x and $\bar{x}$ is the geometric point over it, the sequence splits canonically and gives

$$[x] : \mathrm{Gal}(\bar{\eta}/\eta) \to \mathrm{Aut}(\pi_1^{(p')}(X_{\bar{\eta}}, \bar{x})). \qquad (5.2.3)$$

Theorem 5.3. The image of wild inertia P under (5.2.2), or equivalently under (5.2.3), is finite.

The proof first reduces to the case where X_η is geometrically normal and integral. Let X' be the normalization and let $X'' = X' \times_X X'$. After a finite extension of $k(\eta)$, the connected components of X' may be assumed geometrically normal and integral with rational base points, and those of X'' geometrically connected with rational base points. Choosing prime-to-p path classes invariant under P, the Van Kampen formulas show that if a finite-index subgroup $P' \subset P$ acts trivially on the fundamental groups of the components, then P' acts trivially on $\pi_1^{(p')}(X_{\bar{\eta}})$.

For X_η normal and $U \subset X_\eta$ open, $\pi_1(U)$ maps onto $\pi_1(X_\eta)$. Hence one may suppose X_η smooth and affine. By cutting with generic hyperplane sections and using Bertini, one reduces to X_η smooth, geometrically connected, and of dimension one. After finite extension of $k(\eta)$ one may suppose that X_η is the complement, in a smooth projective curve C, of finitely many rational points; applying semistable reduction, one reduces to a stable curve over S. It remains to carry out the stable-curve calculation.

Theorem 5.4

Let S be a strictly local scheme, let $f : \bar{X} \to S$ be a proper and flat morphism whose special fibre is a curve whose only points of non-smoothness are ordinary double points, and let Y be a subscheme of $\bar{X}$ which is the union of a finite number of disjoint sections contained in the smooth locus. Put

$$X = \bar{X} - Y.$$

Let x be a rational point of X, let U be the open subset of S over which X is smooth, and let ξ be a geometric point of U. Then the image of

$$\pi_1(U, \xi)$$

in

$$\mathrm{Aut}(\pi_1^{(p')}(X_\xi, x_\xi))$$

is abelian and prime to p.

One must keep in mind that one cannot make $\pi_1(U, \xi)$ act on $\pi_1^{(p')}(X_\xi, x_\xi)$ except because the groups $\pi_1^{(p')}$ satisfy the specialization theorem.

The proof reduces first to the case where S is complete, and then to the universal case in which S is a formal moduli scheme for the special fibre equipped with its sections. Then S is a formal power series ring over a Cohen ring,

$$S = \mathrm{Spec}\, W[[t_1, \dots, t_n]],$$

and U is the complement of a normal-crossings divisor with equation

$$t_1 \cdots t_m = 0.$$

In this universal case, by Abhyankar's lemma, $\pi_1(U, \xi)$ is abelian and prime to p, whence the theorem.

Remark 5.5. One can prove a result analogous to 5.4 in arbitrary relative dimension by considering a generic plane section of dimension one and applying Bertini.

6. Appendix: arithmetic proof of the stable reduction theorem

by P. Deligne

Let S be a trait and let A_η be an abelian variety over $k(\eta)$, of dimension $d \neq 0$. For a local morphism of traits $S' \to S$, denote by $A_{\eta'}$ the abelian variety over $k(\eta')$ deduced from A_η by extension of scalars. Let $\mathcal{A}_{S'}$ be the Neron model of $A_{\eta'}$ over S', let $\mathcal{A}_{S',s'}$ be its special fibre, and let $\mathcal{A}^0_{S',s'}$ be the neutral component of $\mathcal{A}_{S',s'}$. The stable reduction theorem is the following.

Theorem 6.1. For a suitable S', $A_{\eta'}$ has stable reduction, i.e. $\mathcal{A}^0_{S',s'}$ is an extension of an abelian variety by a torus.

It follows rather easily from 6.1 that one may choose S' so that $k(\eta')$ is a finite separable extension of $k(\eta)$. Suppose first that the residue field of S is of finite type over the prime field; the same argument would work when $k(s)$ is radicial over such a field.

Let ℓ be a prime number prime to the residual characteristic exponent p, let $T_\ell(A)$ be the Tate module, let

$$V_\ell(A) = T_\ell(A) \otimes \mathbb{Q}_\ell,$$

and let

$$\rho : \mathrm{Gal}(\bar{\eta}/\eta) \longrightarrow \mathrm{GL}(T_\ell(A))$$

be the natural representation. A preliminary extension of scalars reduces us to the case where the inertia group I acts unipotently, by 1.2, hence through its largest pro-ℓ quotient $\mathbb{Z}_\ell(1)$, cf. 0.3. Let T be the image in $\mathrm{GL}(T_\ell(A))$ of a generator of $\mathbb{Z}_\ell(1)$, and let r be the largest integer ≥ 0 such that $(T-1)^r \neq 0$.

Lemma 6.2. The map

$$\sigma \in I, \quad v \in V_\ell(A) \longmapsto (\rho(\sigma) - 1)^r(v)$$

induces morphisms and an isomorphism

$$\text{(6.2.1)} \qquad \begin{array}{ccc} V_\ell(A)_I \otimes \mathbb{Q}_\ell(r) & \xrightarrow{\quad M \quad} & V_\ell(A)^I \\ \downarrow & & \uparrow \\ V_\ell(A)/\operatorname{Ker}(T-1)^r \otimes \mathbb{Q}_\ell(r) & \xrightarrow[\sim]{\quad} & \operatorname{Im}(T-1)^r \end{array} \qquad (1)$$

The modules $V_\ell(A)_I$ and $V_\ell(A)^I$ are $\mathrm{Gal}(\bar{s}/s)$-modules, and M is Galois-equivariant.

6.3. An ℓ-adic representation

$$\tau : \mathrm{Gal}(\bar{s}/s) \longrightarrow \mathrm{GL}(V)$$

is said to be of weight n ($n \in \mathbb{Z}$) if the following condition is satisfied: for a suitable model X of $k(s)$, with X an irreducible variety of finite type over the prime field and $k(s) = k(X)$, the representation τ factors through $\pi_1(X, \bar{s})$, and, for x a closed point of X, the eigenvalues of $\tau(F_x)$ are algebraic numbers all of whose complex conjugates have absolute value

$$|k(x)|^{n/2},$$

where F_x denotes geometric Frobenius Φ_x^{-1}.

If V (respectively W) is of weight n (respectively m) and $n \neq m$, then

$$\mathrm{Hom}_{\mathrm{Gal}}(V, W) = 0.$$

6.4. It follows from the universal property of the Neron model that

$$V_\ell(A)^I = V_\ell(\mathcal{A}^0_{S,s}).$$

If a is the dimension of the largest abelian quotient of $\mathcal{A}^0_{S,s}$, and m the dimension of the multiplicative part of $\mathcal{A}^0_{S,s}$, then, by Weil, $V_\ell(A)^I$ is an extension of a representation of dimension $2a$ and weight -1 by a representation of dimension m and weight -2.

6.5. Let A^*_η be the abelian variety dual to A_η. Recall that $V_\ell(A^*)$ and $V_\ell(A)$, hence also $V_\ell(A^*)^I$ and $V_\ell(A)_I$, are in duality with values in $\mathbb{Q}_\ell(1)$. If a^* and m^* are defined for A^* as a and m for A, it follows from 6.4 that $V_\ell(A)_I$ is an extension of a representation of dimension m^* and weight 0 by a representation of dimension $2a^*$ and weight -1.

6.6. The Galois module $\mathbb{Q}_\ell(r)$ has weight $-2r$. Since $\operatorname{Im}(T-1)^r \neq 0$, one deduces from 6.4, 6.5, and the isomorphism 6.2 that $r \leq 1$. If $r = 0$, then

$$\begin{cases} a + m \leq d, \\ 2a + m = 2d, \end{cases}$$

hence $m = 0$, $a = d$, and A_η has good reduction. Suppose that $r = 1$. Then 6.2 implies that $\operatorname{Im}(T-1)$ has weight -2 and that $V_\ell(A)/V_\ell(A)^I$ has weight 0; these spaces have the same dimension $m_1 \leq m$. Hence

$$\dim V_\ell(A)^I = 2d - m_1 = 2a + m,$$

and

$$2\dim \mathcal{A}^0_{S,s} \geq 2(a+m) \geq 2a + m + m_1 = 2d = 2\dim A_{S,s}.$$

Thus A_η has stable reduction, since $\dim \mathcal{A}^0_{S,s} = a + m$. At the same time one obtains $m_1 = m$, i.e.

$$\operatorname{Im}(T-1) \subset V_\ell(A)^I \simeq V_\ell(\mathcal{A}^0_{S,s})$$

is the V_ℓ of the multiplicative part of $\mathcal{A}^0_{S,s}$.

6.7. To treat the general case, we shall use 0.5, cf. 1.3. We reduce to the case where

a) the conditions of 0.5.2 or 0.5.3 are satisfied;

b) $\operatorname{Gal}(\bar\eta/\eta)$ acts trivially on A_ℓ;

c) the action of I on $T_\ell(A)$ is unipotent.

Under these hypotheses we shall prove that A_η has stable reduction. Hypothesis a) permits one to apply 0.5.1 and to write

$$S = \varprojlim S_i,$$

as in 1.3, whose notation we resume. Let s_i be the closed point of S_i. For i sufficiently large, the neutral component $\mathcal{A}^0_S$ of the Neron model $\mathcal{A}_S$ comes from a smooth group scheme with connected fibres $\mathcal{A}_i$ over S_i, and $(\mathcal{A}_i)_\ell$ is constant on $S_i - D_i$. The group $\pi_1(S_i - D_i, \bar\eta)$ acts on $T_\ell(A)$, and acts trivially on

$$T_\ell(A)/\ell T_\ell(A) \simeq A_\ell.$$

As in 1.3, one sees that P_i acts trivially. Then, by (1.3.1),

$$T_\ell(A)_I = T_\ell(A)_{I_i}, \qquad T_\ell(\mathcal{A}_{i,s_i}) = T_\ell(\mathcal{A}^0_S) = T_\ell(A)^I = T_\ell(A^{I_i}),$$

and (6.2.1) is a diagram of $\operatorname{Gal}(\bar s_i/s_i)$-modules to which one can apply the arguments of 6.4–6.6.

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Expose II. Finiteness properties of the fundamental group

by Michele Raynaud

Let k be a separably closed field of characteristic $p \geq 0$, let X be a connected scheme of finite type over k, and let

$$\pi_1^{(p')}(X)$$

be the largest quotient "prime to p" of $\pi_1(X)$. We show in this expose that, if X is a surface, $\pi_1^{(p')}(X)$ is topologically of finite presentation, and that the same is true for arbitrary X if one admits resolution of singularities.

In the case of a surface, the method of proof, due to J.-P. Murre, consists in reducing to the case where one has a fibration over $\mathbb{P}^1$ having the properties (i), (ii), and (iii) of the following proposition.

Proposition 2.1. Let k be an algebraically closed field and let S be a projective, irreducible, smooth surface over k. Then one can find a blow-up

$$g : S' \longrightarrow S,$$

where S' is a regular surface equipped with a fibration

$$f : S' \longrightarrow \mathbb{P}^1,$$

such that f has a section σ and the following conditions are satisfied:

(i) f is smooth at the points of $\sigma(\mathbb{P}^1)$;

(ii) the fibres of f are geometrically integral curves having only ordinary singularities;

(iii) there exists a non-empty open U of $\mathbb{P}^1$ such that the fibres of f at the points of U are smooth.

The proposition follows from the existence of a projective embedding $S \to \mathbb{P}^r$ such that one can find a Lefschetz pencil for that embedding. Let

$$D = \{H_t\}_{t\in\mathbb{P}^1}$$

be such a pencil of hyperplanes. Recall that it has the following properties: its axis Δ cuts S transversally, so that the closed subscheme $\Delta \cap S$ is reduced and consists of a finite set of closed points $x_1, \ldots, x_n$ of S; there exists a non-empty open $U \subset \mathbb{P}^1$ such that, for every $t \in U$, H_t cuts S transversally; and, for every $t \in \mathbb{P}^1 - U$, H_t cuts S transversally except at one point, which is an ordinary quadratic singular point.

Through every point x of S not belonging to Δ there passes a unique hyperplane H_t; the application that, to x, associates $t \in \mathbb{P}^1$ is a rational map $a : S \dashrightarrow \mathbb{P}^1$ whose domain of definition is $S - (\Delta \cap S)$. Consider the blow-up

$$g : S' \longrightarrow S$$

of the closed subscheme $S \cap \Delta$ of S. The scheme S' is regular and the fibre S'_{x_i} over a point x_i is identified with the projective fibre defined by the tangent space to S at x_i, hence is isomorphic to $\mathbb{P}^1$. Moreover the rational map $f = ag$ is everywhere defined. Finally, to each point x_i one associates the section of f which, to a point $t \in \mathbb{P}^1$, makes correspond the tangent vector at x_i to $H_t \cap S$.

Let us verify (i), (ii), and (iii). For each $t \in \mathbb{P}^1$, the curves S'_t and $S_t = H_t \cap S$ are isomorphic: the morphism $S'_t \to S_t$ is an isomorphism except perhaps at the points x_i, but S_t is regular at those points; moreover S'_t is a complete intersection in S', hence integral, and therefore isomorphic to S_t. Conditions (i) and (iii) are just conditions a) and b) of the Lefschetz pencil. Finally f is smooth at the points of σ, since f is flat and the fibres S'_t are geometrically regular at the points $\sigma(t)$.

Corollary 2.2. With the notation of 2.1, let Y be a normal-crossings divisor in S, and let $Y' = Y \times_S S'$. Then one can choose f, g, U so that $Y' \times_{\mathbb{P}^1} U$ is a normal-crossings divisor relative to U.

Indeed, if there exists a Lefschetz pencil for an embedding $S \to \mathbb{P}^r$, the Lefschetz pencils form an open subset of the variety of lines in $\mathbb{P}^r$. The same holds for the set of Lefschetz pencils whose axis does not meet Y and for which there exists a non-empty open $U_1 \subset \mathbb{P}^1$ such that $Y \cap H_t$ is smooth for $t \in U_1$. It then suffices to construct f and g from a pencil satisfying these two conditions, and to replace U by $U \cap U_1$.

2.3.0. Let G be a profinite group. Recall that G is topologically of finite generation if there exists an integer n such that G is isomorphic to a quotient of the free pro-group on n generators, $F(n)$. Let $K = \mathrm{Ker}(F(n) \to G)$. One says that G is topologically of finite presentation if, in addition, one can find finitely many elements $k_1, \ldots, k_r$ of K such that the smallest closed invariant subgroup of K containing $k_1, \ldots, k_r$ is equal to K.

Theorem 2.3.1. Let k be a separably closed field of characteristic $p \geq 0$, and let X be a connected scheme of finite type over k. Assume that schemes of finite type and of dimension $\leq \dim(X)$ over an algebraic closure of k are strongly desingularizable (SGA 5 I 3.1.5); this condition is satisfied when $\dim X = 2$, by [1], or when $k = 0$, by [2]. Then the fundamental group

$$\pi_1^{(p')}(X)$$

is topologically of finite presentation.

By SGA 1 IX 4.10, one may suppose k algebraically closed.

1) Reduction to the case where X is a regular surface. If $f : X' \to X$ is a morphism of finite type, set $X'' = X' \times_X X'$, and denote by $(X'_a)_{a \in A}$ and $(X''_b)_{b \in B}$ the sets of connected components of X' and X'', respectively. When f is a morphism of effective descent for etale coverings, the fundamental group $\pi_1(X)$ can be computed explicitly in terms of the groups $\pi_1(X'_a)$ and $\pi_1(X''_b)$. It follows from SGA 1 IX 5.2, 5.3 that, if the $\pi_1(X'_a)$ are topologically of finite generation, so is $\pi_1(X)$; and, if the $\pi_1(X'_a)$ are topologically of finite presentation and the $\pi_1(X''_b)$ topologically of finite generation, then $\pi_1(X)$ is topologically of finite presentation.

Since X is desingularizable, one can find a regular scheme X' and a proper birational morphism $f : X' \to X$; since f is a morphism of effective descent for etale coverings, it is enough to prove the theorem under the additional hypothesis that X is regular. Indeed one first deduces that $\pi_1(X)$ is topologically of finite generation; knowing that $\pi_1(X')$ is topologically of finite presentation and the $\pi_1(X''_b)$ topologically of finite generation, one deduces that $\pi_1(X)$ is topologically of finite presentation.

Let now $(U_i)_{i \in I}$ be a finite covering of X by affine open subsets, and let

$$f : \coprod_i U_i \longrightarrow X$$

be the canonical morphism. Since f is a morphism of effective descent for etale coverings, one sees that it is enough to prove the theorem for the U_i; one is therefore reduced to the case where X is affine and smooth.

Assume from now on that X is affine, smooth over k, and $\dim X > 2$. One may then apply the Lefschetz theorem for quasi-projective schemes: if Y is a sufficiently general hyperplane section of X, one has an isomorphism

$$\pi_1(Y) \simeq \pi_1(X).$$

It is therefore enough to prove the theorem for Y, which reduces us to the case $\dim X = 2$.

2) Case where X is a smooth surface over k. By the theorem of strong desingularization of surfaces, one can find a smooth, integral, projective surface S and an open immersion $i : X \to S$ such that the divisor $Y = S - X$ has normal crossings. Apply Corollary 2.2 and keep its notation.

It is enough to prove that the fundamental group of $X' = X \times_S S'$ is topologically of finite presentation. Since X and X' are regular and since

$$\operatorname{codim}\left(X' - \bigcup_{1 \le i \le n} \{x_i\}, X\right) \ge 2,$$

every etale covering of X' induces on $X - \bigcup_{1 \le i \le n} \{x_i\}$ an etale covering which extends uniquely to an etale covering of X (SGA 2 1.11). This shows that one has an isomorphism

$$\pi_1(X') \simeq \pi_1(X).$$

Consider the Cartesian diagram

$$\begin{tikzcd} X' & X'_U \arrow[l] \\ \mathbb{P}^1 \arrow[u, "h"] & U \arrow[l] \arrow[u, "h_U"'] \end{tikzcd}$$

where $h = f|_{X'}$. Let L be the set of prime numbers distinct from p, and let $\bar\eta$ be a geometric point above the generic point η of $\mathbb{P}^1$. Let us verify that the hypotheses of SGA 1 XIII 4.8 are satisfied. The morphism h has a section σ, because f does. Condition a) of SGA 1 XIII 4.7 is proved as follows: the fibres of h being geometrically integral, the morphism h is 0-acyclic and locally 0-acyclic; it remains to show that, for every locally constant sheaf of sets F on X', the pair (F, h) is cohomologically proper in dimension ≤ 0. If ξ is a closed point of $\mathbb{P}^1$ and Z denotes the integral closure of $\operatorname{Spec} \mathcal{O}_{\mathbb{P}^1,\xi}$ in $\bar\eta$, the scheme $X'_Z = X' \times_{\mathbb{P}^1} Z$ satisfies condition (S_2) at the closed points of X' and is regular at every other point, hence is normal. Therefore an etale covering of X'_Z whose restriction to $X'_{\bar\eta}$ is connected is necessarily connected. This proves condition a). Since X'_U is the complement in S'_U of a normal-crossings divisor relative to U, h_U is locally 1-aspherical for L (SGA 4 XV 2.1), and, for every finite constant sheaf F of L-groups on X'_U, the pair (F, h_U) is cohomologically proper in dimension ≤ 1 (SGA 1 XIII 2.4); this is condition b). Since $\mathbb{P}^1$ is normal and $T = \mathbb{P}^1 - U$ has codimension 1, one has

$$\mathrm{prof}_T^{\mathrm{et}}(S) \geq 2,$$

which gives condition c). The condition

$$\mathrm{prof}_x^{\mathrm{top}}(X') \geq 3$$

holds at every closed point of X' by purity, and gives condition e). Finally, for every point $t \in \mathbb{P}^1$, h is smooth at $\sigma(t)$, hence is locally 1-aspherical for L at $\sigma(t)$, giving condition d).

By SGA 1 XIII 4.7, if a is a geometric point of $X'_{\bar{\eta}}$, one has an exact sequence

$$1 \longrightarrow \pi_1^{(p')}(X'_{\bar{\eta}}, a) \longrightarrow \pi'_1(X'_U, a) \longrightarrow \pi_1(U, a) \longrightarrow 1,$$

and the choice of the section σ defines a splitting of this exact sequence. Let K denote the image of $\pi_1(U, a)$ in

$$\mathrm{Aut}(\pi_1^{(p')}(X'_{\bar{\eta}}, a)).$$

The group of coinvariants under K,

$$\pi_1^{(p')}(X'_{\bar{\eta}}, a)_K,$$

is the quotient of $\pi_1^{(p')}(X'_{\bar{\eta}}, a)$ by the closed invariant subgroup generated by the elements of the form $x^{-1}q(x)$, where $x \in \pi_1^{(p')}(X'_{\bar{\eta}}, a)$ and $q \in K$. By loc. cit. 4.7.4, one has an isomorphism

$$\pi_1^{(p')}(X', a) \simeq \pi_1^{(p')}(X'_{\bar{\eta}}, a)_K.$$

Let

$$\mathbb{P}^1 - U = \coprod_{1 \leq i \leq n} \{t_i\},$$

and, for each i, let $\bar{T}_i$ be the strict localization of $\mathbb{P}^1$ at a geometric point $\bar{t}_i$ over t_i, and let s_i be the generic point of $\bar{T}_i$. An etale covering of U whose restriction to each s_i is trivial extends to an etale covering of $\mathbb{P}^1$, hence is trivial. Thus the assertion above amounts to saying that $\pi_1(U, a)$ is the closed invariant subgroup generated by the images of the Galois groups $\mathrm{Gal}(\bar{k}(s_i), k(s_i))$ in $\pi_1(U, a)$. By I 5.3, the image K_i of $\mathrm{Gal}(\bar{k}(s_i), k(s_i))$ in $\mathrm{Aut}(\pi_1^{(p')}(X'_{\bar{\eta}}, a))$ is finite. Since K is the smallest closed invariant subgroup containing all the K_i, and since $\pi_1^{(p')}(X'_{\bar{\eta}}, a)$ is topologically of finite presentation (SGA 1 XIII 3.2), this proves that $\pi_1^{(p')}(X'_{\bar{\eta}}, a)_K$, and therefore also $\pi_1^{(p')}(X)$, to which it is isomorphic, is topologically of finite presentation.

Remark 2.3.2. Resolution of singularities is not necessary for proving Theorem 2.3.1 in the case where X is the open complement of a normal-crossings divisor in a projective scheme smooth over k.

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SGA 7-I, Exposé VI

Formal Deformation Theory

by D. S. Rim

1. Formal existence theorem

We recall the notion of cofibered category (SGA 1, IV).

Let C be a fixed category. A cofibered category A over C is, by definition, a category A together with a functor

$$P : A \longrightarrow C$$

such that:

Ax. 1. For any map f in C and any object a in A with $P(a)$ equal to the source of f, there exists an f-morphism

$$\alpha : a \longrightarrow b$$

which is cocartesian, i.e. for any f-morphism $\alpha' : a \to b'$ there exists a unique T-morphism $\tau : b \to b'$ in A such that

$$\alpha' = \tau\alpha,$$

where T is the target of f.

Ax. 2. A composite of cocartesian morphisms in A is again cocartesian.

The following properties of a cofibered category $P : A \to C$ follow immediately from the definitions.

Cancellation. Let $\alpha, \beta : a \to a'$ be two cocartesian maps such that $P(\alpha) = P(\beta)$. If there exists a cocartesian map $\gamma : b \to a$ such that $\alpha\gamma = \beta\gamma$, then $\alpha = \beta$.

Factorization. Given cocartesian maps $\alpha : a \to a'$ and $\beta : a \to a''$, there exists a cocartesian map $\gamma : a' \to a''$ such that $\gamma\alpha = \beta$ if and only if there exists $f : P(a') \to P(a'')$ in C such that $fP(\alpha) = P(\beta)$. Furthermore γ is unique.

Let A be a cofibered category over C. For each object S in C we denote by $A(S)$ the subcategory of A whose objects are the objects a in A such that $P(a) = S$, and

$$\mathrm{Hom}_{A(S)}(a, a') = \{\alpha \in \mathrm{Hom}_A(a, a') \mid P(\alpha) = \mathrm{id}_S\}.$$

A cofibered groupoid over C is a cofibered category A over C such that $A(S)$ is a groupoid for every object S in C. We recall that a groupoid is a category in which every morphism is an isomorphism. Thus, a cofibered groupoid over C is a cofibered category $P : A \to C$ such that every map in A is cocartesian. Furthermore, for every map $\alpha \in \mathrm{Fl}(A)$, α is an isomorphism in A if and only if $P(\alpha)$ is an isomorphism.

Throughout the rest of this section, we fix a complete noetherian local ring Λ with residue field k, and we set $\mathcal{C}_\Lambda$ to be the category of artinian local Λ-algebras with the same residue field k, where the maps are local homomorphisms over Λ. From 1.2 on, a cofibered groupoid A over $\mathcal{C}_\Lambda$ is always assumed to have the property that $A(k) = \{e\}$, where $\{e\}$ denotes the category in which $\mathrm{Hom}_{\{e\}}(a, b)$ consists of one and only one element for any two objects a, b in $\{e\}$. A map $a' \xrightarrow{\alpha} a$ in A will be called simply a $\mathcal{C}_\Lambda$-map if $P(a') = P(a)$ and $P(\alpha) = 1_{P(a)}$. A map $a' \xrightarrow{\alpha} a$ in A will be called surjective, by abuse of language, if $P(\alpha) : P(a') \to P(a)$ is surjective in $\mathcal{C}_\Lambda$.

Example 1.1(a). Let $F : C \to$ (categories) be a functor. Define the category $\int F$ as follows: an object in $\int F$ is a pair (S, a) where $S \in \mathrm{ob}(C)$ and $a \in \mathrm{ob}\, F(S)$, and a map

$$(S, a) \longrightarrow (S', a')$$

is a pair (f, u') where $f : S \to S'$ is a map in C and $u' : F(f)a \to a'$ is a map in $F(S')$. Composition of two maps

$$(S, a) \xrightarrow{(f,u')} (S', a') \xrightarrow{(g,u'')} (S'', a'')$$

is given by

$$(gf, u'' \cdot F(g)u') : (S, a) \longrightarrow (S'', a'').$$

With respect to the functor $P : \int F \to C$ given by $(S, a) \mapsto S$, $\int F$ becomes a cofibered category over C. If $F(S)$ is a groupoid for every $S \in \mathrm{ob}(C)$, then $\int F$ is a cofibered groupoid over C. In particular, a functor $F : C \to$ (groups) yields a cofibered group, and a functor $F : C \to$ (Ens) yields a cofibered discrete groupoid. We also note that a natural transformation $\varphi : F \to G$, where $F, G : C \to$ (categories) are functors, induces a cocartesian functor $\varphi : \int F \to \int G$ over C.

Example 1.1(b). Let X, Y be schemes over Λ together with a fixed isomorphism $X \otimes_\Lambda k \xrightarrow{\sim} Y \otimes_\Lambda k$. Then the functor

$$\mathrm{Isom}_{X,Y} : \mathcal{C}_\Lambda \longrightarrow \text{(groupoids)}$$

given by

$$\mathrm{Isom}_{X,Y}(S) = \{\text{isomorphisms } X \otimes_\Lambda S \xrightarrow{\sim} Y \otimes_\Lambda S \text{ inducing the fixed one over } k\}$$

yields a cofibered groupoid $P : \int \mathrm{Isom}_{X,Y} \to \mathcal{C}_\Lambda$.

Example 1.1(c). Let X_0 be a fixed scheme over k, and let M_{X_0} be the category consisting of deformations of X_0 over $\mathcal{C}_\Lambda$ (see §4). Thus an object in M_{X_0} is a pair (R, X) where R is in $\mathcal{C}_\Lambda$ and X is a deformation of X_0 to R. Then the functor

$$P : M_{X_0} \longrightarrow \mathcal{C}_\Lambda, \qquad (R, X) \longmapsto R$$

defines a cofibered groupoid over $\mathcal{C}_\Lambda$.

Example 1.1(d). For any groupoid X we denote by $[X]$ the discrete groupoid consisting of the isomorphism classes of objects in X. If $P : F \to C$ is a cofibered groupoid, we obtain a functor $[F] : C \to$ (Ens) given by $[F](S) = [F(S)]$, and in turn a cofibered discrete groupoid, which is, by abuse of notation, denoted by $[F]$.

Definition 1.2. A cofibered groupoid A over $\mathcal{C}_\Lambda$ is called semi-homogeneous if the following properties hold.

(1) Every diagram of solid arrows in A where $a' \to a$ is surjective can be completed into a commutative diagram including the dotted arrow:

$$\begin{tikzcd} a'' \arrow[r, dashed] \arrow[d] & a' \arrow[d] \\ a_1 \arrow[r] & a. \end{tikzcd}$$

(2) Let

$$R \times_k k[\varepsilon] \rightrightarrows R$$

be the projection maps, where $k[\varepsilon]$ is the ring of dual numbers over k, and let

$$a' \rightrightarrows a$$

be a pair of π_1-, π_2-maps in A. Then there exists a $\mathcal{C}_\Lambda$-map $\gamma : a' \to a''$ with $\alpha = \beta\gamma$ if and only if $(\pi_1)_*(a') = (\pi_2)_*(a'')$.

Remark 1.3(a). A reformulation of the notion in terms of 2-categories will be given later in 2.6(b). We note that, if A is semi-homogeneous, then so is $[A]$; in particular 1.2(2) entails that the canonical map

$$[A(R \times_k k[\varepsilon])] \longrightarrow [A(R)] \times [A(k[\varepsilon])]$$

is bijective.

Remark 1.3(b). Consider a commutative diagram in a cofibered category A over $\mathcal{C}_\Lambda$:

$$\begin{tikzcd} & a' \arrow[dl] \arrow[dr] & \\ a_1 \arrow[dr] & & a_2 \arrow[dl] \\ & a & \end{tikzcd}$$

Set $R = P(a)$, $R' = P(a')$, and $R_i = P(a_i)$ for $i = 1, 2$. If we set a'' to be the direct image under the map

$$(P(\nu_1), P(\nu_2)) : R' \longrightarrow R_1 \times_R R_2,$$

we obtain a commutative diagram. Thus 1.2(1) can be replaced by:

(1)' Every diagram in A above $R_1 \leftarrow R' \rightarrow R_2$ in $\mathcal{C}_\Lambda$, where

$$R' \longrightarrow R_1 \times_R R_2$$

is surjective, can be completed to a commutative diagram, where the maps μ_i $(i = 1, 2)$ are projections.

Remark 1.3(c). If A is a semi-homogeneous cofibered groupoid over $\mathcal{C}_\Lambda$, then $[A(k[\varepsilon])]$ is canonically provided with a vector-space structure over k, where $k[\varepsilon]$ is the ring of dual numbers over k. Indeed, let V be the category of finite-dimensional vector spaces over k. Then any functor

$$H : V \longrightarrow (\text{Ens})$$

preserving the product is canonically factored through the forgetful functor $V \to (\text{Ens})$. On the other hand, the functor

$$G : V \longrightarrow \mathcal{C}_\Lambda, \qquad W \longmapsto k[W] = k \oplus W, \quad w^2 = 0,$$

transforms products in V into fibre products over k in $\mathcal{C}_\Lambda$. If A is a semi-homogeneous cofibered groupoid over $\mathcal{C}_\Lambda$, then

$$[A(k[V] \times_k k[W])] \longrightarrow [A(k[V])] \times [A(k[W])]$$

is bijective by 1.3(a), and the functor $V \mapsto [A(k[V])]$ commutes with products, hence the result.

Definition 1.4. (1) Let $R \in \mathrm{ob}(\mathcal{C}_\Lambda)$. A small extension of R in $\mathcal{C}_\Lambda$ is a surjective map $R' \to R$ in $\mathcal{C}_\Lambda$ such that $(\mathrm{Ker}\, f)^2 = 0$ and $\mathrm{Ker}\, f \simeq k$. In other words, $R' \xrightarrow{f} R$ in $\mathcal{C}_\Lambda$ is a small extension of R if

$$0 \longrightarrow k \xrightarrow{t} R' \xrightarrow{f} R \longrightarrow 0$$

is exact for some $t \in R'$ with $t^2 = 0$.

(2) Let A be a cofibered groupoid over $\mathcal{C}_\Lambda$, and let a be an object in A. A small prolongation of a is an arrow $a' \to a$ in A such that $P(a') \to P(a)$ is a small extension in $\mathcal{C}_\Lambda$. An arrow $a' \to a$

in A will be called versal for small prolongations of a if, for every small prolongation $a_1 \to a$ in A, there exists a commutative diagram

$$\begin{tikzcd} a' \arrow[rr] \arrow[dr] & & a_1 \arrow[dl] \\ & a & \end{tikzcd}$$

Proposition 1.5. Let A be a semi-homogeneous cofibered groupoid over $\mathcal{C}_\Lambda$.

(1) Let $R \times_k k[\varepsilon] \rightrightarrows R$ be the projection maps in $\mathcal{C}_\Lambda$, and let $a' \xrightarrow{\alpha} a$ be a map. Then there exists a map $a \xrightarrow{\beta} a'$ such that $\alpha\beta = 1_a$ if and only if there exists a map $a \to (\pi_2)_* a'$.

(2) Let $R' \xrightarrow{f} R$ be a small extension in $\mathcal{C}_\Lambda$, and consider a commutative diagram

$$\begin{tikzcd} R' \times_R R' \arrow[r] \arrow[d] & R' \arrow[d, "f"] \\ R' \arrow[r, "f"'] & R \end{tikzcd} \qquad \text{above} \qquad \begin{tikzcd} a'' \arrow[r] \arrow[d] & a' \arrow[d] \\ a' \arrow[r] & a. \end{tikzcd}$$

Define the canonical map

$$\mu : R' \times_R R' \longrightarrow k[\operatorname{Ker} f], \qquad \mu(r_1, r_2) = r_1(0) + (r_2 - r_1).$$

If there exists a map $a' \to (\mu)_*(a'')$, then there exists a map $\gamma : a' \to a$ such that $\alpha = a\gamma$.

Proof. (1) The "only if" part is clear. Now assume that there exists a map $\varphi : a \to (\pi_2)_* a'$. If we set

$$h : R \longrightarrow R \times_k k[\varepsilon], \qquad h = 1 \times P(\varphi),$$

then the composite map $R \xrightarrow{h} R \times_k k[\varepsilon] \xrightarrow{\pi_1} R$ is the identity, and therefore we get maps $a \xrightarrow{\gamma} a'$ such that $\pi_1\gamma = 1_a$, where $P(\gamma) = h$ and $P(a') = \pi_1$. On the other hand, $\pi_2 h = P(\varphi)$ and hence there exists a π_2-map $h^* a \to (\pi_2)_* a'$. This means that $(\pi_1)^* h^* a = (\pi_2)^* \varphi$, and therefore by 1.2(2) we have a $\mathcal{C}_\Lambda$-map $u : h^* a \to a'$ such that $\alpha u = a_1$. Define $\beta : a \to a'$ by $\beta = u\gamma$. Then $\alpha\beta = \alpha u \gamma = a_1 \gamma = 1_a$.

(2) The map $1 \times f : R' \times_R R' \to R' \times_k k[\operatorname{Ker} f]$ is an isomorphism compatible with the projections on the first factor, and $\pi_2(1 \times f) = \mu$, where $\pi_2 : R' \times_k k[\operatorname{Ker} f] \to k[\operatorname{Ker} f]$ is the projection map. We note that $k[\operatorname{Ker} f] \simeq k[\varepsilon]$ since $R' \to R$ is a small extension. Therefore, if there exists a map $a' \to (\mu)_* a'' = (\pi_2)_*(1 \times f)_* a''$, then by (1) above there exists a map $\gamma' : a' \to a''$ such that $\alpha'\gamma' = 1_{a'}$. Define $\gamma : a' \to a$ by $\gamma = \beta\gamma'$. We then have $\alpha\gamma = a$.

Corollary 1.6. Let A be a semi-homogeneous cofibered groupoid over $\mathcal{C}_\Lambda$ and consider a diagram

$$\begin{tikzcd} a' \arrow[dr, "\alpha"'] & & a_1 \arrow[dl, "\alpha_1"] \\ & a & \end{tikzcd}$$

in A, where a_1 is a small prolongation. Assume that a is versal for small prolongations of e. Then there exists a map $a' \xrightarrow{\rho} a_1$ such that $\alpha = \alpha_1 \rho$ in A if and only if there exists a map $f : P(a') \to P(a_1)$ such that $P(\alpha) = P(\alpha_1) \cdot f$ in $\mathcal{C}_\Lambda$.

Proof. Assume that $P(\alpha) = P(\alpha_1) \cdot f$, where $f : P(a') \to P(a_1)$. Replacing a' by its direct image under f, we may and shall assume that $P(\alpha) = P(\alpha_1)$. Set $R' = P(a_1)$ and $R = P(a)$. Since A is semi-homogeneous, we get a commutative diagram

$$\begin{tikzcd} & a'' \arrow[dl, "\beta"'] \arrow[dr, "\beta_1"] & \\ a' \arrow[dr, "\alpha"'] & & a_1 \arrow[dl, "\alpha_1"] \\ & a & \end{tikzcd} \qquad \begin{tikzcd} & R' \times_R R' \arrow[dl] \arrow[dr] & \\ R' \arrow[dr] & & R' \arrow[dl] \\ & R & \end{tikzcd}$$

above the corresponding diagram of rings. Since a is versal for small prolongations of e, so is a', and therefore our statement follows from 1.5(2).

Definition 1.7. For each object R in $\mathcal{C}_\Lambda$ we put

$$\bar{\Omega}_R = M/(\mathfrak{m}_\Lambda R + M^2),$$

where $\mathfrak{m}_\Lambda$ and M are the maximal ideals of Λ and R respectively. It is a finite-dimensional vector space over k, and $R \mapsto \bar{\Omega}_R$ defines a functor

$$\bar{\Omega} : \mathcal{C}_\Lambda \longrightarrow V,$$

where V is the category of finite-dimensional vector spaces over k. If A is a cofibered category over $\mathcal{C}_\Lambda$, then the composite functor

$$A \xrightarrow{P} \mathcal{C}_\Lambda \xrightarrow{\bar{\Omega}} V$$

is, by abuse of notation, denoted by $\bar{\Omega}$ again.

Proposition 1.8. Let A be a semi-homogeneous cofibered groupoid over $\mathcal{C}_\Lambda$. Then, given an object a in A versal for small prolongations of e, there exists $a' \to a$ such that:

(i) a' is versal for small prolongations of a;

(ii) $P(a') \to P(a)$ is surjective, and the kernel is annihilated by the maximal ideal of $P(a)$;

(iii) $\bar{\Omega}_{a'} \to \bar{\Omega}_a$ is an isomorphism.

Proof. Set $R = P(a)$ and consider an exact sequence

$$0 \longrightarrow J \longrightarrow S \longrightarrow R \longrightarrow 0$$

where S is a formal power-series ring over Λ in a finite number of variables, and $J \subset M^2$, where M is the maximal ideal of S. Let Σ be the set of ideals J_1 in S such that (i) $J \supset J_1 \supset MJ$ and (ii) there exists a π_1-morphism $a_1 \to a$ in A, where $\pi_1 : S/J_1 \to R$ is the canonical map. The property 1.2(1) of A, together with the fact that J/MJ is a finite-dimensional vector space over k, entails that there exists a smallest ideal in the family Σ. Let it be J' and choose a π'-morphism $a' \to a$, where $\pi' : S/J' \to R$ is the canonical map. Note that $\bar{\Omega}_{a'} \to \bar{\Omega}_a$ is an isomorphism since $J' \subset J \subset M^2$. We claim that $a' \to a$ is versal for small prolongations of a. Indeed, let $a_1 \to a$ be a small prolongation. If we put $R_1 = P(a_1)$, we obtain an exact commutative diagram

$$\begin{tikzcd} 0 \arrow[r] & J/MJ \arrow[r] \arrow[d, "h"'] & S/MJ \arrow[r] \arrow[d, "h_1"] & R \arrow[r] \arrow[d, equal] & 0 \\ 0 \arrow[r] & k \arrow[r] & R_1 \arrow[r, "\pi_1"'] & R \arrow[r] & 0. \end{tikzcd}$$

If $h = 0$, then $\pi_1 : R_1 \to R$ admits a cross-section so that h_1 induces $f : S/J' \to R_1$. If $h \neq 0$, then $h_1 : S/MJ \to R_1$ is surjective and hence h_1 induces $f : S/J' \to R_1$ by the definition of the ideal J'. Therefore in either case we obtain a commutative diagram

$$\begin{tikzcd} S/J' \arrow[r, "f"] \arrow[dr, "\pi'"'] & R_1 \arrow[d, "\pi_1"] \\ & R. \end{tikzcd}$$

Since a is versal for small prolongations of e, it follows from 1.6 that we obtain a commutative diagram

$$\begin{tikzcd} a' \arrow[r, "\rho"] \arrow[dr] & a_1 \arrow[d] \\ & a. \end{tikzcd}$$

Let $P : A \to \mathcal{C}_\Lambda$ be a cofibered category (in groupoids). It is then clear that we obtain a cofibered category

$$\mathrm{pro}(P) : \mathrm{pro}(A) \longrightarrow \mathrm{pro}(\mathcal{C}_\Lambda)$$

(in groupoids). For the definition of the category of pro-objects $\mathrm{pro}(A)$, see A. Grothendieck, Séminaire Bourbaki 195.

$$\begin{tikzcd} A \arrow[r] \arrow[d, "P"'] & \mathrm{pro}(A) \arrow[d, "\mathrm{pro}(P)"] \\ \mathcal{C}_\Lambda \arrow[r] & \mathrm{pro}(\mathcal{C}_\Lambda). \end{tikzcd}$$

Now let $\widehat{\mathcal{C}_\Lambda}$ be the full subcategory of $\mathrm{pro}(\mathcal{C}_\Lambda)$ in which

$$\mathrm{ob}(\widehat{\mathcal{C}_\Lambda}) = \{(R_i)_{i \in \mathbb{N}},\ R_i \in \mathrm{ob}(\mathcal{C}_\Lambda),\ \varphi_i : R_i \to R_{i-1} \text{ is surjective for all } i,$$

and induces isomorphisms

$$\mathfrak{m}_i/(\mathfrak{m}_\Lambda + \mathfrak{m}_i^2) \xrightarrow{\sim} \mathfrak{m}_{i-1}/(\mathfrak{m}_\Lambda + \mathfrak{m}_{i-1}^2)$$

for all sufficiently large $i\}$. Here $\mathbb{N}$ stands for the ordered category of natural numbers, and $\mathfrak{m}_i$ is the maximal ideal of R_i. We set $\widehat{A}$ to be the full subcategory of $\mathrm{pro}(A)$ in which

$$\mathrm{ob}(\widehat{A}) = \{a \in \mathrm{ob}(\mathrm{pro}(A)) \mid \mathrm{pro}(P)(a) \in \mathrm{ob}(\widehat{\mathcal{C}_\Lambda})\}.$$

We then obtain a cofibered category

$$\widehat{P} : \widehat{A} \longrightarrow \widehat{\mathcal{C}_\Lambda}$$

(in groupoids) with a commutative diagram

$$\begin{tikzcd} A \arrow[r, hook] \arrow[d, "P"'] & \widehat{A} \arrow[d, "\widehat{P}"] \\ \mathcal{C}_\Lambda \arrow[r, hook] & \widehat{\mathcal{C}_\Lambda}. \end{tikzcd}$$

We note that the category $\widehat{\mathcal{C}_\Lambda}$ is canonically equivalent to the category of complete local Λ-algebras of formally finite type over Λ with fixed residue field k. Henceforth we shall make such identification.

Definition 1.9. Let A be a cofibered groupoid over $\mathcal{C}_\Lambda$. An object ξ in $\widehat{A}$ is called a quasi-initial object of $\widehat{A}$ if for every object a in $\widehat{A}$ there exists a map $\xi \to a$ in $\widehat{A}$. A quasi-initial object ξ of $\widehat{A}$ is called minimal if the morphism $\xi : h_R \to [\widehat{A}]$ induces a bijection

$$\xi(k[\varepsilon]) : \mathrm{Hom}_{\Lambda\text{-}\mathrm{alg}}(R, k[\varepsilon]) \longrightarrow [A(k[\varepsilon])],$$

where $R = P(\xi)$. An object ξ in $\widehat{A}$ is called a versal formal object of A if every surjective map $\alpha : a' \to a$ in A induces a surjective map

$$\mathrm{Hom}_{\widehat{A}}(\xi, a') \longrightarrow \mathrm{Hom}_{\widehat{A}}(\xi, a).$$

A versal formal object ξ of A is called minimal if the morphism $\xi : h_R \to [\widehat{A}]$ induces a bijection

$$\xi(k[\varepsilon]) : \mathrm{Hom}_{\Lambda\text{-}\mathrm{alg}}(R, k[\varepsilon]) \longrightarrow [A(k[\varepsilon])],$$

where $R = P(\xi)$.

Proposition 1.10. Let A be a cofibered groupoid over $\mathcal{C}_\Lambda$.

(i) Every (minimally) versal formal object of A is a quasi-initial (minimal) object of $\widehat{A}$.

(ii) If ξ is an object in $\widehat{A}$ such that

$$\xi(k[\varepsilon]) : \mathrm{Hom}_{\Lambda\text{-}\mathrm{alg}}(R, k[\varepsilon]) \longrightarrow [A(k[\varepsilon])]$$

is injective, where $P(\xi) = R$, then every endomorphism of ξ in $\widehat{A}$ is an automorphism. In particular a quasi-initial minimal object of $\widehat{A}$, if it exists, is unique up to non-canonical isomorphism.

Proof. (i) Let ξ be a versal formal object of A. We must show that, for any object a in $\widehat{A}$, we have a map $\xi \to a$. By the definition of $\widehat{A}$, there exists a sequence of surjective maps in A

$$\cdots \longrightarrow a_n \xrightarrow{\alpha_n} a_{n-1} \longrightarrow \cdots \longrightarrow a_1 \xrightarrow{\alpha_1} a_0$$

such that $a = \lim a_n$. Assume that there exist maps $\varphi_i : \xi \to a_i$ such that $\alpha_i \varphi_i = \varphi_{i-1}$ for all $i < n$. Since ξ is a versal formal object of A and $\alpha_n : a_n \to a_{n-1}$ is surjective in A, we obtain a map $\xi \xrightarrow{\varphi_n} a_n$ such that $\alpha_n \varphi_n = \varphi_{n-1}$. If we set $\varphi = \lim \varphi_n$, we have $\varphi : \xi \to a$.

(ii) Let $\xi \xrightarrow{\alpha} \xi$ be a map in $\widehat{A}$. Then, for every $\lambda \in \mathrm{Hom}_{\Lambda\text{-}\mathrm{alg}}(R, k[\varepsilon])$, we have

$$\lambda\xi = \lambda\alpha\xi = (\lambda \cdot P(\alpha))\xi,$$

and hence by the hypothesis on ξ we have $\lambda = \lambda \cdot P(\alpha)$ for all $\lambda \in \mathrm{Hom}_{\Lambda\text{-}\mathrm{alg}}(R, k[\varepsilon])$. Therefore $P(\alpha) : R \to R$ is surjective. However, every surjective endomorphism of a noetherian ring is an automorphism; hence $P(\alpha)$ is an automorphism of R, and therefore α is an automorphism of ξ.

Theorem 1.11 (Schlessinger). Let A be a cofibered groupoid over $\mathcal{C}_\Lambda$. Then A admits a minimally versal formal object if and only if:

(i) A is semi-homogeneous;

(ii) $[A(k[\varepsilon])]$ is a finite-dimensional vector space over k (cf. 1.3(c)).

Proof. Necessity. Let ξ in $\widehat{A}$ be a minimally versal formal object of A, and consider a diagram

$$\begin{tikzcd} a'' \arrow[r, dashed] \arrow[d] & a' \arrow[d] \\ a_1 \arrow[r] & a \end{tikzcd}$$

in A where q is surjective. Since ξ is a quasi-initial object of $\widehat{A}$, there exists a map $\xi \to a_1$. Since q is surjective, the composite map $\beta = \beta_1\rho$ can be lifted to a', i.e. we obtain a commutative diagram

$$\begin{tikzcd}
& \xi \arrow[dl] \arrow[d] \\
a_1 \arrow[r] & a' \arrow[d] \\
& a.
\end{tikzcd}$$

This proves condition 1.2(1).

We now prove condition 1.2(2). Let $S \times_k k[\varepsilon] \rightrightarrows S$ be the projections, and let

$$\begin{tikzcd}
a' \arrow[r, shift left, "u"] \arrow[r, shift right, "v"'] & b'
\end{tikzcd}$$

be the two maps in A. By the definition of ξ, we obtain a commutative diagram

$$\begin{tikzcd}
& \xi \arrow[dl] \arrow[dr] & \\
a' \arrow[rr, shift left, "u"] \arrow[rr, shift right, "v"'] & & b'.
\end{tikzcd}$$

In particular $\pi_1 P(u) = \pi_1 P(v)$. Assume now that $(\pi_2)^* a' = (\pi_2)^* b'$. Then $\pi_2 P(u) = \pi_2 P(v)$ because of the minimality of ξ, and hence

$$P(u) = \pi_1 P(u) \times \pi_2 P(u) = \pi_1 P(v) \times \pi_2 P(v) = P(v).$$

Since A is a cofibered groupoid, we obtain a unique isomorphism $a' \xrightarrow{\sim} b'$ in $A(S \times_k k[\varepsilon])$ such that $\beta u = v$. Then $\alpha u = \alpha v$ entails that $\beta\alpha = \alpha$, since A is a cofibered category.

Sufficiency. Let $e_1, \dots, e_n$ be objects in $A(k[\varepsilon])$ which form a k-basis of the vector space $[A(k[\varepsilon])]$, and choose a_1 in $A(k[\varepsilon] \times_k \cdots \times_k k[\varepsilon])$ such that $(\pi_i)_* a_1 = e_i$. Then

$$a_1(k[\varepsilon]) : \mathrm{Hom}_{\Lambda\text{-}\mathrm{alg}}(R_1, k[\varepsilon]) \longrightarrow [A(k[\varepsilon])]$$

is bijective, where $R_1 = P(a_1)$; in particular $a_1 \to e$ is versal for small prolongations of e. Choose $a_2 \to a_1$ which is versal for small prolongations of a_1, and such that $\bar{\Omega}_{a_2} \to \bar{\Omega}_{a_1}$ is bijective. We then choose $a_3 \to a_2$, versal for small prolongations of a_2, such that $\bar{\Omega}_{a_3} \to \bar{\Omega}_{a_2}$ is bijective, and so on, and set

$$\xi = \varprojlim_n a_n.$$

We claim that ξ is a minimally versal formal object of A. Indeed, since $\bar{\Omega}_{a_n} \to \bar{\Omega}_{a_{n-1}}$ is bijective for all n, $\xi(k[\varepsilon])$ is bijective. Now consider a diagram

$$\begin{tikzcd}
& a' \arrow[d, "q"] \\
\xi \arrow[ur, dashed] \arrow[r, "h"'] & a
\end{tikzcd}$$

where q is surjective in A. We must show that h can be lifted to a'. For this, one may assume, by induction, that a' is a small prolongation of a. Since $P(a)$ is an artinian ring, $h : \xi \to a$ is factored as a composite map $\xi \to a_n \to a$ for some a_n. Since A is semi-homogeneous, we get a diagram

$$\begin{tikzcd}
a'' \arrow[r] \arrow[d, "q'"'] & a' \arrow[d, "q"] \\
a_n \arrow[r] & a
\end{tikzcd}$$

where q' is also a small prolongation. We then obtain a commutative diagram

$$\begin{tikzcd} & a'' \arrow[d] \\ \xi \arrow[r] \arrow[ur, dashed] & a_n, \end{tikzcd}$$

and therefore we are done.

Remark 1.12. Let

$$\cdots \longrightarrow a_n \longrightarrow \cdots \longrightarrow a_2 \longrightarrow a_1 \longrightarrow e$$

be an infinite sequence of arrows in A such that a_i is versal for small prolongations of a_{i-1}, $P(a_i) \to P(a_{i-1})$ is surjective, and $\bar{\Omega}_{a_i} \to \bar{\Omega}_{a_{i-1}}$ is bijective for $i \gg 0$. Then the above proof shows that $\lim a_i$ is a versal formal object of A.

Remark 1.13. Let A be a cofibered groupoid over $\mathcal{C}_\Lambda$. If A has the properties 1.11(i) and (ii), then so does $[A]$, and $\xi \in \mathrm{ob}(\widehat{A})$ is a minimally versal formal object of A if and only if it is so for $[A]$, by 1.10(ii).

Remark 1.14. Let A be a cofibered groupoid over $\mathcal{C}_\Lambda$, admitting a minimally versal formal object ξ. Then an invariant of $R = P(\xi)$, which is a complete local Λ-algebra of formally finite type over Λ, is automatically an invariant of A over $\mathcal{C}_\Lambda$. Therefore, for example, we shall say that A is formally smooth over $\mathcal{C}_\Lambda$ if R is formally smooth over Λ, i.e. a formal power-series algebra over Λ, or we shall say that A is irreducible over $\mathcal{C}_\Lambda$ if $\mathrm{Spec}(R)$ is irreducible, etc. In the next section we introduce the notion of smoothness of a functor $A \to B$ over $\mathcal{C}_\Lambda$, where A, B are cofibered categories in groupoids over $\mathcal{C}_\Lambda$. It turns out (cf. 2.4) that "A is formally smooth over $\mathcal{C}_\Lambda$" is equivalent to "$P : A \to \mathcal{C}_\Lambda$ is a smooth functor", and therefore no confusion arises. We also define the dimension of A over $\mathcal{C}_\Lambda$ by

$$\dim_\Lambda A = \dim_k R.$$

For this we note the following fact. Let $\mathcal{C}_k$ be the category of artinian local k-algebras with residue field k. Then $\mathcal{C}_k \to \mathcal{C}_\Lambda$ is a fully faithful embedding whose left adjoint functor $\mathcal{C}_\Lambda \to \mathcal{C}_k$ is given by

$$R \longmapsto k \otimes_\Lambda R.$$

Let $A|\mathcal{C}_k$ be the pull-back of $P : A \to \mathcal{C}_\Lambda$ under $\mathcal{C}_k \to \mathcal{C}_\Lambda$. Then it is trivial to see that if ξ is a minimally versal formal object for A over $\mathcal{C}_\Lambda$, then $k \otimes_\Lambda \xi$ is a minimally versal formal object for $A|\mathcal{C}_k$ over $\mathcal{C}_k$, where $\xi \to k \otimes_\Lambda \xi$ is a cocartesian arrow above $R \to k \otimes_\Lambda R$. It follows that

$$\dim_\Lambda A = \dim_k(A|\mathcal{C}_k).$$

1.15. In the remainder of this paragraph, we briefly explain how to modify the preceding theory in order to allow residue-field extensions.

Let Λ and K be complete noetherian local rings, with K a Λ-algebra. We assume that:

(a) the natural map from Λ to K is a local homomorphism;

(b) the residue field K' of K is an extension of finite type of the residue field k of Λ.

The most interesting case is when $K = K'$ and K is a finite inseparable extension of k.

We denote by $\widehat{\mathcal{C}}_{\Lambda,K}$ the category whose objects are the complete noetherian local rings R provided with a structure of Λ-algebra and with a Λ-epimorphism $s : R \to K$.

We denote by $\mathcal{C}_{\Lambda,K}$ the subcategory of $\widehat{\mathcal{C}}_{\Lambda,K}$ consisting of those (R,s) such that $\mathrm{Ker}(s)$ is an R-module of finite length. The category $\mathcal{C}_{\Lambda,K}$ can be identified with a full subcategory of $\mathrm{pro}(\mathcal{C}_\Lambda)$. If K is artinian, so is any object in $\mathcal{C}_{\Lambda,K}$.

For any finite-dimensional vector space V over K', we denote by $D_V(K)$ the algebra of dual numbers over K defined by V; its elements are of the form $\kappa + v\varepsilon$ with $\kappa \in K$, $v \in V$ and $\varepsilon^2 = 0$, and

$$(\kappa + v\varepsilon)(\kappa' + v'\varepsilon) = \kappa\kappa' + (\kappa v' + \kappa' v)\varepsilon.$$

The map $e : D_V(K) \to K$, $e(\kappa + v\varepsilon) = \kappa$, turns $D_V(K)$ into an object of $\mathcal{C}_{\Lambda,K}$. The role played before by $k[\varepsilon]$ will be played here by $D_{K'}(K)$.

As explained in 1.8, any cofibered groupoid A over $\mathcal{C}_{\Lambda,K}$ extends in a natural way to a cofibered category $\widehat{A}_K$ over $\widehat{\mathcal{C}}_{\Lambda,K}$. A map in A or in $\widehat{A}_K$ will be called surjective if the corresponding homomorphism of algebras in $\mathcal{C}_{\Lambda,K}$ or in $\widehat{\mathcal{C}}_{\Lambda,K}$ is surjective.

Continuation of 1.15: residue-field extensions

We assume that $A(K) = \{e\}$.

Definition 1.16. A cofibered groupoid A over $\mathcal{C}_{\Lambda,K}$ is called *semi-homogeneous* if the following properties hold.

(I) Every diagram of solid arrows in A

$$\begin{tikzcd} a'' \arrow[r, dashed] \arrow[d, dashed] & a' \arrow[d] \\ a_1 \arrow[r] & a \end{tikzcd}$$

where $a' \to a$ is surjective, can be completed to a commutative diagram including the dotted arrows.

(II) For every R, the natural map

$$[A(R \times_K D_{K'}(K))] \longrightarrow [A(R)] \times [A(D_{K'}(K))]$$

is bijective, where the right side is the corresponding fibre product of isomorphism classes.

If A is semi-homogeneous, so is $[A]$.

1.17. Let us first restrict A to the full subcategory of $\mathcal{C}_{\Lambda,K}$ whose objects are the $D_V(K)$. Let

$$\Omega = \Omega^1_{K/\Lambda} \otimes_K K',$$

and let $d : K \to \Omega$ be the derivation. A morphism $f : D_V(K) \to D_W(K)$ can be written

$$f(k + v\varepsilon) = k + (u(v) + D\,dk)\varepsilon,$$

with $u \in \mathrm{Hom}_{K'}(V, W)$ and $D \in \mathrm{Hom}_{K'}(\Omega, W)$. If f is identified with the pair (u, D), the composition law is

$$(u, D) \circ (u', D') = (uu',\ D + uD').$$

Condition (II) entails the following condition:

$$(\text{II'}) V \longmapsto [A(D_V(K))]$$

commutes with products as a functor from K'-vector spaces to sets.

Lemma 1.18. Assume that condition (II$'$) is satisfied by A. Then:

(1) $[A(D_{K'}(K))]$ carries a unique K'-vector-space structure such that, functorially in V,

$$[A(D_V(K))] \simeq V \otimes_{K'} [A(D_{K'}(K))].$$

(2) There is a linear map

$$\gamma : \mathrm{Hom}_{K'}(\Omega, K') \longrightarrow [A(D_{K'}(K))]$$

such that, for any $D \in \mathrm{Hom}_{K'}(\Omega, K')$, the endomorphism induced by $(1, D) : D_{K'}(K) \to D_{K'}(K)$ is

$$x \longmapsto x + \gamma(D).$$

(3) More generally, for any $(u, D) : D_V(K) \to D_W(K)$, the corresponding map

$$[A(D_V(K))] \simeq V \otimes [A(D_{K'}(K))] \longrightarrow [A(D_W(K))] \simeq W \otimes [A(D_{K'}(K))]$$

is

$$x \longmapsto u(x) + \gamma(D).$$

Proof. Put

$$F(V) = [A(D_V(K))], \qquad E = F(K').$$

Condition (II$'$), together with the fact that K' is a vector-space object in the category of finite-dimensional K'-vector spaces, gives the vector-space structure on $F(K')$ and the functorial identification $F(V) \simeq V \otimes E$.

We write $F(u, D)$ for the map induced by (u, D). Since

$$(u, D) = (1, D) \circ (u, 0),$$

one has

$$F(u, D) = F(D) \circ (u \otimes 1_E).$$

The functoriality of F amounts to

$$F(0) = \mathrm{id}, \qquad F(D) \circ F(D') = F(D + D'), \qquad (u \otimes 1_E) \circ F(D) = F(uD) \circ (u \otimes 1_E).$$

For $V = K'^n$, using the maps $i : V \to V \oplus K'^n$ and $p : V \oplus K'^n \to K'^n$, The commutative diagram used in this reduction is

$$\begin{tikzcd}
V \otimes E \arrow[r] \arrow[d, "F(D)"'] & (V \oplus K'^n) \otimes E \arrow[r] \arrow[d, "F(i,D)"] & K'^n \otimes E \arrow[d, equal] \\
V \otimes E \arrow[r] & (V \oplus K'^n) \otimes E \arrow[r] & K'^n \otimes E.
\end{tikzcd}$$

one sees that $F(D)$ is a translation:

$$F(D)(y) = y + \alpha(D).$$

The identities above imply

$$\alpha(D + D') = \alpha(D) + \alpha(D'), \qquad \alpha(uD) = u\,\alpha(D),$$

and hence $\alpha(D)$ is defined by a unique linear map $\gamma : \mathrm{Hom}(\Omega, K') \to E$. This proves the assertions.

Definition 1.19. Let ξ be a formal object of A.

(1) The object ξ is called *versal* if every surjective map $a' \to a$ in A induces a surjective map

$$\mathrm{Hom}_A(\xi, a') \longrightarrow \mathrm{Hom}_A(\xi, a).$$

(2) A versal formal object ξ, with image (R, ξ), is called *minimal* if, for any two maps $\varphi, \psi : R \to D_V(K)$, whenever $\varphi(\xi)$ and $\psi(\xi)$ are isomorphic there exists a derivation $D : K \to V$ such that

$$(\psi, D) \circ \xi = \xi,$$

or equivalently $\psi = \varphi - D \circ d$.

Theorem 1.20. If A is semi-homogeneous and

$$\dim_{K'}[A(D_{K'}(K))] < \infty,$$

then A admits a minimally versal formal object, and any two such objects are isomorphic.

Proof. With the notation of Lemma 1.18, let H^* be a subspace of $[A(D_{K'}(K))]$ supplementary to the image of γ, and let H be the dual of H^*. Put

$$R_1 = D_H(K).$$

Choose $\xi_1 \in A(R_1)$ whose image in

$$[A(R_1)] = H^* \otimes [A(D_{K'}(K))] = \mathrm{Hom}(H^*, [A(D_{K'}(K))])$$

is the inclusion of H^* in $[A(D_{K'}(K))]$. Then

$$\mathrm{Hom}_{\mathcal{C}_{\Lambda,K}}(D_{K'}(K), D_V(K)) \longrightarrow [A(D_V(K))]$$

is surjective, and the minimality condition of Definition 1.19 is valid for (R_1, ξ_1).

Choose $S \in \widehat{\mathcal{C}}_{\Lambda,K}$, formally smooth over Λ, and an epimorphism

$$\pi : S \longrightarrow R_1$$

such that $\mathrm{Ker}(\pi) \cap \mathfrak{m}_S^2 = 0$. If $\mathfrak{m}$ is the maximal ideal of S, define inductively, for $n \geq 1$,

$$R_n = S/\mathfrak{m}^n, \qquad \xi_n \in A(R_n),$$

by requiring that:

(a) $\mathfrak{m}^n$ be the intersection of all ideals $I \supset \mathfrak{m}^n$ of S such that ξ_n extends to an object of $A(S/I)$;

(b) ξ_n extend ξ_{n-1}.

For

$$R = \varprojlim R_n, \qquad \xi = \varprojlim \xi_n,$$

one verifies that (R, ξ) is a minimally versal formal object. The construction also gives uniqueness up to isomorphism.

2. Pro-representable cofibered groupoids

Let A be a cofibered groupoid over $\mathcal{C}_\Lambda$, admitting a minimally versal formal object ξ. In general, ξ alone does not allow one to recover A, up to $\mathcal{C}_\Lambda$-equivalence.

Many cofibered groupoids which occur in practice satisfy a stronger property than semi-homogeneity. To study them, it is convenient first to rephrase the preceding construction in terms of smooth functors. The notation and terminology of §1 remain in force.

Definition 2.1. Let

$$\varphi : A \longrightarrow B$$

be a functor of cofibered groupoids over $\mathcal{C}_\Lambda$. We say that φ is *smooth* if every surjective map $\beta : b' \to \varphi(a)$ in B can be completed to a commutative diagram

$$\begin{tikzcd} \varphi(a') \arrow[r] \arrow[d] & b' \arrow[d, "\beta"] \\ \varphi(a) \arrow[r, equal] & \varphi(a) \end{tikzcd} \qquad (*)$$

for some $a' \to a$ in A. We say that φ is *minimally smooth* if it is smooth and the induced map

$$[A(k[\varepsilon])] \longrightarrow [B(k[\varepsilon])]$$

is bijective.

Remark 2.2.

(a) The condition that φ be smooth is equivalent to asking that, in the preceding diagram, $\varphi(a') \to b'$ may be chosen to be a $\mathcal{C}_\Lambda$-isomorphism; it is also equivalent to imposing the lifting condition only for small prolongations $b' \to \varphi(a)$.

(b) For every cofibered groupoid A, the natural functor

$$A \longrightarrow [A]$$

is minimally smooth.

(c) If $R \in \widehat{\mathcal{C}}_\Lambda$, put

$$h_R(S) = \mathrm{Hom}_{\Lambda\text{-alg}}(R, S) \qquad (S \in \mathcal{C}_\Lambda).$$

This is a discrete cofibered groupoid over $\mathcal{C}_\Lambda$. A map $R \to S$ in $\widehat{\mathcal{C}}_\Lambda$ induces $h_S \to h_R$; the latter is smooth if and only if S is a formal power-series ring over R.

(d) Given $\xi \in A$ with $P(\xi) = R$, choose, for each $f : R \to S$, a cocartesian image $\xi(f)$. This defines a functor

$$\xi^* : h_R \longrightarrow A.$$

It is smooth precisely when ξ is a versal formal object; it is minimally smooth precisely when ξ is minimally versal.

Proposition 2.3. For functors of cofibered groupoids

$$A \xrightarrow{\varphi} B \xrightarrow{\psi} C$$

over $\mathcal{C}_\Lambda$:

(a) if φ and ψ are smooth, then $\psi\varphi$ is smooth;

(b) if ψ and $\psi\varphi$ are smooth, then φ is smooth;

(c) if $\psi\varphi$ is smooth and φ is surjective on isomorphism classes of objects, then ψ is smooth.

Remark 2.4. If A has a versal formal object ξ, then $\xi^* : h_R \to A$ is smooth, where $R = P(\xi)$. Hence

$$A \text{ is formally smooth over } \mathcal{C}_\Lambda$$

is equivalent to saying that R is a formal power-series algebra over Λ, and also to saying that the structural functor $A \to \mathcal{C}_\Lambda$ is smooth.

Definition 2.5. A cofibered groupoid A over $\mathcal{C}_\Lambda$ is called *homogeneous* if every diagram

$$\begin{tikzcd} a' \arrow[r] \arrow[d] & a \arrow[d] \\ b' \arrow[r] & b \end{tikzcd}$$

where $a' \to a$ is surjective admits a fibre product $a' \times_a b'$ in A such that

$$P(a' \times_a b') = P(a') \times_{P(a)} P(b').$$

Remark 2.6.

(a) Homogeneity implies semi-homogeneity. Indeed, for any solid diagram in A above a diagram in $\mathcal{C}_\Lambda$, the fibre-product condition supplies the missing object and the missing arrows whenever the relevant map in $\mathcal{C}_\Lambda$ is surjective.

(b) The usual fibre product of categories $X \times_Z Y$ has objects (x, y) with $\Phi(x) = \Psi(y)$. The fibre product in the sense of 2-categories has objects $(x, y; u)$, where $u : \Phi(x) \simeq \Psi(y)$ is an isomorphism, and arrows are pairs which make the evident square commute:

$$\begin{tikzcd} \Phi(x) \arrow[r, "u"] \arrow[d] & \Psi(y) \arrow[d] \\ \Phi(x') \arrow[r, "u'"'] & \Psi(y'). \end{tikzcd}$$

Thus the notions of homogeneous and semi-homogeneous groupoids, and of smooth functors, may be expressed by requiring certain functors to such 2-categorical fibre products to be equivalences, or to be essentially surjective on isomorphism classes.

(c) If A is homogeneous, then for finite-dimensional k-vector spaces V, W, the functor

$$A(k[V \oplus W]) \longrightarrow A(k[V]) \times A(k[W])$$

is an equivalence. In particular $\mathrm{Aut}(1_V)$ and $[A(k[\varepsilon])]$ have canonical k-vector-space structures.

(d) For every $R \in \mathcal{C}_\Lambda$, the discrete groupoid h_R is homogeneous. If A is semi-homogeneous, then $[A]$ is semi-homogeneous; homogeneity of A does not, in general, imply homogeneity of $[A]$.

Proposition 2.7. Let A be a homogeneous cofibered groupoid over $\mathcal{C}_\Lambda$. The following conditions are equivalent:

(i) $[A]$ is homogeneous, that is,

$$[A(R' \times_R S)] \longrightarrow [A(R')] \times_{[A(R)]} [A(S)]$$

is bijective for every small extension $R' \to R$.

(ii) For every small prolongation $a' \to a$ in A, the map

$$\mathrm{Aut}_{A(R')}(a') \longrightarrow \mathrm{Aut}_{A(R)}(a)$$

is surjective, where $R' = P(a')$ and $R = P(a)$.

Proof. The implication (i)⇒(ii) is obtained by applying homogeneity of $[A]$ to the two objects $a' \times_a a'$ and $a' \times_a a'_\tau$, where $\tau \in \mathrm{Aut}_A(a)$. The reverse implication follows by comparing two objects over $R' \times_R S$ whose images over R' and S are isomorphic; the assumed lifting of automorphisms permits the comparison isomorphisms to be modified so that the two squares commute simultaneously.

The comparison is represented schematically by the solid-arrow diagram

$$\begin{array}{ccccc} & a_2'' & & R' \times_R S & \\ \swarrow & \downarrow & \searrow & \swarrow & \searrow \\ a_2' & a_1'' & b_1 \xrightarrow{\tau} b_2 & R' & S \\ & \swarrow & \searrow & \searrow & \swarrow \\ & a_1' & a_1 & & R, \end{array}$$

with the left part in A and the right diamond in $\mathcal{C}_\Lambda$.

Let $R' \xrightarrow{f} R$ be a small extension. There is a canonical map

$$\mathrm{Ker}(f) \longrightarrow R' \times_R R', \qquad (r_1, r_2) \longmapsto r_1(0) + (r_2 - r_1),$$

and the corresponding map

$$D_f : \mathrm{Ker}(f) \longrightarrow R' \times_R R'[\mathrm{Ker}(f)]$$

is an isomorphism. If $a_i \to a$ $(i = 1, 2)$ are two f-maps in A, we write

$$\tau(a_1, a_2) \in A(k[\mathrm{Ker}(f)])$$

for the object obtained by transporting $a_1 \times_a a_2$ through this identification. Then

$$a_1 \times_a a_2 \simeq a_1 \times \tau(a_1, a_2),$$

compatibly with the projection onto the first factor.

Lemma 2.8. Let A be homogeneous over $\mathcal{C}_\Lambda$, and let $a_i \to a$ $(i = 1, 2)$ be f-maps in A.

(1) There exists $p : a_1 \to a_2$ in A, with $a_2 p = a_1$, if and only if there exists an arrow $a_1 \to \tau(a_1, a_2)$ in A.

(2) There exists such a p with $P(p) = \mathrm{id}$ if and only if

$$\tau(a_1, a_2) = 0 \quad \text{in } [A(k[\mathrm{Ker}(f)])].$$

Proof. This is the precise form of the preceding fibre-product description. Such a p is equivalent to a section of

$$a_1 \times \tau(a_1, a_2) \longrightarrow a_1,$$

and the condition $P(p) = \mathrm{id}$ is exactly the condition that the corresponding arrow to $k[\mathrm{Ker}(f)]$ factor through the zero section.

Proposition 2.9. Let

$$A \xrightarrow{\varphi} B \xrightarrow{\psi} C$$

be functors of cofibered groupoids over $\mathcal{C}_\Lambda$.

(a) Assume that B is semi-homogeneous, that ψ is homogeneous, and that

$$[A(k[\varepsilon])] \longrightarrow [C(k[\varepsilon])]$$

is injective. Then $\psi\varphi$ is smooth if and only if φ and ψ are both smooth.

(b) If $\varphi : A \to B$ is smooth and A is semi-homogeneous, then B is semi-homogeneous.

Proof. For (a), assume $\psi\varphi$ is smooth. Since smoothness descends as in Proposition 2.3, it remains to lift a small prolongation $b' \to \varphi(a)$. Smoothness of $\psi\varphi$ gives a lift in A after applying ψ, and the semi-homogeneity of B, together with the injectivity on tangent classes, removes the obstruction.

In the proof of smoothness one completes, by dotted arrows, the diagram

$$\begin{array}{ccc} \varphi(a'_{n+1}) & \dashrightarrow & \varphi(a'_n) \\ \downarrow & \searrow & \downarrow \\ \varphi(a_{n+1}) & \longrightarrow & \varphi(a_n) \end{array} \qquad \begin{array}{ccc} b'_{n+1} & \longrightarrow & b'_n \\ \downarrow \beta_{n+1} & & \downarrow \beta_n \\ \varphi(a_{n+1}) & \longrightarrow & \varphi(a_n). \end{array}$$

After using the semi-homogeneity of A, the corresponding diagram in B is a fibre product; this gives the required lift.

Conversely, if φ and ψ are smooth, their composite is smooth. Part (b) is obtained by applying the lifting condition defining smoothness to the semi-homogeneous diagrams in A.

Corollary 2.10.

(a) Let $\varphi : A \to B$ be a smooth functor of cofibered groupoids over $\mathcal{C}_\Lambda$. Let ξ be a versal formal object of A, and let η be a minimally versal formal object of B. If B is homogeneous, then for any map

$$\beta : \eta \longrightarrow \varphi(\xi)$$

in $\widehat{B}$, the ring $P(\xi)$ is a formal power-series ring over $P(\eta)$ via

$$P(\beta) : P(\eta) \longrightarrow P(\varphi(\xi)) = P(\xi).$$

(b) Let A be a homogeneous cofibered groupoid over $\mathcal{C}_\Lambda$, and let ξ, η be versal formal objects of A. If η is minimal, then for any map

$$\alpha : \eta \longrightarrow \xi$$

in $\widehat{A}$, the ring $P(\xi)$ is a formal power-series ring over $P(\eta)$ through $P(\alpha)$.

Proof. A map $\beta : \eta \to \varphi(\xi)$ induces the commutative square

$$\begin{tikzcd} h_{P(\xi)} \arrow[r, "\widetilde{\beta}"] \arrow[d, "\widetilde{\xi}"'] & h_{P(\eta)} \arrow[d, "\widetilde{\eta}"] \\ A \arrow[r, "\varphi"'] & B. \end{tikzcd}$$

Since $\widetilde{\xi}$ and φ are smooth, $\widetilde{\eta}\,\widetilde{\beta} = \varphi\widetilde{\xi}$ is smooth. Applying Proposition 2.9 and the minimality of η shows that $\widetilde{\beta}$ is smooth; equivalently $P(\xi)$ is a formal power-series ring over $P(\eta)$. Part (b) follows from (a) by taking $A = B$.

We now consider representability of a cofibered groupoid. Let C be a category. Each object $R \in C$ defines the functor

$$h_R : C \longrightarrow (\mathrm{Ens}), \qquad h_R(S) = \mathrm{Hom}_C(R, S),$$

and the Yoneda functor $h : C^\circ \to \mathrm{Hom}(C, \mathrm{Ens})$ is fully faithful.

A *cocategory* in C is a diagram

$$\begin{tikzcd} h_{R_1} \times_{h_{R_0}} h_{R_1} \arrow[r, "c"] & h_{R_1} \arrow[r, shift left, "s_1"] \arrow[r, shift right, "s_2"'] & h_{R_0} \arrow[l, bend left=60, "\varepsilon"] \end{tikzcd}$$

with $s_i\varepsilon = \mathrm{id}$ for $i = 1, 2$, such that c is associative and compatible with the s_i. Equivalently, the two composites

$$h_{R_1} \xrightarrow{\varepsilon} h_{R_0} \times_{h_{R_0}} h_{R_1} \xrightarrow{\varepsilon\times 1} h_{R_1} \times_{h_{R_0}} h_{R_1} \xrightarrow{c} h_{R_1}$$

and

$$h_{R_1} \xrightarrow{\varepsilon} h_{R_1} \times_{h_{R_0}} h_{R_0} \xrightarrow{1\times\varepsilon} h_{R_1} \times_{h_{R_0}} h_{R_1} \xrightarrow{c} h_{R_1}$$

are both identities.

If $(R_0, R_1, s_1, s_2, \varepsilon, c)$ is a cocategory in C, then for every $S \in C$ one obtains a category $h_{(R_0,R_1,\dots)}(S)$ by putting

$$\mathrm{Ob}\, h_{(R_0,R_1,\dots)}(S) = \mathrm{Hom}_C(R_0, S), \qquad \mathrm{Fl}\, h_{(R_0,R_1,\dots)}(S) = \mathrm{Hom}_C(R_1, S),$$

with composition induced by

$$c : h_{R_1} \times_{h_{R_0}} h_{R_1} \longrightarrow h_{R_1}.$$

This defines a functor

$$h_{(R_0,R_1,\dots)} : C \longrightarrow (\text{categories}),$$

hence a cofibered category over C. A cogroupoid is a cocategory for which each $h_{(R_0,R_1,\dots)}(S)$ is a groupoid.

In practice, cocategories and cogroupoids arise as follows. Let $P : F \to C$ be a cofibered category and let ξ be a fixed object of F. Put $R_0 = P(\xi)$, let (R_0, C) be the category of arrows in C with source R_0, and choose a cocartesian section $\xi^{\#}$ of $F' \to (R_0, C)$ with $\xi^{\#}(\mathrm{id}_{R_0}) = \xi$. The functor

$$\mathrm{Hom}_\xi : (R_0 \amalg R_0, C) \longrightarrow (\mathrm{Ens})$$

is given by

$$X = (f_1, f_2) \longmapsto \mathrm{Hom}_{F(S)}(\xi^{\#}(f_2), \xi^{\#}(f_1)),$$

where S is the target of f_1 and f_2. If this functor is represented by

$$X_0 = (R_0 \overset{s_1}{\underset{s_2}{\rightrightarrows}} R_1),$$

then R_1, together with the evident source, target, identity, and composition maps, defines a cocategory $(R_0, R_1, \ldots)$. The resulting functor

$$h_{(R_0,R_1,\ldots)} \longrightarrow F$$

is a C-equivalence if and only if ξ is quasi-initial in F over C.

Definition and Proposition 2.11. A cogroupoid $(R_0, R_1, \ldots)$ in $\widehat{\mathcal{C}}_\Lambda$ is called *smooth* if either of the following equivalent conditions holds:

(i) R_1 is a formal power-series ring over R_0 via s_1, and also via s_2;

(ii) the functor

$$h_{R_0} \longrightarrow h_{(R_0,R_1,\ldots)}$$

is smooth over $\mathcal{C}_\Lambda$.

It is called *minimally smooth*, or *normalized smooth*, if it is smooth and $h_{(R_0,R_1,\ldots)}(k[\varepsilon])$ is totally disconnected, i.e.

$$s_1(k[\varepsilon]) = s_2(k[\varepsilon]).$$

Proof. The smoothness of $\varepsilon : h_{R_0} \to h_{(R_0,R_1,\ldots)}$ means precisely that, for every surjective map $(S', f') \to (S, f)$, a lifting of f to S', together with an isomorphism over S, can be lifted after replacing the isomorphism by one over S'. This is exactly the smoothness of $s_1 : h_{R_1} \to h_{R_0}$, and similarly for s_2.

Proposition 2.12. Let $(R_0, R_1, \ldots)$ be a cogroupoid in $\widehat{\mathcal{C}}_\Lambda$.

(a) If $(R_0, R_1, \ldots)$ is smooth, then $h_{(R_0,R_1,\ldots)}$ is homogeneous.

(b) If $s_1(k[\varepsilon]) = s_2(k[\varepsilon])$, then the following statements are equivalent:

   (i) $(R_0, R_1, \ldots)$ is smooth;

   (ii) $h_{(R_0,R_1,\ldots)}$ is homogeneous;

   (iii) $h_{(R_0,R_1,\ldots)}$ is semi-homogeneous.

(c) If $h_{(R_0,R_1,\ldots)}$ is semi-homogeneous, then it is equivalent to $h_{(S_0,S_1,\ldots)}$ for some minimally smooth cogroupoid S_*.

Proof. For (a), consider a diagram

$$\begin{tikzcd}
(S_1, f_1) \arrow[dr, "{(h_1,u_1)}"'] & & (S_2, f_2) \arrow[dl, "{(h_2,u_2)}"] \\
& (S, f) &
\end{tikzcd}$$

in $h_{(R_0,R_1,\ldots)}$, with $h_2 : S_2 \to S$ surjective. Since $s_1 : h_{R_1} \to h_{R_0}$ is smooth, $u_1^{-1}u_2$ lifts to some $v \in h_{R_1}(S_2)$ such that $s_1(v) = f_2$. This defines the object

$$(S_1 \times_S S_2,\ f_1 \times_S s_2(v))$$

and the commutative diagram

$$\begin{tikzcd}
& (S_1 \times_S S_2,\ f_1 \times s_2(v)) \arrow[dl, "\pi_1"'] \arrow[dr, "{(\pi_2,v^{-1})}"] & \\
(S_1, f_1) \arrow[dr, "{(h_1,u_1)}"'] & & (S_2, f_2) \arrow[dl, "{(h_2,u_2)}"] \\
& (S, f) &
\end{tikzcd}$$

which is a fibre product. This proves homogeneity.

For (b), it remains to show (iii)⇒(i). If $h_{(R_0,R_1,\dots)}$ is semi-homogeneous, then $h_{(R_0,R_1,\dots)}$ admits a minimally versal formal object. On the other hand $(R_0, 1_{R_0})$ is a quasi-initial minimal object, and the equality on $k[\varepsilon]$-points makes it minimally versal. Hence

$$h_{R_0} \longrightarrow h_{(R_0,R_1,\dots)}$$

is minimally smooth, which is the required smoothness.

For (c), let (S_0, ξ_0) be a minimally versal formal object of $h_{(R_0,R_1,\dots)}$. For any object $\xi : R_0 \to T$, the functor $\mathrm{Isom}(\xi, \xi)$ on $\mathcal{C}_\Lambda/(T \times T)$ is represented by

$$R_1 \otimes_{R_0 \otimes_\Lambda R_0} (T \otimes_\Lambda T).$$

Passing to the inverse limit, $\mathrm{Isom}(\xi_0, \xi_0)$ is pro-represented by some S_1, and $h_{(R_0,R_1,\dots)}$ is equivalent to $h_{(S_0,S_1,\dots)}$. By (b), S_* is minimally smooth.

Definition 2.13. A cofibered groupoid A over $\mathcal{C}_\Lambda$ is called *pro-representable* (respectively *minimally pro-representable*) if it is $\mathcal{C}_\Lambda$-equivalent to

$$h_{(R_0,R_1,\dots)}$$

for some cogroupoid (respectively normalized cogroupoid) $(R_0, R_1, \dots)$ in $\widehat{\mathcal{C}}_\Lambda$.

Proposition 2.14. If $(R_0, R_1, \dots)$ is a normalized cogroupoid in $\widehat{\mathcal{C}}_\Lambda$, then every homomorphism

$$(\sigma_0, \sigma_1) : (R_0, R_1, \dots) \longrightarrow (R_0, R_1, \dots)$$

which induces a $\mathcal{C}_\Lambda$-equivalence

$$h_{(\sigma_0,\sigma_1)} : h_{(R_0,R_1,\dots)} \longrightarrow h_{(R_0,R_1,\dots)}$$

is an isomorphism.

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2. Pro-representable groupoids, continued

Proof of Proposition 2.14. We keep the notation of the preceding statement. Suppose that a homomorphism (σ_0, σ_1) of a normalized cogroupoid induces a $\mathcal{C}_\Lambda$-equivalence on the associated cofibered groupoids. Applying the equivalence to the normalized base object gives a diagram of the form

$$\begin{tikzcd}
B_{R_0}(x,e) \arrow[r] \arrow[d] & h_{R_0}(x,e) \arrow[r] \arrow[d] & h_{(R_0,R_1,\dots)}(x,e) \arrow[d, "h_{(\sigma_0,\sigma_1)}"] \\
B_{R_0}(x,e) \arrow[r] & h_{R_0}(x,e) \arrow[r] & h_{(R_0,R_1,\dots)}(x,e).
\end{tikzcd}$$

Since, after passage to $k[\varepsilon]$, the relevant groupoid is totally disconnected, the induced tangent map is bijective. Hence the endomorphism $R_0 \to R_0$ is a surjective endomorphism of a complete local noetherian ring, and therefore an automorphism. The defining pull-back formula for R_1 then implies that $R_1 \to R_1$ is an automorphism as well. Thus (σ_0, σ_1) is an isomorphism.

Lemma 2.15. Let A, B be discrete groupoids over $\mathcal{C}_\Lambda$, and let $\varphi : A \to B$ be a minimally smooth functor. If A and B are homogeneous, then φ is a $\mathcal{C}_\Lambda$-equivalence.

Proof. We must prove that $\varphi(S) : A(S) \to B(S)$ is bijective for every S. It is surjective by minimal smoothness, and bijective for $S = k[\varepsilon]$. We argue by induction on the length of S. If $S' \twoheadrightarrow S$ is a small extension, homogeneity gives

$$S' \times_S S' \simeq S' \times_k k[\operatorname{Ker} f],$$

and therefore the commutative diagram

$$\begin{tikzcd} A(S') \times_{A(S)} A(S') \arrow[r] \arrow[d, "\varphi\times\varphi"'] & A(S') \times A(k[\operatorname{Ker} f]) \arrow[d, "\varphi\times\varphi"] \\ B(S') \times_{B(S)} B(S') \arrow[r] & B(S') \times B(k[\operatorname{Ker} f]). \end{tikzcd}$$

Thus $A(S')$ and $B(S')$ are principal homogeneous sets under tangent groups identified by φ. If $\varphi(S)$ is bijective, then $\varphi(S')$ is injective and hence bijective. □

Corollary 2.16. Let A be homogeneous over $\mathcal{C}_\Lambda$, with finite-dimensional tangent space. For every object ξ of A, the functor

$$\operatorname{Isom}_\xi : \mathcal{C}_\Lambda/R \longrightarrow \mathrm{Ens}, \qquad R = P(\xi),$$

is pro-representable.

Proof. The functor Isom_ξ is homogeneous: for a diagram $T_1 \to T \leftarrow T_2$ with one arrow surjective,

$$\operatorname{Isom}_\xi(T_1 \times_T T_2) \simeq \operatorname{Isom}_\xi(T_1) \times_{\operatorname{Isom}_\xi(T)} \operatorname{Isom}_\xi(T_2).$$

The formal existence theorem gives a minimally smooth functor from a pro-representable functor to Isom_ξ; Lemma 2.15 makes it an equivalence. □

Theorem 2.17. A cofibered groupoid A over $\mathcal{C}_\Lambda$ is pro-representable by a normalized smooth cogroupoid if and only if

(1) A is homogeneous;

(2) $[A(k[\varepsilon])]$ and $\operatorname{Aut}(1_{k[\varepsilon]})$ are finite-dimensional vector spaces over k.

Proof. Smooth cogroupoids are homogeneous, and the tangent and automorphism spaces are kernels and cokernels of maps between spaces $\operatorname{Hom}(R_i, k[\varepsilon])$, hence finite-dimensional. Conversely, choose a minimally versal formal object ξ represented by R_0. Corollary 2.16 pro-represents the functor of isomorphisms between the two pull-backs of ξ over $R_0 \widehat{\otimes}_\Lambda R_0$ by an object R_1, carrying a universal isomorphism. Composition of isomorphisms gives a cogroupoid $(R_0, R_1, \ldots)$. The induced functor $h_{(R_0,R_1,\ldots)} \to A$ is minimally smooth, hence, by Lemma 2.15, an equivalence. □

Corollary 2.18. A functor $F : \mathcal{C}_\Lambda \to \mathrm{Ens}$ is pro-representable if and only if it is homogeneous and $\dim_k F(k[\varepsilon]) < \infty$.

Corollary 2.19. Every smooth cogroupoid in $\widehat{\mathcal{C}}_\Lambda$ can be normalized, up to isomorphism.

For a cogroupoid $(R_0, R_1, \ldots)$, let $\overline{R}_1 = R_1/J$, where J is generated by $s_1(r) - s_2(r)$, $r \in R_0$. Then $(R_0, \overline{R}_1, \ldots)$ defines a formal group scheme over R_0.

Corollary 2.20. Let A be homogeneous and pro-represented by a normalized smooth cogroupoid $(R_0, R_1, \ldots)$. Then $(R_0, \overline{R}_1, \ldots)$ and $(k, k\widehat{\otimes}_{R_0}\overline{R}_1, \ldots)$ pro-represent the automorphism functors Aut_ξ and Aut^0. Moreover the following are equivalent:

(i) the formal group scheme is formally smooth over R_0;

(ii) for every surjection $a' \to a$ in A, the map $\mathrm{Aut}(a') \to \mathrm{Aut}(a)$ is surjective;

(iii) $\overline{R}_1$ pro-represents the functor of isomorphism classes over $\mathcal{C}_\Lambda$.

This is the combination of Proposition 2.7 with Lemma 2.15.

Remarks 2.21–2.22. Pulling a pro-representation back to $\mathcal{C}_\Lambda/k$ replaces the cogroupoid by its base change to k, preserving normalization and relative dimension. If A has two smooth pro-representations and one is normalized, then there is a comparison morphism of cogroupoids; the non-normalized base is a formal power-series extension of the normalized one, and the comparison becomes an isomorphism after this smooth base change.

3. The coherent sheaves D^i

We now recall the elementary obstruction theory of commutative algebra extensions in the language of Grothendieck cohomology on a site. Rings are commutative with unit. For an R-algebra A, let $\mathcal{C}_{A|R}$ be the category of arrows $B \to A$ in R-algebras, endowed with the topology whose coverings are surjective arrows. It has final object $A \to A$, initial object $R \to A$, fibre products and coproducts.

A sheaf F of A-modules on $\mathcal{C}_{A|R}$ is a functor such that, for every cover $B' \twoheadrightarrow B$,

$$0 \to F(B) \to F(B') \rightrightarrows F(B' \times_B B')$$

is exact. Additive sheaves preserve coproducts. An A-module M gives the additive sheaf

$$\mathrm{Der}_R(-, M) : B \longmapsto \mathrm{Der}_R(B, M),$$

where M is viewed as a B-module through $B \to A$.

Let $\widetilde{\mathcal{C}}_{A|R}$ and $\widehat{\mathcal{C}}_{A|R}$ denote sheaves and presheaves. The inclusion of sheaves into presheaves has right derived functors $\mathcal{H}^q_{A|R}$, and

$$H^q(A|R, F) = [\mathcal{H}^q_{A|R}(F)](A).$$

An object $B \in \mathcal{C}_{A|R}$ and a ring homomorphism $S \to R$ give the functorial diagrams

$$\begin{array}{ccccccc} \widehat{\mathcal{C}}_{B|R} & \longrightarrow & \widehat{\mathcal{C}}_{A|R} & \qquad & \widehat{\mathcal{C}}_{A|S} & \longrightarrow & \widehat{\mathcal{C}}_{A|R} \\ \uparrow & & \uparrow & & \uparrow & & \uparrow \\ \widetilde{\mathcal{C}}_{B|R} & \longrightarrow & \widetilde{\mathcal{C}}_{A|R} & & \widetilde{\mathcal{C}}_{A|S} & \longrightarrow & \widetilde{\mathcal{C}}_{A|R}. \end{array}$$

For a covering $B' \to B$, the groups $H^q(\{B' \to B\}, F)$ are the cohomology groups of the Čech semi-simplicial module

$$F(B') \rightrightarrows F(B' \times_B B') \rightrightarrows F(B' \times_B B' \times_B B') \rightrightarrows \cdots .$$

For a presheaf F, set

$$\mathcal{H}^q_{A|R}(F)(B) = \varinjlim_{\{B' \to B\}} H^q(\{B' \to B\}, F).$$

There is the usual spectral sequence

$$H^p(A|R, \mathcal{H}^q(F)) \Rightarrow H^{p+q}(A|R, F),$$

and hence, in low degree,

$$\begin{aligned}
&H^1(A|R,F) = \mathcal{H}^1(A|R,F),\\
&\qquad\qquad\qquad 0 \to \mathcal{H}^2(A|R,F) \to H^2(A|R,F) \to H^1(A|R,\mathcal{H}^1(F))\\
&\qquad\qquad\qquad\quad \to \mathcal{H}^3(A|R,F).
\end{aligned}$$

Lemma 3.2. If $P \twoheadrightarrow B$ is a cover by a polynomial R-algebra, then

$$H^n(\{P \to B\}, F) \xrightarrow{\sim} \mathcal{H}^n_{A|R}(F)(B).$$

Polynomial covers dominate all coverings, which proves the lemma.

Corollary 3.3. (a) If A is noetherian and F is of finite type, then $\mathcal{H}^n(F)$ is of finite type. (b) If P is polynomial over R, then $H^n(P|R,F) = 0$ for all $n > 0$. (c) Exactness of sheaves may be checked on polynomial R-algebras. Indeed, the polynomial subsite is cofinal and every surjection in it admits a section.

Lemma 3.5. For a diagram of R-algebras

$$\begin{array}{ccc}
B' & \longrightarrow & C\\
\big\downarrow\!\!\!\!\downarrow & & \big\downarrow\\
B & \longrightarrow & A
\end{array}$$

and an R-algebra S:

(i) if $\mathrm{Tor}_1^R(B,S) = 0$, then

$$(B' \times_B C) \otimes_R S \simeq (B' \otimes_R S) \times_{B \otimes_R S} (C \otimes_R S);$$

(ii) if B' is R-flat and $\mathrm{Tor}_i^R(B,S) = \mathrm{Tor}_i^R(C,S) = 0$ for $0 < i \le n$, then $\mathrm{Tor}_i^R(B' \times_B C, S) = 0$ for $0 < i \le n$.

The proof compares the exact rows obtained from $0 \to J \to B' \to B \to 0$:

$$\begin{array}{ccccccccc}
0 & \longrightarrow & J \otimes_R S & \longrightarrow & (B' \times_B C) \otimes_R S & \longrightarrow & C \otimes_R S & \longrightarrow & 0\\
 & & \Big\| & & \Big\downarrow & & \Big\| & & \\
0 & \longrightarrow & J \otimes_R S & \longrightarrow & (B' \otimes_R S) \times_{B \otimes_R S} (C \otimes_R S) & \longrightarrow & C \otimes_R S & \longrightarrow & 0.
\end{array}$$

Corollary 3.6. If $\mathrm{Tor}_1^R(A,S) = 0$, then

$$H^n(A \otimes_R S|S,G) \simeq H^n(A|R,SG).$$

If S is R-flat, this gives $S\mathcal{H}^n(G) \simeq \mathcal{H}^n(SG)$. Polynomial covers remain Tor-acyclic under the hypotheses of Lemma 3.5.

Proposition 3.7. (a) If $\mathrm{Tor}_i^R(A,S) = 0$ for $0 < i \le n$, then

$$\mathcal{H}^i(A \otimes_R S|S,F) \xrightarrow{\sim} \mathcal{H}^i(A|R,SF), \quad i \le n.$$

(b) If P is polynomial over R, then for an additive sheaf F on $\mathcal{C}_{A \otimes_R P|R}$,

$$H^n(A \otimes_R P|R,F) \simeq H^n(A|R,F), \quad n > 0.$$

(c) For $S \to R \to A$ and additive F on $\mathcal{C}_{A|S}$, there is an exact sequence

$$0 \to H^0(A|R, F_R) \to H^0(A|S, F) \to H^0(R|S, F) \to H^1(A|R, F_R) \to \cdots,$$

where $F_R(X) = \mathrm{Ker}(F(X) \to F(R))$. The proof compares the spectral sequences

$$\begin{tikzcd} H^p(A \otimes_R S|S, \mathcal{H}^q(F)) \arrow[r, Rightarrow] \arrow[d] & H^{p+q}(A \otimes_R S|S, F) \arrow[d] \\ H^p(A|R, \mathcal{H}^q(SF)) \arrow[r, Rightarrow] & H^{p+q}(A|R, SF) \end{tikzcd}$$

and then uses the Leray sequence for the forgetful functor.

Corollary 3.8. If $\mathrm{Tor}_i^R(A, B) = 0$ for $0 < i \le n$, then for additive F

$$H^i(A \otimes_R B|R, F) \simeq H^i(A|R, F) \oplus H^i(B|R, F), \quad i \le n.$$

This follows from the splitting diagram

$$\begin{tikzcd} H^i(A \otimes_R B|A, F_A) \arrow[r] \arrow[d, "\sim"'] & H^i(A \otimes_R B|R, F) \\ H^i(B|R, F_A) \arrow[r] & H^i(B|R, F). \end{tikzcd}$$

If moreover $A \otimes_R A \simeq A$ and $\mathrm{Tor}_i^R(A, A) = 0$, then $H^i(A|R, F) = 0$ for $i > 0$.

Corollary 3.9. For a multiplicative subset $S \subset A$, if F and F_S are additive, then

$$H^i(S^{-1}A|R, F) \simeq H^i(A|R, F).$$

For an A-module M, define

$$D^i(A|R, M) = H^i(A|R, \mathrm{Der}_R(-, M)).$$

Choose $P \twoheadrightarrow A$ polynomial over R, with kernel I.

Lemma 3.10.

$$D^1(A|R, M) = \mathrm{Coker}(\mathrm{Der}_R(P, M) \to \mathrm{Hom}_A(I/I^2, M)),$$
$$D^2(A|R, M) = H^1(A|R, \mathcal{H}^1(\mathrm{Der}_R(-, M))).$$

The proof uses the simplicial algebra $P^{(n)} = P \times_A \cdots \times_A P$ and the exact sequence

$$0 \to \mathrm{Der}_R(A, M) \to \mathrm{Der}_R(P, M) \to \mathrm{Hom}_A(I/I^2, M) \to H^1(\{P \to A\}, \mathrm{Der}_R(-, M)) \to 0.$$

Let $F \twoheadrightarrow I$ be a free P-module mapping onto I, and let $K_\bullet$ be the associated Koszul complex.

Lemma 3.11.

$$D^2(A|R, M) = \mathrm{Coker}(H^1(K_\bullet, M) \to \mathrm{Hom}_A(H_1(K_\bullet), M)).$$

This follows by replacing the cover $P \to A$ by the polynomial algebra $P[F]$; the obstruction term is

$$(F) \cap (F')/((F)(F')) \simeq \mathrm{Tor}_1^P(A, A) = H_1(K_\bullet).$$

Theorem 3.12. The groups D^i satisfy:

(1) a long exact sequence in $D^i(A|R,-)$ for every exact sequence of A-modules;

(2) for $S \to R \to A$, an exact sequence

$$0 \to D^0(A|R,M) \to D^0(A|S,M) \to D^0(R|S,M) \to D^1(A|R,M) \to \cdots;$$

(3) Tor-independent base change $D^i(A \otimes_R S|S,M) \simeq D^i(A|R,M)$;

(4) localization $D^i(S^{-1}A|R,S^{-1}M) \simeq S^{-1}D^i(A|R,M)$;

(5) the product formula $D^i(A \otimes_R B|R,M) \simeq D^i(A|R,M) \oplus D^i(B|R,M)$ under Tor-independence;

(6) compatibility with tensoring by a flat A-module, when A is essentially of finite type over a noetherian R;

(7) finiteness of $D^i(A|R,M)$ for M finite;

(8) the explicit formulas

$$D^0(A|R,M) = \mathrm{Der}_R(A,M),$$
$$D^1(A|R,M) = \mathrm{Coker}(\mathrm{Der}_R(P,M) \to \mathrm{Hom}_A(I/I^2,M)) = \mathrm{Exalcomm}(A|R,M),$$
$$D^2(A|R,M) = \mathrm{Coker}(H^1(K_\bullet,M) \to \mathrm{Hom}_A(H_1(K_\bullet),M)).$$

The extension interpretation of D^1 is given by the push-out diagram

$$\begin{tikzcd}
0 \arrow[r] & I \arrow[r] \arrow[d] & P \arrow[r] \arrow[d] & A \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & M \arrow[r] & A' \arrow[r] & A \arrow[r] & 0.
\end{tikzcd}$$

Two maps $I \to M$ differing by a derivation of P give isomorphic extensions, and every extension arises this way.

Corollary 3.13.

(a) A is formally smooth over R iff $D^1(A|R,M) = 0$ for every M.

(b) If R is noetherian and A finite type over R, then A is a relative complete intersection iff $D^2(A|R,M) = 0$ for every M.

(c) Localization commutes with D^i.

(d) If A is local, essentially of finite type over noetherian R, and $\widehat{A}$ is its completion, then

$$D^i(\widehat{A}|R, \widehat{A} \otimes_A E) \simeq \widehat{A} \otimes_A D^i(A|R,E)$$

for every finite A-module E.

For (d), the final step compares algebra extensions and uses Artin–Rees. The completion comparison is expressed by the exact commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & \widehat{E} \arrow[r] \arrow[d, equal] & B \arrow[r] \arrow[d] & A \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \widehat{E} \arrow[r] & \widehat{B} \arrow[r] & \widehat{A} \arrow[r] & 0,
\end{tikzcd}$$

which proves the needed injectivity and surjectivity for D^1, and hence the assertion.

3. The coherent sheaves D^i, conclusion

Corollary 3.14. Let R be noetherian and let A be an R-algebra of finite type, smooth everywhere except at one closed point $x \in \operatorname{Spec}(A)$. Then there are natural isomorphisms

$$D^1(A \mid R, A) \simeq D^1(A_x \mid R, A_x) \simeq D^1(\widehat{A}_x \mid R, \widehat{A}_x).$$

Indeed $D^1(A \mid R, A)$ is an A-module of finite type whose support is $\{x\}$. Hence localization and completion at x do not change it, and the assertion follows from 3.13.

Remark 3.15. Let R be a topological ring and A a topological R-algebra. For an A-module E set

$$D^i_{\text{top}}(A \mid R, E) = \varinjlim_{\alpha} D^i(A_\alpha \mid R_\alpha, A_\alpha \otimes_A E),$$

where $A_\alpha = A/J_\alpha$, $R_\alpha = R/I_\alpha$, and J_α, I_α range through open ideals of A, respectively R. With this convention, saying that A is a formally smooth topological R-algebra means that

$$D^1_{\text{top}}(A \mid R, E) = 0$$

for every A-module E.

Let now X be a scheme over a ring R, and let E be a quasi-coherent $\mathcal{O}_X$-module. The sheaves

$$D^i_{X|R}(E) = D^i(X \mid R, E)$$

are defined on affine opens by

$$\big[D^i_{X|R}(E)\big](U) = D^i\big(\Gamma(U, \mathcal{O}_X) \mid R, \Gamma(U, E)\big).$$

If R is noetherian and X is of finite type over R, 3.13(a) gives this construction canonically as a quasi-coherent $\mathcal{O}_X$-module. When no ambiguity is possible we write

$$D^i_{X|R} = D^i_{X|R}(\mathcal{O}_X).$$

The preceding affine statements globalize to these sheaves. The following are the properties needed below.

Corollary 3.15. Let R be noetherian and X an R-scheme of finite type.

(a) For every coherent $\mathcal{O}_X$-module E, the sheaf $D^i_{X|R}(E)$ is coherent for all i, and

$$\operatorname{Supp} D^i_{X|R}(E) \subset X^{\text{Sing}} \quad (i > 0),$$

where X^{Sing} is the set of points at which X is not smooth over R.

(b) If X is flat over R, then for every R-algebra S there is the canonical base-change isomorphism

$$D^i_{X_S|S}(E_S) \simeq D^i_{X|R}(E)_S$$

for all i.

(c) If X is a relative complete intersection over R, then, for every surjection of rings $S \twoheadrightarrow R$, there is an exact sequence

$$0 \to D^1_{X|R}(E) \to D^1_{X|S}(E) \to \mathcal{H}om_{\mathcal{O}_X}(I/I^2 \otimes_R \mathcal{O}_X, E) \to 0,$$

where $I = \operatorname{Ker}(S \to R)$. If, moreover, R is local with residue field k and X is R-flat, then

$$0 \to D^1_{X_0|k}(\mathcal{O}_{X_0}) \to D^1_{X|S}(\mathcal{O}_X) \otimes_R k \to D^1(R \mid S, k) \otimes_k \mathcal{O}_{X_0} \to 0,$$

where $X_0 = X \otimes_R k$.

The proof is exactly the passage from the affine statements 3.12 and 3.13 to sheaves. In (c), relative complete-intersection hypotheses give $D^i_{X|R}(E) = 0$ for $i \geq 2$, and the exact sequence of 3.12(2), applied to $S \to R$, becomes the displayed exact sequence after identifying

$$D^1_{R|S}(E) = \mathcal{H}om_{\mathcal{O}_X}(I/I^2 \otimes_R \mathcal{O}_X, E).$$

The final sequence follows by reducing modulo the maximal ideal of R and using the base-change isomorphism.

Remark 3.16. The quasi-coherent sheaves $D^i_{X|R}(E)$ should occur as the $E_2^{0,i}$-terms of a spectral sequence relating the local theory to the global theory. L. Illusie has obtained such a theory by homotopical algebra, generalizing methods of M. Andre and D. Quillen rather than only cohomological algebra. The present approach should also globalize by using Grothendieck cohomology on the category of schemes with a suitable topology extending the affine one.

4. Formal moduli of deformations

The cofibered categories of deformation problems are among the useful examples in which isomorphism classes of objects are often not pro-representable, but nevertheless admit formal versal objects. We return to the notation of paragraph 1. Thus Λ is a fixed complete local noetherian ring with residue field k, and $\mathcal{C}_\Lambda$ is the category of artinian local Λ-algebras with residue field k.

Let X_0 be a fixed scheme over k. An infinitesimal deformation of X_0 is a pair (R, X), where $R \in \mathrm{ob}(\mathcal{C}_\Lambda)$, X is a flat R-scheme, and there is a cartesian diagram

$$\begin{tikzcd}
X_0 \arrow[r, "i_X"] \arrow[d] & X \arrow[d, "\text{flat}"] \\
\operatorname{Spec} k \arrow[r] & \operatorname{Spec} R.
\end{tikzcd}$$

The induced morphism $X_0 \to X \times_{\operatorname{Spec} R} \operatorname{Spec} k$ is required to be an isomorphism. Since R is artinian local, i_X is a closed immersion and a homeomorphism on underlying spaces.

For two deformations (R, X) and (R', X'), an arrow $(R', X') \to (R, X)$ consists of a morphism $\varphi : R' \to R$ in $\mathcal{C}_\Lambda$ and a morphism $\xi : X \to X'$ fitting into the diagram

$$\begin{tikzcd}
X \arrow[d] \arrow[rr, "\xi"] & X_0 \arrow[l, "i_X"'] \arrow[r, "i_{X'}"] \arrow[d] & X' \arrow[d] \\
\operatorname{Spec} R & \operatorname{Spec} k \arrow[l] \arrow[r] & \operatorname{Spec} R'.
\end{tikzcd}$$

This gives a category $\mathcal{M}_{X_0}$ and a functor

$$p : \mathcal{M}_{X_0} \to \mathcal{C}_\Lambda, \qquad (R, X) \mapsto R.$$

If (R, X) is an object and $R \to S$ is a morphism in $\mathcal{C}_\Lambda$, then

$$X_S = X \times_{\operatorname{Spec} R} \operatorname{Spec} S$$

is a deformation of X_0 over S, and the cartesian square

$$\begin{tikzcd}
X_S \arrow[r, "p_1"] \arrow[d, "\text{flat}"'] & X \arrow[d, "\text{flat}"] \\
\operatorname{Spec} S \arrow[r, "\operatorname{Spec}(\varphi)"'] & \operatorname{Spec} R
\end{tikzcd}$$

shows that $\mathcal{M}_{X_0}$ is cofibered over $\mathcal{C}_\Lambda$.

4.1. The following elementary properties will be used repeatedly.

(a) Every arrow over the identity of R is an isomorphism. Thus $\mathcal{M}_{X_0}$ is cofibered in groupoids.

(b) If $X_0 = \operatorname{Spec} A_0$ is affine, then every deformation (R, X) is affine. The fiber $\mathcal{M}_{X_0}(R)$ is equivalent to the category whose objects are flat R-algebras A, together with an R-algebra map $j : A \to A_0$ inducing $A \otimes_R k \simeq A_0$.

(c) The corresponding formal category $\widehat{\mathcal{M}}_{X_0}$ has, over R, formal schemes $\mathfrak{X}$ adic over $\operatorname{Spf} R$, flat modulo every power of the maximal ideal. If X_0 is noetherian, this is the expected category of formal deformations of X_0.

(d) If $U_0 \subset X_0$ is open, restriction defines $\mathcal{M}_{X_0} \to \mathcal{M}_{U_0}$. Likewise a point $x_0 \in X_0$ gives functors to the deformation categories of $\operatorname{Spec} \mathcal{O}_{X_0,x_0}$ and $\operatorname{Spec} \widehat{\mathcal{O}}_{X_0,x_0}$.

Lemma 4.2. Consider a diagram of rings

$$\begin{tikzcd} R' \arrow[r] & R \\ R'' \arrow[u] \arrow[r, two heads] & R_0 \arrow[u] \end{tikzcd}$$

in which the indicated map is surjective. If E and E' are flat modules over the two upper rings and are identified after base change to R_0, then the fiber product module is flat over the fiber product ring. Equivalently, if

$$0 \to I \to R \to R' \to 0$$

is exact, the sequence

$$0 \to I \otimes E \to E \times E' \to E' \to 0$$

is exact and gives the Tor-vanishing needed for flatness.

Corollary 4.3. For a diagram

$$(R, X) \longleftarrow (R_0, X_0) \longrightarrow (R', X')$$

in $\mathcal{M}_{X_0}$, with the ring map $R' \to R_0$ surjective, the amalgamated sum $X \amalg_{X_0} X'$ is flat over $R \times_{R_0} R'$. Hence $\mathcal{M}_{X_0}$ is a homogeneous cofibered category in groupoids over $\mathcal{C}_\Lambda$.

Lemma 4.4. There is a canonical exact sequence

$$0 \to H^1(X_0, D^0_{X_0}) \to [\mathcal{M}_{X_0}(k[\varepsilon])] \to H^0(X_0, D^1_{X_0}).$$

If $X_0 = \operatorname{Spec} A_0$, the middle set is identified with isomorphism classes of k-algebra extensions of A_0 by A_0, hence with $D^1(A_0 \mid k, A_0)$. In general the map is obtained by restricting to affine opens. Its kernel is the set of locally trivial deformations, and the sheaf of germs of automorphisms of the trivial deformation $X_0 \times \operatorname{Spec} k[\varepsilon]$ is $D^0_{X_0}$; the usual Cech argument gives the displayed exact sequence.

Theorem 4.5. Let X_0 be a scheme of finite type over k. Suppose either that X_0 is proper over k, or that X_0 is affine and

$$X_0^{\mathrm{Sing}} = \{x \in X_0 \mid X_0 \text{ is not smooth over } k \text{ at } x\}$$

is finite. Then:

(a) $\mathcal{M}_{X_0}$ admits a formal versal object over $\mathcal{C}_\Lambda$. If $(R, \mathfrak{X})$ is such a formal versal deformation and (S, Y) is an infinitesimal deformation, there is a morphism $R \to S$ and a cartesian diagram

$$\begin{tikzcd} Y \arrow[r] \arrow[d] & \mathfrak{X} \arrow[d] \\ \operatorname{Spf} S \arrow[r] & \operatorname{Spf} R. \end{tikzcd}$$

Moreover $(R, \mathfrak{X})$ pro-represents the functor of isomorphism classes if and only if, for every surjection $R' \to R$ in $\mathcal{C}_\Lambda$ and every deformation (R', X'), the homomorphism

$$\operatorname{Aut}_{R'}(X' \mid X_0) \to \operatorname{Aut}_R(X' \otimes_{R'} R \mid X_0)$$

is surjective.

(b) If X_0 is proper over k, then $\mathcal{M}_{X_0}$ is smoothly pseudo-representable over $\mathcal{C}_\Lambda$.

The proof uses homogeneity from 4.3 and the tangent exact sequence 4.4. In the proper case, the coherent sheaves D^i have finite-dimensional cohomology. In the affine isolated-singularity case, $H^1(X_0, D^0_{X_0}) = 0$ and $H^0(X_0, D^1_{X_0})$ is finite-dimensional because the support is finite. The criterion for pro-representability is the automorphism criterion from 2.7.

Remarks 4.6. (a) If X_0 is affine, a formal versal deformation exists exactly when $D^1_{X_0}$ has finite support; this need not mean that the non-smooth locus itself is finite. (b) From now on, "deformation of X_0" means an object of $\mathcal{M}_{X_0}$; when ambiguity is possible, an object in the fiber over an artinian ring is called an infinitesimal deformation. (c) Although Λ is fixed, the number $\dim_k \mathfrak{m}_R/\mathfrak{m}_R^2$ for a formal versal deformation depends only on X_0, not on the choice of Λ.

The same method applies to a pair $Y_0 \subset X_0$, where Y_0 is a closed subscheme. A deformation of the pair is a diagram

$$\begin{tikzcd} Y_0 \arrow[r] \arrow[d] & Y \arrow[d, "\text{flat}"] \\ X_0 \arrow[r] \arrow[d] & X \arrow[d, "\text{flat}"] \\ \operatorname{Spec} k \arrow[r] & \operatorname{Spec} R, \end{tikzcd}$$

which becomes the given pair after base change to k. These deformations form a homogeneous cofibered category $\mathcal{M}_{X_0,Y_0}$ over $\mathcal{C}_\Lambda$.

Lemma 4.7. Let I be the ideal sheaf of Y_0 in X_0. There is an exact sequence

$$0 \to H^0(Y_0, \mathcal{H}om_{\mathcal{O}_{Y_0}}(I/I^2, \mathcal{O}_{Y_0})) \to [\mathcal{M}_{X_0,Y_0}(k[\varepsilon])] \to [\mathcal{M}_{X_0}(k[\varepsilon])].$$

The proof identifies the kernel with deformations of Y_0 inside the trivial first-order deformation $X_0 \times \operatorname{Spec} k[\varepsilon]$. Such an object is equivalently a diagram

$$\begin{tikzcd} 0 \arrow[r] & I \arrow[r] \arrow[d] & \mathcal{O}_{X_0} \arrow[r] \arrow[d] & \mathcal{O}_{Y_0} \arrow[r] \arrow[d, equal] & 0 \\ 0 \arrow[r] & \mathcal{O}_{Y_0} \arrow[r, "\varepsilon"] & \mathcal{O}_Y \arrow[r] & \mathcal{O}_{Y_0} \arrow[r] & 0, \end{tikzcd}$$

and hence is determined by a morphism $I/I^2 \to \mathcal{O}_{Y_0}$.

Remark 4.8. A more general formula, implying Lemma 4.7, is given in SGA 5, III, 4.4. One can also define, for any S-scheme X, quasi-coherent sheaves $D^i_{X|S}$, extending the affine case, and one has a spectral sequence relating the global Ext-groups for the cotangent complex to the cohomology of these sheaves. In particular the low-degree terms recover the tangent, indeterminacy, and obstruction groups used in the preceding deformation arguments.

Theorem 4.9. Let X_0 be a k-scheme of finite type and let $Y_0 \subset X_0$ be a closed subscheme. Assume either that X_0 is proper over k, or that X_0 is affine, smooth outside a finite set of points, and Y_0 is proper over k. Then $\mathcal{M}_{X_0,Y_0}$ admits a formal versal deformation over $\mathcal{C}_\Lambda$, with the same pro-representability criterion by surjectivity on automorphism groups. If, moreover,

$$H^0(Y_0, \mathcal{H}om(I/I^2, \mathcal{O}_{Y_0}))$$

is finite-dimensional over k, then $\mathcal{M}_{X_0,Y_0}$ is smoothly pseudo-representable. This follows at once from 4.4, 4.6, 4.7 and the criteria 1.11, 2.7, and 2.13.

We now consider passage to localization and completion in the deformation theory of schemes. If $x_0 \in X_0$ is fixed, one has natural functors

$$\mathcal{M}_{X_0} \longrightarrow \mathcal{M}_{\mathrm{Spec}(\mathcal{O}_{X_0,x_0})} \longrightarrow \mathcal{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_0})},$$

and after completing formal deformations one obtains

$$\widehat{\mathcal{M}}_{X_0} \longrightarrow \widehat{\mathcal{M}}_{\mathrm{Spec}(\mathcal{O}_{X_0,x_0})} \longrightarrow \widehat{\mathcal{M}}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_0})}.$$

If $(R, \mathfrak{X})$ is a formal deformation and

$$\mathfrak{X} = \varprojlim_{\nu} X_\nu, \qquad X_\nu = \mathfrak{X} \times_{\mathrm{Spf}\, R} \mathrm{Spec}(R/\mathfrak{m}^{\nu+1}),$$

then the two images at x_0 are

$$\mathfrak{X}_{(x_0)} = \varprojlim_{\nu} \mathrm{Spec}(\mathcal{O}_{X_\nu,x_0}), \qquad \widehat{\mathfrak{X}}_{(x_0)} = \varprojlim_{\nu} \mathrm{Spec}(\widehat{\mathcal{O}}_{X_\nu,x_0}).$$

Theorem 4.10. Let $X_0 = \mathrm{Spec}(A_0)$ be an affine scheme of finite type over k.

(a) If X_0 is locally a complete intersection, then $\mathcal{M}_{X_0}$ is smooth over $\mathcal{C}_\Lambda$.

(b) Assume that, for a fixed closed point $x \in X_0$, the complement $X_0 - \{x\}$ is smooth over k. Then the functors

$$\mathcal{M}_{X_0} \longrightarrow \mathcal{M}_{\mathrm{Spec}(\mathcal{O}_{X_0,x})} \longrightarrow \mathcal{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x})}$$

are both minimally smooth. In particular a formal versal deformation of X_0 induces formal versal deformations of $\mathrm{Spec}(\mathcal{O}_{X_0,x})$ and $\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x})$.

Proof. For (a), it is enough to prove smoothness of the functor $[\mathcal{M}_{X_0}] \to \mathcal{C}_\Lambda$. Let $(R, \mathrm{Spec}\, A)$ be a deformation of $\mathrm{Spec}\, A_0$, and let $R' \to R$ be a small extension. The exact sequence for the composite $R' \to R \to A$, with coefficients in A_0, gives

$$\cdots \to D^1(A \mid R, A_0) \to D^1(A \mid R', A_0) \to D^1(R \mid R', k) \otimes_k A_0 \to D^2(A \mid R, A_0) \to \cdots.$$

Because A is R-flat,

$$D^i(A \mid R, A_0) = D^i(A_0 \mid k, A_0) \qquad (i > 0),$$

and the last group vanishes for $i = 2$ when A_0 is locally a complete intersection. Thus one finds an R'-algebra extension A' of A by A_0 mapping to the class of R', i.e. a lifting $(R', \mathrm{Spec}\, A')$ of $(R, \mathrm{Spec}\, A)$.

For (b), Corollary 3.14 gives canonical identifications of tangent spaces

$$D^1(A_0 \mid k, A_0)i \simeq D^1((A_0)_x \mid k, (A_0)_x) \simeq D^1(\widehat{(A_0)_x} \mid k, \widehat{(A_0)_x}).$$

Hence

$$[\mathcal{M}_{X_0}(k[\varepsilon])]i \simeq [\mathcal{M}_{\mathrm{Spec}(\mathcal{O}_{X_0,x})}(k[\varepsilon])]i \simeq [\mathcal{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x})}(k[\varepsilon])].$$

To prove minimal smoothness it remains, by 2.7(a), to show smoothness of the functors. Given $(R,$
$SpecA)$ and a small extension $R' \to R$, the same exact commutative diagram for $R' \to R \to A$, after localization and completion at x, shows that any local or completed local lifting may be lifted globally, because the groups $D^i(A \mid R, A_0)$ agree with their local and completed counterparts in positive degree.

Theorem 4.11. Let X_0 be a separated k-scheme of finite type, smooth outside the finite closed set $\{x_1, \ldots, x_r\}$. Let $U_0^1, \ldots, U_0^r$ be affine open neighbourhoods of these points such that $U_0^i \cap U_0^j$ is smooth over k for $i \neq j$. If

$$H^2(X_0, D^0_{X_0}) = 0,$$

then the natural functor

$$\mathcal{M}_{X_0} \to \mathcal{M}_{U_0^1} \times \cdots \times \mathcal{M}_{U_0^r}$$

is smooth. *Proof.* Complete $U_0^1, \ldots, U_0^r$ to an affine open covering by choosing smooth affine opens U_0^j, $j > r$, such that $U_0^i \cap U_0^j$ is affine for all i, j. Let $R' \to R$ be a small extension. Suppose that a deformation (R, X) of X_0 and liftings (R', U'^i) of the prescribed deformations over U_0^i, $1 \leq i \leq r$, are given. For $j > r$, since U_0^j is smooth, 4.9(a) lifts $X|_{U_0^j}$ to a deformation (R', U'^j).

For each intersection $U_0^i \cap U_0^j$, smoothness gives isomorphisms

$$\tau_{ij} : U'^i|_{U_0^i \cap U_0^j} \xrightarrow{\sim} U'^j|_{U_0^i \cap U_0^j}$$

which reduce to the identity over R. The defect $\tau_{ij}\tau_{jk}\tau_{ik}^{-1}$ is a Cech 2-cocycle with values in $D^0_{X_0}$. Its cohomology class lies in $H^2(X_0, D^0_{X_0})$; this class vanishes by hypothesis. Therefore the local liftings can be modified so that the cocycle condition holds, and then they glue to a deformation (R', X') with $X' \otimes_{R'} R = X$.

Remark 4.12. The preceding proof unnecessarily assumes that X_0 is separated. In fact, for a fixed deformation (R, X) and compatible deformations (R', U'^j) on an open covering, the deformations (R', V') of any open subset $V \subset X$, compatible with the already given data, form a gerbe in the sense of Giraud. Its band is $D^0_{X_0}$. Hence the obstruction to a section of this gerbe, i.e. to a global deformation compatible with the local data, is again in $H^2(X_0, D^0_{X_0})$.

Corollary 4.13. Let X_0 be a k-scheme of finite type, smooth outside a finite set

$$X_0^{\mathrm{Sing}} = \{x_1, \ldots, x_r\},$$

and assume $H^2(X_0, D^0_{X_0}) = 0$. Then

$$\mathcal{M}_{X_0} \to \mathcal{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_1})} \times \cdots \times \mathcal{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_r})}$$

is smooth. In particular, if $(R, \mathfrak{X})$ is a formal versal deformation of X_0, and if $(R_i, \mathfrak{X}_i)$ are formal versal deformations of the completed local schemes, then

$$\dim \mathcal{M}_{X_0} = \dim H^1(X_0, D^0_{X_0}) + \sum_{i=1}^{r} \dim \mathcal{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_i})}.$$

If moreover X_0 is locally a complete intersection, then $\mathcal{M}_{X_0}$ is smooth. *Proof.* The first assertion is obtained from 4.10 and 4.9(b). Choose formal versal deformation rings R and R_i. The diagram

$$\begin{tikzcd} h_R \arrow[r] \arrow[d] & h_{R_1 \widehat{\otimes} \cdots \widehat{\otimes} R_r} \arrow[d] \\ \mathcal{M}_{X_0} \arrow[r] & \prod_i \mathcal{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_i})} \end{tikzcd}$$

shows, by 2.7(a), that $h_R \to h_{R_1 \widehat{\otimes} \cdots \widehat{\otimes} R_r}$ is smooth after allowing $d = \dim_k H^1(X_0, D^0_{X_0})$ formal parameters. Taking tangent spaces gives the displayed dimension formula.

4.14. Explicit complete-intersection construction. Suppose

$$A_0 = k[x_1, \dots, x_n]/(f_1, \dots, f_r),$$

where $f_1, \dots, f_r$ is a regular sequence. The exact sequence

$$\cdots \to D^1(k[x_1, \dots, x_n] \mid k, A_0) \to \mathrm{Hom}_{A_0}(J/J^2, A_0) \to D^1(A_0 \mid k, A_0) \to 0$$

allows one to choose maps P_{uj} representing a k-basis of $D^1(A_0 \mid k, A_0)$. Put

$$R = \Lambda[[t_1, \dots, t_d]], \qquad B = R[[x_1, \dots, x_n]]/(G_1, \dots, G_r),$$

where $G_j = F_j - \sum_\nu t_\nu P_{uj}$, and where F_j is obtained from f_j by lifting its coefficients from k to Λ. Since the sequence remains regular modulo the maximal ideal, B is flat over R, and $k \otimes_R B = A_0$. The induced map

$$\mathrm{Hom}_{\Lambda\text{-alg}}(R, k[\varepsilon]) \to D^1(A_0 \mid k, A_0)$$

is bijective by construction. Thus $(R, \mathrm{Spf}\, B)$ is a formal versal object. The same formula applies to the completed local complete-intersection algebra by 4.10(b).

Remarks 4.15–4.16. If X_0 is affine and locally a complete intersection with only isolated singularities, the construction shows that a formal versal deformation is the formalization of an R-scheme of finite type. If X_0 is not necessarily affine but $H^2(X_0, D^0_{X_0}) = 0$, the same statement globalizes.

The global theory of the relative cotangent complex gives a uniform formulation. Let $S_0 \subset S' \subset S$ be defined by ideals $J \supset I$ with $JI = 0$, let X' be flat over S', and put $L = L_{X_0/S_0}$. For the category F of flat schemes over S extending X', one has:

(a) the automorphism group of an object is $\mathrm{Ext}^0(X_0, L, I_{X_0})$;

(b) the set of isomorphism classes of objects, if non-empty, is a torsor under $\mathrm{Ext}^1(X_0, L, I_{X_0})$;

(c) there is a canonical obstruction class in $\mathrm{Ext}^2(X_0, L, I_{X_0})$, whose vanishing is necessary and sufficient for non-emptiness.

Using

$$\mathrm{Ext}^i(X_0, L, I_{X_0}) = H^i(X_0, \mathrm{RHom}(L, I_{X_0})),$$

and the fact that the cohomology sheaves of $\mathrm{RHom}(L, I_{X_0})$ are the $D^q_{X_0}(I_{X_0})$, one obtains the spectral sequence

$$E_2^{p,q} = H^p(X_0, D^q_{X_0}(I_{X_0})) \Longrightarrow \mathrm{Ext}^{p+q}(X_0, L, I_{X_0}).$$

The first obstruction lies in $H^0(X_0, D^2)$; when it vanishes, the local solutions form a torsor under D^1, with obstruction in $H^1(X_0, D^1)$; after choosing a coherent system of local isomorphism classes, the remaining obstruction lies in a quotient of $H^2(X_0, D^0)$. This is the intrinsic form of the gluing argument used above.

5. Formal Jacobian subschemes

Let X_0 be a scheme of finite type over a field k, and let $(R, \mathfrak{X})$ be a formal versal deformation of X_0. Since $\mathfrak{X}$ is an R-adic formal scheme rather than an ordinary scheme over R, we must say what is meant by its non-smooth locus.

We first recall Fitting ideals. Let A be a commutative ring and E an A-module of finite type. Choose a presentation

$$K \to F \to E \to 0,$$

where F is locally free of constant rank n. Define

$$I^p(E) = \operatorname{Im}\Big(\bigwedge^{n-p} K \otimes \bigwedge^{p} F^\vee \to A\Big) \qquad (p = 0, 1, 2, \ldots).$$

These ideals are independent of the chosen presentation and are the Fitting ideals of E. They satisfy:

(a) For every ring map $A \to B$, base change gives

$$B \otimes_A A/I^p(E) \simeq B/I^p(B \otimes_A E).$$

In particular localization commutes with I^p.

(b) The ideals are increasing, and $I^p(E)_x = A_x$ whenever $p > \dim_{k(x)} E(x)$. The support of $A/I^0(E)$ is $\operatorname{Supp} E$.

(c) Direct sums satisfy the usual convolution formula for Fitting ideals.

(d) If A is noetherian adic, $A = \varprojlim A_n$, and $E = \varprojlim E_n$, then

$$I^p(E) = \varprojlim I^p(E_n).$$

Thus for a scheme X and a quasi-coherent module E of finite presentation one obtains quasi-coherent ideals $I^p_X(E)$ by the affine formula

$$I^p_X(E)(U) = I^p(\Gamma(U, E))$$

on affine opens U. If $X \to Y$ is essentially of finite presentation, the relative Kahler differentials $\Omega^1_{X/Y}$ are of finite presentation, so the ideals $I^p_X(\Omega^1_{X/Y})$ define closed subschemes. This is the starting point for the formal Jacobian subschemes.

5. Formal Jacobian subschemes, continued

We use the notation introduced above for the Fitting ideals $I^p(E)$. For a morphism

$$f : X \longrightarrow Y$$

which is essentially of finite presentation, we denote by

$$I^p(X \mid Y) = I^p_X(\Omega^1_{X/Y})$$

the corresponding ideal sheaf on X, and by $J^p(X \mid Y)$ the closed subscheme which it defines. Thus $J^0(X \mid Y) \supset J^1(X \mid Y) \supset \cdots$.

Theorem 5.3. Let $f : X \to Y$ be essentially of finite presentation. Then:

(1) For every base change $Y' \to Y$, with $X' = X \times_Y Y'$, the canonical morphism

$$J^p(X' \mid Y') \longrightarrow J^p(X \mid Y) \times_Y Y'$$

is an isomorphism, for every p.

(2) For every $x \in X$, $y = f(x)$, the canonical local comparison is an isomorphism:

$$I_X^p(\Omega^1_{X/Y})_x \simeq I^p(\Omega^1_{\mathcal{O}_{X,x}/\mathcal{O}_{Y,y}}).$$

(3) The $J^p(X \mid Y)$ form a decreasing sequence of closed subschemes. A point x does not belong to $J^p(X \mid Y)$ once p is larger than

$$\dim_{\kappa(x)}(\Omega^1_{X/Y} \otimes \kappa(x)).$$

(4) One has $x \notin J^p(X \mid Y)$ if and only if, after replacing X by an open neighbourhood U of x, there exists a closed immersion

$$i : U \hookrightarrow U'$$

into a Y-scheme U' which is smooth over Y at $x' = i(x)$, and whose relative embedding dimension at x' is p.

(5) If X_1 and X_2 are two Y-schemes essentially of finite presentation and $X = X_1 \times_Y X_2$, then

$$I^p(X \mid Y) = \sum_{p_1+p_2=p} \operatorname{pr}_1^* I^{p_1}(X_1 \mid Y)\, \operatorname{pr}_2^* I^{p_2}(X_2 \mid Y).$$

Proof. Assertion (3) is the elementary monotonicity of Fitting ideals. Assertion (1) follows from the base-change formula for Fitting ideals and from

$$\Omega^1_{X'/Y'} \simeq \operatorname{pr}^* \Omega^1_{X/Y}.$$

Assertion (2) is the same statement after localization at x. Assertion (5) follows from the direct-sum formula for Fitting ideals applied to

$$\Omega^1_{X_1 \times_Y X_2/Y} \simeq \operatorname{pr}_1^* \Omega^1_{X_1/Y} \oplus \operatorname{pr}_2^* \Omega^1_{X_2/Y}.$$

For (4), suppose first that such an immersion $U \hookrightarrow U'$ exists. Since U' is smooth over Y at x', $\Omega^1_{U'/Y,x'}$ is free of rank p, and the exact conormal sequence

$$I/I^2 \longrightarrow \Omega^1_{U'/Y} \otimes \mathcal{O}_U \longrightarrow \Omega^1_{U/Y} \longrightarrow 0$$

shows that the p-th Fitting ideal is the unit ideal at x. Hence $x \notin J^p(U \mid Y)$, and therefore $x \notin J^p(X \mid Y)$.

Conversely, the question is local. Write $X = \operatorname{Spec} B$, $Y = \operatorname{Spec} A$, and choose a presentation

$$0 \longrightarrow J \longrightarrow P \longrightarrow B \longrightarrow 0, \qquad P = A[x_1, \ldots, x_n].$$

The exact sequence

$$J/J^2 \longrightarrow B \otimes_P \Omega^1_{P/A} \longrightarrow \Omega^1_{B/A} \longrightarrow 0$$

has middle term free of rank n. The condition $x \notin J^p(X \mid Y)$ means that a suitable $(n-p)$-minor of the Jacobian matrix of chosen generators of J is invertible at x. Thus there are elements

$$f_1, \ldots, f_{n-p} \in J$$

such that

$$U' = \operatorname{Spec}(P/(f_1, \dots, f_{n-p}))$$

is smooth over Y at the image x' of x, and $U = \operatorname{Spec} B$ embeds as a closed subscheme of U'. This gives the required local presentation.

The following result is the corresponding Jacobian criterion.

Theorem 5.4. Let $f : X \to Y$ be essentially of finite presentation. Suppose that there is an integer $n \geq 0$ such that either:

(1) f is flat and all fibers have dimension n;

(2) f is a relative complete intersection of virtual dimension n in the sense of SGA 6, VIII 1.9.

Then

$$x \notin J^n(X \mid Y) \quad \Longleftrightarrow \quad f \text{ is smooth at } x.$$

Proof. By Theorem 5.3(4), $x \notin J^n(X \mid Y)$ if and only if locally X embeds into a Y-scheme U', smooth over Y at x', with relative dimension n. It remains to see that such an embedding is an isomorphism at x.

Let

$$0 \longrightarrow I \longrightarrow B' \longrightarrow B \longrightarrow 0$$

be the corresponding exact sequence of local rings, with B' the local ring of U' at x' and B the local ring of X at x. Put $A = \mathcal{O}_{Y,y}$. In case (1), flatness gives the exact sequence on the closed fiber

$$0 \longrightarrow I/\mathfrak{m}I \longrightarrow B'/\mathfrak{m}B' \longrightarrow B/\mathfrak{m}B \longrightarrow 0,$$

where $\mathfrak{m}$ is the maximal ideal of A. The two fiber local rings have the same dimension n, and $B'/\mathfrak{m}B'$ is regular; hence $I/\mathfrak{m}I = 0$, whence $I = 0$.

In case (2), the conormal sequence

$$I/I^2 \longrightarrow B \otimes_{B'} \Omega^1_{B'/A} \longrightarrow \Omega^1_{B/A} \longrightarrow 0$$

is exact. Since X is a relative complete intersection and U' is smooth, the first map is injective and the two modules which occur have the ranks prescribed by the virtual dimension. These ranks force $I/I^2 = 0$, hence $I = 0$.

Definition 5.5. Under the hypotheses of Theorem 5.4, one writes simply

$$I(X \mid Y) = I^n(X \mid Y), \qquad J(X \mid Y) = J^n(X \mid Y).$$

Thus $J(X \mid Y)$ is the Jacobian subscheme measuring failure of smoothness.

We now extend the construction to noetherian formal schemes. Let $\mathfrak{X}$ be a noetherian formal scheme and E a coherent $\mathcal{O}_{\mathfrak{X}}$-module. Define

$$I^p_{\mathfrak{X}}(E)(U) = I^p(\Gamma(U, E))$$

for every affine open $U \subset \mathfrak{X}$. If $\mathfrak{X} = \varprojlim X_v$ and $E = \varprojlim E_v$, then

$$I^p_{\mathfrak{X}}(E) = \varprojlim_v I^p_{X_v}(E_v).$$

Let S be a noetherian formal scheme and let $\mathfrak{X}$ be an S-adic formal scheme, essentially of finite type. Let

$$\mathfrak{X} = \varprojlim X_v, \qquad S = \varprojlim S_v, \qquad X_v = \mathfrak{X} \times_S S_v.$$

Then

$$\Omega^1_{\mathfrak{X}/S} \simeq \varprojlim_v \Omega^1_{X_v/S_v}$$

is a coherent $\mathcal{O}_{\mathfrak{X}}$-module and is compatible with finite-type formal base change.

Definition 5.6. Define

$$I^p(\mathfrak{X} \mid S) = I^p_{\mathfrak{X}}(\Omega^1_{\mathfrak{X}/S})$$

and let $J^p(\mathfrak{X} \mid S)$ be the closed formal subscheme of $\mathfrak{X}$ defined by this ideal. Equivalently

$$I^p(\mathfrak{X} \mid S) = \varprojlim_v I^p(X_v \mid S_v), \qquad J^p(\mathfrak{X} \mid S) = \varprojlim_v J^p(X_v \mid S_v).$$

There is the natural cartesian comparison with the closed fiber:

$$\begin{tikzcd} J^p(X_0 \mid S_0) \arrow[r] \arrow[d] & J^p(\mathfrak{X} \mid S) \arrow[d] \\ X_0 \arrow[r] & \mathfrak{X}. \end{tikzcd}$$

If x is an isolated point of $J^p(\mathfrak{X} \mid S)$, then the formal completion of $J^p(\mathfrak{X} \mid S)$ at x is canonically the corresponding connected component. In particular, if

$$|J^p(X_0 \mid S_0)| = \{x_1, \ldots, x_m\}$$

is finite and discrete, then

$$J^p(\mathfrak{X} \mid S) \simeq \coprod_{1 \leq i \leq m} J^p(\mathfrak{X} \mid S)_{(x_i)}.$$

Corollary 5.7. Let S be noetherian and formal, and let $\mathfrak{X}$ be an S-adic formal scheme essentially of finite type.

(1) The construction $J^p(\mathfrak{X} \mid S)$ commutes with formal base change:

$$J^p(\mathfrak{X}' \mid S') = J^p(\mathfrak{X} \mid S) \times_S S', \qquad \mathfrak{X}' = \mathfrak{X} \times_S S'.$$

(2) The connected components of $J^p(\mathfrak{X} \mid S)$ are in bijection with those of $J^p(X_0 \mid S_0)$, and for each isolated point x of $J^p(X_0 \mid S_0)$ there is a canonical isomorphism of completed local pieces

$$J^p(\mathfrak{X} \mid S)_{(x)} \simeq J^p(X_0 \mid S_0)_{(x)}.$$

In particular, in the finite discrete case one has the above disjoint-union decomposition.

(3) If S is an ordinary noetherian scheme, $S_0 \subset S$ a closed subscheme, $\widehat{S}$ the formal completion of S along S_0, and

$$\widehat{X} = X \times_S \widehat{S},$$

then

$$J^p(\widehat{X} \mid \widehat{S}) = J^p(X \mid S) \times_S \widehat{S}.$$

Thus the formal Jacobian subscheme is the completion of $J^p(X \mid S)$ along $J^p(X_0 \mid S_0)$.

(4) If a fixed integer n satisfies the hypotheses of Theorem 5.4 for all $X_v \to S_v$, then $x \notin J^n(\mathfrak{X} \mid S)$ if and only if $\mathfrak{X}$ is formally smooth over S at x.

Proof. The assertions follow directly from the corresponding statements for ordinary schemes, the compatibility of $\Omega^1_{\mathfrak{X}/S}$ with passage to X_v/S_v, and the base-change formula for Fitting ideals.

Remark 5.8. Let $\mathfrak{X}$ be S-adic, essentially of finite type, and let $x \in \mathfrak{X}$ with image $y \in S$. One can also consider the local formal $S_{(y)}$-scheme

$$\mathfrak{X}_{(x)} = \varprojlim_v \operatorname{Spec}(\mathcal{O}_{X_v,x_v}),$$

with the natural morphism $\mathfrak{X}_{(x)} \to \mathfrak{X}$. Although $\mathfrak{X}_{(x)}$ need not itself be essentially of finite type over $S_{(y)}$, the completed module of differentials is still coherent in the sense needed here:

$$\Omega^1_{\mathfrak{X}_{(x)}/S_{(y)}} \simeq \varprojlim_v \Omega^1_{\widehat{\mathcal{O}_{X_v,x_v}/\mathcal{O}_{S_v,y_v}}}.$$

This gives local formal subschemes $J^p(\mathfrak{X}_{(x)} \mid S)$, and if x is an isolated point of $J^p(\mathfrak{X} \mid S)$, the natural map

$$J^p(\mathfrak{X}_{(x)} \mid S) \longrightarrow J^p(\mathfrak{X} \mid S)$$

identifies the source with the completed local component at x.

We shall also need the image of the Jacobian subscheme in the base. The set-theoretic image may be described by kernels

$$\operatorname{Ker}\big(\mathcal{O}_S \to f_*\mathcal{O}_{J^p(\mathfrak{X}|S)}\big),$$

but the Fitting-ideal formulation is more useful.

Definition 5.9. Let S be noetherian and formal, and let

$$f : \mathfrak{X} \longrightarrow S$$

be S-adic, essentially of finite type. Suppose $J^p(\mathfrak{X} \mid S)$ is proper over S. Then $f_*\mathcal{O}_{J^p(\mathfrak{X}|S)}$ is coherent, and we define an ideal sheaf on S by

$$I^p_S(\mathfrak{X} \mid S) = I^0_S\big(f_*\mathcal{O}_{J^p(\mathfrak{X}|S)}\big).$$

Let $J^p_S(\mathfrak{X} \mid S)$ be the closed formal subscheme of S defined by this ideal. If the fixed-dimension hypotheses of Theorem 5.4 are in force, one writes

$$I_S(\mathfrak{X} \mid S) = I^n_S(\mathfrak{X} \mid S), \qquad J_S(\mathfrak{X} \mid S) = J^n_S(\mathfrak{X} \mid S).$$

Corollary 5.10. Under the hypotheses of Definition 5.9:

(1) The underlying set of $J^p_S(\mathfrak{X} \mid S)$ is the image of $J^p(\mathfrak{X} \mid S)$ under $f : \mathfrak{X} \to S$.

(2) The formation of $J^p_S(\mathfrak{X} \mid S)$ commutes with base change.

(3) If

$$J^p(\mathfrak{X} \mid S) = \coprod_i J^p(\mathfrak{X} \mid S)_{(x_i)}$$

is the disjoint union of its isolated formal local components, then

$$I^p_S(\mathfrak{X} \mid S) = \prod_i I^p_{S,(x_i)}(\mathfrak{X} \mid S).$$

(4) In the case of completion of an ordinary noetherian S-scheme X, the formal closed subscheme $J^p_S(\widehat{X} \mid \widehat{S})$ is the completion of $J^p_S(X \mid S)$.

Proof. Assertion (1) is the usual support property of the zeroth Fitting ideal. Assertions (2) and (4) follow from base change for Fitting ideals. Assertion (3) follows from the product formula for I^0 applied to the decomposition of the coherent algebra $f_*\mathcal{O}_{J^p(\mathfrak{X}|S)}$.

Remark 5.11. The Jacobian ideal above is classical: it commutes with base change and, under the mild dimension hypotheses of Theorem 5.4, gives the smoothness criterion. There are also more intrinsic ways of defining a canonical non-smooth locus in complete generality. For an A-algebra B, set

$$I_{B/A} = \bigcap_E \operatorname{Ann}_B D^1(B \mid A, E),$$

where E runs through the finite-type B-modules. This ideal commutes with localization by Corollary 3.13(c). Hence for a morphism $X \to Y$ one obtains an ideal sheaf $I_{X/Y}$. If $X \to Y$ is essentially of finite type, then f is smooth at x if and only if

$$x \notin \operatorname{Supp}(\mathcal{O}_X/I_{X/Y})$$

by 3.13(a). The same construction carries over to adic formal schemes essentially of finite type and is therefore also suited to deformation theory.

6. Non-degenerate quadratic singularities

The purpose of this section is to study deformations of a k-scheme whose singularities are isolated non-degenerate quadratic singularities. In view of 4.13, under the hypotheses considered here the question becomes a local one.

We return to the notation of Section 4. Thus A is a fixed complete noetherian local ring with residue field k, and $\mathcal{C}_A$ is the category of local artinian A-algebras with residue field k. Let X be a k-scheme of finite type and let x be a closed point of X. Recall that x is a non-degenerate quadratic singularity, rational over k, if

$$\widehat{\mathcal{O}}_{X,x} \simeq k[[x_1, \dots, x_n]]/(f)$$

and

$$\left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n}\right) = (x_1, \dots, x_n).$$

One may then arrange that f itself is a quadratic form.

Lemma 6.1. Let

$$f \in k[[x_1, \dots, x_n]]$$

and suppose

$$\left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n}\right) = \mathfrak{m} = (x_1, \dots, x_n).$$

Then there are parameters $x'_1, \dots, x'_n$, with

$$x_i \equiv x'_i \pmod{\mathfrak{m}^2},$$

such that

$$f = q(x'_1, \dots, x'_n)$$

for a quadratic form q.

Proof. Write $f = f_2 + h_1$, where f_2 is quadratic and $h_1 \in \mathfrak{m}^3$. Since the partial derivatives generate $\mathfrak{m}$, there are elements $e_i^{(1)} \in \mathfrak{m}^2$ such that

$$\sum_i e_i^{(1)} \frac{\partial f}{\partial x_i} \equiv h_1 \pmod{\mathfrak{m}^4}.$$

After replacing x_i by $x_i - e_i^{(1)}$, the remainder is pushed into $\mathfrak{m}^4$. Repeating this construction, one obtains $e_i^{(v)} \in \mathfrak{m}^{v+1}$ and parameters

$$x_i' = x_i - \sum_{v \geq 1} e_i^{(v)}$$

for which all terms of degree at least 3 disappear. Thus f becomes its quadratic part in the new variables.

Lemma 6.2. Let

$$A_0 \simeq k[[x_1, \dots, x_n]]/(f),$$

where f is a non-degenerate quadratic form, and let $(R, \mathfrak{X})$ be a formal versal deformation of $X_0 = \operatorname{Spec} A_0$. Then:

(1) The natural maps

$$J(\mathfrak{X} \mid R) \longrightarrow J_R(\mathfrak{X} \mid R) \longrightarrow \operatorname{Spec} A$$

are isomorphisms.

(2) The base Jacobian ideal is principal:

$$I_R(\mathfrak{X} \mid R) = (t), \qquad R = A[[t]],$$

where t is a free formal generator of R over A.

(3) The morphism $\mathfrak{X} \to \operatorname{Spec} A$ is formally smooth.

Proof. By 4.10(b), the formal versal deformation has the explicit form

$$R = A[[t]], \qquad \mathfrak{X} = \operatorname{Spf}\big(R[[x_1, \dots, x_n]]/(F - t)\big),$$

where F is obtained from f by lifting the coefficients to A. The exact sequence

$$A_0 \xrightarrow{dF} \bigoplus_i A_0 \, dx_i \longrightarrow \Omega^1_{A_0/R} \longrightarrow 0$$

shows that

$$I(\mathfrak{X} \mid R) = (\partial F/\partial x_1, \dots, \partial F/\partial x_n) = (x_1, \dots, x_n)$$

on the special fiber. Hence the Jacobian subscheme is obtained by setting $x_1 = \cdots = x_n = 0$, and the remaining equation is $t = 0$. Thus the maps in (1) are isomorphisms and $I_R(\mathfrak{X} \mid R) = (t)$. Finally

$$R[[x_1, \dots, x_n]]/(F - t) \simeq A[[x_1, \dots, x_n]]$$

as an A-algebra, proving formal smoothness over A.

Corollary 6.3. Let $x \in X_0$ be a non-degenerate quadratic singularity rational over k.

(1) Let $(R, \mathfrak{X}_1)$ and $(R, \mathfrak{X}_2)$ be two deformations of X_0 over an object R of $\mathcal{C}_A$, with the same point above x. Then the completed local R-adic formal schemes $(\mathfrak{X}_1)_{(x)}$ and $(\mathfrak{X}_2)_{(x)}$ are isomorphic over R if and only if

$$I_R((\mathfrak{X}_1)_{(x)} \mid R) = I_R((\mathfrak{X}_2)_{(x)} \mid R).$$

(2) Let R be a regular local ring and $(R, \mathfrak{X})$ a deformation of X_0. Then $\mathcal{O}_{\mathfrak{X},x}$ is regular if and only if

$$I_R(\mathfrak{X}_{(x)} \mid R) \not\subset \mathfrak{m}_R^2,$$

that is, if and only if this Jacobian ideal defines a regular divisor in $\operatorname{Spf} R$.

Proof. Let the formal versal deformation be

$$A[[t]][[x_1, \dots, x_n]]/(F - t).$$

Two induced deformations over R are given by two elements $a, b \in R$:

$$R[[x_1, \dots, x_n]]/(F - a), \qquad R[[x_1, \dots, x_n]]/(F - b).$$

Their Jacobian ideals are (a) and (b). Equality of these ideals gives $a = vb$ for a unit v. Since R is complete and v has a square root after the standard lifting argument used here, the change of variables $x_i \mapsto v^{1/2} x_i$ identifies the two equations, because F is quadratic. The converse is immediate. The second assertion follows from the fact that the quotient of the regular local ring $R[[x_1, \dots, x_n]]$ by $F - a$ is regular precisely when $a \notin \mathfrak{m}_R^2$.

Theorem 6.4. Let X_0 be a k-scheme of finite type with only isolated non-smooth points, and let $(R, \mathfrak{X})$ be a formal versal deformation of X_0. Let

$$\{z_1, \dots, z_m\}$$

be the set of non-smooth points of X_0. Suppose

$$H^2(X_0, D^0) = 0$$

and suppose that all z_i are non-degenerate quadratic singularities, rational over k. Then:

(1) R is formally smooth over A, i.e. R is a formal power-series ring over A.

(2) The Jacobian subscheme decomposes as a disjoint union

$$J(\mathfrak{X} \mid R) = \coprod_{i=1}^{m} J(\mathfrak{X}_{(z_i)} \mid R),$$

and for each i the natural maps from the local Jacobian component to its image in $\operatorname{Spf} R$ are isomorphisms.

(3) The ideal $I_R(\mathfrak{X}_{(z_i)} \mid R)$ is principal. If

$$I_R(\mathfrak{X}_{(z_i)} \mid R) = (t_i),$$

then $t_1, \dots, t_m$ can be extended to a system of free formal generators of R over A, and

$$I_R(\mathfrak{X} \mid R) = \prod_{i=1}^{m} (t_i).$$

(4) The morphism $\mathfrak{X} \to \operatorname{Spec} A$ is formally smooth.

(5) If X_0 is a complete irreducible curve, then

$$\dim_k \mathfrak{m}_R/\mathfrak{m}_R^2 = 3g - 3 + a,$$

where

$$g = \dim_k H^1(X_0, \mathcal{O}_{X_0}), \qquad a = \dim_k H^0(X_0, D^1).$$

Proof. For each singular point z_i, let $(R_i, \mathfrak{U}_i)$ be the formal versal deformation of $\operatorname{Spec}(\mathcal{O}_{X_0,z_i})$. Since $(R, \mathfrak{X})$ induces a deformation of this local scheme, there is a continuous local map $R_i \to R$ and a cartesian diagram

$$\begin{tikzcd} \mathfrak{X}_{(z_i)} \arrow[r] \arrow[d] & \mathfrak{U}_i \arrow[d] \\ \operatorname{Spf} R \arrow[r] & \operatorname{Spf} R_i. \end{tikzcd}$$

Lemma 6.2 gives

$$J(\mathfrak{X}_{(z_i)} \mid R) \simeq J(\mathfrak{U}_i \mid R_i) \times_{\operatorname{Spf} R_i} \operatorname{Spf} R$$

and

$$I_R(\mathfrak{X}_{(z_i)} \mid R) = (t_i),$$

where t_i is the image in R of the corresponding local parameter. Since $H^2(X_0, D^0) = 0$, the local-to-global criterion of 4.13 implies that

$$\operatorname{Spf} R \longrightarrow \prod_i \operatorname{Spf} R_i$$

is formally smooth in the complementary directions; equivalently $t_1, \ldots, t_m$ extend to a system of free formal generators of R. This proves (1), (2), and (3). Assertion (4) follows because each local singular piece is formally smooth over $\operatorname{Spec} A$ by Lemma 6.2(3), while away from the z_i the original scheme is already smooth.

For (5), formal smoothness of R over A, together with the exact sequence obtained in 4.4 and 4.11,

$$0 \to H^1(X_0, D^0) \to \mathfrak{m}_R/\mathfrak{m}_R^2 \to H^0(X_0, D^1) \to 0,$$

gives

$$\dim_k \mathfrak{m}_R/\mathfrak{m}_R^2 = \dim_k H^1(X_0, D^0) + \dim_k H^0(X_0, D^1).$$

Thus it remains to compute the Euler characteristic of D^0 on the curve. After extension of the base field to an algebraic closure, each non-degenerate quadratic singularity of a curve has completed local ring

$$k[[x, y]]/(xy).$$

If $A = \mathcal{O}_{X_0,z}$ and A' is its normalization, then

$$\ell(A'/A) = 1, \qquad \operatorname{Der}_k(A, \mathfrak{m}_A) = \operatorname{Der}_k(A, A),$$

and the conductor C of A in A' is $\mathfrak{m}_A$. Consequently

$$\operatorname{Der}_k(A, A) = \operatorname{Der}_k(A', A')$$

as A-submodules of $\operatorname{Der}_k(\operatorname{Frac} A, \operatorname{Frac} A)$, and one obtains an exact sequence

$$0 \to D^0_{X_0} \to D^0_{\widetilde{X}_0} \to \mathcal{O}_{\widetilde{X}_0}/C \to 0,$$

where $\widetilde{X}_0$ denotes the normalization. Since X_0 is locally a complete intersection,

$$\ell(\mathcal{O}_{\widetilde{X}_0}/C) = 2m.$$

Riemann–Roch on the normalization gives

$$\chi(D^0_{\widetilde{X}_0}) = 3 - 3(g - m).$$

Therefore

$$\chi(D^0_{X_0}) = \chi(D^0_{\widetilde{X}_0}) - \ell(\mathcal{O}_{\widetilde{X}_0}/C) = 3 - 3g + m,$$

which is equivalent to the stated formula.

Remark 6.5. In the formula of Theorem 6.4(5), the integer

$$a = \dim_k H^0(X_0, D^1)$$

is easily computed by Riemann–Roch. Namely

$$a = \begin{cases} 0, & g \geq 2, \\ 1, & g = 1, \\ 3, & g = 0. \end{cases}$$

Finally, in characteristic 2 there is no non-degenerate quadratic form in an odd number of variables. Equivalently, there is no non-degenerate quadratic singularity of even dimension when $\operatorname{char} k = 2$. In that case one uses quadratic forms of maximum possible non-degeneracy.

Definition 6.6. Let X be a k-scheme of finite type. A closed point $z \in X$ is called an ordinary quadratic singularity, rational over k, if:

(i) either $\operatorname{char} k \neq 2$, or $\operatorname{char} k = 2$ and $\dim \mathcal{O}_{X,z}$ is odd; in this case z is required to be a non-degenerate quadratic singularity rational over k;

(ii) or $\operatorname{char} k = 2$ and $\dim \mathcal{O}_{X,z}$ is even; after extension to an algebraic closure $\bar{k}$, the completed local ring has the form

$$\bar{k}[[x_0, x_1, \ldots, x_{2n}]]/(x_0^2 + x_1x_2 + \cdots + x_{2n-1}x_{2n}).$$

Lemma 6.7. Assume $\operatorname{char} k = 2$, and put

$$A_0 = k[[x_0, x_1, \ldots, x_{2n}]]/(f), \qquad f = x_0^2 + x_1x_2 + \cdots + x_{2n-1}x_{2n}.$$

Let $(R, \mathfrak{X})$ be a formal versal deformation of $X_0 = \operatorname{Spec} A_0$. Then

$$R = A[[t_1, t_2]],$$

where t_1, t_2 are free formal generators over A,

$$I_R(\mathfrak{X} \mid R) = (t_1^2),$$

and $\mathfrak{X} \to \operatorname{Spec} A$ is formally smooth.

Proof. Since

$$\left(\frac{\partial f}{\partial x_0}, \frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_{2n}}\right) = (x_2, x_1, \ldots, x_{2n}, x_{2n-1}),$$

the deformation space has two formal parameters. By 4.14 the versal deformation may be written

$$R = A[[t_1, t_2]], \qquad \mathfrak{X} = \operatorname{Spf}\big(R[[x_0, \ldots, x_{2n}]]/(F)\big),$$

with

$$F = f + t_1 + t_2x_0.$$

The exact sequence for differentials gives

$$I(\mathfrak{X} \mid R) = (t_2, x_1, \ldots, x_{2n}),$$

and the residual equation on the Jacobian locus is t_1. Thus the induced base ideal is (t_1^2). Moreover the equation can be rewritten so that the total formal scheme is formally smooth over A, giving the last assertion.

Corollary 6.8. Let X_0 be a k-scheme of finite type with isolated non-smooth points only, and let $(R, \mathfrak{X})$ be a formal versal deformation of X_0. Suppose

$$H^2(X_0, D^0) = 0,$$

$\operatorname{char} k = 2$, $\dim X_0$ is even, and all non-smooth points $z_1, \ldots, z_m$ are ordinary quadratic singularities. Then:

(1) R is formally smooth over A.

(2) Each local base Jacobian ideal is principal of the form

$$I_R(\mathfrak{X}_{(z_i)} \mid R) = (t_i^2),$$

and

$$I_R(\mathfrak{X} \mid R) = \prod_i (t_i^2).$$

Moreover $t_1, \ldots, t_m$ can be completed to a system of free formal generators of R over A.

(3) The morphism $\mathfrak{X} \to \operatorname{Spec} A$ is formally smooth.

(4) If k is algebraically closed and X_0 is a complete irreducible curve, then

$$\dim_k \mathfrak{m}_R/\mathfrak{m}_R^2 = 3g - 3 + a,$$

where

$$g = \dim_k H^1(X_0, \mathcal{O}_{X_0}), \qquad a = \dim_k H^0(X_0, D^1).$$

The proof is the same as that of Theorem 6.4, using Lemma 6.7 in place of Lemma 6.2 for the local calculation in characteristic 2.

Exposé VII. Biextensions of sheaves of groups

by A. Grothendieck

0. Introduction

The notion of a *biextension* was introduced by D. Mumford, in the context of formal groups, in order to express conveniently the relations between the formal groups associated with an abelian scheme and with the dual abelian scheme. In the present exposé we study this notion systematically in the general context of sheaves of groups on a fixed site. We show in particular that it fits, in a remarkably simple way, into the classical cohomological formalism of $\otimes^L$ and $R\mathrm{Hom}$. In the next exposé we shall spell out the most interesting special cases, in particular the case of biextensions of group schemes by $\mathbb{G}_m$, working with sites such as the Zariski site, the fpqc site, or an intermediate site. One will see, in particular, that if A and B are two abelian schemes over a scheme S, then biextensions of (A, B) by $\mathbb{G}_m$ have a remarkable interpretation as, essentially, the classical divisor correspondences on A and B.

Thus the cohomological meaning of the notion of divisor correspondence is made quite explicit; this meaning is already implicit in Tate duality for abelian varieties over p-adic fields. This interpretation is also convenient for following the fate of such a correspondence when one lets A and B, supposed defined over the fraction field of a discrete valuation ring, degenerate by means of Néron's theory. The results of Exposé VIII show, for example, that one obtains a biextension of the pair of neutral components of the Néron models, hence a biextension of the connected special fibres of the Néron models, playing the role of a specialization of the original divisor correspondence. These results will be used in Exposé IX, together with the monodromy theorem of Exposé I, 3, to prove results on Néron models of abelian varieties, the most important being the semi-stable reduction theorem for the Néron model.

Throughout what follows, unless explicitly stated otherwise, we work in a fixed $\mathcal{U}$-topos, denoted $\mathcal{T}$. Thus, in the terminology of SGA 4 IV, $\mathcal{T}$ is equivalent to the category of sheaves on a suitable $\mathcal{U}$-site, which may be taken to be $\mathcal{T}$ itself endowed with its canonical topology. We shall often identify, in terminology, the objects of $\mathcal{T}$ with sheaves of sets on $\mathcal{T}$. The Groups, extensions of Groups, and so on which occur below are all relative to this topos $\mathcal{T}$.

1. Complements on extensions of Groups

1.1. Extensions and bitorsors

Let P and G be two Groups of $\mathcal{T}$, and let

$$(1.1.1) 1 \longrightarrow G \xrightarrow{i} E \xrightarrow{j} P \longrightarrow 1 \tag{2}$$

be an extension of P by G. Thus E is a Group, $j : E \to P$ is an epimorphism of Groups, and i identifies G with $\operatorname{Ker} j$. We shall reinterpret the datum of such an extension in different terms, useful for defining and studying biextensions.

First, the homomorphism i makes G act on E on the left and on the right by

$$g\, x\, g' = i(g)\, x\, i(g')$$

for $g, g' \in G(S)$, $x \in E(S)$, and any object S of $\mathcal{T}$. If, via j, one regards E as an object above P, i.e. as an object of the induced topos $\mathcal{T}/P$, then this says equivalently that the induced group

$$G_P = G \times P$$

over P acts on E on the left and on the right. Moreover each of these actions makes E a torsor over P, with structural group G_P, and more precisely a bitorsor under (G_P, G_P).

Recall that this means that E is a right torsor E_d for the right action of G_P, and that the left action induces an isomorphism from G_P to the Group of automorphisms of E_d, i.e. the sheaf of groups of automorphisms of E commuting with the right action of G_P. This condition is equivalent to the symmetric condition saying that the left action of G_P makes E a left torsor E_s, and the right action induces an isomorphism from the opposite Group G_P° of G_P to the Group of automorphisms of E_s.

When $\mathcal{T}$ is the punctual topos, so that P and G are ordinary groups, the datum of a bitorsor E under (G_P, G_P) is plainly equivalent to the datum, for every $p \in P$, of a bitorsor E_p under (G, G), namely the fibre of E over p. In the general case the analogous interpretation is: for every point $p : S \to P$, where S is an arbitrary object of $\mathcal{T}$, one is given a bitorsor E_p under the pair of Groups (G_S, G_S) on S, in such a way that the formation of E_p is compatible with base change. More explicitly, for a morphism $S' \to S$ and the composite $p' : S' \to P$, one is given an isomorphism between the bitorsor $E_{p'}$ and the inverse image of E_p, subject to the usual transitivity condition for composites $S'' \to S' \to S$.

In canonical language this assertion says that the datum of a bitorsor under (G_P, G_P) on P is equivalent to a Cartesian section, over the category $\mathcal{T}/P$, of the fibred category of bitorsors of structural group (G_S, G_S) over a variable base S. More precisely, the functor "value at P" induces an equivalence from the category of such Cartesian sections to the category of bitorsors under (G_P, G_P). This is tautological, since P is the final object of the base category $\mathcal{T}/P$. Nevertheless this interpretation is useful for formulating compatibilities in a language close to the set-theoretic case, using points with values in an object S rather than ordinary points.

The bitorsor structure just described uses only part of the multiplication on E: namely multiplication of pairs of points of which at least one comes from G. We now interpret the full multiplicative structure of E as an additional structure on the bitorsor. Consider two sections p, p' of P. With the preceding notation this gives two bitorsors $E_p, E_{p'}$ under (G, G), namely the inverse images in E of the sections $p, p' : e \to P$, where e denotes the final object of $\mathcal{T}$. The multiplication of E induces a morphism of objects of $\mathcal{T}$

$$\varphi : E_p \times E_{p'} \longrightarrow E_{pp'}.$$

In terms of the left and right structures of the three bitorsors involved, it satisfies

$$(1.1.3.0)\begin{array}{l}\varphi(gx, x') = g\varphi(x, x'), \quad \varphi(x, x'g') = \varphi(x, x')g', \\ \varphi(xg, x') = \varphi(x, gx').\end{array} \tag{3}$$

The third relation can be interpreted by means of the contracted product of bitorsors,

$$E_p \overset{G}{\times} E_{p'},$$

and says that φ factors as

$$E_p \times E_{p'} \longrightarrow E_p \overset{G}{\times} E_{p'} \xrightarrow{\varphi_{p,p'}} E_{pp'}.$$

The two first relations then mean that

$$(1.1.3.1)\varphi_{p,p'} : E_p \overset{G}{\times} E_{p'} \xrightarrow{\sim} E_{pp'} \tag{4}$$

is an isomorphism of bitorsors, the bitorsor structure on the left being the one for which the left operations of G act on the first factor and the right operations act on the second factor. As every homomorphism of bitorsors under a fixed pair of Groups is necessarily an isomorphism, $\varphi_{p,p'}$ is an isomorphism.

Again, the definition extends when p, p' are points of P with values in an arbitrary object S of $\mathcal{T}$, by applying what precedes in the induced topos $\mathcal{T}/S$. The formation of the preceding $\varphi_{p,p'}$'s is compatible with arbitrary base change $S' \to S$. Equivalently, it gives a homomorphism of Cartesian sections over the category of double arrows $p, p' : S \rightrightarrows P$ with target P. The whole system may also be recovered from the universal case, $S = P \times P$ and p, p' the two projections. In that case it is an isomorphism of bitorsors

$$(1.1.3.2)\varphi : \mathrm{pr}_1^*(E) \overset{G}{\times} \mathrm{pr}_2^*(E) \xrightarrow{\sim} \pi^*(E), \tag{5}$$

where $\pi : P \times P \to P$ is multiplication in P. In practice the compatibilities are easiest to state using the data in the form (1.1.3.1), rather than by drawing less intuitive universal diagrams.

Writing the relations (1.1.3.0), which express the composition law of E through the bitorsor isomorphisms (1.1.3.1), uses associativity only when the product involves three points of E of which one at least comes from G. Full associativity is expressed by requiring that, for three points p, p', p'' of P, the following diagram commute:

$$(1.1.4.1)\quad \begin{tikzcd} & E_{pp'}E_{p''} \arrow[dr, "\varphi_{pp',p''}"] & \\ E_pE_{p'}E_{p''} \arrow[ur, "\varphi_{p,p'}\times\mathrm{id}"] \arrow[dr, "\mathrm{id}\times\varphi_{p',p''}"'] & & E_{pp'p''} \\ & E_pE_{p'p''} \arrow[ur, "\varphi_{p,p'p''}"'] & \end{tikzcd} \tag{6}$$

where the contracted product signs have been omitted. The convention is that the contracted product $Q.R$ of two bitorsors uses, for the quotient in $Q \times R$, the right action of the structural group G on Q and the left action of G on R. This contracted product is associative up to canonical isomorphism, and we henceforth make this identification, writing both $(E_pE_{p'})E_{p''}$ and $E_p(E_{p'}E_{p''})$ simply as $E_pE_{p'}E_{p''}$. Similar evident conventions will be used without further mention.

It is enough to verify this associativity in the universal case $S = P \times P \times P$, $p = \mathrm{pr}_1$, $p' = \mathrm{pr}_2$, $p'' = \mathrm{pr}_3$. We leave the reader to write the corresponding diagram in whatever notation he chooses.

Conversely, suppose P and G are given Groups of $\mathcal{T}$, together with a bitorsor

$$(1.1.5.1) j : E \longrightarrow P \tag{7}$$

under (G_P, G_P), and with a system of bitorsor isomorphisms over a variable base S

$$(1.1.5.2) \varphi_{p,p'} : E_pE_{p'} \xrightarrow{\sim} E_{pp'} \qquad (p, p' \in P(S)), \tag{8}$$

compatible with base change $S' \to S$, equivalently an isomorphism as in (1.1.3.2). Suppose moreover that the corresponding associativity diagrams (1.1.4.1) commute. Then there exists on E a unique structure of extension of P by G from which the data just described are obtained by the constructions above.

First take $p = p' = 1$, the unit section of P, in (1.1.5.2). One obtains an isomorphism

$$(1.1.5.3) \varphi_{1,1} : E_1 \xleftarrow{\sim} E_1 \overset{G}{\times} E_1. \tag{9}$$

Taking the contracted product of both sides with the inverse bitorsor E_1^{-1} gives a canonical isomorphism

$$(1.1.5.4) E_1 \xrightarrow{\sim} G_d^s, \tag{10}$$

where G_d^s is the unit bitorsor, with G acting on itself by left and right translations. Equivalently one has a section 1_E of E_1, hence of E,

$$(1.1.5.5) 1_E \in \Gamma(E_1) = \mathrm{Hom}(e, E_1) \subset \Gamma(E), \tag{11}$$

characterized by

$$g1_E = 1_Eg \quad (g \in G(S)),$$

and by

$$(1.1.5.6) \varphi_{1,1}(1_E, 1_E) = 1_E. \tag{12}$$

The system φ is equivalently a multiplication morphism

$$(1.1.5.7) \pi_E : E \times E \longrightarrow E, \qquad (x, y) \longmapsto xy, \tag{13}$$

such that

$$(1.1.5.8) j(xy) = j(x)j(y). \tag{14}$$

Finally define

$$(1.1.5.9) i : G \longrightarrow E, \qquad i(g) = g1_E = 1_E g. \tag{15}$$

Then π_E makes E a Group with unit 1_E, and $j : E \to P$ and $i : G \to E$ are homomorphisms of Groups forming an extension of P by G. Associativity of π_E is exactly the commutativity of (1.1.4.1). The construction by means of the $\varphi_{p,p'}$'s also shows that translations on $E(S)$ by an arbitrary element are injective. Since $1_E 1_E = 1_E$, the element 1_E is a two-sided identity. The inverse of $x \in E(S)$, lying over $p \in P(S)$, is obtained by using the canonical isomorphism

$$(1.1.5.10) E_{p^{-1}} \xrightarrow{\sim} (E_p)^{\circ}. \tag{16}$$

Thus E is a Group; moreover j is multiplicative by (1.1.5.8), and i is multiplicative by the calculation

$$i(g)i(g') = (g1_E)(g'1_E) = (g1_E g')1_E = g(1_E g')1_E = g(g'1_E) = i(gg').$$

Since j is the structural projection of a torsor, it is an epimorphism of objects of $\mathcal{T}$, and i identifies G with $\operatorname{Ker} j = E_1$.

We shall use the preceding arguments as proving that, for a fixed topos $\mathcal{T}$, the constructions above give an equivalence between the category of extensions (1.1.1) and the category of systems (P, G, E, φ) where P and G are Groups of $\mathcal{T}$, E is a bitorsor under (G_P, G_P) over P, and φ is an isomorphism of bitorsors as in (1.1.3.2), equivalently a base-change-compatible system (1.1.3.1), satisfying associativity (1.1.4.1). We also leave to the reader the evident description of inverse and direct images of extensions under a morphism of Groups

$$P' \longrightarrow P, \qquad \text{respectively} \qquad G \longrightarrow G',$$

and of the compatibility with inverse image f^* for a morphism of topoi $f : \mathcal{T}' \to \mathcal{T}$.

1.2. The commutative case

From 1.4 onward we shall be concerned only with extensions of commutative Groups. Thus P and G are assumed commutative. We spell out the preceding description in terms of bitorsors in the case where E is a commutative Group. A first necessary condition, expressing the fact that $i(G)$ is a central sub-Group of E, is that the bitorsors E_p be ordinary torsors; equivalently, the left and right actions of G on E_p coincide. Once this condition is imposed, commutativity of E is expressed by the commutativity, for $p, p' \in P(S)$, of the square

$$(1.2.1)\quad \begin{tikzcd} E_p E_{p'} \arrow[r, "\varphi_{p,p'}"] \arrow[d, "\mathrm{sym}"'] & E_{pp'} \arrow[d, "\mathrm{id}"] \\ E_{p'} E_p \arrow[r, "\varphi_{p',p}"'] & E_{p'p}. \end{tikzcd} \tag{17}$$

The first vertical arrow is the canonical commutativity isomorphism of the contracted product, defined because G is now commutative; and $pp' = p'p$ because P is commutative. The universal version over $P \times P$ is equivalent.

1.3. Rigidified and birigidified torsors

Let P be an object of $\mathcal{T}$ endowed with a section e_P, let G be a Group of $\mathcal{T}$, and let E be a torsor under G_P. A *rigidification* of E, relative to e_P, is a trivialization of the torsor

$$E_1 = e_P^*(E)$$

under G, i.e. a section of this torsor. The terminology is inspired by the case where G is commutative and

$$\text{(1.3.1.1)}\, \mathrm{Hom}_{\mathrm{pt}}(P, G) \simeq (e), \tag{18}$$

where the left-hand side denotes the set of morphisms transforming e_P into the unit section e_G of G. This condition is equivalent to the statement that the natural homomorphism

$$\text{(1.3.1.2)}\Gamma(G) \xrightarrow{\sim} \mathrm{Hom}(P, G) \tag{19}$$

which associates to a section g of G the constant morphism $P \to G$ with value g, is an isomorphism. Under this condition, every automorphism of a torsor respecting the rigidification is the identity.

Let Q also be an object of $\mathcal{T}$ with a section e_Q, and let E be a torsor under $G_{P\times Q}$. The section e_Q defines a section $e_{P\times Q/P}$ of $P \times Q$ over P; hence one may consider rigidifications α of E along $P \times e_Q$. Interchanging P and Q, one similarly has rigidifications β along $e_P \times Q$. Two such partial rigidifications are called *compatible* if the sections α and β coincide on $e_P \times e_Q$, i.e. if they define the same section of the induced torsor $E_{1,1}$ under G. A pair of compatible partial rigidifications is called a *birigidification* of E.

If G is commutative, a birigidification exists if and only if partial rigidifications exist, and one of the two partial rigidifications can be chosen in advance. Indeed one then corrects the other by a morphism $Q \to G$. Moreover two birigidified torsors under $G_{P\times Q}$ are isomorphic as torsors if and only if they are isomorphic as birigidified torsors, provided morphisms $P \times Q \to G$ with prescribed restrictions are controlled as above. Explicitly, if an isomorphism $u : E \to E'$ of $G_{P\times Q}$-torsors changes the two rigidifications by morphisms $f : P \to G$ and $g : Q \to G$, with the same value at the marked point, then the correction

$$h(p, q) = f(p)g(q)a^{-1}$$

gives an isomorphism respecting both rigidifications.

The same terminology extends to bitorsors. A rigidification of a bitorsor E under (G_P, G_P) is an isomorphism of $e_P^*(E)$ with the trivial bitorsor G_d^s, equivalently a central section e_E of $e_P^*(E)$, i.e. one satisfying

$$ge_E = e_E g.$$

Birigidifications for bitorsors are defined similarly.

Returning to two Groups P and G, choose the unit section e_P. There are natural functors

$$\text{(1.3.4.1)}\, \mathrm{EXT}(P; G) \xrightarrow{i} \mathrm{BITORSRIG}(P, G) \xrightarrow{\delta} \mathrm{BITORSBIRIG}(P, P; G), \tag{20}$$

where the three expressions denote respectively the category of extensions of Groups of P by G, the category of bitorsors under (G_P, G_P) rigidified at e_P, and the category of bitorsors under $(G_{P\times P}, G_{P\times P})$ birigidified at (e_P, e_P). The functor i assigns to an extension its corresponding bitorsor over P, rigidified by the unit of E. The functor δ is given by

$$\text{(1.3.4.2)}\delta(E) = \pi^*(E)\, \mathrm{pr}_1^*(E)^{-1}\, \mathrm{pr}_2^*(E)^{-1}, \tag{21}$$

where $\pi : P \times P \to P$ is multiplication and $\mathrm{pr}_1, \mathrm{pr}_2$ are the projections. There is a canonical birigidification of $\delta(E)$, obtained by restricting to $P \times e_P$ and $e_P \times P$, using the given rigidification of E.

The composite δi is canonically isomorphic to the trivial functor: the multiplication isomorphism φ of (1.1.3.2) trivializes $\delta i(E)$ compatibly with its canonical birigidification.

Proposition 1.3.5. Let P and Q be two Groups of $\mathcal{T}$.

a) Suppose that every morphism of sheaves of sets $P \times P \to G$ which is trivial on $P \times e_P$ and on $e_P \times P$ is trivial. Then the functor i in (1.3.4.1) is fully faithful. In particular, on a bitorsor E under (G_P, G_P), birigidified relative to e_P, there is at most one extension structure of P by G compatible with the bitorsor structure and admitting e_E as unit.

b) Suppose moreover that every morphism $P \times P \times P \to G$ trivial on the three partial products $P \times P \times e_P$, $P \times e_P \times P$, and $e_P \times P \times P$, is trivial. Let E be a torsor under G_P. An extension structure of P by G compatible with the torsor structure exists if and only if the birigidified torsor

$$\delta(E) = \pi^*(E)\, \mathrm{pr}_1^*(E)^{-1}\, \mathrm{pr}_2^*(E)^{-1}$$

is trivial. When this is so, every central section e_E of $E_1 = e_P^*(E)$, i.e. every rigidification of E, determines a unique extension structure on E, compatible with its bitorsor structure and admitting e_E as unit. Equivalently, the rigidified bitorsor lies in the essential image of i. In addition there is a unique trivialization of the birigidified torsor $\delta(E)$.

In the language of (1.3.4.1), this says that the essential image of the fully faithful functor i is the kernel of δ; the diagram is therefore an exact sequence of pointed categories, and in fact of categories endowed with the pseudo-group structures introduced below.

For (a), let E, E' be two extensions of P by G, and let $f : E \to E'$ be a morphism of the underlying rigidified bitorsors. We show that f is a morphism of extensions, equivalently a homomorphism of Groups. It is enough to prove $f(xy) = f(x)f(y)$. Since both sides have the same image in P, there is a well-defined morphism

$$u : P \times P \longrightarrow G$$

such that

$$f(xy) = u(j(x), j(y))\, f(x)f(y).$$

Putting $x = 1$ and $y = 1$, and using $f(1) = 1$, gives

$$u(p, 1) = u(1, p) = 1.$$

By the hypothesis, u is trivial; hence $f(xy) = f(x)f(y)$.

For (b), the necessity of the condition is the triviality of δi. Conversely, choose a trivialization φ of $\delta(E)$. Pulling it back by the unit section of $P \times P$ gives a rigidification e_E of E. A chosen trivialization of $\delta(E)$ is not necessarily compatible with this birigidification; its restrictions to $P \times e_P$ and $e_P \times P$ are expressed by central morphisms

$$f : P \to G, \qquad g : P \to G,$$

with common value at the unit section. The central morphism

$$h : P \times P \to G, \qquad h(p, q) = f(p)g(q)a^{-1}$$

has these prescribed restrictions. Replacing φ by the corrected trivialization gives one compatible with the birigidification. The ambiguity is a central morphism $P \times P \to G$ trivial on $P \times e_P$ and $e_P \times P$, hence trivial by hypothesis; thus the compatible trivialization is unique. It remains only to verify associativity. The two composites in (1.1.4.1) differ by a central morphism

$$u : P \times P \times P \to G,$$

given by

$$(xy)z = u(jx, jy, jz)\, x(yz).$$

Specializing successively $y = z = 1$, $z = x = 1$, and $x = y = 1$, and using the fact that 1 is a two-sided identity, shows that u is trivial on the three partial products; by the hypothesis it is trivial. This proves (b).

Corollary 1.3.6. Suppose every morphism of sheaves of sets $P \times G \to G$ trivial on $P \times e_G$ and on $e_P \times G$ is trivial. Then, if G is commutative, G is central in every extension of Groups of P by G. If P is also commutative and the hypothesis of Proposition 1.3.5(a) is satisfied, then every extension of P by G is commutative.

The first assertion follows by observing that the action of P on G defined by an extension is represented by a morphism $u : P \times G \to G$ such that the corresponding morphism $(p, g) \mapsto u(p, g)g^{-1}$ is trivial on $P \times e_G$ and on $e_P \times G$. Hence $u(p, g) = g$. For the second assertion, compare the extension E with its opposite E°. If the extension is central and $P = P^\circ$, then the uniqueness assertion in Proposition 1.3.5(a) gives $E = E^\circ$.

A particularly interesting case where all the hypotheses just used are satisfied is

$$(1.3.7.1)\, \mathrm{Hom}_{\mathrm{pt}}(P, G) = e \tag{22}$$

(the final sheaf), where the left-hand side denotes morphisms of pointed sheaves from P to G. Equivalently,

$$(1.3.7.2) G \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{ens}}(P, G), \tag{23}$$

where the second member denotes morphisms of the underlying sheaves of sets. This condition implies that, for every object Q of $\mathcal{T}$, the map

$$(1.3.7.3)\, \mathrm{Hom}_{\mathrm{ens}}(Q, G) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{ens}}(P \times Q, G), \qquad u \longmapsto u \circ \mathrm{pr}_2 \tag{24}$$

is bijective. Hence the hypotheses of 1.3.5 and 1.3.6 are satisfied; the categories in (1.3.4.1) are rigid. Since the condition is stable under all base change $S \to e$, descent and the uniqueness part of 1.3.5(a) imply that a rigidified bitorsor E over P comes from an extension of P by G as soon as it does so locally over e. In view of 1.3.5(b), this is expressed by the vanishing of the section

$$\delta(E) \in \Gamma R^1 h_*(G_{P \times P}),$$

where

$$h : P \times P \to e$$

is the structural morphism.

Let $p : P \to S$ be a coherent, that is quasi-compact and quasi-separated, flat morphism of schemes, and suppose

$$(1.3.8.1) p_*(\mathcal{O}_P) \xleftarrow{\sim} \mathcal{O}_S. \tag{25}$$

Because p is coherent, the formation of $p_*(\mathcal{O}_P)$ commutes with all flat base change. It follows that the conditions just considered for the relative scheme P/S are stable under Cartesian products, hence hold for $P \times P$, $P \times P \times P$, and so on, and are stable under flat base change. If G is affine over S, then (1.3.8.1) gives (1.3.7.2), since for $G = \operatorname{Spec} A$,

$$\operatorname{Hom}_S(P, G) = \operatorname{Hom}_{\text{alg}}(A, p_*\mathcal{O}_P) \simeq \operatorname{Hom}_{\text{alg}}(A, \mathcal{O}_S).$$

The same assertion applies to the Cartesian products, and, if G is flat over S, to the relative scheme P_G over G_G. Hence, when P and G are group schemes over S, viewed as sheaves on a topology on (Sch)$/S$ no finer than the canonical topology, every morphism from P, $P \times P$, $P \times P \times P$, and so on to G, compatible with the marked sections, is trivial. Also every S-morphism $P \times G \to G$ compatible with marked sections is trivial. Thus the hypotheses of 1.3.5 and 1.3.6 apply, and if G is flat the same is true for the corresponding relative statement.

The most interesting case in which (1.3.8.1) holds, and remains true after all base extension, is that of an abelian scheme P over S. Thus the conclusions of 1.3.5 and 1.3.6 are valid for every affine S-group G. This is the case first considered by J.-P. Serre, and the preceding developments are visibly inspired by it. Another interesting case in which (1.3.8.1) holds, though not universally, is when P is the Néron model, over a discrete valuation base S, of an abelian scheme over the fraction field.

1.4. Extensions of a free commutative Group

Let I be an object of $\mathcal{T}$, and put

$$(1.4.1) P = \mathbb{Z}[I], \tag{26}$$

the free $\mathbb{Z}$-Module generated by I. To an extension E of P by a given $\mathbb{Z}$-Module G we associate the corresponding torsor under G_P, and then its restriction to I via the canonical morphism

$$I \longrightarrow P = \mathbb{Z}[I].$$

This gives a canonical functor

$$(1.4.2)\, \operatorname{EXT}(P, G) \longrightarrow \operatorname{TORS}(I, G_I). \tag{27}$$

This functor is an equivalence of categories. Equivalently, for every G the maps

$$\operatorname{Hom}_{\mathbb{Z}}(P, G) \longrightarrow H^0(I, G_I) = G(I), \qquad (1.4.3) \tag{28}$$

$$\operatorname{Ext}^1_{\mathbb{Z}}(P, G) \longrightarrow H^1(I, G_I) \qquad (1.4.4) \tag{29}$$

are isomorphisms. More generally, for any ring object A of $\mathcal{T}$, not necessarily commutative, and $P = A[I]$ the free A-Module generated by I, there are functorial canonical isomorphisms

$$(1.4.5)\, \operatorname{Ext}^i_A(A[I], G) \xrightarrow{\sim} H^i(I, G_I), \tag{30}$$

which for $A = \mathbb{Z}$ and $i = 0, 1$ are (1.4.3) and (1.4.4). Indeed, the first member is the derived functor of

$$G \longmapsto \operatorname{Hom}_A(A[I], G),$$

which is, by the definition of $A[I]$, isomorphic to $G \mapsto G(I) = H^0(I, G_I)$. Since $G \mapsto G_I$ sends injective Modules to injective Modules of $\mathcal{T}/I$, one obtains the asserted result. The functor (1.4.2) is compatible with inverse images under a morphism of topoi

$$f : \mathcal{T}' \longrightarrow \mathcal{T}.$$

Exposé VII. Biextensions of sheaves of groups: section 2 and opening of section 3

2. The notion of biextension of abelian sheaves

2.0. Notation

Throughout this section, P, Q, G denote commutative Groups of the topos $\mathcal{T}$. We shall often use the Cartesian square

$$(2.0.1)\quad \begin{array}{ccc} P \times Q & \longrightarrow & P \\ \downarrow & & \downarrow \\ Q & \longrightarrow & e. \end{array} \tag{31}$$

The Groups G_P, G_Q, and $G_{P\times Q}$ are the inverse images of G on P, Q, and $P \times Q$. We shall identify $G_{P\times Q}$ equally with $(G_P)_{P\times Q}$ via pr_1, and with $(G_Q)_{P\times Q}$ via pr_2. In this notation the group structures of $P, Q, P \times Q$ do not intervene. We shall almost never need the product group structure on $P \times Q$. What will be useful is to consider $P \times Q$ as the inverse-image P-Group of Q,

$$(2.0.2) P \times Q = Q_P, \tag{32}$$

and symmetrically as the inverse-image Q-Group of P,

$$(2.0.3) P \times Q = P_Q. \tag{33}$$

Thus, over P, one has the two Groups G_P and Q_P; over Q, one has G_Q and P_Q. If E is a torsor on $P \times Q$ with group $G_{P\times Q}$, then E may be regarded either as a torsor on P of group $(G_P)_{Q_P}$, or as a torsor on Q of group $(G_Q)_{P_Q}$. It therefore makes sense to speak of a commutative extension structure of Q_P by G_P on E, compatible with its torsor structure, and symmetrically of an extension structure of P_Q by G_Q on E.

Applying the dictionary of 1.6 in the induced topoi, a structure of extension of Q_P by G_P on E is described by the following data. For every object S of $\mathcal{T}$, for $p, p' \in P(S)$ and $q \in Q(S)$, one is given an isomorphism of G_S-torsors

$$(2.0.4) \lambda_{p,p';q} : E_{p,q}\, E_{p',q} \xrightarrow{\sim} E_{pp',q}. \tag{34}$$

Here $E_{p,q}$ denotes the inverse image of E under the point $(p, q) : S \to P \times Q$. These isomorphisms are functorial in S, i.e. compatible with pullback along morphisms $S' \to S$ of $\mathcal{T}$, and satisfy the associativity condition expressed by the commutativity of

$$(2.0.5)\quad \begin{array}{ccccc} & & E_{pp',q}E_{p'',q} & & \\ & \nearrow^{\lambda_{p,p';q}\times \mathrm{id}} & & \searrow^{\lambda_{pp',p'';q}} & \\ E_{p,q}E_{p',q}E_{p'',q} & & & & E_{pp'p'',q} \\ & \searrow_{\mathrm{id}\times\lambda_{p',p'';q}} & & \nearrow_{\lambda_{p,p'p'';q}} & \\ & & E_{p,q}E_{p'p'',q} & & \end{array} \tag{35}$$

and the commutativity condition

$$
(2.0.6)\quad
\begin{tikzcd}
E_{p,q}E_{p',q} \arrow[r, "\lambda_{p,p';q}"] \arrow[d, "\mathrm{sym}"'] & E_{pp',q} \arrow[d, "\mathrm{id}"] \\
E_{p',q}E_{p,q} \arrow[r, "\lambda_{p',p;q}"'] & E_{p'p,q}.
\end{tikzcd}
\tag{36}
$$

Symmetrically, a structure of extension of P_Q by G_Q on E is described by isomorphisms

$$
(2.0.7)\mu_{p;q,q'} : E_{p,q}\, E_{p,q'} \xrightarrow{\sim} E_{p,qq'} \tag{37}
$$

for $p \in P(S)$, $q, q' \in Q(S)$, functorial in S, satisfying the analogous associativity and commutativity conditions

$$
(2.0.8)\quad
\begin{tikzcd}
 & E_{p,qq'}E_{p,q''} \arrow[dr, "\mu_{p;qq',q''}"] & \\
E_{p,q}E_{p,q'}E_{p,q''} \arrow[ur, "\mu_{p;q,q'}\times\mathrm{id}"] \arrow[dr, "\mathrm{id}\times\mu_{p;q',q''}"'] & & E_{p,qq'q''} \\
 & E_{p,q}E_{p,q'q''} \arrow[ur, "\mu_{p;q,q'q''}"'] &
\end{tikzcd}
\tag{38}
$$

and

$$
(2.0.9)\quad
\begin{tikzcd}
E_{p,q}E_{p,q'} \arrow[r, "\mu_{p;q,q'}"] \arrow[d, "\mathrm{sym}"'] & E_{p,qq'} \arrow[d, "\mathrm{id}"] \\
E_{p,q'}E_{p,q} \arrow[r, "\mu_{p;q',q}"'] & E_{p,q'q}.
\end{tikzcd}
\tag{39}
$$

Definition 2.1. Let P, Q, G be abelian Groups of $\mathcal{T}$, and let E be a torsor under $G_{P\times Q}$. Suppose that E is endowed with a structure of extension of Q_P by G_P, given by the λ's above, and with a structure of extension of P_Q by G_Q, given by the μ's above. The two structures are said to be *compatible* if, for every object S of $\mathcal{T}$ and every $p, p' \in P(S)$, $q, q' \in Q(S)$, the following square of contracted products commutes:

$$
(2.1.1)\quad
\begin{tikzcd}
(E_{p,q}E_{p,q'})E_{p',q'} \arrow[r, "\mu_{p;q,q'}\times\mathrm{id}"] \arrow[d, "\mathrm{id}\times\lambda_{p,p';q'}"'] & E_{p,qq'}E_{p',q'} \arrow[d, "\lambda_{p,p';qq'}"] \\
E_{p,q}E_{pp',q'} \arrow[r, "\mu_{pp';q,q'}"'] & E_{pp',qq'}.
\end{tikzcd}
\tag{40}
$$

Equivalently, in the partial-composition notation below, this is the identity

$$
(2.1.1.2)\lambda\big(\mu(x,y),\mu(z,t)\big) = \mu\big(\lambda(x,z),\lambda(y,t)\big), \tag{41}
$$

for x, y, z, t lying respectively in $E_{p,q}(S), E_{p,q'}(S), E_{p',q}(S), E_{p',q'}(S)$, with the evident notation.

A *biextension* of (P, Q) by G is a torsor E under $G_{P\times Q}$ endowed with these two extension structures, compatible in the preceding sense.

In more elementary language, a biextension structure on E is given by two partially defined laws of composition, denoted $\lambda(x, y)$ and $\mu(x, y)$. The first is defined when $\mathrm{pr}_2\, j(x) = \mathrm{pr}_2\, j(y)$, the second when $\mathrm{pr}_1\, j(x) = \mathrm{pr}_1\, j(y)$, where $j : E \to P \times Q$ is the structural morphism. The axioms are the associativity and commutativity of these two partial laws, the relations

$$(2.1.1.1)\lambda(gx, y) = \lambda(x, gy) = g\lambda(x, y) \tag{42}$$

and the analogous relations for μ, together with the mixed relation (2.1.1.2).

2.2. Unit sections

There are unit sections for E considered as an extension of Q_P by G_P,

$$(2.2.1)e_{E/P} \in \Gamma(E/P) = \mathrm{Hom}_P(P, E), \tag{43}$$

and, symmetrically, for E considered as an extension of P_Q by G_Q,

$$(2.2.2)e_{E/Q} \in \Gamma(E/Q) = \mathrm{Hom}_Q(Q, E). \tag{44}$$

Thus the torsor E over $P \times Q$ is canonically trivialized over $1_P \times Q$, by $e_{E/P}$, and over $P \times 1_Q$, by $e_{E/Q}$. These two trivializations agree over $1_P \times 1_Q$. They therefore define one section

$$(2.2.3)e_E \in \Gamma(E), \tag{45}$$

called, by abuse of language, the unit or neutral section of E. The equality follows by applying the compatibility relation to the four sections e, e', e', e over $1_P \times 1_Q$: the two sides are respectively e and e'. Consequently the two partial laws λ and μ coincide on the fibre over $1_P \times 1_Q$, both being transported from the group law of G.

Restricting the extension of Q_P by G_P to the unit section of P gives an extension of Q by G which is canonically split by $e_{E/P}$. The symmetric statement holds for the restriction over the unit section of Q.

2.3. Morphisms of biextensions

Let E be a biextension of (P, Q) by G, and let E' be a biextension of (P', Q') by G'. A morphism of biextensions from E to E' is a system (u, v, w, f), where

$$(2.3.1)u : P \to P', \qquad v : Q \to Q', \qquad w : G \to G' \tag{46}$$

are morphisms of Groups and

$$(2.3.2)f : E \to E' \tag{47}$$

is a morphism of the underlying sheaves of sets such that f is both a homomorphism of the relative extensions over P and P', associated with the relative homomorphisms $u \times v$ and $u \times w$, and a homomorphism of the relative extensions over Q and Q', associated with $u \times v$ and $v \times w$.

The maps u, v, w are determined by f: one recovers $u \times v$ by passing to the quotient, hence u and v by restriction to $P \times 1_Q$ and $1_P \times Q$, and one recovers w by restricting f to $G \cdot e_E$. We shall therefore often identify the morphism with f itself. Adding source and target to morphisms in the usual canonical way, biextensions form a category. There is a canonical functor

$$(2.3.3)\ \mathrm{BIEXT}(\mathcal{T}) \longrightarrow \mathrm{GRAB}(\mathcal{T}) \times \mathrm{GRAB}(\mathcal{T}) \times \mathrm{GRAB}(\mathcal{T}), \tag{48}$$

where $\mathrm{GRAB}(\mathcal{T})$ denotes abelian Groups of $\mathcal{T}$. It is useful to decompose it as

$$(2.3.4)\ \mathrm{BIEXT}(\mathcal{T}) \longrightarrow \mathrm{GRAB}(\mathcal{T}) \times \mathrm{GRAB}(\mathcal{T}), \tag{49}$$

by $E \mapsto (P, Q)$, and

$$(2.3.5)\ \mathrm{BIEXT}(\mathcal{T}) \longrightarrow \mathrm{GRAB}(\mathcal{T}), \tag{50}$$

by $E \mapsto G$.

2.4. The categories $\mathrm{BIEXT}(P, Q; G)$ with variables

The functor (2.3.4) is fibred, while the functor (2.3.5) is cofibred. Thus, given a biextension E of (P, Q) by G, and morphisms $u : P' \to P$, $v : Q' \to Q$, there is an inverse-image biextension E' of (P', Q') by G, together with a universal morphism $E' \to E$ compatible with u, v, id_G. Similarly, given $w : G \to G'$, there is a direct-image biextension E' of (P, Q) by G', together with a universal morphism $E \to E'$ compatible with $\mathrm{id}_P, \mathrm{id}_Q, w$. These are the usual constructions for extensions of Groups, paraphrased in the present setting.

For fixed P, Q, G, write

$$(2.4.1)\, \mathrm{BIEXT}(P, Q; G) \tag{51}$$

for the fibre category over (P, Q, G). This category is covariant in G and contravariant in P, Q, and additively dependent on each of its three arguments. For the argument G, for example, this says that $\mathrm{BIEXT}(P, Q; 0)$ is equivalent to the punctual category, and that for two abelian Groups G, G' of $\mathcal{T}$, the functor

$$(2.4.2)\, \mathrm{BIEXT}(P, Q; G \times G') \longrightarrow \mathrm{BIEXT}(P, Q; G) \times \mathrm{BIEXT}(P, Q; G') \tag{52}$$

induced by the two projections is an equivalence. In the language of the cited reference, for fixed P, Q the category of biextensions of (P, Q) by variable Groups G is an additive cofibred category over $\mathrm{GRAB}(\mathcal{T})$.

Remark 2.4.3. With the terminology of loc. cit., this cofibred category is not only additive but left exact. We shall not verify this here, since it will follow below, in 3.5, from more precise facts. The general theory would then describe this cofibred category, up to equivalence, in terms of a chain complex $L_\bullet$ in $\mathrm{GRAB}(\mathcal{T})$, depending only on P and Q. Below we shall give a direct construction of this complex as $P \otimes^L Q$, the tensor product being taken in a derived category.

2.5. The structure of $\mathrm{BIEXT}(P, Q; G)$

The formal facts for additive cofibred categories give useful consequences. For fixed G, two biextensions E, E' of (P, Q) by G have a product biextension $E.E'$. This construction is the analogue of the product of two extensions of one commutative Group by another, and is compatible with viewing E, E' either as extensions of Q_P by G_P, or as extensions of P_Q by G_Q. As a $G_{P\times Q}$-torsor, $E.E'$ is the product of the torsors underlying E and E'. This law is associative and commutative up to canonical isomorphism and has a two-sided unit, the trivial biextension. This unit can be described as the trivial torsor

$$(2.5.1.1) P \times Q \times G, \tag{53}$$

with its two relative group structures induced from the product group structures on $Q \times G$ and $P \times G$. Every biextension E also has an inverse E^{-1}, characterized up to unique isomorphism by an isomorphism $E.E^{-1} \simeq 1$. Thus $\mathrm{BIEXT}(P, Q; G)$ is a pseudo-group in the 2-category of categories.

It follows that, for any biextension E, its endomorphism monoid is canonically isomorphic to the endomorphism monoid of the trivial biextension. This latter monoid is a commutative group, denoted

$$(2.5.2.1)\, \mathrm{Biext}^0(P, Q; G). \tag{54}$$

Moreover the product law on biextensions makes the set

$$(2.5.2.2)\, \mathrm{Biext}^1(P, Q; G) \tag{55}$$

of isomorphism classes of biextensions of (P, Q) by G into a commutative group.

For a homomorphism $G \to G'$, the induced functor

$$(2.5.3.1)\,\mathrm{BIEXT}(P, Q; G) \longrightarrow \mathrm{BIEXT}(P, Q; G') \tag{56}$$

is compatible with products and inverses up to canonical isomorphism. Hence it induces homomorphisms

$$(2.5.3.2)\,\mathrm{Biext}^i(P, Q; G) \longrightarrow \mathrm{Biext}^i(P, Q; G') \qquad (i = 0, 1). \tag{57}$$

For $i = 0$, this is the natural map on automorphism groups of a biextension and its image by $G \to G'$.

The group $\mathrm{Biext}^0(P, Q; G)$ is explicit. An automorphism u of a biextension E is first an automorphism of the underlying $G_{P\times Q}$-torsor, and is therefore given by a morphism

$$(2.5.4.0)u : P \times Q \longrightarrow G, \tag{58}$$

where $u(x) = u(j(x))\,x$ in torsor notation. Compatibility with the first extension structure is equivalent to additivity in the second variable; compatibility with the second extension structure is equivalent to additivity in the first variable. Thus u is an automorphism of the biextension structure if and only if it is bilinear. This gives the canonical bijection, in fact isomorphism of groups,

$$(2.5.4.1)\,\mathrm{Biext}^0(P, Q; G) \simeq \mathrm{Hom}(P \otimes Q, G), \tag{59}$$

functorial in P, Q, G. Below we shall determine $\mathrm{Biext}^1(P, Q; G)$ as an Ext^1-group.

2.6. Variance in P, Q, and G

The contravariance in P, Q commutes with the covariance in G. Hence the categories just considered can be regarded as fibres of a tri-additive cofibred category

$$(2.6.1)\,\mathrm{BIEXT}'(\mathcal{T}) \longrightarrow \mathrm{GRAB}(\mathcal{T})^\circ \times \mathrm{GRAB}(\mathcal{T})^\circ \times \mathrm{GRAB}(\mathcal{T}), \tag{60}$$

where the structural functor is $E \mapsto (P, Q, G)$. The pseudo-group structures obtained by using the additivity in any one of the three variables are canonically isomorphic. Consequently the abelian groups

$$(2.6.2)\,\mathrm{Biext}^i(P, Q; G) \qquad (i = 0, 1) \tag{61}$$

are trilinear functors: contravariant in P, Q, covariant in G, and valued in abelian groups. The isomorphism (2.5.4.1) is functorial for this full variance, and the same will hold for the later description of Biext^1 by an Ext^1.

2.7. The symmetric biextension

If E is a biextension of (P, Q) by G, define its symmetric biextension ${}^sE$, a biextension of (Q, P) by G, as follows. The underlying sheaf of sets and the G-action are the same. The structural morphism is the composite of $E \to P \times Q$ with the symmetry $P \times Q \to Q \times P$. The left and right extension structures of ${}^sE$ are respectively the right and left extension structures of E. The operation $E \mapsto {}^sE$ is an involutive automorphism of $\mathrm{BIEXT}(\mathcal{T})$:

$$(2.7.1)^s({}^sE) = E. \tag{62}$$

Thus every assertion concerning biextensions has a symmetric assertion obtained by interchanging left and right. In practice only one of the two symmetric statements will usually be written out.

2.8. Change of topos and localization

Let

$$(2.8.0) f : \mathcal{T}' \longrightarrow \mathcal{T} \quad (63)$$

be a morphism of topoi. The inverse-image functor f^* sends a biextension E of (P, Q) by G to a biextension f^*E of (f^*P, f^*Q) by f^*G. This is another instance of the fact that such a structure is expressible by finite projective limits and arbitrary inductive limits. Compatibility of the two extension structures on f^*E follows from the universal form of the commutative diagram (2.1.1). Hence one obtains a canonical functor

$$(2.8.1) f^* : \mathrm{BIEXT}(\mathcal{T}) \longrightarrow \mathrm{BIEXT}(\mathcal{T}'). \quad (64)$$

All constructions with biextensions used later are compatible with inverse images by morphisms of topoi; in particular this includes the product $E.E'$ and inverse E^{-1} of 2.5.

Taking $\mathcal{T}' = \mathcal{T}/S'$ and $f = i_{S'} : \mathcal{T}/S' \to \mathcal{T}$, one obtains the biextension induced by E on S', denoted $E|S'$ or $E_{S'}$. The usual transitivity of restriction implies that the categories $\mathrm{BIEXT}(\mathcal{T}/S)$, for variable $S \in \mathcal{T}$, are the fibres of a fibred category over $\mathcal{T}$. This fibred category is a stack on $\mathcal{T}$ in Giraud's sense: morphisms and objects can be glued. Thus a biextension on S, up to canonical isomorphism, is obtained from a covering family $S_i \to S$, biextensions E_i on S_i, and descent isomorphisms on $S_i \times_S S_j$ satisfying the usual cocycle condition. This stack is the stack of biextensions on $\mathcal{T}$. The functors (2.3.3)–(2.3.5), and the operations $E.E'$ and E^{-1}, are morphisms of stacks or stack operations in this sense.

2.9. Relation with birigidified torsors

We saw in 2.2 that every biextension E of (P, Q) by G has an underlying $G_{P\times Q}$-torsor canonically birigidified with respect to the unit sections of P and Q. This gives a canonical functor

$$(2.9.1)\, \mathrm{BIEXT}(P, Q; G) \longrightarrow \mathrm{TORSBIRIG}(P, Q; G), \quad (65)$$

where the target is the category of birigidified torsors on $P \times Q$, with structural group $G_{P\times Q}$.

Using the unit sections, for every finite product of factors isomorphic to P or Q one has a canonical inclusion of each factor. We shall therefore say that a morphism from such a product to G is trivial on a given factor. Consider the products

$$(2.9.2) P\times Q, \quad P\times Q\times Q, \quad P\times Q\times Q\times Q, \quad P\times P\times Q, \quad P\times P\times P\times Q, \quad P\times P\times Q\times Q. \quad (66)$$

Proposition 2.9.3. Suppose that every morphism from one of the products (2.9.2) to G, trivial on each of its factors, is trivial. Then:

a) The functor (2.9.1) is fully faithful, and both categories are rigid. In particular, if E is a $G_{P\times Q}$-torsor birigidified for (e_P, e_Q), then there is at most one biextension structure on E compatible with its birigidified torsor structure.

b) Such a structure exists on E if and only if there exists on E a structure of extension of Q_P by G_P, compatible with its G_P-torsor structure and the rigidification on $P \times e_Q$, and a structure of extension of P_Q by G_Q, compatible with its G_Q-torsor structure and the rigidification on $e_P \times Q$. In that case both extension structures are unique and are compatible, hence define a biextension.

For (a), the automorphism group of a birigidified torsor on $P \times Q$ is the group of morphisms $P \times Q \to G$ trivial on the two partial factors; by the hypothesis it is the identity. If E, E' are biextensions and $f : E \to E'$ is a morphism of birigidified torsors, Proposition 1.3.5(a), applied to the two relative extension structures, shows that f is a morphism of biextensions. For (b), Proposition 1.3.5(b) gives the unique relative extension structures; the two composites in the compatibility diagram (2.1.1) differ by a morphism $P \times P \times Q \times Q \to G$, trivial on each factor, hence trivial by the hypothesis.

Corollary 2.9.4. If P, Q, G satisfy

$$(2.9.4.1)\, \mathrm{Hom}_{\mathrm{pt}}(P, G) = e, \qquad \mathrm{Hom}_{\mathrm{pt}}(Q, G) = e, \tag{67}$$

then the hypotheses of Proposition 2.9.3 hold. Moreover, for a birigidified $G_{P\times Q}$-torsor E, the existence of a compatible biextension structure is equivalent to the condition that the two pointed morphisms which it defines,

$$(2.9.4.2) Q \longrightarrow R^1p_*(G_P), \qquad P \longrightarrow R^1q_*(G_Q), \tag{68}$$

where $p : P \to e$ and $q : Q \to e$ are the structural morphisms, are morphisms of Groups.

The first assertion follows from 1.3.7. For the second, multiplicativity of the first morphism says that the rigidified torsor locally satisfies the condition of 1.3.5(b) over P; by 1.3.7 it then satisfies it globally. The symmetric argument applies to the second morphism, and Proposition 2.9.3(b) gives the conclusion.

Example 2.9.5. Let S be a scheme, let P, Q be abelian schemes over S, and let $G = \mathbb{G}_m$. As noted in 1.3.8, condition (2.9.4.1) holds. A birigidified $\mathbb{G}_m$-torsor on $P \times Q$ is the same as a birigidified invertible Module on $P \times Q$. A divisor correspondence on (P, Q) is the isomorphism class of such a birigidified invertible Module. With the fppf topology, the two morphisms (2.9.4.2) are the morphisms

$$(2.9.5.1) Q \longrightarrow \mathrm{Pic}(P/S), \qquad P \longrightarrow \mathrm{Pic}(Q/S) \tag{69}$$

defined by the divisor correspondence; by the theorem of the square these are necessarily compatible with the group structures. Thus the category of biextensions of (P, Q) by $\mathbb{G}_m$ is equivalent to the category of birigidified invertible Modules on $P \times Q$. This category is rigid, and its set of isomorphism classes is the set of divisor correspondences on (P, Q). The identification respects the natural group laws.

Remark 2.9.6. Additivity of the first morphism in (2.9.4.2) is equivalent to saying that the second morphism factors through the subsheaf $\mathrm{Ext}^1(Q, G)$; the canonical map

$$(2.9.6.0)\, \mathrm{Ext}^1(Q, G) \longrightarrow R^1q_*(G_Q) \tag{70}$$

is injective by 1.3.5. Symmetrically, additivity of the second morphism is equivalent to the factorization of the first through $\mathrm{Ext}^1(P, G)$. Hence one obtains group isomorphisms

$$(2.9.6.1)\, \mathrm{Biext}(P, Q; G) \simeq \mathrm{Hom}(Q, \mathrm{Ext}^1(P, G)) \simeq \mathrm{Hom}(P, \mathrm{Ext}^1(Q, G)). \tag{71}$$

In the situation of Example 2.9.5 these become the classical isomorphisms

$$(2.9.6.2)\, \mathrm{Corr}(P, Q) \simeq \mathrm{Hom}(P, Q') \simeq \mathrm{Hom}(Q, P'), \tag{72}$$

where $P' = \mathrm{Ext}^1(P, \mathbb{G}_m)$ and $Q' = \mathrm{Ext}^1(Q, \mathbb{G}_m)$ are the dual abelian schemes, identified with $\mathrm{Pic}^0_{P/S}$ and $\mathrm{Pic}^0_{Q/S}$.

2.10. Various generalizations of biextensions

We mention these only for completeness.

First, one can define biextensions of (P,Q) by G without assuming that P and Q are commutative, while still assuming G commutative. Such an object is a torsor under $G_{P\times Q}$, endowed with central extension structures of Q_P by G_P and of P_Q by G_Q, compatible with the torsor structure and with one another by diagrams of the form (2.1.1). Most of the preceding development applies mutatis mutandis. When P and Q are also commutative, this is more general than Definition 2.1, which corresponds to the case in which both relative extension structures are commutative. In the non-commutative case the category still depends additively on G, but not on P and Q.

Second, for a fixed integer $n \geq 1$, one defines in the same way an n-extension of a sequence of n commutative Groups $(P_1, \dots, P_n)$ by a commutative Group G: it is a torsor over $P_1 \times \cdots \times P_n$ under the inverse image of G, with n partial extension structures on the products obtained by omitting one factor, and a compatibility relation of type (2.1.1) for every pair of distinct indices. When $n = 3$, the P_i are abelian schemes over S, and $G = \mathbb{G}_m$, every such n-extension is trivial.

Third, biextensions of A-Modules may be defined. Let A be a commutative ring object of $\mathcal{T}$, and let P, Q, G be A-Modules. A biextension of A-Modules of (P,Q) by G is a biextension in the sense of 2.1, in which the two partial extension structures are enriched to extensions of relative A_P- and A_Q-Modules, compatible with the relative Module structures on G_P, Q_P and G_Q, P_Q. The two relative Module structures on E amount to two actions of the multiplicative monoid underlying A on the sheaf underlying E, which are required to commute. Each action must be distributive with respect not only to its corresponding partial composition law on E, but also with respect to the other partial composition law.

The dictionary of 1.1.6 also has an A-linear variant: an extension of the A-Module P by G corresponds to a family of torsors E_p under G_S, parameterized by points $p \in P(S)$, which varies A-linearly in p. In addition to the isomorphisms $\varphi_{p,p'}$ of (1.1.3.1), one has isomorphisms

$$(2.10.3.1)\mu_{p,a} : a_*(E_p) \xrightarrow{\sim} E_{ap} \qquad (p \in P(S),\ a \in A(S)), \tag{73}$$

where the first member denotes the image of the G_S-torsor E_p by extension of scalars $g \mapsto ag : G_S \to G_S$. The conditions on φ and μ are the evident ones ensuring that they come from an extension of Modules.

Exposé VII, §3. Complements on homological algebra

The principal aim of this number is to define the isomorphism announced above,

$$(3.0.1)\,\mathrm{Biext}^1(P,Q;G) \simeq \mathrm{Ext}^1(P \otimes^L Q, G). \tag{74}$$

For this purpose we need some homological algebra, useful independently, occupying §§3.1–3.4. These developments may be regarded as complements to the theory of additive cofibred categories used earlier.

3.1. The additive cofibred category attached to a chain complex

Let $\mathcal{C}$ be an abelian category and let

$$(3.1.1) \cdots \longrightarrow L_2 \xrightarrow{d_2} L_1 \xrightarrow{d_1} L_0 \tag{75}$$

be a chain complex in $\mathcal{C}$, written with lower indices and differential of degree -1. Thus $L_i = 0$ for $i < 0$. We define a cofibred category $\ominus_L$ over $\mathcal{C}$. An object of the fibre $\ominus_L(A)$ is a pair (X, a) consisting of an extension

$$(3.1.2) 0 \longrightarrow A \longrightarrow X \xrightarrow{j} L_0 \longrightarrow 0 \tag{76}$$

and a partial trivialization of its inverse image by $d_1 : L_1 \to L_0$. Equivalently it is a lifting

$$(3.1.3) \quad \begin{tikzcd} & X \arrow[d, "j"] \arrow[dl] & \\ A \arrow[r] & L_0 & L_1 \arrow[l, "d_1"'] \arrow[ul, "a"'] \end{tikzcd} \qquad \text{with} \qquad a\, d_2 = 0. \tag{77}$$

This description depends only on the truncation of $L_\bullet$ to order 1, obtained from $L_\bullet$ by killing the homology in degrees ≥ 2. A morphism $(X, a) \to (X', a')$ is a morphism of extensions over L_0 compatible with the identity on A and satisfying $ua = a'$.

The obvious functor

$$(3.1.4) \ominus_L \longrightarrow \mathcal{C}, \qquad (X, a) \longmapsto A, \tag{78}$$

is cofibred. For a morphism $u : A \to B$ and an object $(X, a) \in \ominus_L(A)$, one takes the push-forward extension $u_*(X)$ of L_0 by B, equipped with the induced partial splitting. The canonical map $(X, a) \to (u_*(X), u_*a)$ is cocartesian by the universal property of push-forward of extensions.

The cofibred category $\ominus_L$ is additive: $\ominus_L(0)$ is equivalent to the punctual category and, for any A, B,

$$(3.1.6) \ominus_L(A \oplus B) \longrightarrow \ominus_L(A) \times \ominus_L(B) \tag{79}$$

is an equivalence. We shall use freely the consequences of additivity recalled earlier. The trivial object of $\ominus_L(A)$ is obtained from the split extension $L_0 \oplus A$ and the lifting $(d_1, 0) : L_1 \to L_0 \oplus A$. The automorphism group of any object in $\ominus_L(A)$ is canonically the automorphism group of the trivial object; explicitly

$$(3.1.8) \ominus_L^0(A) \simeq \mathrm{Ker}\big(\mathrm{Hom}(L_0, A) \to \mathrm{Hom}(L_1, A)\big) \simeq \mathrm{Hom}(H_0(L_\bullet), A). \tag{80}$$

The set of isomorphism classes in $\ominus_L(A)$ is likewise a commutative group, denoted $\ominus_L^1(A)$, additive in A.

3.2. Determination of $\ominus_L^1(A)$

Let $(X, a) \in \ominus_L(A)$. Consider the cochain complex of length one

$$(3.2.1) K^\bullet = [X \xrightarrow{j} L_0] \tag{81}$$

with X in degree 0. It is a resolution of A, hence represents A in $\mathrm{D}(\mathcal{C})$. The diagram defining a gives a degree-one morphism of complexes

$$(3.2.2) u : L_\bullet \longrightarrow K^\bullet[-1], \tag{82}$$

which may be displayed as the anticommutative square diagram

$$(3.2.3) \quad \begin{tikzcd} L_2 \arrow[r, "d_2"] \arrow[d, "0"'] & L_1 \arrow[r, "d_1"] \arrow[d, "a"] & L_0 \arrow[r] \arrow[d, "\mathrm{id}"] & 0 \arrow[d] \\ 0 \arrow[r] & X \arrow[r, "j"] & L_0 \arrow[r] & 0. \end{tikzcd} \tag{83}$$

Thus one obtains, in the derived category, a degree-one morphism

$$(3.2.4) c(X,a) : L_\bullet \longrightarrow A[1], \tag{84}$$

and hence a canonical map

$$(3.2.4') \ominus_L^1(A) \longrightarrow \mathrm{Ext}^1(L_\bullet, A) = \mathrm{Hom}_{\mathrm{D}(\mathcal{C})}(L_\bullet, A[1]). \tag{85}$$

Theorem 3.2.5. The map $(3.2.4')$ is an isomorphism of groups, functorial in A.

We recall the proof. Additivity and functoriality in A are immediate from the construction. For injectivity, suppose that the morphism of complexes u in (3.2.3) is zero in $\mathrm{D}(\mathcal{C})$. Then it is already zero in the homotopy category $\mathrm{K}(\mathcal{C})$: there is a morphism $s : L_0 \to X$ such that

$$(3.2.5.1) js = \mathrm{id}_{L_0}, \qquad sd_1 = -a. \tag{86}$$

Indeed, after composing with a suitable quasi-isomorphism to another resolution of A, one may suppose the target resolution has no terms in degrees ≥ 2, and the required homotopy factors uniquely through the fibre product defining X. Such an s is precisely a splitting of the extension X of L_0 by A, compatible with the partial splitting a, hence (X,a) is trivial in $\ominus_L^1(A)$.

For surjectivity, represent an element of $\mathrm{Ext}^1(L_\bullet, A)$ by a degree-one morphism to a resolution $C^\bullet$ of A, which may be taken with $C^i = 0$ for $i \geq 2$:

$$(3.2.5.2) \quad \begin{array}{ccccccc} L_2 & \longrightarrow & L_1 & \longrightarrow & L_0 & \longrightarrow & 0 \\ \downarrow & & \downarrow{\scriptstyle -u^0} & & \downarrow{\scriptstyle u^1} & & \\ 0 & \longrightarrow & C^0 & \longrightarrow & C^1 & \longrightarrow & 0. \end{array} \tag{87}$$

The object C^0 is an extension of C^1 by A; pulling it back by $L_0 \to C^1$ gives an extension X of L_0 by A. The anticommutativity of the square forces $-u^0$ to factor through a map $a : L_1 \to X$ lifting d_1, and the relation on L_2 gives $ad_2 = 0$. Then $(X,a) \in \ominus_L(A)$, and the original class is exactly $c(X,a)$. This proves the theorem.

3.3. Properties of $\ominus_L$

The cofibred additive category $\ominus_L$ is left exact in the sense of the preceding theory; it admits a quasi-maximal object; every object has a typical complex; and the typical pro-complex of $\ominus_L$ is canonically isomorphic in $\mathrm{D}(\mathcal{C})$ to the truncation of $L_\bullet$ used in §3.1.

Left exactness means the following. If

$$(3.3.2.1) 0 \longrightarrow A' \longrightarrow A \xrightarrow{v} A'' \tag{88}$$

is exact, then the natural functor

$$(3.3.2.2) \ominus_L(A') \longrightarrow \ominus_L([A \to A'']) \tag{89}$$

is an equivalence. Here $\ominus_L([A \to A''])$ is the category of pairs consisting of an object over A and a trivialization of its image over A''. For $\ominus_L$, such an object is a triple (X,a,p), with (X,a) as above and $p : X \to A''$ satisfying $pi = v$ and $pa = 0$. Then $X' = \mathrm{Ker}(p)$ is an extension of L_0 by $A' = \mathrm{Ker}\, v$, and a factors through X'.

A quasi-maximal object is constructed as follows. Put

$$(3.3.3.1) L_1' = \mathrm{Coker}(d_2 : L_2 \to L_1). \tag{90}$$

Let $E_0 = (X_0, a_0)$, where $X_0 = L_0 \oplus L'_1$ is the split extension of L_0 by L'_1, and where $a_0 : L_1 \to X_0$ has components d_1 and the quotient map $L_1 \to L'_1$. Every object (X, a) maps, after a monomorphism on the base object, to an object whose underlying extension is split, and every such object is dominated by E_0.

If $E = (X, a)$ lies above A, the functor

$$(3.3.4.1) B \longmapsto \mathrm{Hom}(E, B) \tag{91}$$

is represented by $\mathrm{Coker}(a)$, and the canonical map from A is the composite

$$(3.3.4.2) A \longrightarrow X \longrightarrow \mathrm{Coker}(a). \tag{92}$$

The complex $[A \to \mathrm{Coker}(a)]$ is the typical complex of E. Applying the general theory to a quasi-maximal object gives the typical complex of $\ominus_L$. In the present construction it is canonically the truncation of $L_\bullet$ itself.

For every short exact sequence as above, the Ext long exact sequence

$$(3.3.6.1) \begin{array}{l} 0 \to \mathrm{Ext}^0(L, A') \to \mathrm{Ext}^0(L, A) \to \mathrm{Ext}^0(L, A'') \\ \quad \to \mathrm{Ext}^1(L, A') \to \mathrm{Ext}^1(L, A) \to \mathrm{Ext}^1(L, A'') \to \cdots \end{array} \tag{93}$$

corresponds, under (3.1.8) and (3.2.5), to the geometric exact sequence obtained from the left exactness of $\ominus_L$. The boundary morphism differs from the geometric one by the conventional sign described in the source; the editor explicitly notes that these sign questions will not be used in the subsequent exposés.

Finally, for variable L, the theory also describes the variance of $\ominus_L$. The class of cocartesian functors $\ominus_L \to \ominus_{L'}$ corresponds canonically to isomorphisms $L' \to L$ in $\mathrm{D}(\mathcal{C})$. We shall use only the elementary form developed next.

3.4. Variance with respect to L

Let

$$(3.4.1) f : L' \longrightarrow L \tag{94}$$

be a morphism of chain complexes, given in low degrees by

$$(3.4.2) \quad \begin{tikzcd} \cdots \arrow[r] & L'_2 \arrow[r] \arrow[d, "f_2"'] & L'_1 \arrow[r] \arrow[d, "f_1"'] & L'_0 \arrow[r] \arrow[d, "f_0"'] & 0 \\ \cdots \arrow[r] & L_2 \arrow[r] & L_1 \arrow[r] & L_0 \arrow[r] & 0. \end{tikzcd} \tag{95}$$

There is a canonical cocartesian functor

$$(3.4.3) f^* : \ominus_L \longrightarrow \ominus_{L'}. \tag{96}$$

If $(X, a) \in \ominus_L(A)$, then $f^*(X, a) = (X', a')$, where X' is the inverse image of the extension X of L_0 by A along $f_0 : L'_0 \to L_0$, and a' is induced from a by the commutative square involving f_1.

The functor (3.4.3) gives, for $i = 0, 1$, homomorphisms

$$(3.4.4) f^i : \ominus^i_L \longrightarrow \ominus^i_{L'}, \tag{97}$$

and the morphism of complexes gives

$$(3.4.5) f^i : \mathrm{Ext}^i(L, -) \longrightarrow \mathrm{Ext}^i(L', -). \tag{98}$$

For $i = 0, 1$, these are compatible with the identifications (3.1.8) and (3.2.5):

$$\text{(3.4.6)} \qquad \begin{tikzcd} \ominus^i_L(A) \arrow[r, "\sim"] \arrow[d] & \mathrm{Ext}^i(L, A) \arrow[d] \\ \ominus^i_{L'}(A) \arrow[r, "\sim"'] & \mathrm{Ext}^i(L', A). \end{tikzcd} \qquad (99)$$

Proposition 3.4.7. The functor (3.4.3) is faithful, respectively fully faithful, respectively an equivalence of categories, if and only if

$$\text{(3.4.7.1)} H_0(L') \longrightarrow H_0(L) \qquad (100)$$

is an epimorphism, respectively $H_0(L') \to H_0(L)$ is an isomorphism and $H_1(L') \to H_1(L)$ is an epimorphism, respectively both H_0 and H_1 maps are isomorphisms. This is just the preceding compatibility plus the elementary criterion for a homomorphism of additive cofibred categories to be faithful, fully faithful, or an equivalence.

Now take a short exact sequence of complexes

$$\text{(3.4.8.1)} 0 \longrightarrow L' \longrightarrow L \longrightarrow L'' \longrightarrow 0. \qquad (101)$$

The expected transitivity of functors of the form (3.4.3) gives an equivalence of categories

$$\text{(3.4.8.2)} \ominus_{L'}(A) \xrightarrow{\sim} \ominus_L(A) \times_{\ominus_{L''}(A)} \{0\}, \qquad (102)$$

where 0 is the trivial object of $\ominus_{L''}(A)$. In concrete terms, an object on the right is an object $(X, a) \in \ominus_L(A)$ together with a trivialization of its image in $\ominus_{L''}(A)$; the usual left exactness for extensions shows that this is the same as an object of $\ominus_{L'}(A)$.

The long exact sequence in Ext associated with (3.4.8.1)

$$\text{(3.4.9.1)} \begin{array}{l} 0 \to \mathrm{Ext}^0(L'', A) \to \mathrm{Ext}^0(L, A) \to \mathrm{Ext}^0(L', A) \\ \to \mathrm{Ext}^1(L'', A) \to \mathrm{Ext}^1(L, A) \to \mathrm{Ext}^1(L', A) \to \mathrm{Ext}^2(L'', A) \to \cdots \end{array} \qquad (103)$$

also has a geometric interpretation through $\ominus_L$. The boundary map $\mathrm{Ext}^0(L', A) \to \mathrm{Ext}^1(L'', A)$ sends a morphism $u : L'_1 \to A$, with $ud = 0$, to the class of the object of $\ominus_{L''}(A)$ obtained by lifting the trivial object and twisting its automorphism by u. The compatibility is reduced in the source to the corresponding statement of the general theory, after passing to the opposite category; the editor again warns that one must be careful with signs, but these signs do not enter the later exposés.

3.5. A canonical partial flat resolution of a $\mathbb{Z}$-module

We apply the preceding material to the abelian category $\mathrm{GrAb}(\mathcal{T})$ of abelian groups, or $\mathbb{Z}$-modules, in the topos $\mathcal{T}$. Let P be such a group. We define functorially an augmented chain complex

$$\text{(3.5.1)} 0 \longrightarrow L_2(P) \xrightarrow{d_2} L_1(P) \xrightarrow{d_1} L_0(P) \xrightarrow{\epsilon} P \longrightarrow 0 \qquad (104)$$

with $L_i(P) = 0$ for $i \geq 3$, where

$$\text{(3.5.1')} L_0(P) = \mathbb{Z}[P], \qquad L_1(P) = \mathbb{Z}[P \times P], \qquad L_2(P) = \mathbb{Z}[P \times P \times P] \oplus \mathbb{Z}[P \times P]. \quad (105)$$

For $p, q, r \in P(S)$, write $[p]$, $[p, q]$, $[p, q, r]$ for the corresponding generators. The maps are

$$\begin{aligned}(3.5.2)\epsilon([p]) &= p, &(106)\\ d_1([p, q]) &= [p + q] - [p] - [q], &(107)\\ d_2([p, q, r]) &= [p + q, r] - [p, q + r] + [p, q] - [q, r], &(108)\\ d_2'([p, q]) &= [p, q] - [q, p]. &(109)\end{aligned}$$

The relations $\epsilon d_1 = 0$ and $d_1 d_2 = 0$ are just the commutativity and associativity of the addition law on P. The terms $\mathbb{Z}[P^n]$ are flat $\mathbb{Z}$-modules in the topos.

Proposition 3.5.3. One has $H_i(L_\bullet(P)) = 0$ for $i > 0$, and the augmentation induces $H_0(L_\bullet(P)) \simeq P$.

The proof reduces to the following criterion supplied by Proposition 3.4.7. It is enough to show that, for every abelian group G of the topos, the functor

$$(3.5.3.1)\ominus_P(G) \longrightarrow \ominus_{L(P)}(G) \tag{110}$$

is an equivalence. The left-hand side is the category of extensions of the $\mathbb{Z}$-module P by G. On the right, using §1.4, extensions of the free modules $\mathbb{Z}[P]$, $\mathbb{Z}[P \times P]$, and $\mathbb{Z}[P^3] \oplus \mathbb{Z}[P^2]$ by G are described as torsors over the corresponding bases. Thus an object is a torsor E under G_P, together with a trivialization of

$$(3.5.3.2)\ \mathrm{pr}_1^* E \ \ \mathrm{pr}_2^* E\ (+)^* E^{-1} \tag{111}$$

over $P \times P$, satisfying the commutativity and associativity conditions encoded by the two summands of $L_2(P)$. This is precisely the bitorsor dictionary of §1.1 for a commutative extension of P by G.

Remark 3.5.6. It would be useful to strengthen the preceding partial resolution to a full functorial flat resolution of P, with terms functorially sums of free modules generated by Cartesian powers of P. Such a resolution would give a spectral sequence whose first term is expressed in terms of cohomology groups of powers P^n, and which would calculate all $\mathrm{Ext}^i(P, A)$, not only the partial information just obtained. S. Kleiman constructed such a resolution, according to the proof-corrector's note in the source.

There is also a variant for A-modules, where A is a fixed ring of the topos. The shape of the truncated complex is more complicated because the corresponding extension dictionary is more complicated:

$$\begin{aligned}(3.5.9)L_0(P) &= A[P], &(112)\\ L_1(P) &= A[P \times P] \oplus A[P], &(113)\\ L_2(P) &= A[P^3] \oplus A[P^2] \oplus A[P^2] \oplus A[P]. &(114)\end{aligned}$$

The formulas (3.5.2) are supplemented by relations expressing compatibility of scalar multiplication:

$$\begin{aligned}(3.5.10)d_1([a, p]) &= [ap] - a[p], &(115)\\ d_2([a, p, q]) &= [a, p + q] - [a, p] - [a, q], &(116)\\ d_2([a, b, p]) &= [ab, p] - a[b, p] - [a, bp], &(117)\end{aligned}$$

and the remaining formulas say exactly that the internal addition on the torsor and the external action of A make it an A-module extension of P by G.

3.6. Application to biextensions

Let P, Q be two $\mathbb{Z}$-modules of the topos $\mathcal{T}$. Choose flat resolutions $L_\bullet(P)$ and $L_\bullet(Q)$, agreeing in dimensions ≤ 2 with the partial complexes just constructed. There are canonical maps

$$(3.6.1) L_\bullet(P) \to P, \qquad L_\bullet(Q) \to Q, \tag{118}$$

and hence

$$(3.6.2) L_\bullet(P,Q) := L_\bullet(P) \otimes L_\bullet(Q) \longrightarrow P \otimes^L Q. \tag{119}$$

This map is an isomorphism in degrees ≤ 2, for the purposes of the preceding construction.

Theorem 3.6.4. The additive cofibred category, over $\mathrm{GrAb}(\mathcal{T})$, whose fibre over G is the category of biextensions of (P, Q) by G, is canonically equivalent to the additive cofibred category

$$(3.6.4.1) \ominus_{L(P,Q)}. \tag{120}$$

Together with Theorem 3.2.5, this gives the announced result.

Corollary 3.6.5. For $P, Q, G \in \mathrm{GrAb}(\mathcal{T})$, there is a canonical isomorphism, functorial in G,

$$(3.6.5.1)\ \mathrm{Biext}^1(P, Q; G) \xrightarrow{\sim} \mathrm{Ext}^1(P \otimes^L Q, G). \tag{121}$$

The proof of Theorem 3.6.4 is obtained by writing explicitly the first three terms of $L(P,Q) = L(P) \otimes L(Q)$. They are

$$(3.6.6) L_0(P,Q) = \mathbb{Z}[P \times Q], \tag{122}$$
$$L_1(P,Q) = \mathbb{Z}[P \times Q \times Q] \oplus \mathbb{Z}[P \times P \times Q], \tag{123}$$
$$L_2(P,Q) = \mathbb{Z}[P \times Q^3] \oplus \mathbb{Z}[P \times Q^2] \oplus \mathbb{Z}[P^2 \times Q^2] \tag{124}$$
$$\oplus\, \mathbb{Z}[P^3 \times Q] \oplus \mathbb{Z}[P^2 \times Q]. \tag{125}$$

Using the dictionary of §1.4, an extension of these free modules by G is described as a system of torsors over the corresponding Cartesian products. Thus an object of $\ominus_{L(P,Q)}(G)$ is a torsor E on $P \times Q$, together with two partial composition laws: one in the Q-direction and one in the P-direction. The five components of $L_2(P,Q)$ express the associativity in each variable, commutativity in each variable, and compatibility of the two partial laws. These are exactly the axioms of a biextension of (P,Q) by G. The verification of compatibility of morphisms is formal and is left, as in the source, to the reader.

Remark 3.6.7. Each of the generalizations of biextensions mentioned in §2.10 gives a corresponding variant of Theorem 3.6.4 and Corollary 3.6.5. The case of any number of factors is formal; the case of A-modules follows from the variant of the construction of $L_\bullet(P)$ in Remark 3.5.6. The non-commutative case is more interesting homologically: to a Group P, not necessarily commutative, one attaches a two-truncated complex $K_\bullet(P)$, analogous to $L_\bullet(P)$ but with $K_2(P) = \mathbb{Z}[P \times P \times P]$ only. Then $H_0(K_\bullet(P))$ is the abelianization $P/[P,P]$, identified with $H_1(P, \mathbb{Z})$, and $H_1(K_\bullet(P))$ is identified with $H_2(P, \mathbb{Z})$. Thus $K_\bullet(P)$ is the truncation to order two of the usual chain complex computing group homology. The same argument gives the corresponding cofibred-category statement.

Exposé VII, continuation: compatibilities and trivializations

3.7. Various compatibilities

The preceding discussion first gives, for variable commutative Groups, an equivalence between central extensions of a commutative Group P by variable commutative Groups G, and the cofibred category $\Psi_{K(P)}$, where $K(P)$ denotes the complex attached to P. Similarly, for two Groups P, Q, the cofibred category of biextensions of (P, Q) is equivalent to the cofibred category

$$\Psi_{L(P,Q)}, \qquad L(P,Q) = K(P) \otimes K(Q).$$

These remarks suggest that, for P commutative, one may perhaps find a canonical resolution of P of the kind desired in 3.5.4(a), by a judicious modification of the chain complex $C(P, \mathbb{Z})$.

In the present section we examine the behaviour of the equivalence of cofibred categories 3.6.4 when P and Q vary. The variance of the cofibred category of biextensions of (P, Q) by variable G, with respect to the variables P, Q, was defined in principle in 2.4, and may also be formulated by saying that one has a cofibred functor (2.6.1). Notice that the variance in P, Q is contravariant. The variance of the cofibred category $\Psi_{L(P,Q)}$, with respect to P, Q, is equally clear, because $L(P,Q) = L(P) \otimes L(Q)$ depends functorially on the pair (P, Q), whereas Ψ_L depends contravariantly on L (3.4). Thus a first compatibility says that, for every triple of morphisms

$$(3.7.1) P' \longrightarrow P, \qquad Q' \longrightarrow Q, \qquad G \longrightarrow G', \tag{126}$$

one has a commutative square of functors, up to the canonical isomorphism,

$$(3.7.2) \quad \begin{tikzcd} \Psi_{L(P,Q)}(G) \arrow[r] \arrow[d] & \Psi_{L(P',Q')}(G') \arrow[d] \\ \mathrm{Biext}(P,Q;G) \arrow[r] & \mathrm{Biext}(P',Q';G'). \end{tikzcd} \tag{127}$$

We shall admit this compatibility, since writing down the commutativity isomorphism is no less tedious than evident; it has the same character as the complete explicit construction involved in the equivalence of cofibred categories 3.6.4.

It follows in particular that the isomorphism (3.6.5.1) is functorial not only in G, but also in P and Q. Similarly, and more trivially, the isomorphism defined through (3.1.8), as (3.6.5.1) was defined through (3.2.4), gives

$$(3.7.4)\ \mathrm{Biext}^0(P,Q;G) \xrightarrow{\sim} \mathrm{Ext}^0(P \otimes^L Q, G) \simeq \mathrm{Hom}(P \otimes Q, G). \tag{128}$$

This is the same isomorphism already obtained in (2.5.4.1), where it was noted in (2.6) that it is functorial in the three arguments.

The interest of (3.6.5) is that it expresses the geometrically defined group $\mathrm{Biext}^1(P,Q;G)$ by the simple homological invariant $\mathrm{Ext}^1(P \otimes^L Q, G)$, which belongs to the usual sequence of invariants $\mathrm{Ext}^i(P \otimes^L Q, G)$. The latter sequence is a cohomological functor in each of the three arguments. Thus a short exact sequence

$$(3.7.5.1) 0 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 0 \tag{129}$$

or, respectively,

$$(3.7.5.2) 0 \longrightarrow P' \longrightarrow P \longrightarrow P'' \longrightarrow 0, \quad \text{or} \quad 0 \longrightarrow Q' \longrightarrow Q \longrightarrow Q'' \longrightarrow 0, \tag{130}$$

gives rise to an infinite exact sequence in the Ext^i's. Its first six terms, involving only Ext^0's and Ext^1's, may be interpreted in terms of biextensions by 3.6.5 and 3.7.4. We have already indicated the geometric interpretation of the exact sequence deduced from (3.7.5.1), and in particular of the boundary operator

$$(3.7.5.3) \partial : \mathrm{Biext}^0(P,Q;G'') \longrightarrow \mathrm{Biext}^1(P,Q;G'), \tag{131}$$

in the framework of left exact cofibred categories (3.3.6). One may likewise ask for a geometric interpretation of the six-term exact sequence attached to either of the short exact sequences (3.7.5.2), say the first; equivalently, by 3.7.3, one seeks an interpretation of the boundary homomorphism

$$(3.7.5.4)\partial : \mathrm{Biext}^0(P', Q; G) \longrightarrow \mathrm{Biext}^1(P'', Q; G). \tag{132}$$

Proposition 3.7.6. The cofibred category, over $\mathrm{GRAB}(\mathcal{T})^\circ \times \mathrm{GRAB}(\mathcal{T})^\circ \times \mathrm{GRAB}(\mathcal{T})$, of biextensions of (P, Q) by G, with P, Q, G variable, is left exact not only in the argument G, but also in each of the arguments P and Q. Moreover the boundary homomorphism (3.7.5.4) deduced from the exact sequence of the Ext^i's attached to (3.7.5.2), through 3.6.5 and 3.7.4, is the boundary homomorphism of the theory of left exact cofibred categories over $\mathcal{C} = \mathrm{GRAB}(\mathcal{T})^\circ$ recalled in 3.3.6, applied to the cofibred category of biextensions of (P, Q) by G, where Q and G are fixed and P is variable, up to the sign convention of 3.4.9.

We sketch the proof. By 3.6.4 and the compatibility (3.7.2), biextensions of (P, Q) by G are interpreted as objects of $\Psi_{L(P,Q)}(G)$. Consider the morphism

$$L(P, Q) = L(P) \otimes L(Q) \longrightarrow P \otimes L(Q)$$

induced by the augmentation $L(P) \to P$. From 3.5.3 and the flatness of $L(Q)$ this morphism induces isomorphisms on H_0 and H_1, hence by (3.4.7) an equivalence

$$\Psi_{P\otimes L(Q)} \xrightarrow{\sim} \Psi_{L(P,Q)}.$$

Thus biextensions of (P, Q) by G may be interpreted as objects of $\Psi_{P\otimes L(Q)}(G)$. The exact sequence (3.7.5.2) then gives an exact sequence of chain complexes

$$(3.7.6.1) 0 \longrightarrow P' \otimes L(Q) \longrightarrow P \otimes L(Q) \longrightarrow P'' \otimes L(Q) \longrightarrow 0. \tag{133}$$

The asserted left exactness follows from 3.4.8 applied to this exact sequence. The direct verification is immediate. Finally, the homomorphism (3.7.5.4) is the boundary $\mathrm{Ext}^0 \to \mathrm{Ext}^1$ obtained from this exact sequence, and the last assertion is therefore a special case of 3.4.9.

3.8. Biextensions of (P, Q) trivialized on extensions of P and Q

Consider two exact sequences of abelian Groups

$$(3.8.1) 0 \longrightarrow L_1 \xrightarrow{i} L_0 \xrightarrow{u} P \longrightarrow 0, \qquad 0 \longrightarrow M_1 \xrightarrow{j} M_0 \xrightarrow{v} Q \longrightarrow 0. \tag{134}$$

We wish to describe the category of biextensions E of (P, Q) by an abelian Group G whose inverse image $(u, v)^*(E)$ on (L_0, M_0) is trivial. Thus one is given a homomorphism s making the triangle

$$(3.8.2) \quad \begin{tikzcd} E \arrow[d] \arrow[dr, "s"] & \\ P \times Q & L_0 \times M_0 \arrow[l, "u\times v"'] \end{tikzcd} \tag{135}$$

commute, and satisfying the additivity condition in each of the two arguments. Such an s identifies $(u, v)^*(E)$ with the trivial biextension $E_0 = L_0 \times M_0 \times G$ of (L_0, M_0) by G.

By the left exactness in P, Q of the cofibred category of biextensions of (P, Q) by a fixed G (3.7.6), the datum of E together with the trivialization s of its inverse image is essentially equivalent to the datum of two partial trivializations of the trivial biextension E_0 of (L_0, M_0) by G, on (L_1, M_0) and (L_0, M_1), respectively, coinciding on (L_1, M_1). Since the biextensions induced by E_0 on (L_1, M_0), (L_0, M_1), and (L_1, M_1) are trivial, such partial trivializations are equivalent to homomorphisms, or pairings,

$$(3.8.3) f : L_1 \otimes M_0 \longrightarrow G, \qquad g : L_0 \otimes M_1 \longrightarrow G, \tag{136}$$

and the compatibility condition says simply that these two homomorphisms coincide on $L_1 \otimes M_1$:

$$(3.8.4) f(x_1 \otimes y_1) = g(x_1 \otimes y_1). \tag{137}$$

In terms of the homomorphism s of (3.8.2), the forms f, g are characterized by

$$(3.8.5) \begin{aligned} s(x_1, y_0) &= f(x_1 \otimes y_0) e_{E/P(y_0)}, \\ s(x_0, y_1) &= g(x_0 \otimes y_1) e_{E/P(x_0)}. \end{aligned} \tag{138}$$

If another trivialization s' is used, then

$$(3.8.6) s'(x_0, y_0) = s(x_0, y_0) + h(x_0 \otimes y_0) \tag{139}$$

for an arbitrary homomorphism $h : L_0 \otimes M_0 \to G$, and the two forms change to

$$(3.8.7) \begin{aligned} f'(x_1 \otimes y_0) &= f(x_1 \otimes y_0) + h(x_1 \otimes y_0), \\ g'(x_0 \otimes y_1) &= g(x_0 \otimes y_1) + h(x_0 \otimes y_1). \end{aligned} \tag{140}$$

Thus one obtains the following description.

Proposition 3.8.8. The subgroup of $\mathrm{Biext}^1(P, Q; G)$ formed by the classes of biextensions whose inverse image by (u, v) is a trivial biextension of (L_0, M_0) - equivalently, the kernel of

$$\mathrm{Biext}^1(P, Q; G) \longrightarrow \mathrm{Biext}^1(L_0, M_0; G)$$

- is canonically isomorphic, by the preceding description, to the quotient of the group of pairs of forms (f, g) as in (3.8.3), satisfying (3.8.4), by the subgroup consisting of those pairs obtained by restricting a form $h : L_0 \otimes M_0 \to G$.

We now interpret this quotient homologically, taking into account the canonical isomorphism $\mathrm{Biext}^1(P, Q; G) \simeq \mathrm{Ext}^1(P \otimes^L Q, G)$ of 3.6.5. Regard the data (3.8.1) as two resolution complexes of P and Q:

$$(3.8.9) L_\bullet : 0 \longrightarrow L_1 \longrightarrow L_0 \longrightarrow 0, \qquad M_\bullet : 0 \longrightarrow M_1 \longrightarrow M_0 \longrightarrow 0. \tag{141}$$

Their tensor product is the complex

$$0 \longrightarrow L_1 \otimes M_1 \xrightarrow{d_2} L_1 \otimes M_0 + L_0 \otimes M_1 \xrightarrow{d_1} L_0 \otimes M_0 \longrightarrow 0.$$

There is a canonical morphism in the derived category

$$P \otimes^L Q \longrightarrow L_\bullet \otimes M_\bullet,$$

hence a canonical homomorphism

$$(3.8.10) H^1(\mathrm{Hom}^\bullet(L_\bullet \otimes M_\bullet, G)) \longrightarrow \mathrm{Ext}^1(P \otimes^L Q, G). \tag{142}$$

The group of 1-cocycles of the complex $\mathrm{Hom}^\bullet(L_\bullet \otimes M_\bullet, G)$ identifies with the group of pairs (f, g) as in (3.8.3) satisfying (3.8.4), by assigning to such a pair the homomorphism

$$(3.8.11) F : L_1 \otimes M_0 + L_0 \otimes M_1 \longrightarrow G, \qquad F(x_1 \otimes y_0 + x_0 \otimes y_1) = f(x_1 \otimes y_0) + g(x_0 \otimes y_1). \tag{143}$$

Moreover, the coboundary of a 0-cochain $h : L_0 \otimes M_0 \to G$ is the pair of its restrictions to $L_1 \otimes M_0$ and $L_0 \otimes M_1$. Therefore the quotient described in Proposition 3.8.8 is canonically the first member of (3.8.10).

Proposition 3.8.12. Let $\xi \in \operatorname{Ker}(\operatorname{Biext}^1(P, Q; G) \to \operatorname{Biext}^1(L_0, M_0; G))$ be described by a pair (f, g) as in 3.8.8. Under the isomorphism (3.6.5.1), the image of ξ in $\operatorname{Ext}^1(P \otimes^L Q, G)$ is the image by (3.8.10) of the class represented by the cocycle (3.8.11).

The verification of this compatibility is left to the reader.

We shall use this especially with

$$L_0 = \mathbb{Z}[P], \qquad M_0 = \mathbb{Z}[Q],$$

where L_1 and M_1 are the kernels of the canonical homomorphisms $\mathbb{Z}[P] \to P$ and $\mathbb{Z}[Q] \to Q$. There are canonical isomorphisms

$$L_0 \otimes^L M_0 \simeq L_0 \otimes M_0 \simeq \mathbb{Z}[P \times Q]$$

since L_0, M_0 are flat, and therefore, by (1.4.5), canonical isomorphisms

$$\operatorname{Ext}^i(L_0 \otimes^L M_0, G) \simeq H^i(P \times Q, G_{P \times Q}).$$

Consequently, the functor sending a biextension of (L_0, M_0) by G to the torsor which it induces over $P \times Q$ is an equivalence of categories. Thus, for a biextension E of (P, Q) by G, giving a trivialization of its inverse image on (L_0, M_0) is the same as giving a section of E as an object over $P \times Q$, i.e. a trivialization of E as a torsor under $G_{P \times Q}$.

Biextensions endowed with this supplementary structure are therefore described by pairs (f, g) satisfying the compatibility (3.8.4), modulo the relation described in 3.8.8. With the notation of 3.5, one may view L_1 as the cokernel of $L_2(P) \to L_1(P)$, and similarly for M_1. This expresses (f, g) as a system of homomorphisms

$$\begin{cases} f' : L_1(P) \otimes L_0(Q) = \mathbb{Z}(P \times P \times Q) \longrightarrow G, \\ g' : L_0(P) \otimes L_1(Q) = \mathbb{Z}(P \times Q \times Q) \longrightarrow G, \end{cases}$$

i.e. as homomorphisms of sheaves of sets

$$P \times P \times Q \longrightarrow G, \qquad P \times Q \times Q \longrightarrow G,$$

satisfying five compatibility conditions: four expressing the factorization through f and g, and the condition (3.8.4). These f' and g' are precisely the partial multiplication data defining a biextension, and the five conditions are the five axioms of biextensions considered in 2.1.

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Exposé VIII. Complements on biextensions; general properties of biextensions of group schemes

by A. Grothendieck

Summary

0. Introduction

We continue here the general study of biextensions begun in the preceding exposé. We first give complements in the case of an arbitrary base topos, developing in particular the formalism of the pairings attached to biextensions, whose importance in the theory of abelian schemes is well known. We then begin the study of biextensions of general group schemes, the most interesting case for us being biextensions by the multiplicative group $\mathbb{G}_m$. As a technical intermediate step we shall also have to study biextensions by other types of group schemes.

The most important result for later use is 7.1. It gives conditions under which, for P and Q smooth over the generic point η of a trait, a biextension of (P_η, Q_η) by $\mathbb{G}_m$ extends to a biextension of (P, Q) by $\mathbb{G}_m$, determined up to unique isomorphism. This will be convenient in the next exposé, which concerns Néron models of abelian varieties. It is likely that these developments will be useful elsewhere as well, although the later part of the seminar will use them only sparingly.

When working with group schemes over a scheme S, we shall tacitly use the fppf topos of S, attached to the $\mathcal{U}$-site of schemes locally of finite presentation over S, endowed with the fppf topology. Most statements would remain valid, and essentially equivalent, for other topologies whose underlying category contains the schemes locally of finite presentation over S, such as the Zariski, étale, or fpqc topology. The reason is descent: the category of extensions of a group scheme P, locally of finite presentation over S, by $\mathbb{G}_m$, and similarly the notion of biextension, do not depend, up to equivalence, on this choice. Technically, the fppf topology, or a finer topology such as fpqc, is preferred because the homomorphism $n \cdot \mathrm{id}$ of $\mathbb{G}_m$ is an fppf epimorphism for every $n > 0$, whereas it is not in general a Zariski or étale epimorphism.

Finally, most statements on extensions or biextensions by $\mathbb{G}_m$ remain true if $\mathbb{G}_m$ is replaced by a torus G. Since such a G is locally isomorphic, for the fppf and even for the étale topology, to $\mathbb{G}_m^r$, this is essentially a trivial generalization.

1. Various special cases on an arbitrary topos

As in Exposé VII, let $\mathcal{T}$ be a topos, and let P, Q, G be three abelian Groups of $\mathcal{T}$. We use the canonical isomorphism

$$(1.1.1)\, \mathrm{Biext}^1(P, Q; G) \simeq \mathrm{Ext}^1(P \otimes^L Q, G). \tag{144}$$

Recall also the Cartan isomorphisms

$$(1.1.2)\, \mathrm{Ext}^1(P \otimes^L Q, G) \simeq \mathrm{Ext}^1(P, \mathrm{RHom}(Q, G)) \simeq \mathrm{Ext}^1(Q, \mathrm{RHom}(P, G)), \tag{145}$$

functorial in the three arguments. From (1.1.1) one obtains the exact sequence

$$(1.1.3)\begin{gathered} 0 \to \mathrm{Ext}^1(P \otimes Q, G) \to \mathrm{Biext}^1(P, Q; G) \to \mathrm{Hom}(\mathrm{Tor}_1(P, Q), G) \to \\ \mathrm{Ext}^2(P \otimes Q, G) \to \mathrm{Ext}^2(P \otimes^L Q, G) \to \cdots , \end{gathered} \tag{146}$$

because the homology objects of $P \otimes^L Q$ are the $\mathrm{Tor}_i(P, Q)$, zero for $i \geq 1$. Similarly, since the homology objects of $\mathrm{RHom}(Q, G)$ are the $\mathrm{Ext}^i(Q, G)$, the first isomorphism in (1.1.2) gives a five-term exact sequence

$$(1.1.4)\begin{gathered} 0 \to \mathrm{Ext}^1(P, \mathrm{Hom}(Q, G)) \to \mathrm{Biext}^1(P, Q; G) \to \mathrm{Hom}(P, \mathrm{Ext}^1(Q, G)) \to \\ \mathrm{Ext}^2(P, \mathrm{Hom}(Q, G)) \to \mathrm{Ext}^2(P, \mathrm{RHom}(Q, G)). \end{gathered} \tag{147}$$

This sequence is associated with the usual spectral sequence

$$\mathrm{Ext}^*(P, \mathrm{RHom}(Q, G)) \Longleftarrow \mathrm{Ext}^p(P, \mathrm{Ext}^q(Q, G)).$$

A fully faithful functor may be used to describe the first arrow in (1.1.3), respectively in (1.1.4), namely

$$(1.1.5)\, \mathrm{EXT}(P \otimes Q, G) \to \mathrm{BIEXT}(P, Q; G), \tag{148}$$

or

$$(1.1.6)\, \mathrm{EXT}(P, \mathrm{Hom}(Q, G)) \to \mathrm{BIEXT}(P, Q; G). \tag{149}$$

The symmetric Cartan isomorphism yields the analogous functor

$$(1.1.7)\, \mathrm{EXT}(Q, \mathrm{Hom}(P, G)) \to \mathrm{BIEXT}(P, Q; G). \tag{150}$$

If $\mathrm{Tor}_1(P, Q) = 0$, then (1.1.3) gives a canonical isomorphism

$$(1.2.2)\, \mathrm{Biext}^1(P, Q; G) \simeq \mathrm{Ext}^1(P \otimes Q, G). \tag{151}$$

This is the case, for example, if P or Q is a flat $\mathbb{Z}$-Module; more precisely, under the same hypothesis the functor (1.1.5) is an equivalence of categories.

If $P \otimes Q = 0$, then (1.1.3) gives a canonical isomorphism

$$(1.3.2)\, \mathrm{Biext}^1(P, Q; G) \simeq \mathrm{Hom}(\mathrm{Tor}_1(P, Q), G). \tag{152}$$

A particularly useful case occurs when there is an integer $n > 0$ such that

$$(1.3.3) nP = P, \qquad Q = \varinjlim_{i} {}_{n^i}Q, \tag{153}$$

that is, P is divisible by n and Q is a filtered limit of subgroups killed by powers of n. Put

$$T_n(P) = ({}_{n^i}P)_{i \geq 0}, \qquad {}_{n^\infty}P = \varinjlim_{i} {}_{n^i}P.$$

Then

$$(1.3.4) P \otimes Q = 0, \qquad \mathrm{Tor}_1(P,Q) \simeq T_n(P) \otimes Q \simeq {}_{n^\infty}P \otimes T_n(Q), \tag{154}$$

and hence

$$(1.3.5)\, \mathrm{Biext}^1(P,Q;G) \simeq \mathrm{Hom}(T_n(P), D(Q)) \simeq \mathrm{Hom}(Q, D(T_n(P))). \tag{155}$$

Here D denotes $\mathrm{Hom}(-,G)$. The transition map in the inductive system appearing in (1.3.4) is obtained by identifying the first term with ${}_{n^j}P \otimes {}_{n^i}Q$, using n^{j-i} id on the first factor and the inclusion on the second. For a group M killed by n^i one defines

$$T_n(P) \otimes M \text{ dfn } {}_{n^i}P \otimes M,$$

and $D(Q) = (\mathrm{Hom}({}_{n^i}Q,G))_{i\geq 0}$. The remaining identifications are those of morphisms of projective systems.

The calculation of $\mathrm{Tor}_1(P,Q)$ used above is reduced to the case in which Q is killed by n^i. It suffices to use the exact sequence

$$0 \to P \xrightarrow{m} P \to P/mP \to 0$$

and the standard connecting morphism to get, when $mP = P$ and $mQ = 0$,

$$(1.3.6)\, \mathrm{Tor}_1(P,Q) \xleftarrow{\sim} {}_mP \otimes_{\mathbb{Z}/m\mathbb{Z}} Q. \tag{156}$$

The choice of this isomorphism amounts only to a sign convention.

If

$$(1.4.1)\, \mathrm{Hom}(Q,G) = 0, \tag{157}$$

then (1.1.4) gives a canonical isomorphism

$$(1.4.2)\, \mathrm{Biext}^1(P,Q;G) \simeq \mathrm{Hom}(P, \mathrm{Ext}^1(Q,G)). \tag{158}$$

This case occurs, for example, when Q is an abelian scheme and $G = \mathbb{G}_m$. Then the functor in P, $P \mapsto \mathrm{Biext}^1(P,Q;G)$, is representable by

$$Q^* = \mathrm{Ext}^1(Q,G),$$

and one obtains a canonical biextension E^Q of (Q^*,Q) by G, uniquely determined up to unique isomorphism. This biextension, or the symmetric biextension ${}^sE^Q$ of (Q,Q^*) by G, may be called the Weil biextension of (Q^*,Q), or of (Q,Q^*), by G.

If

$$(1.5.1)\, \mathrm{Hom}(P, \mathrm{Ext}^1(Q,G)) = 0, \tag{159}$$

then (1.1.4) yields the canonical isomorphism

$$(1.5.2)\, \mathrm{Biext}^1(P,Q;G) \simeq \mathrm{Ext}^1(P, \mathrm{Hom}(Q,G)). \tag{160}$$

More precisely, the fully faithful functor (1.1.6) is then an equivalence of categories. If both

$$(1.6.1)\, \mathrm{Hom}(P,G) = 0, \qquad \mathrm{Hom}(P, \mathrm{Ext}^1(Q,G)) = 0 \tag{161}$$

hold, then the two isomorphisms of 1.4 combine into a duality formula

$$(1.6.2)\, \mathrm{Ext}^1(P, \mathrm{Hom}(Q,G)) \simeq \mathrm{Hom}(Q, \mathrm{Ext}^1(P,G)) \simeq \mathrm{Biext}^1(P,Q;G). \tag{162}$$

The first condition in (1.6.1) implies $\mathrm{Hom}(P \otimes Q, G) = 0$. Hence the category of commutative extensions of P by $\mathrm{Hom}(Q,G)$ is rigid, and the same is true of $\mathrm{BIEXT}(P,Q;G)$. Therefore their structure is known once the group of isomorphism classes is known; by (1.6.2) this group is canonically $\mathrm{Hom}(Q, \mathrm{Ext}^1(P, G))$.

The map (1.6.2) is actually defined without assuming (1.6.1). It may be described directly, without passing through $\mathrm{Biext}^1(P, Q; G)$, in terms of the natural pairing

$$Q \times Q' \longrightarrow G, \qquad Q' = \mathrm{Hom}(Q, G),$$

which is equivalently a homomorphism $Q \to \mathrm{Hom}(Q', G)$. It gives the composite

$$(1.6.3)\, \mathrm{Ext}^1(P, Q') \times Q \longrightarrow \mathrm{Ext}^1(P, Q') \times \mathrm{Hom}(Q', G) \longrightarrow \mathrm{Ext}^1(P, G), \tag{163}$$

recovering (1.6.2).

2. Pairings defined by a biextension

Let P, Q be two abelian Groups of $\mathcal{T}$, and let $n > 0$ be an integer. We write ${}_nP$ for the kernel of multiplication by n on P. We define a canonical homomorphism

$$(2.1.1)\alpha = \alpha^n_{P,Q} : {}_nP \otimes {}_nQ \longrightarrow \mathrm{Tor}_1(P, Q). \tag{164}$$

It is the composite

$$(2.1.2)_nP \otimes {}_nQ \longrightarrow \mathrm{Tor}_1({}_nP, {}_nQ) \longrightarrow \mathrm{Tor}_1(P, Q), \tag{165}$$

where the first arrow is defined first in the case in which P and Q are killed by n. Suppose for instance that ${}_nQ = Q$. Taking a flat resolution $L_\bullet$ of P, one has canonical isomorphisms

$$P \otimes^L_{\mathbb{Z}} Q \simeq L_\bullet \otimes Q \simeq (L_\bullet \otimes \mathbb{Z}/n\mathbb{Z}) \otimes_{\mathbb{Z}/n\mathbb{Z}} Q,$$

and this gives a homological Kunneth spectral sequence

$$(2.1.4)\, \mathrm{Tor}^{\mathbb{Z}}_*(P, Q) \Longleftarrow E^2_{p,q} = \mathrm{Tor}^{\mathbb{Z}/n\mathbb{Z}}_p(\mathrm{Tor}^{\mathbb{Z}}_q(P, \mathbb{Z}/n\mathbb{Z}), Q). \tag{166}$$

The usual flat resolution

$$0 \longrightarrow \mathbb{Z}_{\mathcal{T}} \xrightarrow{n} \mathbb{Z}_{\mathcal{T}} \longrightarrow 0$$

of $(\mathbb{Z}/n\mathbb{Z})_{\mathcal{T}}$ gives

$$\mathrm{Tor}^{\mathbb{Z}}_0(P, \mathbb{Z}/n\mathbb{Z}) = P_n = P/nP, \qquad \mathrm{Tor}^{\mathbb{Z}}_1(P, \mathbb{Z}/n\mathbb{Z}) = {}_nP,$$

and higher Tor_i's vanish. Therefore (2.1.4) reduces essentially to the exact sequence

$$(2.1.5)0 \to \mathrm{Tor}^{\mathbb{Z}/n\mathbb{Z}}_2(P_n, Q) \to {}_nP \otimes Q \xrightarrow{\alpha} \mathrm{Tor}^{\mathbb{Z}}_1(P, Q) \to \mathrm{Tor}^{\mathbb{Z}/n\mathbb{Z}}_1(P_n, Q) \to \cdots . \tag{167}$$

This defines the desired map and shows that it is functorial in P and Q. In particular it is an isomorphism when ${}_nP$ is a flat $\mathbb{Z}/n\mathbb{Z}$-Module (for example when $nP = P$ and ${}_nP$ is flat), or when Q is flat over $\mathbb{Z}/n\mathbb{Z}$.

The map α may be described more explicitly. To know $\alpha(p, q)$ for sections $p \in {}_nP$ and $q \in {}_nQ$, one reduces by functoriality to the case $Q = (\mathbb{Z}/n\mathbb{Z})_{\mathcal{T}}$ and $q = 1$. In that case $\mathrm{Tor}^{\mathbb{Z}}_1(P, Q) \simeq {}_nP$, and α is the identity. Equivalently, one further reduces to $P = Q = (\mathbb{Z}/n\mathbb{Z})_{\mathcal{T}}$, where

$$(2.1.8.1)(\mathbb{Z}/n\mathbb{Z})_{\mathcal{T}} \xrightarrow{\sim} \mathrm{Tor}^{\mathbb{Z}}_1((\mathbb{Z}/n\mathbb{Z})_{\mathcal{T}}, (\mathbb{Z}/n\mathbb{Z})_{\mathcal{T}}) \tag{168}$$

is induced by the canonical flat resolution

$$(2.1.8.2)0 \longrightarrow \mathbb{Z}_{\mathcal{T}} \xrightarrow{n} \mathbb{Z}_{\mathcal{T}} \longrightarrow 0 \tag{169}$$

of the first factor.

The definition of α is not symmetric. By exchanging P and Q in the construction, one obtains an analogous map

$$(2.1.9.1)\beta^n_{P,Q} : {}_nP \otimes {}_nQ \longrightarrow \mathrm{Tor}_1(P,Q). \tag{170}$$

The two maps may also be characterized by the commutative square

$$(2.1.9.2) \quad \begin{tikzcd} {}_nP \otimes {}_nQ \arrow[r, "\sim"] \arrow[d, "\beta^n_{P,Q}"'] & {}_nQ \otimes {}_nP \arrow[d, "\alpha^n_{Q,P}"] \\ \mathrm{Tor}_1(P,Q) \arrow[r, "\sim"'] & \mathrm{Tor}_1(Q,P), \end{tikzcd} \tag{171}$$

where the horizontal arrows are the symmetry isomorphisms. Their relation is

$$(2.1.9.3)\beta^n_{P,Q} = -\alpha^n_{P,Q}. \tag{172}$$

Lemma 2.1.10. Let P, Q be two commutative Groups of $\mathcal{T}$, and let $n > 0$. Then the square

$$(2.1.10.1) \quad \begin{tikzcd} {}_nP \otimes {}_nQ \arrow[r, "\sim"] \arrow[d, "\alpha^n_{P,Q}"'] & {}_nQ \otimes {}_nP \arrow[d, "\alpha^n_{Q,P}"] \\ \mathrm{Tor}_1(P,Q) \arrow[r, "\sim"'] & \mathrm{Tor}_1(Q,P) \end{tikzcd} \tag{173}$$

is anti-commutative, the horizontal arrows being the symmetry isomorphisms.

Proof. The procedure of 2.1.8 reduces the question to the case $P = Q = (\mathbb{Z}/n\mathbb{Z})_{\mathcal{T}}$. It is enough then to work in the punctual topos, i.e. in ordinary commutative groups. Take

$$L_\bullet : 0 \longrightarrow \mathbb{Z} \xrightarrow{n} \mathbb{Z} \longrightarrow 0$$

as a flat resolution of $P = Q = \mathbb{Z}/n\mathbb{Z}$. Using $L_\bullet \otimes L_\bullet$, the isomorphism defining α is represented by the cycle

$$-1 \otimes l_1 + l_1 \otimes 1,$$

whereas the isomorphism defining β is represented by

$$1 \otimes l_1 - l_1 \otimes 1.$$

This proves the lemma.

SGA 7-I, Exposé VIII. Complements on biextensions; general properties of biextensions of group schemes

2.1. Compatibilities of the torsion-to-Tor homomorphism

We continue the comparison, begun in 2.1, between torsion in the two variables and the group $\mathrm{Tor}_1(P,Q)$. Let $n \mid n'$. With the notation already fixed, the homomorphisms

$${}_nP \otimes {}_nQ \longrightarrow \mathrm{Tor}_1(P,Q)$$

are compatible with the transition maps in both variables. More precisely, the diagram

$$\begin{array}{ccc} {}_{n'}P \otimes {}_{n'}Q & \longrightarrow & \mathrm{Tor}_1(P,Q) \\ \big\downarrow & & \big\downarrow \\ {}_{n}P \otimes {}_{n}Q & \longrightarrow & \mathrm{Tor}_1(P,Q) \end{array} \qquad (2.1.11.1)$$

commutes, where the left vertical map is induced by multiplication and the natural inclusions of kernels, and where the right vertical map is the corresponding transition map on the Tor group. The symmetric diagram, obtained by interchanging P and Q, commutes as well. By the reduction used in 2.1.8, it is enough to check this for the punctual topos and for cyclic modules. One identifies $\mathrm{Tor}_1(P,Q)$ from the standard resolution

$$0 \longrightarrow \mathbb{Z} \xrightarrow{n'} \mathbb{Z} \longrightarrow \mathbb{Z}/n'\mathbb{Z} \longrightarrow 0,$$

and the assertion then becomes the evident commutativity of the diagram obtained by passing from n'-torsion to n-torsion.

The same argument gives the following useful form. With the preceding notation, one has a commutative diagram

$$\begin{array}{ccc} {}_{nn'}P \otimes {}_{nn'}Q & \longrightarrow & \mathrm{Tor}_1(P,Q) \\ \big\downarrow & & \big\downarrow{\scriptstyle m} \\ {}_{n'}P \otimes {}_{n'}Q & \longrightarrow & \mathrm{Tor}_1(P,Q), \end{array} \qquad (2.1.12.1)$$

where the horizontal arrows are the torsion-to-Tor maps and m denotes the endomorphism induced by multiplication. Indeed, for sections $x \in {}_{nn'}P$, $y \in {}_{nn'}Q$, the class attached to (mx, my) is the same as the class attached to (x, my) multiplied by m; this is exactly the commutativity of the diagram.

Let now ℓ be a prime. For any commutative group E in the topos write

$$\mathrm{T}_\ell(E) = \varprojlim_{\nu} {}_{\ell^\nu}E.$$

The compatibilities just recalled give, functorially in P and Q, a homomorphism of projective systems

$$\mathrm{T}_\ell(P) \otimes \mathrm{T}_\ell(Q) \longrightarrow \mathrm{T}_\ell(\mathrm{Tor}_1(P,Q)). \qquad (2.1.13.2)$$

The family of these maps, as ℓ varies, is equivalent to the family of the finite-level maps for all integers n. The anti-commutativity of 2.1.10 becomes the ℓ-adic anti-commutativity

$$\begin{array}{ccc} \mathrm{T}_\ell(P) \otimes \mathrm{T}_\ell(Q) & \xrightarrow{\mathrm{sym}} & \mathrm{T}_\ell(Q) \otimes \mathrm{T}_\ell(P) \\ \big\downarrow & & \big\downarrow \\ \mathrm{T}_\ell(\mathrm{Tor}_1(P,Q)) & \xrightarrow[-\,\mathrm{sym}]{} & \mathrm{T}_\ell(\mathrm{Tor}_1(Q,P)). \end{array} \qquad (2.1.13.3)$$

The construction also has an evident version over a commutative ring object A of the topos. If $\mathfrak{a}$ and $\mathfrak{b}$ are ideals of A, and if P, Q are A-modules, one obtains a functorial homomorphism

$${}_{\mathfrak{a}}P \otimes {}_{\mathfrak{b}}Q \longrightarrow \mathrm{Hom}_A(\mathfrak{a} \cap \mathfrak{b}/\mathfrak{a}\mathfrak{b}, \mathrm{Tor}_1^A(P,Q)). \qquad (2.1.14.0)$$

It is the composite of

$$ {}_{\mathfrak{a}}P \otimes {}_{\mathfrak{b}}Q \longrightarrow \mathrm{Hom}_A(A/\mathfrak{a}, P) \otimes \mathrm{Hom}_A(A/\mathfrak{b}, Q) \longrightarrow \mathrm{Hom}_A(\mathrm{Tor}_1^A(A/\mathfrak{a}, A/\mathfrak{b}), \mathrm{Tor}_1^A(P, Q)), $$

together with the canonical isomorphism

$$ \mathrm{Tor}_1^A(A/\mathfrak{a}, A/\mathfrak{b}) \simeq \mathfrak{a} \cap \mathfrak{b}/\mathfrak{a}\mathfrak{b}. \qquad (2.1.14.2) $$

The sign of 2.1.10 is hidden precisely in the choice of this last isomorphism, according to whether one resolves $A/\mathfrak{a}$ or $A/\mathfrak{b}$. Thus, under transposition, the resulting square is anti-commutative. In particular, when $\mathfrak{a} = \mathfrak{b}$, the symmetry automorphism on $\mathrm{Tor}_1^A(A/\mathfrak{a}, A/\mathfrak{a})$ is $-\mathrm{id}$.

2.2. Pairings attached to a biextension

Let E be a biextension of (P, Q) by an abelian group G, and let

$$ \xi(E) \in \mathrm{Biext}^1(P, Q; G) \simeq \mathrm{Ext}^1(P \otimes^L Q, G) $$

be its class. Composing this class with the canonical map from torsion to $\mathrm{Tor}_1(P, Q)$ produces pairings on torsion subgroups. For every integer $n > 0$ one obtains a functorial pairing

$$ {}_nP \times {}_nQ \longrightarrow \mathrm{Coker}(G \xrightarrow{n} G). \qquad (2.2.1) $$

Equivalently, it is the homomorphism

$$ {}_nP \otimes {}_nQ \longrightarrow \mathrm{Coker}(n : G \to G) $$

which is obtained by applying the connecting homomorphism associated with multiplication by n on G to the image in $\mathrm{Tor}_1(P, Q)$. The construction is bilinear, functorial in P, Q, G, compatible with inverse images in the first two variables and with push-out in the third, and anti-symmetric with the Koszul sign under interchange of P and Q.

In concrete terms, if $p \in {}_nP(S)$ and $q \in {}_nQ(S)$, restrict the biextension to the subgroup generated by p and q. The two relations $np = 0$ and $nq = 0$ give two ways of returning to the neutral section after n additions. Their quotient is an element of G/nG, and this element is the value of the pairing at (p, q). The biextension axiom is exactly what makes the element independent of the choices.

Passing to the projective limit over the powers of a prime ℓ gives an ℓ-adic pairing

$$ \mathrm{T}_\ell(P) \times \mathrm{T}_\ell(Q) \longrightarrow \mathrm{T}_\ell(G). \qquad (2.2.2) $$

For $G = \mathbb{G}_m$ this is customarily written with values in $\mathbb{Z}_\ell(1)$, or after tensoring with $\mathbb{Q}_\ell$, in $\mathbb{Q}_\ell(1)$. These pairings are compatible with the transition maps in n, with change of topos, and with the transposition of a biextension.

2.3. Relation with Cartier duality

When P and Q are finite locally free commutative group schemes, and when $G = \mathbb{G}_m$, the preceding pairing is the usual Cartier-duality pairing associated with the biextension. Fixing a section p of P, the restriction of the biextension to $\{p\} \times Q$ is an extension of Q by $\mathbb{G}_m$, hence a point of the Cartier dual $Q^\vee$. Thus the biextension defines a morphism

$$ P \longrightarrow Q^\vee, $$

or, equivalently, a bilinear pairing $P \times Q \to \mathbb{G}_m$. The comparison with the construction by Tor_1 is local for the flat topology and reduces to the explicit cyclic calculation in 2.1. The commutator element in the restricted biextension is then exactly the value obtained from the Tor class.

Consequently the homological construction, the Cartier-duality construction, and the torsion-pairing construction agree. For powers of ℓ, this gives the usual pairings of Tate modules attached to finite flat group schemes or to abelian schemes through their torsion subgroups.

3. Biextensions of group schemes by $\mathbb{G}_m$: generalities

The principal example is the Poincare biextension. If A is an abelian scheme over S, and A' is the dual abelian scheme, the Poincare invertible sheaf on $A \times_S A'$, rigidified along the two zero sections, is a biextension of (A, A') by $\mathbb{G}_m$. The partial extension structures are the theorem of the square in each variable, and their compatibility is the theorem of the cube. The universal property identifies A' with the sheaf of extensions of A by $\mathbb{G}_m$, and the Poincare biextension is the geometric representative of the identity of A'.

If one of the two variables, say Q, is an abelian scheme, a biextension of (P, Q) by $\mathbb{G}_m$ may be read as a morphism from P to the sheaf of extensions of Q by $\mathbb{G}_m$. Under the usual representability and rigidity hypotheses this gives

$$\mathrm{Biext}^1(P, Q; \mathbb{G}_m) \simeq \mathrm{Hom}(P, Q'), \qquad (3.2.1)$$

where Q' is the dual abelian scheme. Conversely a morphism $P \to Q'$ pulls back the Poincare biextension on $Q \times Q'$. This description is compatible with base change and with the torsion pairings of Section 2.

When one of the variables is finite locally free, Cartier duality gives a parallel description. If Q is finite locally free, the restriction of a biextension to $\{p\} \times Q$ defines an extension of Q by $\mathbb{G}_m$, that is a point of $Q^\vee$. Thus biextensions are described by morphisms from the other variable to the Cartier dual, subject to the additivity imposed by the biextension law. If the finite group scheme is killed by n, the torsion pairing of Section 2 recovers this morphism after applying the standard identification with μ_n.

In the unipotent case, and more generally when the sheaf of extensions by $\mathbb{G}_m$ is trivial by rigidity, the corresponding group of biextensions vanishes. This will be used below to isolate the toric and component-group contributions to the obstruction for prolongation. Thus the preceding special cases can be summarized as follows: abelian schemes contribute through their duals; finite flat groups contribute through Cartier duals; tori contribute through their character groups; and unipotent groups do not contribute in the same way to biextensions by $\mathbb{G}_m$.

For two abelian schemes A, B over S, the formula specializes to

$$\mathrm{Biext}^1(A, B; \mathbb{G}_m) \simeq \mathrm{Hom}(A, B') \simeq \mathrm{Hom}(B, A'), \qquad (3.6.1)$$

where the equality of the two descriptions is induced by the transposition of biextensions. A biextension of (A, B) is therefore the same data as a divisor correspondence from A to B. With the Poincare biextension this recovers the classical correspondence between homomorphisms of abelian schemes and divisor correspondences modulo the rigidifications along the origins.

The same discussion extends to semi-abelian schemes, after separating the abelian and toric parts. In that setting the toric part is controlled by character groups, and the abelian part by the dual abelian scheme. The compatibility between these two pieces is the compatibility of the biextension with the extension structures defining the semi-abelian schemes.

4. Smooth connected group schemes over a field

Let k be a field. For a smooth connected commutative k-group P, extensions by $\mathbb{G}_m$ are strongly constrained by Chevalley's theorem. The affine part is built from tori and unipotent groups, while the quotient by the affine part is an abelian variety. Extensions and biextensions by $\mathbb{G}_m$ therefore decompose into contributions coming from these three types. For unipotent groups the relevant groups vanish under the hypotheses used here. For tori, extensions and biextensions are described by character lattices. For abelian varieties, they are described by the dual abelian variety and the Poincare biextension.

The following consequences are the ones used later. If P and Q are smooth connected commutative groups over k, and if one of the unipotent contributions vanishes, a biextension of (P, Q) by $\mathbb{G}_m$ is determined by its restrictions to the maximal abelian quotients and maximal tori. If P and Q are abelian varieties, this is simply the description by a homomorphism $P \to Q'$. If one of them is a torus, the biextensions are read on the character group of the torus and the corresponding extension of the other variable.

One also needs the behavior of invertible sheaves under translation. On a smooth connected group scheme, the theorem of the square and the theorem of the cube identify those invertible sheaves which define extensions or biextensions. Rigidifications along the unit section eliminate the ambiguity by pullback from the base. These facts are repeatedly used to turn statements about biextensions into statements about rigidified line bundles on products of group schemes.

5. Biextensions by constant truncated torsion-free group schemes

Let Z be a constant group scheme associated with a torsion-free abelian group. For smooth connected group schemes P, Q over a base, extensions and biextensions by Z are very restrictive. Over a connected base, a morphism from a smooth connected group to such a constant group is constant. This rigidity yields the vanishing of many Hom and Ext groups, and consequently full-faithfulness assertions for restriction functors.

More precisely, if $T \subset S$ is a closed subscheme and if Z_T is a constant torsion-free group on T, then restrictions of extensions by Z_T are controlled by the connected components of the special fibres. The exact sequences obtained from

$$0 \longrightarrow \mathbb{G}_m \longrightarrow N \longrightarrow Z_s \longrightarrow 0 \qquad (5.1.1)$$

for the Neron-Raynaud model N will convert these extension groups into homomorphism groups from component groups to $\mathbb{Q}/\mathbb{Z}$. This is the mechanism that produces the obstruction pairings of Section 7.

6. Biextensions by the Neron model of $\mathbb{G}_m$

Let $S = \operatorname{Spec} V$ be a trait, with generic point η and closed point s. Let N be the Neron model of $(\mathbb{G}_m)_\eta$. The model N fits into an exact sequence

$$0 \longrightarrow \mathbb{G}_m \longrightarrow N \longrightarrow \mathbb{Z}_s \longrightarrow 0, \qquad (6.1.1)$$

where $\mathbb{Z}_s$ is supported on the closed fibre. The Neron mapping property implies that restriction from S to η is an equivalence for extensions and biextensions by N, provided the other group schemes are smooth and satisfy the standard hypotheses.

Thus prolonging an extension or a biextension by $\mathbb{G}_m$ may be separated into two problems. First one prolongs it uniquely by N; this is formal from the Neron mapping property. Then one asks whether the resulting N-extension or N-biextension comes from $\mathbb{G}_m$, i.e. whether its image in the corresponding group with coefficients in $\mathbb{Z}_s$ vanishes. The obstruction is therefore expressed in terms of extensions or biextensions by $\mathbb{Z}_s$.

For a smooth group scheme P over S, write

$$\Phi_P = P_s/P_s^0$$

for the group of connected components of the special fibre. Under the torsion hypotheses used below, one has canonical identifications

$$\mathrm{Ext}^1(P, \mathbb{Z}_s) \simeq \mathrm{Hom}(\Phi_P, \mathbb{Q}/\mathbb{Z}), \qquad (6.2.1)$$

and, for two such groups P, Q,

$$\mathrm{Biext}^1(P, Q; \mathbb{Z}_s) \simeq \mathrm{Hom}(\Phi_P \otimes \Phi_Q, \mathbb{Q}/\mathbb{Z}). \qquad (6.2.2)$$

These identifications are functorial in P and Q and compatible with base change of traits, with a factor given by the ramification index.

7. Canonical prolongations of extensions and biextensions by $\mathbb{G}_m$

Let $S = \operatorname{Spec} V$ be a trait. Let P be a smooth commutative group scheme over S, and set $\Phi = P_s/P_s^0$. Assume that Φ is torsion; this holds, for example, when P is of finite type over S. Then restriction from extensions of P by $\mathbb{G}_m$ over S to extensions of P_η by $\mathbb{G}_m$ over η is fully faithful. If $n\Phi = 0$, every generic extension E_η has a canonical obstruction

$$d(E_\eta) : \Phi \longrightarrow (\mathbb{Q}/\mathbb{Z})_k, \qquad (7.1.1)$$

whose vanishing is necessary and sufficient for E_η to be the generic fibre of an extension of P by $\mathbb{G}_m$. If P_s is connected, restriction is an equivalence.

For biextensions the statement is analogous. Let P, Q be smooth commutative group schemes over S, and put

$$\Phi = P_s/P_s^0, \qquad \Psi = Q_s/Q_s^0.$$

Assume that Φ or Ψ is torsion. Then restriction from biextensions of (P, Q) by $\mathbb{G}_m$ over S to biextensions of (P_η, Q_η) by $\mathbb{G}_m$ over η is fully faithful. If $n\Phi = 0$ or $n\Psi = 0$, each generic biextension E_η has a canonical obstruction pairing

$$d(E_\eta) : \Phi \otimes \Psi \longrightarrow (\mathbb{Q}/\mathbb{Z})_k, \qquad (7.2.1)$$

and this pairing vanishes if and only if E_η extends to a biextension of (P, Q) by $\mathbb{G}_m$ over S. If one of the special fibres P_s, Q_s is connected, restriction is an equivalence.

The proof is the one indicated in Section 6. The Neron-Raynaud model N makes restriction for coefficients N an equivalence. The exact sequence

$$0 \to \mathbb{G}_m \to N \to \mathbb{Z}_s \to 0$$

then produces exact sequences for extensions and for biextensions. Full faithfulness follows from the vanishing of the relevant Hom-groups to $\mathbb{Z}_s$, while essential surjectivity is governed by the obstruction in $\mathrm{Ext}^1(P, \mathbb{Z}_s)$ or $\mathrm{Biext}^1(P, Q; \mathbb{Z}_s)$. The identifications (6.2.1) and (6.2.2) give the displayed forms of the obstruction.

There is also a torsion interpretation. When $n\Phi = 0$, the obstruction of an extension E_η is detected by restricting E_η to the n-torsion of P_η. In the biextension case, the generic biextension defines the torsion pairings of Section 2, and these recover

$$d(E_\eta) : \Phi \otimes \Psi \to (\mathbb{Q}/\mathbb{Z})_k.$$

If $q \in Q(S)$ has image $q_s \in \Psi(k)$, then the extension of P_η obtained by restricting the biextension along q_η has obstruction

$$d_q(x) = d(E_\eta)(x, q_s). \qquad (7.3.1)$$

This characterizes the biextension obstruction by the ordinary extension obstructions obtained by fixing one variable.

The obstruction is symmetric under transposition. If ${}^tE_\eta$ is the transposed biextension of (Q_η, P_η), then the diagram

$$\begin{tikzcd} \Phi \otimes \Psi \arrow[r, "d(E_\eta)"] \arrow[d, "\mathrm{sym}"'] & (\mathbb{Q}/\mathbb{Z})_k \arrow[d, equal] \\ \Psi \otimes \Phi \arrow[r, "d({}^tE_\eta)"'] & (\mathbb{Q}/\mathbb{Z})_k \end{tikzcd} \qquad (7.3.2)$$

commutes. Hence, if $P = Q$ and the biextension is symmetric, the obstruction form on Φ is symmetric.

Finally, if $S' \to S$ is a dominant local morphism of traits, with ramification index $e(S'/S)$, then

$$d(E_{\eta'}) = e(S'/S)\, d(E_\eta) \otimes_k k'. \qquad (7.3.3)$$

For unramified extensions, henselizations, strict henselizations, and completions, the index is one and the obstruction commutes with base change. In general a finite ramified extension may kill the obstruction and allow a prolongation.

One also has the following form over a reduced base. Let S be reduced, and let P be a smooth group scheme over S whose fibres at the maximal points are connected. Then the functor from extensions of P by $\mathbb{G}_m$ to rigidified $\mathbb{G}_m$-torsors on P is fully faithful. If S is normal and P has connected fibres, a rigidified torsor belongs to the essential image if and only if it does so over every maximal point. The analogous statement holds for biextensions of (P, Q): the functor to bitrivialized $\mathbb{G}_m$-torsors on $P \times_S Q$, rigidified along the two unit sections, is fully faithful under the same connectedness hypotheses, and its essential image over a normal base is detected at the maximal points.

As a consequence, if S is normal and P, Q are smooth group schemes over S, with connected maximal fibres and with one of the two schemes having connected fibres, then under the usual hypotheses on the maximal fibres–for instance that one of P_η, Q_η is an extension of an abelian variety by a torus, or has a composition series whose factors are abelian varieties, tori, or additive groups–every bitrivialized torsor of the preceding kind comes from a unique biextension.

Bibliography to Exposé VIII

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Exposé VIII, continuation. Complements on biextensions and group schemes

Original subsequent passages 271–317; completion of Exposé VIII.

4. The case of a base field: divisor correspondences and separation

We continue the discussion begun in the preceding segment. For a birigidified invertible module L on $P \times Q$ to come from a biextension it is necessary and sufficient, in the formulation of 4.3, that its inverse images on $P \times P \times Q$ and on $P \times Q \times Q$, by the two natural multiplication morphisms, satisfy the theorem of the cube. This last condition is simply the translation, in the present notation, of the conditions of VII 2.9.3(b), as made explicit in VII 3.5(b).

4.4. It is not known here whether the cube condition just mentioned is automatic. At least one knows from Raynaud's theorem that, for a given L, there is an integer $n \geq 1$, a power of the characteristic exponent of k, such that $L^{\otimes n}$ satisfies this condition; hence $L^{\otimes n}$ comes from a biextension of (P, Q) by $\mathbb{G}_m$. One can also find a radicial extension k'/k for which the required condition becomes true after base extension. Thus, after this extension, L gives a birigidified invertible module which comes from a biextension. In particular, if k is perfect, the condition of 4.3 is always satisfied. This last assertion, which formally implies the preceding ones by a trace argument, also follows from 4.7 below; see 4.9.

We recall the following lemma, in the form needed below, from EGA, Errata IV, 53, 21.4.13.

Lemma 4.5. Let $f : X \to Y$ be a smooth surjective morphism with integral fibres, with Y normal integral of generic point η. Let L be an invertible module on X whose restriction to X_η is trivial. Then there exists an invertible module M on Y such that $L \simeq f^*M$.

Proposition 4.6. Let P, Q be smooth connected algebraic groups over k, with P affine. Suppose moreover that P admits a composition series whose successive factors are either tori or copies of the additive group $\mathbb{G}_a$, a condition satisfied for example if k is perfect. Suppose, alternatively, that Q is an extension of an abelian variety B by a smooth connected affine algebraic group satisfying the same condition. Under these hypotheses every birigidified invertible module L on $P \times Q$ is trivial. A fortiori every biextension of (P, Q) by $\mathbb{G}_m$ is trivial.

Indeed, by descent and by 4.3 we may pass to a finite separable extension and then to the case where k is separably closed. Then every torus is split. If P has a composition series with quotients isomorphic to $\mathbb{G}_a$ or $\mathbb{G}_m$, repeated use of Lemma 4.5, together with $\mathrm{Pic}(\mathbb{G}_a) = \mathrm{Pic}(\mathbb{G}_m) = 0$, shows that $\mathrm{Pic}(P) = 0$. Applying 4.5 to $P \times Q \to Q$ and using the trivialization on $e_P \times Q$, one obtains that L is trivial.

In the second case one applies Lemma 4.5 to the morphism $P \times Q \to P \times B$. Its generic fibre is a torsor under the affine group occurring in the extension of Q, and it is trivial by the standard relations $H^1(K, \mathbb{G}_a) = 0$ and $H^1(K, \mathbb{G}_m) = 0$. After replacing the resulting line bundle on $P \times B$ by an equivalent birigidified one, one reduces to the case where Q itself is an abelian variety. Then L defines a morphism $P \to Q'$, where Q' is the dual abelian variety of Q. A morphism from a smooth connected affine algebraic group to an abelian variety is constant; since it is pointed, it is zero. Therefore L is pulled back from P, and the rigidification forces this pullback to be trivial.

Proposition 4.7. Let P, Q be smooth connected algebraic groups over k. Assume that P is an extension of an abelian variety A by a smooth connected affine algebraic group L which has a composition series with factors either tori or copies of $\mathbb{G}_a$. Then every birigidified invertible module on $P \times Q$ comes from a biextension of (P, Q) by $\mathbb{G}_m$. Moreover the functor

$$\mathrm{Biext}^1(A, Q; \mathbb{G}_m) \longrightarrow \mathrm{Biext}^1(P, Q; \mathbb{G}_m)$$

induces an isomorphism of groups.

The first assertion is reduced, again by descent, to the case where k is separably closed. The preceding proof shows that the given module is the inverse image of a birigidified invertible module on $A \times Q$. If $P = A$ is abelian, this module defines a morphism $Q \to A'$, and is the pullback of the Weil biextension on $A \times A'$. The result follows from VII 2.9.6, where divisor correspondences on a product of abelian varieties are shown to come from biextensions. For general P, the asserted isomorphism follows from the exact sequence of Biext^1's associated with

$$0 \longrightarrow L \longrightarrow P \longrightarrow A \longrightarrow 0$$

and from

$$\mathrm{Biext}^1(L, Q; \mathbb{G}_m) = 0, \qquad \mathrm{Biext}^2(A, Q; \mathbb{G}_m) = 0,$$

the first equality being a special case of the rigidity assertion of 4.3 and the second the first case of 4.6.

Corollary 4.8. Under the hypotheses of 4.7, suppose that Q is an extension of an algebraic group B by a smooth connected affine group M. Then

$$\mathrm{Biext}^1(A, B; \mathbb{G}_m) \longrightarrow \mathrm{Biext}^1(P, Q; \mathbb{G}_m)$$

is an equivalence of rigid categories, hence an isomorphism on groups of isomorphism classes. This follows as above from the exact Biext-sequence attached to $0 \to M \to Q \to B \to 0$ and from the vanishing supplied by 4.6.

Remark 4.9. If k is perfect and P, Q are smooth connected algebraic groups, each of them is an extension of an abelian variety by a smooth connected affine group, and that affine group is a product of a torus by an iterated extension of groups $\mathbb{G}_a$. Hence the theory of biextensions of (P, Q) by $\mathbb{G}_m$ is reduced, by 4.8, to the classical theory for the pair of abelian varieties (A, B).

Corollary 4.10. Let P, Q be smooth connected algebraic groups over k, let L be a smooth connected affine algebraic subgroup of P, and let E be a biextension of (P, Q) by $\mathbb{G}_m$. The pairing on Tate modules deduced from E,

$$T_\ell(P) \times T_\ell(Q) \longrightarrow T_\ell(\mathbb{G}_m),$$

vanishes on $T_\ell(L) \times T_\ell(Q)$. Over an algebraic closure this is exactly the pairing attached to the restriction of E to (L, Q), which is trivial by 4.6.

4.11. Separating biextensions. Let P, Q be smooth algebraic groups over k, let E be a biextension of (P, Q) by $\mathbb{G}_m$, and let T be the maximal subtorus of P. We seek the left annihilator of the pairing

$$T_\ell(P) \times T_\ell(Q) \longrightarrow T_\ell(\mathbb{G}_m).$$

It always contains $T_\ell(T)$. Choose an extension k'/k over which P and Q are extensions of abelian varieties by affine groups built from tori and additive groups; if the unipotent ranks are zero, one may take $k' = k$, and in any case one may take the perfect closure or a finite radicial extension. Let A' and B' be the abelian quotients of $P_{k'}$ and $Q_{k'}$. By 4.7 the pullback of E comes from a uniquely determined biextension F' of (A', B'). Thus the left annihilator is the inverse image of the left annihilator of the pairing attached to F'. If $C' \subset A'$ is the largest abelian subvariety on which F' is trivial, equivalently the connected reduced kernel of the morphism

$$A' \longrightarrow (B')'$$

associated with F', then the left annihilator is the inverse image of $T_\ell(C')$. We shall say that E is *left separating* if this annihilator is exactly $T_\ell(T)$; equivalently, the associated morphism $A' \to (B')'$ has finite kernel. Dually, E is *right separating* if the dual associated morphism has finite kernel. It is *separating* if both conditions hold, equivalently if the associated morphism is an isogeny. These notions are invariant under arbitrary extension of the base field.

4.12. Let P be a smooth connected group scheme and L an invertible module on P. Put

$$\delta(L) = m^*L \ \mathrm{pr}_1^* L^{-1} \ \mathrm{pr}_2^* L^{-1},$$

with notation as in 4.2. This is a birigidified invertible module on $P \times P$. If it comes from a biextension, that biextension is unique by 4.3 and is symmetric. When P is an extension of an abelian variety A by a group built from tori and additive groups, and when the maximal torus is split, the construction reduces to the abelian case: one may write $L = f^*M$, for $f : A \to P$, and then $\delta(L)$ is the pullback of $\delta(M)$.

Proposition 4.13. In the preceding situation, assume L is ample. If the biextension $\delta(L)$ of (P, P) by $\mathbb{G}_m$ is defined, then it is separating. If P is an extension of an abelian variety A by a smooth connected affine group satisfying 4.7, the biextension of (A, A) from which $\delta(L)$ comes is of the form $\delta(M)$, with M ample on A. This is reduced to the algebraically closed case and then to Raynaud's theorem that ampleness descends to the abelian quotient; Weil's theory then gives that the corresponding homomorphism $A \to A'$ is an isogeny.

Remark 4.14. In the favourable case where P is an extension of an abelian variety A by a group built from tori and additive groups, the composite

$$\mathrm{Pic}(P) \longrightarrow \mathrm{Biext}^1(P, P; \mathbb{G}_m) \xrightarrow{\sim} \mathrm{Biext}^1(A, A; \mathbb{G}_m)$$

identifies the Neron-Severi group $\mathrm{NS}(A)$, or equivalently $\mathrm{NS}(P) = \mathrm{Pic}(P)/\mathrm{Pic}^0(P)$, with the group of symmetric biextensions of (P, P) by $\mathbb{G}_m$. An ample divisor correspondence, or ample biextension, is one coming from an ample element of this Neron-Severi group. Raynaud's theorem says that an element $\xi \in \mathrm{Pic}(P)$ is ample precisely when its image in the group of symmetric biextension classes is ample in this sense.

5. Extensions and biextensions by truncated constant group schemes without torsion

The next two sections record auxiliary results used in Section 7, especially in the case $\ell = p$.

Proposition 5.1. Let S be geometrically unibranch and irreducible with generic point η, and let Z be a torsion-free group, viewed as a constant group scheme Z_S on S. Every torsor under Z_S is trivial, and the functor sending a torsor to its generic fibre is an equivalence with the category of torsors under Z_η.

The proof first uses descent to see that such torsors are represented and étale-locally trivial. It is then enough to prove $H^1(S, Z_S) = 0$. Passing to the strict localization and then to the spectrum of a field reduces the assertion to the elementary fact that, for a profinite Galois group Γ,

$$H^1(\Gamma, Z) = \mathrm{Hom}_{\mathrm{cont}}(\Gamma, Z) = 0$$

when Z is torsion-free and discrete.

Corollary 5.2. With the same hypotheses, let P be smooth over S. Then torsors under the induced constant group on P are controlled by the generic fibres. Indeed, P is a locally finite sum of geometrically unibranch irreducible preschemes whose generic points lie over η, and 5.1 applies component by component. The analogous assertion for bitorsors follows in the same way.

Proposition 5.3. Let S, Z be as in 5.1 and let P be a smooth group scheme over S. Restriction to the generic fibre gives an equivalence from the category of group-scheme extensions of P by Z_S to the corresponding category over η. If P and Z are commutative, it also gives an equivalence for commutative extensions. The proof is the bitorsor description of VII 1.1.6, applied to P, $P \times P$, and $P \times P \times P$.

Corollary 5.4. If P, Q are smooth commutative group schemes over S, restriction to η gives an equivalence between biextensions of (P, Q) by Z_S and biextensions of (P_η, Q_η) by Z_η.

Proposition 5.5. Let k be a field, Z a torsion-free group, and P a group scheme locally of finite type over k. Let P^0 be the neutral connected component and $\Phi = P/P^0$ the group of connected components.

(i) The natural functor from extensions of Φ by Z_k to extensions of P by Z_k, and similarly for commutative extensions, is an equivalence. Hence, if P is connected, every such extension is trivial and the category is rigid.

(ii) If Φ is torsion and P, Z are commutative, the category of extensions of P by Z_k is rigid. If some $n > 0$ kills Φ, then the group of commutative extension classes is canonically

$$\mathrm{Ext}^1(P, Z_k) \simeq \mathrm{Hom}(\Phi, \mathbb{Q}/Z),$$

the isomorphism being the boundary map attached to $0 \to Z \to \mathbb{Q} \to \mathbb{Q}/Z \to 0$.

Corollary 5.6. Let P, Q be commutative group schemes locally of finite type over k, and put $\Phi = P/P^0$, $\Psi = Q/Q^0$. Biextensions of (P, Q) by Z_k are equivalent to biextensions of (Φ, Ψ) by Z_k. If one of Φ, Ψ is torsion, the category is rigid; if one is killed by an integer $n > 0$, then

$$\mathrm{Biext}^1(P, Q; Z_k) \simeq \mathrm{Hom}(\Phi \otimes \Psi, \mathbb{Q}/Z).$$

5.7. Truncated constant schemes. Let S be a scheme, $T \subset S$ a closed subscheme, and $U = S - T$. For a set Z, the *constant scheme on S truncated on T* is obtained by gluing the disjoint sum of copies of S, indexed by Z, along the common open U. It will be denoted $Z_{S,T}$. It is étale over S, and for a variable S-scheme S' one has functorially

$$\mathrm{Hom}_S(S', Z_{S,T}) \simeq \mathrm{Hom}_T(S'_T, Z_T).$$

Equivalently, as a sheaf, $Z_{S,T}$ is the extension by the closed part of the constant sheaf on T. The construction is functorial in Z, commutes with finite projective limits, and hence sends groups to groups and commutative groups to commutative groups. It is compatible with arbitrary base change.

5.8. Assume Z is commutative. Restriction to T gives an equivalence from torsors under $Z_{S,T}$ to torsors under Z_T. Full faithfulness reduces by fpqc descent to the trivial torsor and is the preceding Hom formula; essential surjectivity is reduced to the cohomological vanishing $R^1 i_* Z_T = 0$ for the closed immersion $i : T \to S$.

Proposition 5.9. For a commutative group scheme P over S, restriction to T gives an equivalence between extensions of P by $Z_{S,T}$ and extensions of P_T by Z_T.

Corollary 5.10. If Q is another commutative group scheme over S, restriction to T gives an equivalence between biextensions of (P, Q) by $Z_{S,T}$ and biextensions of (P_T, Q_T) by Z_T.

6. Extensions and biextensions by the Neron-Raynaud model of $\mathbb{G}_m$

6.1. Associated with $Z_{S,T}$ is a canonical surjection from the ordinary constant sheaf Z_S. Let $\widetilde{Z}_{S,T}$ denote its kernel; thus there is an exact sequence, functorial in Z,

$$0 \longrightarrow \widetilde{Z}_{S,T} \longrightarrow Z_S \longrightarrow Z_{S,T} \longrightarrow 0.$$

The kernel is the disjoint sum of pieces S_n, $n \in Z$, with $S_0 = S$ and $S_n = U$ for $n \neq 0$. It follows that for any group G on S, restriction to U induces a bijection

$$\mathrm{Hom}(\widetilde{Z}_{S,T}, G) \simeq \mathrm{Hom}(Z_U, G_U).$$

Lemma 6.2. Let G be a commutative group on S. The category of commutative group extensions of $\widetilde{\mathbb{Z}}_{S,T}$ by G is equivalent to the category of pairs (F, s), where F is a torsor under G and s is a section of $F|_U$. Under this equivalence, the extension corresponding to (F, s) has, over the component indexed by an integer n, the torsor $F^{\otimes n}$, with the trivialization over U deduced from $s^{\otimes n}$.

If T is a sum of connected non-empty components T_i, a section of the resulting group over S can be described by integers n_i indexed by those components and by compatible sections of the powers $F^{\otimes n_i}$. When T is connected and non-empty, a section is simply an integer n and a section of $F^{\otimes n}$ on S. This description also shows that the extension is representable whenever all the torsors $F^{\otimes n}$ are representable, for example when G is represented by an affine group scheme over S.

6.3. We now take $G = \mathbb{G}_m$. A torsor under $\mathbb{G}_m$ is an invertible module L, and a section over U is a trivialization of $L|_U$. Thus the data are a pair (L, s). When U is schematically dense in S, with S locally noetherian, or more generally reduced with locally finite irreducible components, such a pair is the same thing as a divisor D supported on T: one associates

$$(L, s) = (\mathcal{O}_S(-D), \tau|_U),$$

where τ is the canonical rational section. In all cases a divisor D supported on T defines in this way an extension $\mathcal{G}$ of $\widetilde{\mathbb{Z}}_{S,T}$ by $\mathbb{G}_m$. This is called the *Neron-Raynaud model of* $\mathbb{G}_m$ associated with D and T; when $T = \operatorname{supp} D$, it is omitted from the notation. Its sections are the regular meromorphic functions on S whose divisor is locally an integral multiple of D.

6.4. Suppose S is normal with locally finite irreducible components, and that D is positive with all multiplicities equal to one. Put $T = \operatorname{supp} D$, and suppose T is locally irreducible and geometrically unibranch. Then every rational function on S, regular on U, has divisor locally an integral multiple of D; hence

$$\mathcal{G}(S) = \Gamma(U, \mathcal{O}_U^*).$$

Moreover, for every smooth S-scheme S', one obtains functorially

$$\mathrm{Hom}_S(S', \mathcal{G}) \simeq \Gamma(S'_U, \mathcal{O}^*_{S'_U}).$$

This characterizes $\mathcal{G}$ uniquely and justifies the name Neron-Raynaud model.

Theorem 6.5. Let S be normal and locally integral, let D be a positive divisor without multiple components, with geometrically unibranch and locally integral support, and let $\mathcal{G}$ be the associated Neron-Raynaud model. The restriction functor from torsors under $\mathcal{G}$ on S to torsors under $\mathbb{G}_m$ on U is fully faithful. If the local rings of S are factorial, for instance if S is regular, this functor is an equivalence.

Full faithfulness is local for the Zariski topology and reduces to the trivial torsor; then it is exactly the equality $\mathcal{G}(S) = \Gamma(U, \mathcal{O}_U^*)$. Essential surjectivity in the factorial case is the assertion that every invertible module on U extends to S.

Corollary 6.6. Let S be locally noetherian and regular, and let $D \subset S$ be a reduced geometrically unibranch closed subscheme, pure of codimension one. Let $\mathcal{G}$ be the associated Neron-Raynaud model and let P be a smooth commutative group scheme over S. Restriction to $U = S - D$ induces an equivalence between commutative group extensions of P by $\mathcal{G}$ and extensions of P_U by $\mathbb{G}_m$. The non-commutative version is analogous, with bitorsors in place of torsors.

Corollary 6.7. Under the same hypotheses, if P, Q are smooth commutative group schemes over S, restriction to U gives an equivalence between biextensions of (P, Q) by $\mathcal{G}$ and biextensions of (P_U, Q_U) by $\mathbb{G}_m$.

Remarks 6.8. When S is a trait and D is its closed point, U is the spectrum of the fraction field and $\mathcal{G}$ is the usual Neron-Raynaud model of the multiplicative group. The proofs of 6.6 and 6.7 use Theorem 6.5 after smooth base change, so bases more general than traits enter formally. More generally, if U is only a schematically dense open in a normal locally noetherian scheme S, the corresponding Neron-Raynaud model may be a sheaf rather than a representable scheme; it is described by the exact sequence

$$0 \longrightarrow \mathbb{G}_m \longrightarrow \mathcal{G} \longrightarrow \mathcal{D} \longrightarrow 0,$$

where $\mathcal{D}$ is the sheaf of divisors supported in $S - U$. Compatibility with smooth base change follows from EGA IV, Errata 53, 21.4.9.

6.9. Autocritique, after observations of P. Deligne. Deligne points out that Sections 5 and 6 conceal a simple geometric statement behind a hand-made formalism. The core assertions are these: if $j : U \hookrightarrow S$ is a schematically dense open immersion in a normal noetherian scheme, then $\mathbb{G}_{mS}$ is a subsheaf of $j_*\mathbb{G}_{mU}$, and the quotient sheaf of divisors commutes with smooth base change. If $U = S - Y$ with Y a geometrically unibranch divisor, the quotient is represented by the truncated constant group scheme associated with the closed part. Finally, the truncated constant sheaf attached to a closed subscheme is representable and compatible with base change. These statements give, on the smooth site, the Neron-Raynaud model used above and allow the results of Section 6 to be deduced formally.

7. Canonical prolongations of extensions and biextensions by $\mathbb{G}_m$

7.0. Let S be a trait, η its generic point, s its closed point, and $k = k(s)$. Let P be a smooth commutative group scheme over S, and put $P_s = P \times_S s$. In this section all groups are commutative.

Theorem 7.1.

a) Assume that the group of connected components

$$\Phi = P_s / P_s^0$$

is torsion, for example if P is of finite type over S. Then the functor from extensions of P by $\mathbb{G}_m$ to extensions of P_η by $\mathbb{G}_m$ is fully faithful. If some $n > 0$ kills Φ, then every extension E_η of P_η by $\mathbb{G}_m$ has a canonical obstruction

$$d(E_\eta) : \Phi \longrightarrow \mathbb{Q}/Z,$$

whose vanishing is necessary and sufficient for E_η to be the generic fibre of an extension over S. If P_s is connected, the restriction functor is an equivalence.

b) Let P, Q be smooth commutative group schemes over S, put

$$\Phi = P_s/P_s^0, \qquad \Psi = Q_s/Q_s^0,$$

and assume that Φ or Ψ is torsion. Then restriction from biextensions of (P, Q) by $\mathbb{G}_m$ to biextensions of (P_η, Q_η) by $\mathbb{G}_m$ is fully faithful. If an integer $n > 0$ kills Φ or Ψ, a biextension E_η has a canonical obstruction pairing

$$d(E_\eta) : \Phi \otimes \Psi \longrightarrow \mathbb{Q}/Z,$$

and its vanishing is necessary and sufficient for prolongation to a biextension over S. If P_s or Q_s is connected, restriction is an equivalence.

The proof replaces $\mathbb{G}_m$ by the Neron-Raynaud model $\mathcal{G}$. By 6.6 and 6.7, restriction from extensions or biextensions by $\mathcal{G}$ to the generic fibre is an equivalence. The only remaining question is the passage from the kernel $\mathbb{G}_m$ to $\mathcal{G}$, and this is measured by the exact sequence

$$0 \longrightarrow \mathbb{G}_m \longrightarrow \mathcal{G} \longrightarrow Z_{S,T} \longrightarrow 0$$

and by the associated exact sequences of Ext^1 and Biext^1. The groups which control full faithfulness become $\mathrm{Hom}(\Phi, Z)$ or $\mathrm{Hom}(\Phi \otimes \Psi, Z)$, hence vanish under the torsion hypotheses; the obstruction groups become $\mathrm{Hom}(\Phi, \mathbb{Q}/Z)$ and $\mathrm{Hom}(\Phi \otimes \Psi, \mathbb{Q}/Z)$, as in 5.5 and 5.6.

Remark 7.2. The same idea treats a normal connected regular base S. To decide whether an extension over the generic fibre prolongs to S, it suffices to test after base change to the local traits $\mathrm{Spec}(\mathcal{O}_{S,x})$ for codimension-one points x. The condition is the vanishing of the corresponding homomorphisms $\Phi_x \to \mathbb{Q}/Z$. The analogous statement for biextensions is obtained by the same argument and the codimension at least two extension property for line bundles.

7.3. Functorialities of the obstructions. Let $u : P' \to P$ be a morphism of smooth group schemes over S. If E_η is an extension of P_η by $\mathbb{G}_m$, the obstruction to prolonging its inverse image along u_η is the pullback of the obstruction of E_η. In diagrammatic form:

$$\begin{tikzcd} \Phi' \arrow[d, "u_s"'] \arrow[dr, "d(E'_\eta)"] & \\ \Phi \arrow[r, "d(E_\eta)"'] & \mathbb{Q}/Z. \end{tikzcd}$$

Similarly, for morphisms $u : P' \to P$ and $v : Q' \to Q$, the obstruction pairing for the inverse image of a biextension is obtained by composing with $u_s \otimes v_s$:

$$\begin{tikzcd} \Phi' \otimes \Psi' \arrow[d, "u_s \otimes v_s"'] \arrow[dr, "d(E'_\eta)"] & \\ \Phi \otimes \Psi \arrow[r, "d(E_\eta)"'] & \mathbb{Q}/Z. \end{tikzcd}$$

For an extension E_η of P_η there is a largest open subgroup $V \subset P$, containing P_η, over which E_η prolongs. It is defined by

$$V_s/V_s^0 = \mathrm{Ker}\, d(E_\eta) \subset \Phi.$$

For a biextension E_η of (P_η, Q_η), there is similarly a largest open subgroup $V \subset P$ such that the restriction to (V_η, Q_η) prolongs; it is defined by the left annihilator of the pairing

$$d(E_\eta) : \Phi \otimes \Psi \longrightarrow \mathbb{Q}/Z.$$

There are symmetric statements interchanging P and Q.

The obstruction is compatible with symmetry. If ${}^sE_\eta$ denotes the symmetric biextension of (Q_η, P_η), then the square

$$\begin{tikzcd}
\Phi \otimes \Psi \arrow[d, "\mathrm{sym}"'] \arrow[dr, "d(E_\eta)"] & \\
\Psi \otimes \Phi \arrow[r, "d({}^sE_\eta)"'] & \mathbb{Q}/Z
\end{tikzcd}$$

commutes. Hence a symmetric biextension has a symmetric obstruction form, and if the obstruction vanishes, the prolonged biextension is symmetric as well.

The obstruction for a biextension can be reduced to the obstruction for extensions by evaluating on sections of the second group. If $q \in Q(S)$ has image $q_s \in \Psi(k)$, and if q^*E_η denotes the extension of P_η induced by q, then

$$d(q^*E_\eta)(x) = d(E_\eta)(x, q_s).$$

Conversely, the extension case is obtained from the biextension case by taking $Q = \widetilde{\mathbb{Z}}_{S,T}$ and evaluating at the unit section.

If $S' \to S$ is a local dominant morphism of traits, with ramification index

$$e(S'/S) = \mathrm{length}_{V'}(V'/\mathfrak{m}_V V'),$$

then under the natural comparison of component groups the obstruction after base change is multiplied by $e(S'/S)$. In particular, for étale, henselian, strictly henselian, or completed base change, where the ramification index is one, the obstruction commutes with base change. Consequently every extension or biextension over the generic fibre becomes prolongable after a suitable finite extension of traits with separable generic residue field and prescribed ramification divisible by the exponent occurring in Theorem 7.1.

Proposition 7.4. Let S be reduced.

a) Let P be a smooth group scheme over S whose fibres over maximal points are connected. The natural functor

$$\mathrm{Ext}(P, \mathbb{G}_m) \longrightarrow \mathrm{TORSRIG}(P, \mathbb{G}_m)$$

from extensions of P by $\mathbb{G}_m$ to $\mathbb{G}_m$-torsors on P rigidified along the unit section is fully faithful. If S is normal and P has connected fibres, such a rigidified torsor comes from an extension if and only if it does so on the fibres over all maximal points.

b) Let P, Q be smooth group schemes over S whose fibres over maximal points are connected. The natural functor

$$\mathrm{Biext}(P, Q; \mathbb{G}_m) \longrightarrow \mathrm{TORSBIRIG}(P, Q; \mathbb{G}_m)$$

from biextensions to birigidified $\mathbb{G}_m$-torsors on $P \times Q$ is fully faithful. If S is normal and one of P, Q has connected fibres, an object of the right hand side comes from a biextension if and only if this is true over every maximal point of S.

For (a), the full faithfulness follows from the fact that a morphism from a flat product $X \times_S Y$ to the separated group $\mathbb{G}_m$ is trivial as soon as it is trivial on the fibres over the maximal points. This is the same argument as in 4.1. The essential-image assertion is local on S; after reducing to the noetherian normal case, Theorem 7.1 gives an open complement of codimension at least two over which the desired extension exists, and normality then extends the required isomorphism over all of $P \times_S P$. The proof of (b) is identical, with biextension data in place of extension data.

Corollary 7.5. Let S be normal and let P, Q be smooth group schemes over S, with connected fibres over maximal points, and with one of P, Q having connected fibres everywhere. Suppose moreover that for every maximal point η of S, P_η or Q_η is an extension of an abelian variety by a torus, more generally admits a composition series whose factors are abelian varieties, tori, or copies of $\mathbb{G}_a$. Then the functor in 7.4(b) is an equivalence. This is an immediate consequence of 7.4(b) and 4.7.

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Exposé IX. Néron models and monodromy

by A. Grothendieck
with an appendix by M. Raynaud

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14. Transcendental links: the rigid-analytic case.

Guide to the exposé

The first four sections review the ingredients leading to semi-stable reduction; the following sections develop the algebraic study of the Néron model. The guide to the exposé is the directed graph

$$\begin{gathered} 1 \to 2 \to 3, \qquad 3 \to 4, \qquad 3 \to 5, \\ 5 \to 6, \quad 5 \to 7, \quad 5 \to 9, \quad 7 \to 8, \\ 7 \to 13, \quad 7 \to 14, \quad 8 \to 13, \quad 8 \to 14, \\ 9 \to 10, \quad 10 \to 11, \quad 10 \to 12, \quad 10 \to 14. \end{gathered}$$

which records the main dependencies between the numbered sections.

0. Introduction

In this exposé we apply the results of the preceding exposés to the study of the Néron model A of an abelian scheme A_K over the function field K of a henselian trait S. The most important properties of the Néron model can be interpreted in terms of the local monodromy group I acting on the Tate module $T_\ell(A_K(\bar{K}))$, where ℓ is a prime distinct from the residual characteristic p. Since the dual of this Tate module is the ℓ-adic cohomology group $H^1(A_{\bar{K}}, \mathbb{Z}_\ell)$, and since for a smooth projective K-scheme X_K, the group $H^1(X_{\bar{K}}, \mathbb{Z}_\ell)$ identifies with $H^1(\mathrm{Alb}_{X/K,\bar{K}}, \mathbb{Z}_\ell)$, this exposé may also be read as a detailed study of local monodromy phenomena for the first ℓ-adic, or better motivic, cohomology of smooth projective varieties over K. In this perspective the results below should ultimately belong to a Néron theory for motives of arbitrary weight.

The principal result is the semi-stable reduction theorem, Theorem 3.6: after a finite separable extension K'/K, the Néron model of $A_{K'}$ has special fibre whose neutral component is an extension of an abelian variety by a torus. Serre conjectured this result in 1964, and the principle of the proof used here goes back to the same year. The condition is equivalent to saying that the monodromy action on $T_\ell(A_{\bar{K}})$ is essentially unipotent of step two. In that form it is a special case of the monodromy theorem proved earlier. To pass between the group-theoretic and Néron-model formulations one needs the orthogonality theorem 2.4, for which the theory of biextensions developed in the preceding exposés is the decisive tool.

Mumford found a different proof of the semi-stable reduction theorem using theta functions, though, in the form then available, it was restricted to residual characteristic different from 2. The theorem is indispensable not only in the algebraic study of Néron models, but also in the rigid-analytic or theta-function approach to abelian varieties over valued fields, as developed by Tate, Morikawa, Mumford and Raynaud. It is also a key ingredient in the Artin–Deligne–Mumford semi-stable reduction theorem for curves, already used for the monodromy action on the fundamental group and used again below in the proof of the integrality and positivity theorem 10.4.

A special feature of the present exposé is that, in studying $T_\ell(A)$, the case $\ell = p$ is also systematically taken into account. The language of Barsotti–Tate groups, or p-divisible groups, allows one to extend to this case many of the phenomena established when $\ell \neq p$, subject sometimes to Tate's fundamental theorem, presently known only in characteristic zero. In characteristic zero the pro-p-Barsotti–Tate group $T_p(A)$ can still be interpreted by means of the monodromy action on the p-adic Tate module. Nevertheless the analogy with $\ell \neq p$ is deceptive: for example the monodromy action need not be essentially unipotent. With more suitable translations, such as 5.13 below, the study of geometrically defined Barsotti–Tate groups over the function field of a trait belongs to the local monodromy theory of p-adic local systems. For higher cohomology a suitable enlargement of the category of Barsotti–Tate groups should be inspired by crystalline cohomology.

Finally, the exposition is close to the oral lectures up to Section 4 and becomes more detailed from Section 5 onward. Some results below, especially 10.4 and 12.5, use the Picard–Lefschetz formula, which is established only later in the exposé.

1. Néron models of abelian schemes; notation; the canonical pairings

1.0. Poincaré biextension and Tate pairings

Throughout this exposé let S be a trait, with generic point η, closed point s, residue field k, and function field K. Let A_K be an abelian scheme over K, and let A'_K denote its dual. The duality between A_K and A'_K is given by the Poincaré divisor correspondence $\mathcal{W}_K$ on $A_K \times A'_K$, i.e. by an invertible sheaf birigidified along the unit sections. Its class is

$$w_K \in \operatorname{Biext}^1(A_K, A'_K; \mathbb{G}_{mK}). \tag{1.0.1}$$

The corresponding pairing is

$$\psi_\ell : T_\ell(A_K) \times T_\ell(A'_K) \longrightarrow T_\ell(\mathbb{G}_{mK}). \tag{1.0.2}$$

For $\ell \neq p = \operatorname{char} k$, this is equivalently a pairing of free $\mathbb{Z}_\ell$-modules of finite type with Galois action,

$$\psi_\ell : T_\ell(A_K(\bar{K})) \times T_\ell(A'_K(\bar{K})) \longrightarrow \mathbb{Z}_\ell(1), \tag{1.0.3}$$

compatible with $\Gamma = \operatorname{Gal}(\bar{K}/K)$. By the preceding exposé this pairing is perfect.

Sometimes a polarization $\xi \in \operatorname{NS}(A_K/K)$ is chosen. It gives the usual homomorphism

$$t_\xi : A_K \longrightarrow A'_K \tag{1.0.5}$$

and therefore

$$T_\ell(t_\xi) : T_\ell(A_K) \longrightarrow T_\ell(A'_K). \tag{1.0.6}$$

The induced alternating form is

$$\psi_{\ell,\xi}(x, y) = \psi_\ell(x, T_\ell(t_\xi)y) : T_\ell(A_K) \times T_\ell(A_K) \longrightarrow \mathbb{Z}_\ell(1). \tag{1.0.7}$$

For $\ell \neq p$ this is a bilinear form on free $\mathbb{Z}_\ell$-modules with operators. The fact that ξ is a polarization means that t_ξ is an isogeny; equivalently, $T_\ell(t_\xi)$ is injective with finite cokernel, or the form $\psi_{\ell,\xi}$ is separated.

1.1. The Néron model and the component groups

Recall that there is a smooth S-scheme A, the Néron model of A_K, representing the functor

$$S' \longmapsto \mathrm{Hom}_K(S'_K, A_K) \tag{1.1.1}$$

for smooth relative S-schemes S'. Thus, for S' smooth over S, there is a canonical bijection

$$\mathrm{Hom}_S(S', A) = \mathrm{Hom}_K(S'_K, A_K). \tag{1.1.2}$$

Néron constructed A when k is perfect, and Raynaud extended the existence theorem to arbitrary residue fields. We shall use the existence theorem, and the fact that A is quasi-projective over S. The generic fibre of A is canonically A_K, and the restriction map in (1.1.2) is simply restriction to generic fibres.

The functor (1.1.1) is a commutative group functor, so A is a smooth commutative group scheme over S. Let

$$A_s = A \times_S s \tag{1.1.4}$$

be its special fibre, and let A_s^0 be the neutral component. The component group

$$\Phi = A_s / A_s^0 \tag{1.1.5}$$

is a finite ét group scheme over k, equivalently, after choosing a separable closure $\bar{k}$, a finite ordinary group $\Phi(\bar{k})$ with its natural action of $\mathrm{Gal}(\bar{k}/k)$.

The dual abelian scheme A'_K has its Néron model A', with special fibre A'_s, neutral component A'^0_s, and component group

$$\Phi' = A'_s / A'^0_s. \tag{1.1.8}$$

1.2–1.5. The component pairing and the canonical biextension

Applying the preceding theory of prolongation of biextensions to the pair (A, A') and to $\mathcal{W}_K$, one obtains the canonical pairing

$$\Phi \times \Phi' \longrightarrow (\mathbb{Q}/\mathbb{Z})_k, \tag{1.2.1}$$

or equivalently a pairing of $\mathrm{Gal}(\bar{k}/k)$-modules

$$\Phi(\bar{k}) \times \Phi'(\bar{k}) \longrightarrow \mathbb{Q}/\mathbb{Z}. \tag{1.2.2}$$

It is the obstruction to prolonging $\mathcal{W}_K$ to a biextension of (A, A') by $\mathbb{G}_{mS}$.

Conjecture 1.3. The pairing (1.2.1) is a perfect duality: it identifies each component group with the dual of the other, with values in $(\mathbb{Q}/\mathbb{Z})_k$.

It is enough to prove this after replacing S by a complete trait with separably closed residue field. Artin and Mazur proved the conjecture when A_K is the Jacobian of a smooth proper curve over K, by using their general theory of autoduality for the relative Jacobian in the derived-category sense. They also proved it when the residue field is finite, by using Tate duality. Later sections recover the prime-to-p primary components and the semi-stable case, making the conjecture highly plausible. It should in fact be a by-product of a general duality theory for group schemes over fraction fields of complete discrete valuation rings, formally analogous to SGA 5 I and containing local class-field theory in Serre's form.

If $U \subset A$ and $U' \subset A'$ are open subgroup schemes such that $\mathcal{W}_K$ prolongs to a biextension of (U, U') by $\mathbb{G}_{mS}$, the subgroups $U/U^0 \subset \Phi$ and $U'/U'^0 \subset \Phi'$ are orthogonal for (1.2.1). The resulting biextension is denoted by $\mathcal{W}$. Taking $U = A^0$, $U' = A'^0$, one obtains a canonical biextension

$$\mathcal{W}^0 \in \mathrm{Biext}^1(A^0, A'^0; \mathbb{G}_{mS}). \tag{1.4.4}$$

All other prolongations restrict to $\mathcal{W}^0$ on (A^0, A'^0).

Proposition 1.5. The special fibre $\mathcal{W}_s$, a biextension of (A^0_s, A'^0_s) by $\mathbb{G}_{mk}$, is separated, i.e. separated on the left and on the right.

Indeed, choose an ample invertible sheaf L on A^0_s. It defines a homomorphism $A^0_s \to A'^0_s$, and the inverse image of $\mathcal{W}_s$ by $(\mathrm{Id}, A^0_s \to A'^0_s)$ is the biextension attached to the birigidified sheaf L. Since L is ample, the separatedness theorem for biextensions applies. Symmetry gives separatedness on both sides. This will be sharpened below when A^0_s has unipotent rank zero.

2. Fixed part and toric part of $T_\ell(A_K)$; good reduction; orthogonality for $\ell \neq p$

2.1. Toric, unipotent, and abelian ranks

The commutative algebraic group A^0_s has a greatest subtorus, stable under extension of the base field,

$$T_s \subset A^0_s. \tag{2.1.1}$$

If k is perfect, then A^0_s/T_s is an extension of an abelian variety B_s by a smooth connected unipotent group U_s:

$$0 \longrightarrow U_s \longrightarrow A^0_s/T_s \longrightarrow B_s \longrightarrow 0. \tag{2.1.2}$$

More generally, whenever A^0_s is an extension

$$0 \longrightarrow L_s \longrightarrow A^0_s \longrightarrow B_s \longrightarrow 0 \tag{2.1.3}$$

by a smooth connected affine group L_s and an abelian variety B_s, this structure is unique; then T_s is the maximal subtorus of L_s and $U_s = L_s/T_s$. Thus (2.1.2) again applies. The condition that A^0_s have unipotent rank zero is equivalent to saying that A^0_s/T_s is already an abelian variety, i.e. $U_s = 0$.

For $\ell \neq \mathrm{char}\, k$, the inclusion of the torus gives an exact sequence of pro-group schemes

$$0 \longrightarrow T_\ell(T_s) \longrightarrow T_\ell(A^0_s) \longrightarrow T_\ell(A^0_s/T_s) \longrightarrow 0. \tag{2.1.6}$$

When (2.1.2) is available this becomes

$$0 \longrightarrow T_\ell(T_s) \longrightarrow T_\ell(A^0_s) \longrightarrow T_\ell(B_s) \longrightarrow 0. \tag{2.1.8}$$

Put

$$n = \dim A_K, quad \mu = \dim T_s, \quad \lambda = \dim U_s, \quad a = \dim B_s. \tag{2.1.9}$$

Then

$$n = \mu + \lambda + a, \tag{2.1.10}$$

and, for $\ell \neq p$,

$$\mathrm{rk}\, T_\ell(T_s) = \mu, \quad \mathrm{rk}\, T_\ell(B_s) = 2a, \quad \mathrm{rk}\, T_\ell(A^0_s) = \mu + 2a. \tag{2.1.11}$$

If k has characteristic $p > 0$ and A^0_s is p-divisible, equivalently has unipotent rank zero, the same exact sequence holds for $\ell = p$ as an exact sequence of Barsotti–Tate groups, and the same rank formulas hold with rank interpreted as height.

The dual Néron model A' gives analogous objects T'_s, U'_s, B'_s. Later it will follow that the corresponding integers μ, λ, a are the same for A and for A', so no new notation is needed.

2.2. The fixed part

We compare $T_\ell(A_K)$, $T_\ell(A^0_s)$, and $T_\ell(A^0)$. First recall the following lemma.

Lemma 2.2.1. Let $m > 0$. For the multiplication endomorphism of A^0, the following conditions are equivalent:

(i) $m\,\mathrm{Id}_{A^0}$ is flat;

(ii) $m\,\mathrm{Id}_{A^0}$ is surjective;

(iii) $m\,\mathrm{Id}_{A^0}$ is quasi-finite;

(iv) m is prime to $\operatorname{char} k$, or A^0_s has unipotent rank zero.

When these conditions hold for m, they hold for all powers m^ν, and the group schemes

$$ {}_{m^\nu}A^0 = \mathrm{Ker}(m^\nu\,\mathrm{Id}_{A^0}) \tag{2.2.1.1} $$

are quasi-finite, separated and flat over S, with faithfully flat transition morphisms.

This is checked fibrewise by the flatness criterion, since the fibres are smooth and connected; the equivalence with (iv) is the usual structure theorem for smooth connected commutative algebraic groups.

Take now a prime ℓ satisfying the conditions of the lemma and put

$$ T_\ell(A^0) = \varprojlim_{\nu} {}_{\ell^\nu}A^0. \tag{2.2.2.1} $$

After base change to η and to s, this recovers respectively $T_\ell(A_K)$ and $T_\ell(A^0_s) = T_\ell(A_s)$.

Assume from now in this paragraph that S is henselian. If X is quasi-finite and separated over S, then

$$ X = X^f \amalg X' \tag{2.2.3.1} $$

canonically, with X^f finite over S and $X'_s = \emptyset$. If X is a group scheme, X^f is a subgroup scheme; if X is flat or ét, so is X^f. Applying this to the terms of $T_\ell(A^0)$, one obtains a projective subsystem

$$ T_\ell(A^0)^f \subset T_\ell(A^0), \tag{2.2.3.2} $$

called the fixed part of $T_\ell(A^0)$. It satisfies

$$ T_\ell(A^0)^f \times_S s = T_\ell(A^0_s). \tag{2.2.3.3} $$

Its generic fibre defines a canonical subgroup of $T_\ell(A_K)$, again denoted

$$ T_\ell(A_K)^f \subset T_\ell(A_K). \tag{2.2.3.4} $$

For $\ell \neq p$, this is an ℓ-adic locally constant twisted sheaf on S, hence corresponds to a $\mathbb{Z}_\ell$-submodule of $T_\ell(A_K(\bar{K}))$ stable under Γ.

Proposition 2.2.5. Suppose $\ell \neq p$ and S henselian. The submodule of $T_\ell(A_K(\bar{K}))$ corresponding to the fixed part $T_\ell(A_K)^f$ is precisely

$$ T_\ell(A_K(\bar{K}))^I, \tag{2.2.5} $$

where I is the inertia group.

Indeed it is enough to prove, for each ν,

$$({}_{\ell^\nu}A^0)^f(\bar{K}) = {}_{\ell^\nu}A_K(\bar{K})^I. \tag{2.2.5.0}$$

Let $\widetilde{S}$ be the strict localization of S, with fraction field the maximal unramified extension of K in $\bar{K}$. Since $({}_{\ell^\nu}A^0)^f$ is finite ét over S, its $\bar{K}$-points are its $\widetilde{S}$-points. By the Néron mapping property these are the same as the $\widetilde{K}$-points of ${}_{\ell^\nu}A_K$, which are exactly the inertia-invariant $\bar{K}$-points. Passing to the projective limit gives the assertion; the final equality with the special fibre follows because an element infinitely divisible by ℓ in $A_s(k)$ lies in $A_s^0(k)$.

Proposition 2.2.6. Let C_K be another abelian scheme over K, with Néron model C, and let $u_K : C_K \to A_K$ be an isogeny, extending to $u : C \to A$. For every prime ℓ, the induced map on fixed parts

$$T_\ell(C_K)^f \longrightarrow T_\ell(A_K)^f \tag{2.2.6.0}$$

is an isogeny. Also $T_\ell(C_s) \to T_\ell(A_s)$ is an isogeny; the map on the greatest subtori is an isogeny; and, after an extension of k over which the abelian parts are defined, the induced map between abelian parts is an isogeny.

This is formal. The indicated constructions are additive functors of the abelian scheme, and being an isogeny means that there is a morphism back such that both composites are multiplication by some integer.

Corollary 2.2.7. If A_K and C_K are isogenous, the invariants μ, λ, a are the same for both. In particular they are the same for A_K and its dual A'_K.

These assertions about special fibres remain valid without the henselian hypothesis, since formation of the Néron model and of the special fibre invariants commutes with passing to the henselization of S.

Corollary 2.2.9 (good reduction). Let S be a trait, A_K an abelian scheme over K, A its Néron model, and $\ell \neq \operatorname{char} k$. The following conditions are equivalent:

(i) there exists an abelian scheme over S with generic fibre isomorphic to A_K;

(ii) A is an abelian scheme;

(iii) A^0 is an abelian scheme;

(iv) A_s is an abelian scheme;

(v) A_s^0 is an abelian scheme, equivalently $\lambda = \mu = 0$;

(vi) A is proper over S;

(vii) A^0 is proper over S;

(viii) $T_\ell(A_K)$ is unramified over S, i.e. inertia acts trivially;

(ix) equivalently, all ℓ^ν-torsion ${}_{\ell^\nu}A_K$ extends to an ét covering of S.

When these conditions hold, the abelian scheme over S is the Néron model and is unique. The equivalence with unramifiedness is the Néron–Ogg–Shafarevich criterion.

2.3. The toric part

The inclusion

$$T_s \subset A_s^0 \tag{2.3.1}$$

gives a sub-ℓ-adic sheaf

$$T_\ell(A^0)^t \subset T_\ell(A^0)^f, \tag{2.3.2}$$

characterized by the special fibre condition

$$T_\ell(A^0)^t \times_S s = T_\ell(T_s) \subset T_\ell(A_s^0). \tag{2.3.3}$$

The generic fibre is the toric part

$$T_\ell(A_K)^t \subset T_\ell(A_K)^f \subset T_\ell(A_K). \tag{2.3.4}$$

Similarly one has

$$T_\ell(A'_K)^t \subset T_\ell(A'_K)^f \subset T_\ell(A'_K). \tag{2.3.5}$$

The middle terms are, by Proposition 2.2.5, the inertia invariants. The toric parts will be described by orthogonality with respect to the Poincaré pairing.

Theorem 2.4 (orthogonality theorem). Assume S henselian and $\ell \neq p$. For the filtrations (2.3.4) and (2.3.5), the toric part $T_\ell(A_K)^t$ is the intersection of the fixed part $T_\ell(A_K)^f$ with the orthogonal of the fixed part $T_\ell(A'_K)^f$ under the canonical pairing (1.0.3). Symmetrically, $T_\ell(A'_K)^t$ is the intersection of $T_\ell(A'_K)^f$ with the orthogonal of $T_\ell(A_K)^f$.

Proof. The restriction of the pairing to the fixed parts is obtained from the canonical biextension $\mathcal{W}^0$ of (1.4.4). More precisely, $\mathcal{W}^0$ gives an ℓ-adic pairing

$$T_\ell(A^0) \times T_\ell(A'^0) \longrightarrow T_\ell(\mathbb{G}_{mS}), \tag{2.4.1}$$

whose generic fibre is the original pairing. Restricting to fixed parts and passing to special fibres gives the pairing

$$T_\ell(A_s^0) \times T_\ell(A_s'^0) \longrightarrow T_\ell(\mathbb{G}_{mk}) \tag{2.4.3}$$

attached to $\mathcal{W}_s$. By the definition of the toric part, the desired vanishing is equivalent to the vanishing of

$$T_\ell(T_s) \times T_\ell(A_s'^0) \longrightarrow T_\ell(\mathbb{G}_{mk}), \tag{2.4.5}$$

which follows from the general theorem on biextensions of algebraic groups. This proves

$$T_\ell(A_K)^t \subset T_\ell(A_K)^f \cap \left(T_\ell(A'_K)^f\right)^\perp. \tag{2.4.6}$$

To see that equality holds, it is enough, again because the objects are ℓ-adic locally constant sheaves, to check the special fibres. The assertion becomes that the inclusion

$$T_\ell(T_s) \subset T_\ell(A_s^0) \cap T_\ell(A_s'^0)^\perp \tag{2.4.7}$$

is an equality, or equivalently that $\mathcal{W}_s$ is non-degenerate on the left. This is precisely Proposition 1.5, ultimately a consequence of Raynaud's thesis. Symmetry gives the statement for A'_K.

2.5–2.6. Polarized form and remarks

Using a polarization ξ of A_K, the theorem may be expressed as a statement about the alternating form $\psi_{\ell,\xi}$ on $L = T_\ell(A_K)$. Let

$$W = T_\ell(A_K)^t, \qquad V = T_\ell(A_K)^f, \qquad L = T_\ell(A_K). \tag{2.5.2}$$

Then

$$W = V \cap V^\perp. \tag{2.5.3}$$

The ranks of the successive pieces are

quotient	rank
W	μ
V/W	$2a$
$V^\perp/V$	λ
$L/V^\perp$	μ

with $n = \mu + \lambda + a$. These formulas are just (2.1.10), (2.1.11) and the orthogonality theorem.

There is a more arithmetic proof of the inclusion in (2.4.6), modelled on the proof of the monodromy theorem: after reduction to finite residue field, orthogonality follows from a weight argument for Frobenius, using Weil's theorem on its eigenvalues. This proof does not seem to give the equality in (2.4.6), which is used essentially in the proof of semi-stable reduction. Thus the biextension formalism remains indispensable here.

The orthogonality theorem lets one read the most important properties of the Néron model of A_K directly from the inertia representation on $T_\ell(A_K(\bar{K}))$. The good-reduction criterion 2.2.9 is the first example. The next section gives further examples, beginning with the semi-stable reduction criterion.

Exposé IX, continuation. Néron models and monodromy

3. The semi-stable reduction case. The theorem of semi-stable reduction

We first record an auxiliary result in the style of SGA 3.

Proposition 3.1. Let S be a scheme, let G be a commutative S-group scheme, flat, separated, and of finite presentation over S, let H be a quasi-separated S-group scheme locally of finite type, and let

$$u : G \longrightarrow H$$

be a homomorphism of S-group schemes. Put $K = \operatorname{Ker} u$. For $s \in S$, let $\bar{s}$ denote the spectrum of an algebraic closure of $k(s)$, and write $\mu(s)$, respectively $\alpha(s)$, for the reductive rank, respectively the abelian rank, of $K_{\bar{s}}$, i.e. the dimension of the largest subtorus, respectively of the largest abelian variety quotient, of $(K_{\bar{s}})^0_{\mathrm{red}}$. Let $s \in S$, let $t \in S$ be a generalization of s, and let p be the characteristic exponent of $k(s)$.

a) For every integer $n > 0$ prime to p, the group ${}_nK$ is, in a neighbourhood of s, finite, separated and étale over S, and

$$\operatorname{rank}({}_nK_s) \mid \operatorname{rank}({}_nK_t). \tag{3.1.1}$$

b) One has
$$\mu(s) + 2\alpha(s) \leq \mu(t) + 2\alpha(t). \tag{3.1.2}$$

c) If $(K_t)^0_{\mathrm{red}}$ is unipotent, then so is $(K_s)^0_{\mathrm{red}}$.

d) Suppose G_s has unipotent rank zero, that H is smooth over S, and that u_t is an isogeny. Then u is flat and quasi-finite over an open neighbourhood U of s; in particular $K|_U$ is flat and quasi-finite.

e) Under the hypotheses of d), if u_t is moreover a monomorphism, hence an isomorphism of G_t onto an open subgroup of H_t, then u_s is an isomorphism of G_s onto an open subgroup of H_s. If moreover S is irreducible with generic point t, then, after replacing S by a neighbourhood of s, the morphism u induces an isomorphism of $G|_U$ onto an open subgroup of $H|_U$.

The proof is by reduction to standard facts on finite flat group schemes and on the openness of the unit section of a separated formally étale group. For a), one may suppose that n is invertible on S. Then multiplication by n on G is étale fibre by fibre, hence étale; the kernel ${}_nG$ is finite over S, and the kernel ${}_nK$ is an open subgroup of ${}_nG$. After strict henselization at s, ${}_nK$ decomposes into a finite part and a part with empty special fibre, giving the divisibility (3.1.1), and even an injection on geometric fibres

$${}_nK(\bar{s}) \hookrightarrow {}_nK(\bar{t}).$$

Raising the two sides to powers in n gives (3.1.2), since the torsion of a commutative algebraic group detects the sum $\mu + 2\alpha$ away from p. Assertion c) follows formally from b). For d), the generic isogeny implies that the generic kernel is finite; the hypotheses on dimensions and on the unipotent rank of G_s force the special kernel to be finite, and the SGA 3 flatness criterion gives flatness and quasi-finiteness in a neighbourhood. Assertion e) follows from d) and the open immersion criterion.

Lemma 3.1.3. Let K be quasi-finite, of finite presentation, separated and flat over S. If $s \in S$ and t is a generalization of s, then

$$\operatorname{rank} K_s \leq \operatorname{rank} K_t.$$

If $\operatorname{rank} K_\eta \leq 1$ for every maximal point η of S, then $K \to S$ is an open immersion; hence it is an isomorphism when it has a section. The proof is local on S. In the strictly local case one separates off the finite open-and-closed part with non-empty special fibre; flatness and finite presentation make its rank constant.

Remark 3.1.4. In 3.1 c) and in the sequel to Proposition 3.1, no information is supplied about ${}_nK_s$ when n is a power of p. If G is smooth, or at least if the relevant finite-presentation hypotheses are imposed on H, the same method gives an analogous statement for the multiplicative rank. Namely
$$\operatorname{rank}_{\mathrm{mult}}({}_nK_s) \mid \operatorname{rank}_{\mathrm{mult}}({}_nK_t), \tag{3.1.4.1}$$
where $\operatorname{rank}_{\mathrm{mult}}$ denotes the rank of the largest subgroup of multiplicative type. A torsion group of multiplicative type is finite, so this again implies the unipotence assertion.

Proposition 3.2. Let S be a locally noetherian regular integral scheme of dimension 1, let K be its function field, let A_K be an abelian scheme over K, and let A be its Néron model over S. Thus A is a smooth separated group scheme over S, and it is quasi-projective when S is noetherian. The following conditions are equivalent:

(i) for every $s \in S$, A^0_s has unipotent rank zero; equivalently A^0_s is an extension of an abelian scheme by a torus;

(ii) there is a smooth separated finite-type group scheme G over S, extending A_K, whose fibres have unipotent rank zero.

Moreover, if G is as in (ii), the unique morphism $G \to A$ inducing the identity on generic fibres is an isomorphism of G onto an open subgroup of A, and therefore induces an isomorphism on neutral components,

$$G^0 \xrightarrow{\sim} A^0.$$

The definition of the Néron model here is the same as in the trait case recalled earlier; its existence and its separatedness of finite type are local on the base. The implication (i) $\Rightarrow$ (ii) is obtained by taking $G = A$. Conversely, if G is as in (ii), Proposition 3.1 e) says that $G \to A$ is an open immersion.

Corollary 3.3. With the notation of 3.2, suppose that the neutral components of the fibres of the Néron model have unipotent rank zero. If S' satisfies the same hypotheses as S, if K' is its function field, and if $f : S' \to S$ is dominant, then the canonical morphism from the pullback of A to the Néron model A' of $A_K \otimes_K K'$ is an open immersion and induces an isomorphism

$$A^0 \times_S S' \xrightarrow{\sim} A'^0. \tag{3.3.1}$$

This follows from 3.2 applied to S', taking $G = A \times_S S'$.

Definition 3.4. In the notation of 3.2, the abelian scheme A_K over K is said to have *semi-stable reduction over S* when the equivalent conditions (i) and (ii) of Proposition 3.2 hold. Equivalently, this may be tested after localizing at traits, and even after passing to the henselization or strict henselization.

Proposition 3.5 (Galois criterion for semi-stable reduction). Assume that S is a trait. Keep the notation of 2.5.3 after passing to the henselization of S. The following conditions on A_K are equivalent:

(i) A_K has semi-stable reduction over S;

(ii) with the notation of (2.5.3) and (2.5.4),

$$V \subset V^\perp;$$

(iii) for the toric part W of the Tate module,

$$W = V^\perp;$$

(iv) the inertia group I acts on $U = T_\ell(A(\bar{K}))$ unipotently of echelon 2: there exists an I-stable submodule $U' \subset U$ such that I acts trivially on U' and on U/U'.

Equivalently, if g is a topological generator of $\mathbb{Z}_\ell(1) = T_\ell(\bar{K}^*)$, the action of inertia factors through the maximal pro-ℓ finite quotient and

$$(1 - g_U)^2 = 0. \tag{3.5.3}$$

The proof uses the orthogonality theorem of Section 2: condition (ii) says $\lambda = 0$, hence is equivalent to the semi-stability of the neutral component; (ii) and (iii) are equivalent by the orthogonality $W = V \cap V^\perp$, and (iv) is the same assertion expressed through the dual action of inertia.

Theorem 3.6 (semi-stable reduction theorem). Let S be a connected regular noetherian scheme of dimension 1, let K be its function field, and let A_K be an abelian scheme over K. Then there is a finite Galois extension K'/K such that $A_K \otimes_K K'$ has semi-stable reduction over the normalization S' of S in K'.

It suffices to prove this after replacing S by traits at the finitely many closed points where A_K does not have good reduction. The Galois group of a finite normal extension permutes the points above a fixed closed point, so semi-stability at one such point suffices. By the criterion 3.5, the assertion is reduced to the existence of an open subgroup of inertia acting unipotently of echelon 2 on $T_\ell(A(\bar{K}))$, which follows from the monodromy theorem already established.

Corollary 3.7. If S is a trait, there exists an open subgroup I' of the inertia group I such that the action of I' on $U = T_\ell(A(\bar{K}))$ is unipotent of echelon 2. This is equivalently the existence of an open subgroup π' of the fundamental group for which the induced inertia action has this property.

Corollary 3.8. The equivalent conditions of Proposition 3.5 are also equivalent to the condition that the action of I on U be unipotent. Indeed, if an open subgroup acts unipotently of a given echelon on a finite-dimensional $\mathbb{Q}_\ell$-vector space, then the Zariski closures defined by the full group and by the open subgroup coincide, since a unipotent algebraic group in characteristic zero is connected.

Corollary 3.9. In the notation of 3.2, the two conditions of Proposition 3.2 are also equivalent to the following stability property: for every finite étale extension K'/K, if S' is the normalization of S in K' and A' is the Néron model of $A_K \otimes_K K'$, then the canonical morphism

$$A \times_S S' \longrightarrow A'$$

induces an isomorphism on the neutral components of the fibres.

Expose IX. Neron models and monodromy, continued

4. Application to a conjecture of Serre–Tate and to the conductor

4.1. We suppose that S is henselian and keep the notation of 2.5.3, with $\ell \neq p$. In order to study the action of

$$\pi = \mathrm{Gal}(\overline{K}/K) \quad \text{and of its inertia subgroup } I$$

on

$$U = T_\ell(A(\overline{K})),$$

we shall use the fact that there is an open subgroup I' of I, which we may even take normal in π, such that the action of I' on U is unipotent of echelon two:

$$(g-1)^2 = 0 \qquad (g \in I').$$

The submodule $U^{I'}$ is then independent of the particular admissible choice of I'. We shall call it the essentially fixed submodule of U, and denote it by U^{f}. Its orthogonal, under the polarization pairing, is called the essentially toric submodule and is denoted U^{t}. Thus one obtains a three-step filtration

$$U^{\mathrm{t}} \subset U^{\mathrm{f}} \subset U, \tag{4.1.1}$$

which is invariant under π.

There is a largest subgroup of inertia which acts unipotently on U: it is the subgroup of elements acting unipotently, or equivalently the kernel of the action induced on the graded object associated with (4.1.1). By the Galois criterion for semi-stable reduction this subgroup corresponds to the smallest finite Galois extension over which A_K acquires semi-stable reduction. This description also shows that the subgroup does not depend on the auxiliary prime $\ell \neq p$.

Moreover the representation on U/U^{f} is known up to isogeny once the representation on U^{t} is known. The polarization pairing gives an isogeny

$$U/U^{\mathrm{f}} \longrightarrow (U'^{\mathrm{t}})^{\vee}(1), \tag{4.1.2}$$

where the prime refers to the dual abelian variety and (1) is the Tate twist $\mathbb{Z}_\ell(1) = T_\ell(\mathbb{G}_m)$. Thus it is enough to describe carefully the action on U^{t}. We put

$$\pi' = \pi/I'. \tag{4.1.3}$$

4.2. Let K'/K be a finite Galois extension corresponding to an open normal subgroup whose intersection with I is I'. If $G = \mathrm{Gal}(K'/K)$, one has an exact sequence

$$1 \longrightarrow T \longrightarrow G \longrightarrow \Gamma \longrightarrow 1, \tag{4.2.1}$$

where

$$T = I/I' \tag{4.2.2}$$

is the inertia subgroup, and Γ is the Galois group of the largest unramified subextension of K'/K. This sequence is obtained from the analogous exact sequence for π':

$$1 \longrightarrow I/I' \longrightarrow \pi' \longrightarrow \mathrm{Gal}(\bar{k}/k) \longrightarrow 1. \tag{4.2.3}$$

The corresponding field diagram is

$$\overline{K} \longrightarrow K' \longrightarrow K'' \longrightarrow K\,,$$

where K'' is the maximal unramified subextension of K'/K.

Let S' be the normalization of S in K', and let A^* be the Neron model of $A_K \otimes_K K'$. By the Galois criterion for semi-stable reduction, A^* has semi-stable reduction over S', i.e. the neutral component A^{*0} is an extension of an abelian scheme by a torus. By transport of structure the group G acts on A^* and on A^{*0}, compatibly with its action on S'. The action of π' on U^{t} is then the one obtained from this action on the special fibre. More precisely, if k'/k is the residue extension and $A^{*0}_{k'}$ the connected smooth group over k', then

$$U^{\mathrm{t}} \simeq T_\ell(T^*)(\bar{k}) ag4.2.7$$

where T^* is the maximal subtorus of $A^{*0}_{k'}$. If

$$B^* = A^{*0}_{k'}/T^* \tag{4.2.8}$$

is the abelian quotient, then one also has

$$U^{\mathrm{f}}/U^{\mathrm{t}} \simeq T_\ell(B^*)(\bar{k}). \tag{4.2.9}$$

The action of G on $A^{*0}_{k'}$ induces the corresponding actions on T^* and B^*, and hence on the two displayed Tate modules.

Consequently, for the character $g \mapsto \mathrm{Tr}(g \mid U)$ of the action of π', one obtains

$$\mathrm{Tr}(g \mid U) = \mathrm{Tr}(g \mid T_\ell(B^*)) + \mathrm{Tr}(g \mid T_\ell(T^*)) + \chi(g)\ \mathrm{Tr}(g \mid T_\ell(T'^*)), \tag{4.2.10}$$

where T'^* is the analogous torus for the dual abelian variety and

$$\chi : \pi' \longrightarrow \mathbb{Z}_\ell^* \tag{4.2.12}$$

is the cyclotomic character coming from the action on $\mathbb{Z}_\ell(1) = T_\ell(\mathbb{G}_m)$. The character χ is trivial on the wild-inertia subgroup which acts trivially on the residue field.

We are now in a position to prove the following special case of a conjecture of Serre–Tate.

Theorem 4.3. Let S be a henselian trait with residue field k, function field K, and let A_K be an abelian scheme over K. Let

$$U_\ell = T_\ell(A(\overline{K})), \qquad \ell \neq \operatorname{char} k.$$

a) There exists an open subgroup $I' \subset I$ acting unipotently on U_ℓ, so that the character of π on U_ℓ is a function on π/I'. On I/I' this function is integer-valued and independent of ℓ. More generally, for every $g \in I$, the coefficients of the characteristic polynomial of g on U_ℓ are rational integers independent of ℓ.

b) If k is finite with q elements, so that $\pi/I \simeq \widehat{\mathbb{Z}}$ with arithmetic Frobenius as generator, then for every $g \in \pi$ whose image in π/I is a non-negative power of Frobenius, the trace, and more generally every coefficient of the characteristic polynomial of g on U_ℓ, is a rational number independent of ℓ. It is in fact an integer when the image of g is a non-negative integral power of Frobenius.

Proof. We spell out the proof for traces. The assertion for the coefficients of the characteristic polynomial follows either by the same argument, or formally from the trace assertions for the iterates, together with the Dwork–Fatou lemma.

For (a), formula (4.2.10) expresses $\operatorname{Tr}(g \mid U_\ell)$ as a sum of traces on the Tate modules of an abelian variety and of a torus over the residue field, plus the cyclotomic contribution. For the abelian part, the independence of ℓ is Weil's theorem on the characteristic polynomials attached to abelian varieties. For the toric part, it is immediate from the dual character lattice: if $M^* \simeq \operatorname{Hom}(T^*, \mathbb{G}_m)$, then

$$T_\ell(T^*) \simeq \operatorname{Hom}(M^*, \mathbb{Z}_\ell(1)),$$

and the required trace is the contragredient of an integral representation on a free abelian group, with the Tate twist accounted for by χ. This proves the asserted independence on inertia.

For (b), formula (4.2.10) reduces the matter to the following standard lemma.

Lemma 4.3.2. Let k be the algebraic closure of a finite field, let A be a commutative algebraic group of finite type over k, and let f be an automorphism of A compatible with an iterated Frobenius automorphism $x \mapsto x^q$ of $\operatorname{Spec} k$. If g is the automorphism induced on $T_\ell(A(k))$ by transport of structure, then the trace, and indeed the characteristic polynomial, of g has coefficients in $\mathbb{Z}$, independent of ℓ. Its eigenvalues are algebraic integers whose complex absolute values are the expected powers of $q^{1/2}$ on the abelian part and powers of q on the toric part.

Indeed, after composing with Frobenius, the operation is represented by an endomorphism of the algebraic group. Passing to the limit over ℓ^n-torsion identifies the induced operation with the corresponding endomorphism of the Tate module. One then devisses A into its abelian, toric, and finite pieces. For abelian varieties this is Weil's theorem; for tori it follows from the character module; the remaining finite part is harmless. This proves the lemma, hence Theorem 4.3.

Corollary 4.4. Under the hypotheses of 4.3(b), for every element $g \in \pi'$ mapping to a non-negative Frobenius power in π/I, the eigenvalues on the three graded pieces

$$U^{\mathrm{t}}, \qquad U^{\mathrm{f}}/U^{\mathrm{t}}, \qquad U/U^{\mathrm{f}}$$

are algebraic integers whose absolute values are respectively q^n, $q^{n/2}$, and 1. On the two extreme pieces the eigenvalues have the more precise form $q^n\varepsilon_i$ and ε_i^{-1}, where the ε_i are roots of unity.

This follows from the last formula and from (4.1.2): on the quotient dual to the toric part, the eigenvalues are the reciprocal Tate-twisted eigenvalues of the toric piece, and the roots of unity occur in conjugation-stable families.

4.5. We apply 4.3(a) to the local conductor of abelian schemes. Suppose first that the residue field is algebraically closed. For a constructible etale ℓ-group F_K on K, $\ell \neq p$, the wild conductor is defined as follows. Choose a finite Galois extension K'/K, with group Γ, which splits F_K. Associated with K'/K there is a projective $\mathbb{Z}_\ell[\Gamma]$-module

$$\mathrm{Sw}(K'/K, \mathbb{Z}_\ell), \tag{4.5.2}$$

the Swan module. If M is the ℓ-group defined by F_K over K', set

$$a_\ell(F_K) = \mathrm{length}_{\mathbb{Z}_\ell} \mathrm{Hom}_{\mathbb{Z}_\ell[\Gamma]}(\mathrm{Sw}(K'/K, \mathbb{Z}_\ell), M). \tag{4.5.3}$$

This number does not depend on K'. Since the Swan module is projective, the construction is additive in short exact sequences.

For the abelian scheme A_K, put

$$a_s(A_K) = \mathrm{length}_{\mathbb{Z}_\ell} \mathrm{Hom}_{\mathbb{Z}_\ell[\Gamma 93]}(\mathrm{Sw}(K'/K, \mathbb{Z}_\ell), U_\ell), \qquad U_\ell = T_\ell(A(\overline{K})). \tag{4.5.4}$$

This is the wild, or immoderate, conductor exponent of A_K. By passing to the henselization, the definition makes sense without assuming that S is henselian.

Corollary 4.6. The wild conductor exponent $a_s(A_K)$ is independent of the auxiliary prime $\ell \neq p$.

Proof. Replacing the Swan module by any projective $\mathbb{Z}_\ell[\Gamma]$-module L, the filtration (4.1.1) of U_ℓ gives a filtration of $L \otimes U_\ell$, and additivity reduces the length to the three graded pieces. If φ is the character of Γ on L, and if ω_i are the characters on the graded pieces, then

$$\mathrm{length}_{\mathbb{Z}_\ell} H^0(\Gamma, L \otimes U_\ell) = \frac{1}{|\Gamma|} \sum_{g\in\Gamma} \varphi(g)\,\omega(g). \tag{4.6.1}$$

Taking $L = \mathrm{Sw}(K'/K, \mathbb{Z}_\ell)$, the character φ is the usual Artin conductor character, independent of ℓ, and 4.3(a) says the same for ω. Hence $a_s(A_K)$ is independent of ℓ. More explicitly one obtains

$$a_s(A_K) = \frac{1}{|\Gamma|} \sum_{g\in\Gamma} (a(g) - |\Gamma|\delta_e(g) + 1)\,\omega(g), \tag{4.6.2}$$

where $a(g)$ is the Artin conductor character, δ_e the delta function at the identity, and ω the trace function described in 4.3. If the reduction is already semi-stable, one may take $K' = K$, and therefore

$$a_s(A_K) = 0. \tag{4.6.3}$$

Proposition 4.7 (M. Raynaud's criterion for semi-stable reduction). Let S be a trait, A_K an abelian scheme over its function field K, and let $n > 1$ be prime to the residue characteristic p. Suppose that A_K is unramified with respect to n, i.e. inertia acts trivially on ${}_nA(\overline{K})$. Then either A_K has semi-stable reduction over S, or $n = 2$ and, over a strictly local base, A_K acquires semi-stable reduction over a finite Galois extension whose Galois group is isomorphic to $(\mathbb{Z}/2\mathbb{Z})^r$.

The proof reduces to the case where S is strictly local and to the minimal extension over which semi-stable reduction is obtained. The inertia group then acts faithfully on the connected special fibre of the Neron model. The following two lemmas complete the argument.

Lemma 4.7.1 (J.-P. Serre). Let G be a commutative algebraic group over a field, extension of an abelian variety by a torus, and let $n \geq 2$. Every finite-order automorphism of G inducing the identity on ${}_nG$ is the identity if $n \neq 2$, and is either the identity or of order two if $n = 2$.

Indeed the condition says that $1 - u = nv$ in $\mathrm{End}(G)$. Reducing to the case where u has prime-power order, one embeds the torsion-free algebra generated by v into a product of cyclotomic fields. The elementary fact that a primitive root of unity congruent to 1 modulo $n \geq 2$ must be -1 with $n = 2$, or else be 1, gives the assertion.

Lemma 4.7.2. A finite group in which every non-identity element has order two is isomorphic to $(\mathbb{Z}/2\mathbb{Z})^r$ for some $r \geq 0$.

5. The case of residual characteristic p

5.1. We now discuss the prime $\ell = p$, under the hypotheses where the theory of Barsotti–Tate groups is available. Let S be complete and let A_K have semi-stable reduction. Let A be the Neron model, A^0 its neutral component, $T \subset A^0_s$ the maximal torus of the special fibre, and $B = A^0_s/T$ the abelian quotient. The same notation with primes will be used for the dual abelian variety. The orthogonality theorem identifies the toric and fixed pieces of the Tate modules of A_K and its dual, and yields the analogues of (4.1.1) for all primes ℓ, including p.

If M and M' denote the character groups of the tori T and T', then for every integer n one has

$$M/nM \simeq {}_nT, \qquad M'/nM' \simeq {}_nT'. \tag{5.5.5}$$

Passing to Tate modules gives canonical isomorphisms

$$T_\ell(T) \simeq \mathrm{Hom}(M, \mathbb{Z}_\ell(1)), \tag{5.5.6}$$
$$T_\ell(T') \simeq \mathrm{Hom}(M', \mathbb{Z}_\ell(1)). \tag{5.5.7}$$

By the orthogonality theorem and the polarization pairing, one then obtains

$$T_\ell(A_K)/T_\ell(A_K)^{\mathrm{f}} \simeq \mathrm{Hom}(M', \mathbb{Z}_\ell), \tag{5.5.8}$$
$$T_\ell(A_K)^{\mathrm{f}}/T_\ell(A_K)^{\mathrm{t}} \simeq T_\ell(B), \tag{5.5.9}$$

and similarly for the dual abelian variety.

Proposition 5.6. Suppose that S is complete and that A_K has semi-stable reduction over S. For every prime ℓ, possibly equal to the residue characteristic, the pro-ℓ-Barsotti–Tate group $T_\ell(A_K)^{\mathrm{t}}$ is the generic fibre of a Barsotti–Tate group over S of toric type, described by (5.5.7). The quotient $T_\ell(A_K)^{\mathrm{f}}/T_\ell(A_K)^{\mathrm{t}}$ is the generic fibre of a pro-etale Barsotti–Tate group, described by (5.5.8). Finally $T_\ell(A_K)^{\mathrm{f}}$ and $T_\ell(A_K)/T_\ell(A_K)^{\mathrm{t}}$ are the generic fibres of Barsotti–Tate groups over S, respectively $T_\ell(A^0)^{\mathrm{f}}$ and the Cartier dual of the corresponding fixed part for the dual abelian variety.

The same finite etale extension of S splits the toric and cotoric parts; equivalently it splits the twisted constant groups M and M'. There is therefore a smallest such finite etale extension.

Proposition 5.7. Assume that A^0 is ℓ-divisible and that S is henselian. Let F_K be a constructible ℓ-adic sheaf on K which is unramified, i.e. which extends to a twisted constant ℓ-adic sheaf F on S. Every homomorphism of pro-groups

$$u : F_K \longrightarrow T_\ell(A_K)$$

factors through the fixed part $T_\ell(A_K)^{\mathrm{f}}$.

Indeed, evaluating at ℓ^n-torsion gives maps from finite etale group schemes over S to the Neron model. By the Neron mapping property these extend uniquely over S, and by the definition of the fixed part their special fibres land in the infinitely ℓ-divisible part. Passing to the limit gives the desired factorization.

For $\ell \neq p$, this property characterizes the fixed part as the largest unramified sub-ℓ-adic sheaf of $T_\ell(A_K)$. For $\ell = p$ the statement is less intrinsic, and Raynaud's stronger theorem gives the right replacement.

Theorem 5.8 (M. Raynaud). Let A_K be an abelian variety with semi-stable reduction over the function field of a henselian trait S. Let ℓ be a prime, and let

$$X_K = T_\ell(A_K), \qquad X_K^{\mathrm{f}} = T_\ell(A_K)^{\mathrm{f}}.$$

Assume, when $\ell = p$, that $\operatorname{char} K = 0$, so that Tate's theorem on homomorphisms of Barsotti–Tate groups is available. If F is a pro-ℓ-Barsotti–Tate group over S, every homomorphism $F_K \to X_K$ factors through X_K^{f}.

Equivalently, passing to the associated ind-ℓ-groups, one has the following form.

Corollary 5.9. If F is an ind-ℓ-Barsotti–Tate group over S, every homomorphism

$$u_K : F_K \longrightarrow A_K$$

extends uniquely to a homomorphism $u : F \to A$ into the Neron model. Thus the Neron mapping property remains valid for ind-Barsotti–Tate groups, even without the smoothness hypothesis which occurs in the usual definition.

Raynaud proves this in greater generality, without assuming semi-stable reduction and even for Neronian group schemes whose generic fibre need not be abelian. In the present case, one decomposes F into a connected part and an ind-etale quotient. The etale part is handled by the Neron property. For the connected part, compose

$$F_K \longrightarrow X_K \longrightarrow X_K/X_K^{\mathrm{f}}.$$

By Proposition 5.6 the quotient is the generic fibre of an ind-etale Barsotti–Tate group. Tate's theorem extends the composite over S, and a morphism from a connected Barsotti–Tate group to an ind-etale one is zero. Hence the original map factors through X_K^{f}.

The last reduction uses the following elementary lemma.

Lemma 5.9.2. Let S be a trait, let A be a Neronian group scheme over S, and let

$$0 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 0$$

be an extension of group schemes over S, with G'' smooth. Let $u_K : G_K \to A_K$ and $v : G' \to A$ be compatible homomorphisms. If the extension of G'' by A obtained by pushing out through v is representable, then there exists a unique homomorphism $G \to A$ extending both u_K and v. Under the hypotheses indicated in the source, representability holds, for instance when $G \to G''$ has local sections for the finite locally free topology and A is quasi-projective over S.

Corollary 5.10. Let A_K be an abelian variety over the function field K of a trait S, and let ℓ be a prime. If $\ell = p$, assume $\operatorname{char} K = 0$. Then A_K has good reduction over S if and only if the pro-ℓ-Barsotti–Tate group $T_\ell(A_K)$ has good reduction over S, i.e. extends to a Barsotti–Tate group over S.

Necessity is clear. For sufficiency one may suppose S henselian. If the reduction is semi-stable, Theorem 5.8 applied to $F = X$ gives

$$T_\ell(A_K) = T_\ell(A_K)^{\mathrm{f}},$$

so the torus in the special fibre is zero; the neutral component is an abelian scheme, and the Neron model is abelian. In the general case one passes to a finite Galois extension over which semi-stable reduction holds. The uniqueness theorem of Tate for Barsotti–Tate groups shows that the Galois group acts trivially on the special fibre of the extended Barsotti–Tate group, hence trivially on the connected special fibre of the Neron model. Therefore the original variety already has semi-stable, hence good, reduction.

Remark 5.11. Once Tate's theorem is known without a restriction on $\operatorname{char} K$, the preceding results 5.8, 5.9, 5.10, and also 5.13 below, hold without that restriction.

Definition 5.12. Let S be a henselian trait, let ℓ be a prime, and let X_K be a pro-Barsotti–Tate group over the generic point. We say that X_K has good reduction over S if it extends to a Barsotti–Tate group over S. It has virtual good reduction if it admits a filtration by sub-pro-Barsotti–Tate groups whose graded pieces have good reduction. If this can be done with at most n non-zero graded pieces, it has virtual good reduction of echelon n. The corresponding essential notions are obtained after a finite extension of K and normalization of S.

For $\ell \neq p$, these six notions translate into the usual properties of the inertia representation on X_K: trivial, unipotent, unipotent of echelon n, and the corresponding properties after passage to an open subgroup of inertia.

If $X_K = T_\ell(A_K)$, then semi-stable reduction of A_K implies virtual good reduction of echelon two for X_K, and the semi-stable reduction theorem gives essential virtual good reduction of echelon two. This is the natural p-adic formulation of the geometric monodromy theorem for H^1.

Proposition 5.13. Let S be a henselian trait, let A_K be an abelian variety over its fraction field, and let ℓ be a prime. When $\ell = p$, assume $\operatorname{char} K = 0$.

a) A_K has good reduction over S if and only if $T_\ell(A_K)$ has good reduction over S.

b) A_K has essential good reduction over S if and only if $T_\ell(A_K)$ has essential good reduction over S. In that case one may take the extension of K to be finite separable.

c) The following conditions are equivalent: A_K has semi-stable reduction over S; $T_\ell(A_K)$ has virtual good reduction of echelon two; $T_\ell(A_K)$ has virtual good reduction.

Part (a) is Corollary 5.10. Part (b) is immediate from (a) and the definitions. For (c), the implication from semi-stability to virtual good reduction of echelon two has just been explained, and the reverse implication is the Galois criterion for $\ell \neq p$. For $\ell = p$, one compares with a prime $\ell' \neq p$ and uses the independence of trace functions established in 4.3 together with the computation supplied by 6.4 below.

Remark 5.13.1. It is plausible that a pro-ℓ-Barsotti–Tate group with virtual good reduction and essential virtual good reduction of echelon n should already have virtual good reduction of echelon n. This is true for $\ell \neq p$.

Proposition 5.14 (Criterion of essential good reduction). Let S be a trait with function field K, let A_K be an abelian variety over K, and let ℓ be prime to $\operatorname{char} K$.

a) If $\ell \neq p = \operatorname{char} k$, the following are equivalent: A_K has essential good reduction; inertia acts through a finite quotient on $U_\ell = T_\ell(A(\overline{K}))$; the action of inertia on $U_\ell \otimes \mathbb{Q}_\ell$ is semisimple.

b) If $\ell = p$ and $\dim A_K > 0$, the image of inertia in the automorphism group of U_ℓ is not finite; more precisely its image in the automorphism group of $\mathbb{Z}_\ell(1)$ has finite index. Essential good reduction is implied, but not characterized, by semisimplicity of the inertia action on $U_\ell \otimes \mathbb{Q}_\ell$.

The proof uses the semi-stable reduction theorem. In the case $\ell \neq p$, the unipotent inertia action attached to semi-stable reduction is semisimple only when it is trivial, whence good reduction. For $\ell = p$, the cyclotomic character already has infinite image on inertia.

Corollary 5.15. Suppose A_K has semi-stable reduction over S, and that the fixed part U_ℓ^{f} has an inertia-stable complement in U_ℓ. Then A_K has good reduction.

It is enough to prove $U_\ell = U_\ell^{\mathrm{f}}$, for then the torus in the special fibre is zero. After passing to a separably closed residue field, the assumed complement defines a pro-Barsotti–Tate subgroup isogenous to the cotoric quotient. Since this quotient is pro-etale and unramified, Proposition 5.7 forces it to lie in the fixed part. Hence the quotient is zero.

Expose IX. Neron models and monodromy, continued

6.2. On the fixed part outside the semi-stable case

Let us return to the largest p-divisible subgroup $A_o^{\circ\, p\text{-div}}$ of the neutral component of the special fibre, and write

$$F_o = p^\infty(A_o^\circ).$$

It is the largest Barsotti–Tate p-subgroup of A_o, equivalently of A_o°. It is natural to ask whether this group can be lifted to a Barsotti–Tate subgroup of the Neron model A, playing the role which the fixed part $T_p(A)^f$ plays in the prime-to-p theory and in the semi-stable case.

When such a lift exists, it is unique. Moreover the resulting sub-pro-groups in $T_p(A)$ and $T_p(A')$, which deserve again to be called fixed parts, satisfy the same orthogonality theorem as in the semi-stable case. The fixed part can still be characterized by the condition already used above for elements of the Tate module.

However this question does not have an affirmative answer in general. By the Serre–Tate lifting theorem, when S is complete the existence of the fixed part is equivalent to the existence of compatible infinitesimal liftings of the relevant special-fibre Barsotti–Tate group. Equivalently, one would need a smooth group scheme over S, an extension of an abelian scheme by a torus, whose special fibre is the prescribed group and which maps formally to the Neron model extending the inclusion on the special fibre. In particular, when the toric part is zero, this is the same as asking for an abelian scheme lifting the special fibre, together with a homomorphism into the Neron model. Raynaud's examples show that this is not always possible.

Here is the form of the example used in the source. Let $S' \to S$ be the normalization in a finite separable ramified extension K'/K, and let A' be an abelian scheme over S'. The Weil restriction

$$A = \mathrm{Res}_{S'/S}(A')$$

is Neronian over S: it identifies with the Neron model of its generic fibre. The neutral special fibre is an extension of an abelian scheme by a connected unipotent group, non-zero as soon as the extension is ramified and $A' \neq 0$. Taking K'/K quadratic and A' an elliptic curve over S', one arranges that A_K is simple over K by choosing the invariant j in such a way that it does not descend algebraically from K. This gives precisely a case where the desired fixed part cannot be constructed.

6.3. Essential fixed and toric parts

Assume now only that S is henselian. Choose a finite Galois extension K'/K over which A_K acquires semi-stable reduction, and let S' be the normalization of S in K'. The semi-stable theory over S' gives the fixed part and the toric part of the corresponding pro-Barsotti–Tate group. Descending by Galois invariants gives two strict sub-pro-groups of $T_p(A)$, denoted

$$T_p(A)^{ef} \subset T_p(A), \qquad T_p(A)^{et} \subset T_p(A), \tag{6.3.1}$$

called respectively the *essentially fixed part* and the *essentially toric part*. These definitions are independent of the chosen semi-stable extension. The toric part is the orthogonal of the fixed part for the canonical pairing, and the same assertion remains true after descent.

If A_K has essential good reduction, the essentially fixed part is the whole Tate group. If A_K has no such fixed contribution, the essentially toric part controls the monodromy part of the representation. In residue characteristic p, the trace functions attached to these objects are defined by passing, after finite extension, to the Dieudonne module of the special fibre. For a Barsotti–Tate group Y over the residue field one associates a finite free module M over the Witt vectors $W(k')$. The Galois group acts semilinearly on M, and the inertia trace thereby obtained agrees with the trace function previously defined on T_ℓ for $\ell \neq p$, once the usual comparison is made in the semi-stable situation. The construction used in the proof of 4.3.2 also gives the corresponding trace function on the higher inertia quotients.

7. Raynaud's extension attached to an abelian variety with semi-stable reduction

7.1. The basic construction

Let $S = \operatorname{Spec} V$, where V is a noetherian ring complete and separated for the topology defined by an ideal $\mathfrak{m}$. Let A be a smooth group scheme over S, let A° be its neutral component, and suppose that A_o/A_o° is finite over the closed fibre and that A_o° is an extension of an abelian scheme B_o by an isotrivial torus T_o. One associates to A a system of extensions which, in the semi-stable case, is Raynaud's extension of the abelian variety.

The construction begins with a torus T over S, lifting the torus T_o, and with an abelian scheme B over S, lifting B_o, together with an extension

$$0 \longrightarrow T \longrightarrow A^\natural \longrightarrow B \longrightarrow 0$$

whose special fibre maps to the neutral component of A_o. The source constructs this extension by applying the algebraization and extension results established in Expose VIII. For the dual abelian scheme one obtains a second extension

$$0 \longrightarrow T' \longrightarrow A'^\natural \longrightarrow B' \longrightarrow 0,$$

and the two extensions are related by the Poincare biextension.

The first statements establish that, under the stated hypotheses, $A^\natural$ is algebraizable. If S is normal, the corresponding abelian scheme B is projective over S. Indeed, a relatively ample invertible sheaf on A gives a birigidified invertible sheaf on $A \times_S A$; the absence of unipotent rank in the fibres allows the construction of a symmetric biextension, and the amplitude criterion of Raynaud implies projectivity. The normality hypothesis can then be removed by reduction to the normal case.

Proposition 7.1.5. With the preceding notation, if S is normal, the extension $A^\natural$ is algebraizable, and the abelian scheme B from which it comes is projective over S. The same algebraizability conclusion remains true without normality after passage to the normal case.

7.2. The category of semi-stable smooth group schemes

Let $\mathcal{C}$ be the category of smooth finite-type group schemes A over S such that A/A° is finite etale over the closed fibre and A° is an extension of an abelian scheme B by an isotrivial torus T. Raynaud's analysis identifies this category with a category of data consisting of

$$(B, T, Y, \text{extension data and component data}),$$

where Y is the character group of the torus and the component group acts through finite etale data. The exact form is used to compare the group scheme A with its dual construction. The lemma in the source shows that this description is stable under the operations which will be used below: duality, passage to finite etale covers, and base change between complete traits.

The resulting extension $A^\natural$ has generic fibre

$$A_K^\natural = (A^\natural)_K, \tag{7.6.1}$$

and is itself an extension of the finite etale component group by the neutral extension of the abelian variety by a torus. For every prime ℓ one obtains canonical identifications

$$T_\ell(A_K^\natural) \simeq T_\ell(A_K^{\natural\circ}) \simeq T_\ell(A_K)^f, \qquad T_\ell(T_K) \simeq T_\ell(A_K)^t. \tag{7.6.2}$$

The formation of the neutral Raynaud extension commutes with every base change of complete traits.

7.3–7.6. Duality and polarizations

The Poincare biextension on (B, B') pulls back to a biextension on $(A^\natural, A'^\natural)$. It yields a duality between the toric and cotoric pieces, and therefore a pairing between the lattices of characters and cocharacters. If a divisor correspondence W_K on (A_K, A'_K) is a perfect duality, the same is true for the corresponding correspondence $W^\natural$ on Raynaud extensions. If W_K is attached to a polarization of A_K, then $W^\natural$ is attached to a polarization of $A^\natural$.

The formation of $W^\natural$ is compatible with inverse images of correspondences under homomorphisms of abelian schemes

$$\overline{A}_K \to A_K, \qquad \overline{A}'_K \to A'_K.$$

This follows from the intrinsic characterization of the biextension and the compatibility of the Poincare biextension with pullback. The assertion about polarizations is the one used in 7.1.5 to prove algebraizability and projectivity.

8. Passage to the limit over finite extensions

Let K'/K run through finite separable extensions inside a fixed separable closure, and write S' for the corresponding normalization of the trait. Applying Raynaud's construction after base change gives group schemes $A^\natural_{K,K'}$ over K. If K''/K' is another such extension, there is a canonical open immersion

$$A^\natural_{K,K'} \hookrightarrow A^\natural_{K,K''}. \tag{8.3.1}$$

The quotient

$$\Phi_{K,K'} = A^\natural_{K,K'}/A^{\natural\circ}_{K,K'} \tag{8.3.2}$$

is finite etale over K, and the transition morphisms are compatible with the open immersions. Thus one may set

$$A_K^b = \varinjlim_{K'} A^\natural_{K,K'} \tag{8.3.3}$$

in the category of fppf sheaves on K. This limit is in fact represented by a group scheme over K, generally not quasi-compact. With the same notation one has

$$A_K^{b\circ} = A_K^{\natural\circ}, \qquad A_K^b / A_K^{b\circ} \simeq \varinjlim_{K'} \Phi_{K,K'}. \tag{8.3.4}$$

The latter group is identified later with the constant sheaf $M \otimes \mathbb{Q}/\mathbb{Z}$, where M is the monodromy lattice. This prepares the definition of the monodromy pairing in the following section.

SGA 7-I

Exposé IX. Néron Models and Monodromy

Monodromy Pairings and Component Groups

9. Definition of the monodromy pairing

9.1. The setting

In this section the trait S is henselian and the abelian variety A_K has semi-stable reduction over S. Equivalently, the same is true for the dual abelian variety A'_K. We shall use the twisted constant $\mathbb{Z}$-modules M and M' attached to the maximal tori of the special fibres of the connected Neron models of A_K and A'_K. After a suitable finite etale extension of S these become constant free groups, non-canonically isomorphic to $\mathbb{Z}^r$, where r is the common rank of the toric parts.

For each prime ℓ, put

$$M_\ell = M \otimes \mathbb{Z}_\ell, \qquad M'_\ell = M' \otimes \mathbb{Z}_\ell. \tag{9.1.1}$$

The geometric meaning of these sheaves is given by the formulae of the preceding sections, where they occur as the Tate modules of the toric parts. Our aim is to define a canonical bilinear form, the *monodromy pairing*,

$$u_\ell : M_\ell \otimes M'_\ell \longrightarrow \mathbb{Z}_\ell. \tag{9.1.2}$$

It will be defined by means of the Barsotti-Tate pro-ℓ-groups $T_\ell(A^0)$, $T_\ell(A'^0)$, and even by means of the generic fibre $T_\ell(A_K)$ of the first of these groups. When $\ell = p$, this last description uses Tate's theorem for homomorphisms of Barsotti-Tate groups; for the present purposes it is available in the equalities of characteristic to which it is applied below.

When $\ell \neq p$, the same construction may be expressed in purely Galois-theoretic terms. If

$$U = T_\ell(A_K(\bar{K})) \tag{9.1.3}$$

with its action of the inertia group I, then u_ℓ is the datum which measures the unipotent monodromy of I on U.

9.2. The case $\ell \neq p$

Assume first that $\ell \neq p$. We use the notation introduced earlier for the exact sequence of I-modules attached to the semi-stable reduction of A_K. Up to torsion, the relation which defines the toric submodule says that it is generated by elements of the form

$$gx - x \qquad (x \in U,\ g \in I). \tag{9.2.1}$$

Thus for each $g \in I$ one has

$$gx = x + u(g)(x), \tag{9.2.2}$$

where $u(g)$ is a homomorphism from U to the toric part, vanishing on that toric part. Equivalently,

$$u(g) \in \mathrm{Hom}(U/V, V), \tag{9.2.3}$$

where V denotes the submodule on which the transvections are parallel. In the semi-stable case the operation of every g is a transvection parallel to V, and composition of such transvections corresponds to addition of the associated homomorphisms. Hence the action of inertia is described by a homomorphism

$$I \longrightarrow \mathrm{Hom}(U/V, V). \tag{9.2.4}$$

By the orthogonality theorem applied both to A_K and to A'_K, one identifies U/V with the Tate twist of the toric Tate module of A'_K. Consequently the homomorphism (9.2.4) may be interpreted as a canonical homomorphism

$$M_\ell \otimes M'_\ell \longrightarrow \mathbb{Z}_\ell, \tag{9.2.5}$$

or as an element

$$u_\ell \in \mathrm{Hom}(M_\ell \otimes M'_\ell, \mathbb{Z}_\ell). \tag{9.2.6}$$

This is the element announced in (9.1.2) in the case $\ell \neq p$. It is invariant under transport of structure, hence is defined over the original residue field and not merely after passing to a separably closed residue field.

To define (9.1.2) without excluding $\ell = p$, we next recall a purely homological construction.

9.3. Mixed extensions

Let $\mathcal{C}$ be an abelian category and let

$$P, \quad R, \quad Q \tag{9.3.1}$$

be three objects of $\mathcal{C}$. We are interested in objects X of $\mathcal{C}$ endowed with a three-step filtration

$$0 = X^3 \subset X^2 \subset X^1 \subset X$$

with specified identifications of the successive quotients

$$X/X^1 \simeq Q, \qquad X^1/X^2 \simeq R, \qquad X^2 \simeq P.$$

Writing

$$F = X^1, \qquad E = X/X^2,$$

one obtains two ordinary extensions

$$0 \to P \to F \to R \to 0, \qquad 0 \to R \to E \to Q \to 0. \tag{9.3.2}$$

Conversely, suppose that the two extensions F and E in (9.3.2) are fixed. A *mixed extension* of E by F, or in Grothendieck's terminology an *extension panachee*, is such a filtered object X, together with isomorphisms

$$X^1 \simeq F, \qquad X/X^2 \simeq E, \tag{9.3.3}$$

compatible with the filtrations and with the common quotient R.

For fixed P, Q, R, E, F, the mixed extensions of E by F form a groupoid, denoted

$$\mathrm{EXTPAN}(E, F).$$

Every arrow is invertible, and the automorphism group of any object X is canonically

$$\mathrm{Hom}(Q, P). \tag{9.3.4}$$

Indeed an endomorphism of the underlying object of X which is the identity on the graded pieces differs from the identity by the composite

$$X \longrightarrow Q \longrightarrow P \longrightarrow X.$$

The groupoid $\mathrm{EXTPAN}(E, F)$ is naturally a torsor under the category-group $\mathrm{EXT}(Q, P)$ of extensions of Q by P. The action is the mixed analogue of the Baer sum. If G is an extension of Q by P and X is a mixed extension of E by F, then

$$(G, X) \longmapsto G \wedge X \tag{9.3.5}$$

is obtained by applying the Baer composition to the corresponding extensions of Q by F. For each fixed X, the functor

$$G \longmapsto G \wedge X$$

from $\mathrm{EXT}(Q,P)$ to $\mathrm{EXTPAN}(E,F)$ is an equivalence, and one has the associativity relation

$$G \wedge (G' \wedge X) = (G \wedge G') \wedge X. \tag{9.3.7}$$

It follows that

a) $\mathrm{EXTPAN}(E,F)$ is a groupoid, and the automorphism group of each object is canonically $\mathrm{Ext}^0(Q,P) = \mathrm{Hom}(Q,P)$;

b) the set $\mathrm{Extpan}(E,F)$ of isomorphism classes is either empty or is a torsor under $\mathrm{Ext}^1(Q,P)$;

c) there is a canonical obstruction class

$$e(E,F) \in \mathrm{Ext}^2(Q,P), \tag{9.3.9}$$

whose vanishing is necessary and sufficient for the existence of a mixed extension of E by F.

The class is described as follows. A mixed extension of E by F is equivalently an extension of E by P whose image in $\mathrm{Ext}^1(R,P)$ is the class of F. In the exact sequence obtained from the second sequence in (9.3.2),

$$\mathrm{Ext}^1(E,P) \longrightarrow \mathrm{Ext}^1(R,P) \xrightarrow{\partial} \mathrm{Ext}^2(Q,P),$$

one has

$$e(E,F) = \partial([F]). \tag{9.3.10}$$

The construction is functorial for exact functors. If

$$r : \mathcal{C} \longrightarrow \mathcal{C}' \tag{9.3.11}$$

is exact, and if primes denote images under r, then there is a commutative diagram of groupoids, up to canonical isomorphism,

$$\begin{array}{ccc} \mathrm{EXT}(Q,P) \times \mathrm{EXTPAN}(E,F) & \longrightarrow & \mathrm{EXTPAN}(E,F) \\ \downarrow & & \downarrow \\ \mathrm{EXT}(Q',P') \times \mathrm{EXTPAN}(E',F') & \longrightarrow & \mathrm{EXTPAN}(E',F'). \end{array} \tag{9.3.12}$$

Assume $\mathrm{EXTPAN}(E,F) \neq \varnothing$ and choose $X \in \mathrm{EXTPAN}(E,F)$. For a mixed extension Y' of E' by F', one obtains a class

$$e(Y') \in \mathrm{Ext}^1(Q',P')/\Im\,\mathrm{Ext}^1(Q,P). \tag{9.3.13}$$

It is independent of the choice of X, and vanishes precisely when Y' is isomorphic to the image by r of a mixed extension of E by F.

9.4. A finite-level specialization criterion

Let S be a henselian trait with generic point η, let $n > 0$, and put $\Lambda = \mathbb{Z}/n\mathbb{Z}$. Let $\mathcal{C}$ be the category of Λ-modules on S, for the fpqc topology, and let $\mathcal{C}_\eta$ be the corresponding category on η. Let P, R, Q be objects of $\mathcal{C}$, with E an extension of Q by R and F an extension of R by P. Assume that Q is locally free of finite type over Λ_S, and that P is represented by a finite multiplicative-type group over S, say

$$P = D(Q^*), \tag{9.4.0}$$

where D denotes Cartier duality. Suppose a mixed extension X_η of E_η by F_η is given. We shall attach to it a canonical pairing

$$e(X_\eta) : Q_\eta \otimes Q_\eta^* \longrightarrow \Lambda_\eta. \tag{9.4.1}$$

If S is strictly local, then Q and Q^* are constant, and giving (9.4.1) is the same as giving an element of

$$Q^\vee \otimes (Q^*)^\vee. \tag{9.4.2}$$

Apply the construction of 9.3 to the restriction functor $\mathcal{C} \to \mathcal{C}_\eta$. The obstruction group $\operatorname{Ext}^2(Q, P)$ vanishes, so mixed extensions over S exist. The required identifications rest on the following computations.

Lemma 9.4.3. Under the preceding hypotheses, with S strictly local, one has canonical isomorphisms

$$\operatorname{Ext}^i(Q, P) \simeq H^i(S, Q^* \otimes P),$$
$$\operatorname{Ext}^i(Q_\eta, P_\eta) \simeq H^i(\eta, Q^* \otimes P),$$

and

$$Q^* \otimes P \simeq D(Q \otimes Q^*).$$

For every finite constant group H on S,

$$H^i(S, D(H)) = H^i(\eta, D(H)) = 0 \quad (i \geq 2),$$

and there is a canonical isomorphism

$$H^1(\eta, D(H))/\Im H^1(S, D(H)) \simeq H^\vee.$$

The proof is obtained from Cartier duality and the Kummer sequence. In the case $H = \mathbb{Z}/m\mathbb{Z}$, the cohomology sequence associated with

$$0 \longrightarrow \mu_m \longrightarrow \mathbb{G}_m \xrightarrow{m} \mathbb{G}_m \longrightarrow 0$$

reduces the assertion to the vanishing of the relevant cohomology of $\mathbb{G}_m$ over a strictly henselian trait and over its generic point. This also describes explicitly the boundary homomorphism giving the last isomorphism.

Combining these computations with 9.3 gives a canonical isomorphism

$$\operatorname{Ext}^1(Q_\eta, P_\eta)/\Im \operatorname{Ext}^1(Q, P) \simeq Q \otimes Q^*. \tag{9.4.5}$$

The class attached in 9.3 to the generic mixed extension X_η therefore becomes the pairing (9.4.1). Its vanishing is exactly the obstruction to extending X_η to a mixed extension over S. The construction over a merely henselian trait is obtained by descent from the strict henselization.

Proposition 9.4.7. With the same data, without assuming that the trait is henselian, the restriction functor

$$\mathrm{EXTPAN}(E, F) \longrightarrow \mathrm{EXTPAN}(E_\eta, F_\eta)$$

is fully faithful. Its essential image consists of those generic mixed extensions for which the pairings (9.4.1), at all closed points after henselization, vanish.

The proof is local for the etale topology on S. Full faithfulness follows because homomorphisms extend uniquely from the generic point to S, using normality and finiteness of the Cartier-dual group. Essential surjectivity is reduced to the strictly local case, where it is precisely the obstruction computation above.

9.5. Mixed extensions of Barsotti-Tate pro-groups

Fix a prime ℓ. Let

$$P, \quad R, \quad Q \tag{9.5.1}$$

be pro-ℓ-Barsotti-Tate groups on S, and suppose given extensions

$$0 \to P \to F \to R \to 0, \qquad 0 \to R \to E \to Q \to 0. \tag{9.5.2}$$

For every $n \geq 0$ this gives finite-level extensions

$$0 \to P(n) \to F(n) \to R(n) \to 0, \qquad 0 \to R(n) \to E(n) \to Q(n) \to 0. \tag{9.5.3}$$

Assume Q is of etale type and P of multiplicative type. If over η a pro-ℓ-Barsotti-Tate group X_η is given as a mixed extension of E_η by F_η, then each finite level $X_\eta(n)$ gives a pairing of the form

$$Q(n) \otimes Q^*(n) \longrightarrow \mathbb{Z}/\ell^{n+1}\mathbb{Z}. \tag{9.5.4}$$

These pairings are compatible in n, hence define a homomorphism of twisted constant ℓ-adic sheaves

$$e(X_\eta) : Q \otimes Q^* \longrightarrow \mathbb{Z}_\ell. \tag{9.5.5}$$

Proposition 9.5.6. The restriction functor from mixed Barsotti-Tate extensions over S to those over η is fully faithful. Its essential image consists of the generic mixed extensions for which the pairing (9.5.5) vanishes. If $\operatorname{char} K = 0$, so that Tate's theorem on homomorphisms of Barsotti-Tate groups applies, then a generic mixed extension belongs to the essential image as soon as its underlying Barsotti-Tate group extends over S.

The first assertion follows level by level from Proposition 9.4.7. For the last assertion, extend the generic Barsotti-Tate group X_η to X over S. Tate's theorem extends uniquely the homomorphism $X_\eta \to Q_\eta$. Its finite-level maps are epimorphisms, so their kernels form a Barsotti-Tate group F, and the generic isomorphism with the prescribed F_η extends. The quotient by P is similarly identified with E, giving a mixed extension over S.

When $\ell \neq p$, all these Barsotti-Tate groups are equivalently twisted constant ℓ-adic sheaves, or representations of $\mathrm{Gal}(\bar{K}/K)$ on finite free $\mathbb{Z}_\ell$-modules. The homomorphism (9.5.5) is then exactly the homomorphism measuring the inertia action, as described in 9.2.

9.6. Definition in the semi-stable case

We now apply the preceding construction to the semi-stable abelian variety A_K. Let

$$P = T_\ell(T), \qquad Q = M', \qquad R = T_\ell(A^0), \tag{9.6.1}$$

and let F be the resulting extension of R by P. The dual extension attached to A'_K, after Cartier duality and the orthogonality theorem, gives an extension E of Q by R. As generic mixed extension one takes

$$X_\eta = T_\ell(A_K), \tag{9.6.2}$$

with its canonical three-step filtration induced by the toric and fixed parts. We are therefore in the situation of 9.5, and the pairing (9.5.5) takes precisely the form (9.1.2). For $\ell \neq p$ it agrees with the inertia construction of 9.2. Under Tate's theorem, it may be described entirely in terms of the Barsotti-Tate pro-ℓ-group $T_\ell(A_K)$ over the generic point.

9.7. Descent from a finite semi-stable extension

If A_K is not assumed semi-stable, choose a finite Galois extension K'/K such that $A_{K'}$ has semi-stable reduction over the normalization S' of S. Descending the monodromy pairing from K', and normalizing by the ramification index

$$e(S'/S) = \mathrm{length}(V'/\mathfrak{m}V'),$$

one obtains a pairing

$$u_\ell : M_K \otimes M'_K \longrightarrow \mathbb{Q}_{\ell,K} \tag{9.7.1}$$

which is independent of the chosen semi-stable extension. When A_K already has semi-stable reduction, it is the restriction to the generic point of (9.1.2). The integrality theorem below shows that this pairing is in fact independent of ℓ and has values in $\mathbb{Q}_K$. In the elliptic curve case the pairing can be interpreted as an ordinary rational number; Neron's results identify it with the valuation of the absolute invariant j.

10. Properties of the monodromy pairing. Integrality and positivity

This section records the formal properties of the monodromy pairings and states the deeper integrality and positivity theorem. The verifications of the formal properties are straightforward from the definitions and are left mostly to the reader.

10.1. Transport of structure

The definition of (9.1.2) is compatible with transport of structure in the triple (S, A_K, A'_K). Thus every isomorphism of such data carries the corresponding modules M, M' and the pairing u_ℓ to the corresponding objects for the transported data.

10.2. Symmetry

Apply the construction to the dual abelian variety A'_K. Its dual is canonically identified with A_K. The resulting pairing

$$u_\ell^{A'} : M'_\ell \otimes M_\ell \longrightarrow \mathbb{Z}_\ell \tag{10.2.1}$$

is obtained from u_ℓ^A by the transposition isomorphism

$$M'_\ell \otimes M_\ell \xrightarrow{\sim} M_\ell \otimes M'_\ell.$$

In other words,

$$u_\ell^{A'}(x', x) = u_\ell^A(x, x'). \tag{10.2.2}$$

10.3. Change of trait

Let $S' \to S$ be a dominant morphism of traits, and let $A_{K'}$ be obtained from A_K by base change. The monodromy pairing changes by multiplication with the ramification index:

$$u_\ell^{S'} = e(S'/S)\, u_\ell^S. \tag{10.3.4}$$

This is compatible with the description by inertia when $\ell \neq p$, and in the general case follows by the finite-level mixed extension construction.

Theorem 10.4. Let S be a henselian trait and let A_K have semi-stable reduction over S. Let M and M' be the character lattices introduced in 9.1. Then:

a) There exists a unique pairing

$$u : M \otimes M' \longrightarrow \mathbb{Z} \tag{10.4.1}$$

whose ℓ-adic extension is the pairing u_ℓ of (9.1.2), for every prime ℓ. This pairing is separating.

b) If $\xi : A_K \to A'_K$ is a polarization, and if

$$(x, y) = u(x, \xi y) \tag{10.4.2}$$

denotes the induced bilinear form on M, then this form is symmetric, positive, and non-degenerate.

Equivalently, after choosing a geometric point of S, the induced form on the fibre M_0 is a symmetric positive non-degenerate integral form. The pairing (10.4.1) is again called the monodromy pairing.

10.5. Reduction of the proof

The uniqueness in Theorem 10.4 is immediate, and existence may be checked after any dominant change of traits. It is therefore enough to prove the theorem after replacing S by a complete strictly local trait. If A_K is isogenous to a direct factor of an abelian variety for which the theorem is known, then the theorem is known for A_K; this follows from functoriality of the lattices and of the monodromy pairing, together with Poincare reducibility.

After a finite extension of K, every abelian variety is a quotient of the Jacobian of a proper smooth geometrically connected curve. By a theorem of Artin, after another finite extension the curve has a regular proper flat model whose geometric special fibre has only ordinary quadratic singularities, i.e. normal crossings of two branches. Thus Theorem 10.4 is reduced to the case of such a Jacobian. The integral formula and positivity will then be verified geometrically in the Picard-Lefschetz calculation of Section 12.

It remains to check that positivity for one polarization implies positivity for every polarization. If ξ_0 and ξ are two polarizations, their associated homomorphisms satisfy, in $\mathrm{End}(A_K) \otimes \mathbb{Q}$,

$$\xi = \xi_0 a = a^* \xi_0, \qquad a = b^* b$$

for some $b \in \mathrm{End}(A_K) \otimes \mathbb{Q}$. In terms of the biextensions attached to the polarizations, this gives

$$\mathcal{B}(\xi) = \mathcal{B}(\xi_0) \circ (b, b). \tag{10.5.5.1}$$

Consequently

$$u_\xi(x, y) = u_{\xi_0}(bx, by). \tag{10.5.5.2}$$

Symmetry and positivity are therefore inherited from ξ_0, and non-degeneracy is equivalent to the separating property of (10.4.1), which is independent of the polarization.

11. Connected components of the Neron model: duality and asymptotic behavior

11.0. Component groups

Let

$$\Phi = A/A^0 \tag{11.0.1}$$

be the finite etale group of connected components of the Neron model. For each prime ℓ, let $\Phi(\ell)$ be its ℓ-primary component. The knowledge of Φ is equivalent to the knowledge of all the $\Phi(\ell)$. Since formation of the Neron model is compatible with strict henselization, the study of Φ may be reduced to the strictly local case; the Galois action on the component group is then recovered by transport of structure.

11.1. Strictly local description

Assume S strictly local. Then

$$\Phi(k) \simeq A(k)/A^0(k) \simeq A(S)/A^0(S) \simeq A(K)/A(K)^0. \tag{11.1.1}$$

Here the first equality uses smoothness over the residue field, the second is Hensel's lemma for the smooth group scheme A over S, and the third is the equality $A(S) = A(K)$ for the Neron model. We put

$$A(K)^0 = A^0(S). \tag{11.1.2}$$

For n prime to $p = \mathrm{char}\, k$, multiplication by n is etale and surjective on A^0, hence $A(K)^0$ is n-divisible. Therefore

$$\Phi(k)/n\Phi(k) \simeq A(K)/nA(K). \tag{11.1.3}$$

If $\ell \neq p$, let

$$U = T_\ell(A_K(\bar{K})). \tag{11.1.5}$$

Since the fixed part of A_K is described by the inertia invariants of U, (11.1.3) yields, for n a power of ℓ,

$$\Phi(k)/n\Phi(k) \simeq U_n^I/(U^I)_n. \tag{11.1.6}$$

Passing to the direct limit over n gives:

Proposition 11.2. Assume S strictly local and $\ell \neq p$. There is a canonical isomorphism

$$\Phi(\ell) \simeq (U \otimes D_\ell)^I/(U^I \otimes D_\ell), \qquad D_\ell = \mathbb{Q}_\ell/\mathbb{Z}_\ell. \tag{11.2.1}$$

Thus the ℓ-primary component group is determined by the Tate module U and the inertia action.

11.3. The duality pairing for $\ell \neq p$

Let $U' = T_\ell(A'_K(\bar{K}))$, and let

$$U \otimes U' \longrightarrow T_\ell(\mathbb{G}_m) \tag{11.3.2}$$

be the perfect Weil pairing, compatible with the inertia action. The structure of inertia gives an exact sequence

$$1 \longrightarrow R \longrightarrow I \longrightarrow T \longrightarrow 1 \tag{11.3.3}$$

with R pro-prime-to-ℓ. Put

$$\Phi = (U \otimes D_\ell)^I/(U^I \otimes D_\ell), \quad \Phi' = (U' \otimes D_\ell)^I/(U'^I \otimes D_\ell). \tag{11.3.4}$$

We define a perfect pairing

$$\Phi \otimes \Phi' \longrightarrow \mathbb{Q}_\ell/\mathbb{Z}_\ell. \tag{11.3.5}$$

Consider the exact sequence

$$0 \to U \to U \otimes \mathbb{Q}_\ell \to U \otimes D_\ell \to 0 \tag{11.3.6}$$

and the corresponding cohomology exact sequence for I. The group Φ is canonically identified with the kernel of the connecting map into $H^1(I, U)$, hence with the torsion subgroup of $H^1(I, U)$. The same holds for U'. The standard duality for the cohomology of I, using the perfect pairing (11.3.2), gives perfect pairings

$$H^0(I, U \otimes D_\ell) \times H^1(I, U')_{\mathrm{tor}} \longrightarrow D_\ell, \tag{11.3.12}$$

and after passage to the limit,

$$H^0(I, U \otimes D_\ell) \times H^1(I, U' \otimes D_\ell) \longrightarrow D_\ell. \tag{11.3.13}$$

The maps occurring in the two exact sequences are transposes of one another under these pairings. Therefore the cokernel on one side and the kernel on the other are canonically dual, which gives the pairing (11.3.5).

11.4–11.6. The semi-stable formula for all primes

The preceding description is only for $\ell \neq p$. Suppose now that A_K has semi-stable reduction. For every prime ℓ we shall define perfect pairings

$$\Phi(\ell) \otimes \Phi'(\ell) \longrightarrow \mathbb{Q}_\ell/\mathbb{Z}_\ell. \tag{11.4.1}$$

This is done by expressing $\Phi(\ell)$ in terms of the Barsotti-Tate group $T_\ell(A_K)$ and the monodromy pairing.

Theorem 11.5. Assume S henselian and A_K semi-stable. Let ℓ be any prime, and let

$$u_\ell : M_\ell \otimes M'_\ell \longrightarrow \mathbb{Z}_\ell$$

be the monodromy pairing. Let

$$\overline{u}_\ell : M_\ell \longrightarrow \mathrm{Hom}(M'_\ell, \mathbb{Z}_\ell)$$

be the corresponding homomorphism, and put $\Phi_u(\ell) = \operatorname{coker} \overline{u}_\ell$. Then there is a canonical isomorphism

$$\Phi(\ell) \simeq \Phi_u(\ell). \tag{11.5.1}$$

In particular $\overline{u}_\ell$ is an isogeny, including for $\ell = p$. After Theorem 10.4 one may equivalently state the result by means of the integral monodromy pairing $u : M \otimes M' \to \mathbb{Z}$: if Φ_u is the cokernel of $M \to \mathrm{Hom}(M', \mathbb{Z})$, then

$$\Phi \simeq \Phi_u. \tag{11.5.3}$$

To construct the isomorphism, we reduce to the strictly local case. Let T and T' be the maximal tori, and identify

$$M = D(T), \qquad M' = D(T'). \tag{11.6.1}$$

It is enough to define, for each power n of ℓ, compatible isomorphisms

$$\Phi(n) \simeq \mathrm{Ker}(M_n \xrightarrow{\overline{u}_n} M_n'^{\vee}). \tag{11.6.4}$$

Let A_n be the kernel of multiplication by n on the Neron model. It is flat and quasi-finite over S. Its fixed and toric parts give a canonical filtration

$$A_n \supset A_n^f \supset A_n^t \supset 0. \tag{11.6.5}$$

The quotient

$$Y = A_n^f / A_n^t \tag{11.6.6}$$

is flat, separated, and quasi-finite, with generic fibre identified with $(A_K^0)_n^f/(A_K^0)_n^t$. The formulas from the preceding section give a canonical isomorphism on the generic fibre

$$M_n \xrightarrow{\sim} Y_\eta. \tag{11.6.7}$$

Lemma 11.6.8. The group scheme Y is etale over S, and (11.6.7) extends uniquely to a morphism

$$M_n \longrightarrow Y \tag{11.6.8.1}$$

which is an open immersion.

The etaleness follows from the special fibre; normality and separatedness extend the morphism, and an etale monomorphism is an open immersion.

Let

$$\Lambda_n = Y \cap A_n^0. \tag{11.6.9}$$

It is a finite open subgroup of Y, and the composite

$$\Lambda_n \hookrightarrow Y \hookleftarrow M_n \tag{11.6.11}$$

identifies Λ_n with a subgroup of M_n. It is enough to prove:

Lemma 11.6.13. The image of Λ_n in M_n is exactly the kernel of $\overline{u}_n$.

The kernel is the left annihilator of the finite-level monodromy pairing

$$M_n \otimes M_n' \longrightarrow \mathbb{Z}/n\mathbb{Z},$$

which was defined in terms of the two finite extensions and of the generic mixed extension. By Proposition 9.4.7, a subgroup belongs to this annihilator precisely when the corresponding pulled-back mixed extension extends over S. The subgroup Λ_n has such an extension by construction, so $\Lambda_n \subset \mathrm{Ker}\,\overline{u}_n$. Conversely, if a subgroup $K_n \subset \mathrm{Ker}\,\overline{u}_n$ is given, the associated mixed extension extends over S. The induced morphism into A_n factors through A_n^0, hence through Λ_n; this gives $K_n \subset \Lambda_n$. The lemma follows, and with it Theorem 11.5.

Remarks 11.7. The preceding construction shows that the finite mixed extensions (11.6.5), and hence the subgroup $A_n^0 \subset A_n$, can be reconstructed from the structural data of 9.6 and from the generic mixed extension. The annihilator condition determines the open finite subgroup, and gluing along the common open part recovers the extension. Thus the system of finite mixed extensions is encoded by the Barsotti-Tate mixed extension and the monodromy pairing.

11.8–11.10. Asymptotic behavior under finite extensions

Let $\bar{K}$ be the separable closure of K, written as the filtered union of finite extensions K'. For each K', let $S[K']$ be the normalization of S in K', and let

$$\Phi[K'] \tag{11.8.1}$$

be the component group of the Neron model of $A_{K'}$. After a sufficiently large finite extension the reduction is semi-stable, and the transition maps on component groups are injective. Put

$$\Psi = \varinjlim_{K'} \Phi[K']. \tag{11.8.2}$$

For each prime ℓ, Theorem 11.5 gives, for large K', an identification

$$\Phi[K'](\ell) \simeq \mathrm{Ker}(M \otimes D_\ell \xrightarrow{u(K')} M' \otimes D_\ell). \tag{11.8.3}$$

The transition maps are compatible with the inclusions into $M \otimes D_\ell$; hence one obtains a canonical monomorphism

$$\Psi \hookrightarrow M \otimes \mathbb{Q}/\mathbb{Z}. \tag{11.8.5}$$

Theorem 11.9. The homomorphism (11.8.5) is an isomorphism.

It remains only to prove surjectivity. The image is described by the kernels in (11.8.3), and the behavior of the monodromy pairing under ramified extension is given by (10.3.4). Thus it is enough to use the elementary fact that, over a strictly local trait, every integer can be made to divide the ramification degree of a suitable finite separable extension. This makes every element of $M \otimes \mathbb{Q}/\mathbb{Z}$ appear at some finite stage.

Corollary 11.10. Under the hypotheses in which the semi-stable reduction and monodromy constructions apply, the group of connected components of the model after passage through all finite extensions has the form

$$\varinjlim_{K'} \Phi[K'] \simeq M \otimes \mathbb{Q}/\mathbb{Z}. \tag{11.10.1}$$

For $\ell \neq p$ this is already immediate from the inertia description of Proposition 11.2. The delicate part is the p-primary component, where Theorem 11.5 is needed. The existence of the isomorphism in Corollary 11.10, with the right-hand side understood as the group associated with the toric character lattice, was conjectured by J.-P. Serre.

SGA 7-I: Exposé IX, continuation

Néron models, the Picard–Lefschetz formula, and analytic comparison

12. Néron model of a Jacobian and the Picard–Lefschetz formula

We first recall the following general results of M. Raynaud. They are used in the form needed below, where the base is a strictly local trait.

Theorem 12.1 (M. Raynaud). Let S be a strictly local trait and let $f : X \to S$ be proper and flat, with fibres of dimension one, such that

$$H^1(X, \mathcal{O}_X) = 0 \tag{12.1.1}$$

and such that the local rings of X are factorial. For each maximal point x_i of the special fibre, put

$$a_i = \mu_i \, [k(x_i) : k]_{\mathrm{rad}}, \tag{12.1.2}$$

where μ_i is the multiplicity of the component through x_i, and where the radical degree is a power of the characteristic exponent of the residue field. Let d, respectively d^0, be the greatest common divisor of the integers a_i, respectively of their separable parts. Thus $d = d^0$ when the residue field is perfect. Put

$$P = \mathrm{Pic}_{X/S} \colon (\mathrm{Sch})/S^{\mathrm{op}} \longrightarrow (\mathrm{Ab}). \tag{12.1.3}$$

Consider the following conditions:

1) $d = 1$.
2) X has a zero-cycle of degree 1.
3) There exists an invertible module L on $X \times P^0$ whose corresponding homomorphism $P^0 \to \mathrm{Pic}^0_{X/S} = P^0$ is the canonical inclusion.
4) X is cohomologically flat over S; equivalently, it is cohomologically flat in dimension zero, and the canonical map
$$k \longrightarrow H^0(X_s, \mathcal{O}_{X_s})$$
is an isomorphism, with $d = 1$.
5) P is representable, equivalently by a smooth group scheme over S.
6) P^0 is representable, equivalently by a smooth group scheme over S.
7) P^0 is separated over S, i.e. satisfies the valuative criterion of separatedness; equivalently, every section of P^0 over S which is zero on the generic point is zero.
8) $d^0 = 1$.

Then one has the chain of implications displayed in Raynaud's theorem; in particular, when $d = d^0$, for example when k is perfect, the eight conditions are equivalent.

Assume now that $d^0 = 1$, so that P is represented by a smooth group scheme over S. Then the schematic closure E of the unit section of P is an S-étale subgroup scheme. If I denotes the set of reduced irreducible components X_i of the special fibre, one obtains an isomorphism

$$\mathbb{Z}^I/\mathbb{Z} \xrightarrow{\sim} E \tag{12.1.5}$$

by associating to $i \in I$ the section of P defined by the invertible sheaf $\mathcal{O}_X(X_i)$. Moreover

$$P^0 \cap E = \text{the zero section of } P. \tag{12.1.6}$$

The quotient

$$Q = P/E \tag{12.1.7}$$

is represented by a smooth separated group scheme over S, and is Néronian in the sense that it satisfies the Néron mapping property on its special fibre. If, in addition, the geometric generic fibre of X is integral, there is a canonical exact sequence

$$0 \longrightarrow A \longrightarrow Q \xrightarrow{\deg} \mathbb{Z}_S \longrightarrow 0, \tag{12.1.8}$$

where A is smooth, separated, and of finite type over S, and is the Néron model of

$$A_\eta = \mathrm{Pic}^0_{X_\eta/\eta}\,. \tag{12.1.9}$$

The composite $P^0 \to P \to Q$ induces an isomorphism

$$P^0 \xrightarrow{\sim} A^0. \tag{12.1.10}$$

The case of interest for us is the following. Suppose X_η is smooth, so that, in view of (12.1.1), the geometric generic fibre is smooth and connected, and suppose that the special fibre has a zero-cycle of degree 1. Then all the preceding conditions hold. The group $A_\eta = \mathrm{Pic}^0_{X_\eta/\eta}$ is an abelian scheme on the generic point, and (12.1.8) gives a very explicit construction of its Néron model. In particular (12.1.10) gives

$$A^0_s \simeq \mathrm{Pic}^0_{X_s/s}\,. \tag{12.1.12}$$

It follows that A_η has semi-stable reduction exactly when the maximal torus in $\mathrm{Pic}^0_{X_s/s}$ has the expected semi-stable form, equivalently in the regular case when the only singularities of the geometric special fibre are ordinary quadratic points. This last verification follows from the usual description of the Picard scheme of a proper separable curve over a field and from EGA IV, 21.8.5; it is also an immediate consequence of the geometric monodromy theorem together with the Galois criterion for semi-stable reduction.

Raynaud's proof is cited from the papers used above. We shall need the theorem only in the special situation just described, and from now on we restrict to the case in which the singular points of the special fibre are ordinary quadratic points.

12.3. The toric part of the Picard scheme of a singular curve

Let C be a proper separable curve over a separably closed field k, and let $\widetilde{C}$ be its normalization. The Picard scheme $\mathrm{Pic}^0_{\widetilde{C}/k}$ has reductive rank zero, hence the maximal torus of $\mathrm{Pic}^0_{C/k}$ is contained in

$$N = \mathrm{Ker}(\mathrm{Pic}^0_{C/k} \longrightarrow \mathrm{Pic}^0_{\widetilde{C}/k}).$$

By EGA IV, 21.8.5, the group N is an extension of a torus T, described below, by a connected unipotent group. Thus T is canonically the maximal torus of $\mathrm{Pic}^0_{C/k}$.

Let J be the set of irreducible components of $\widetilde{C}$. For $x \in C$, let $J(x)$ be the set of branches of C through x, i.e. the set of points of $\widetilde{C}$ above x. Put

$$R(x) = \mathbb{Z}^{J(x)}/\mathbb{Z}, \tag{12.3.1}$$

where $\mathbb{Z} \to \mathbb{Z}^{J(x)}$ is the diagonal map, and

$$R = \bigoplus_x R(x), \tag{12.3.2}$$

the sum being taken only over points with at least two branches. Then

$$T = \operatorname{Hom}(R, \mathbb{G}_m). \tag{12.3.3}$$

Equivalently, the character group M of T is

$$M = \operatorname{Hom}(T, \mathbb{G}_m) = \operatorname{Ker}\left(\bigoplus_x \mathbb{Z}^{J(x)} \longrightarrow \mathbb{Z}^J\right), \tag{12.3.4}$$

where the map is induced by the natural map sending a branch to the component on which it lies.

We shall use only the case in which at most two branches pass through a point. If I is the set of points with two branches, then, after choosing an order of the two branches at each such point,

$$R \simeq \mathbb{Z}^I, \qquad M = \operatorname{Ker}(\mathbb{Z}^I \longrightarrow \mathbb{Z}^J). \tag{12.3.6}$$

Equivalently one can describe M by the graph $\Gamma(C)$ whose vertices are the irreducible components and whose edges are the unordered pairs of branches meeting at a singular point. Then

$$M \simeq H_1(\Gamma(C), \mathbb{Z}), \qquad R \simeq H^1(\Gamma(C), \mathbb{Z}). \tag{12.3.7}$$

The canonical homomorphism from the branch description to the graph description is induced by the map from the set of branches over the double points to the set of components.

12.4. The pairing attached to ordinary double points

We now place ourselves in the situation fixed at the end of 12.2. By (12.1.12) and by the preceding description applied to $C = X_s$, the character group M of the maximal torus T_s of the special fibre A_s^0 of the Néron model is

$$M = \operatorname{Ker}(\mathbb{Z}^I \longrightarrow \mathbb{Z}^J), \tag{12.4.1}$$

where I is the set of singular points of X_s, and J the set of irreducible components. The map $\mathbb{Z}^I \to \mathbb{Z}^J$ is obtained by assigning, to a double point, the difference of the two components which contain the two branches through that point. Thus, after choosing an order (C_i', C_i'') of the two branches through x_i, one has

$$d(x_i) = C_i' - C_i''. \tag{12.4.2}$$

The operator d, and the isomorphism (12.4.1), depend on these choices. However, if

$$\varphi_{x_i} : M \longrightarrow \mathbb{Z} \tag{12.4.3}$$

is the composite of $M \to \mathbb{Z}^I$ with the projection onto the factor corresponding to x_i, then φ_{x_i} is determined up to sign, and the sign changes exactly when the order of the two branches is reversed. Consequently the bilinear form

$$\varphi_{x_i} \otimes \varphi_{x_i} \in M^\vee \otimes_{\mathbb{Z}} M^\vee \simeq \operatorname{Hom}(M \otimes M, \mathbb{Z}) \tag{12.4.4}$$

is canonically determined by the singular point x_i. We may therefore put

$$u = \sum_i \varphi_{x_i} \otimes \varphi_{x_i} : M \otimes_{\mathbb{Z}} M \longrightarrow \mathbb{Z}. \tag{12.4.5}$$

This pairing is well-defined, independent of all choices of signs. It is positive definite, being the restriction to $M \subset \mathbb{Z}^I$ of the standard quadratic form $\sum_i X_i^2$ on $\mathbb{Z}^I$.

The notation M for the character group of T_s differs from the notation used earlier, where M' denoted the corresponding group for the dual abelian variety. Since here A_η is a Jacobian, it has its canonical principal polarization, giving a canonical isomorphism

$$A_\eta \xrightarrow{\sim} A'_\eta. \tag{12.4.6}$$

We shall use this to identify A_η and A'_η, and hence M and M'. For every prime ℓ, the monodromy pairing may therefore be viewed as a pairing

$$u_\ell : M \otimes \mathbb{Z}_\ell \times M \otimes \mathbb{Z}_\ell \longrightarrow \mathbb{Z}_\ell. \tag{12.4.7}$$

Theorem 12.5 (Picard–Lefschetz formula). Let S be a strictly local trait, and let X be regular, projective and flat over S, with fibres of dimension one. Assume that the geometric generic fibre is smooth and connected, and that the geometric special fibre has no singularities other than ordinary quadratic points. With the notation introduced in 12.4, for every prime ℓ the monodromy pairing (12.4.7) is the scalar extension to $\mathbb{Z}_\ell$ of the integral pairing (12.4.5).

This proves the monodromy formula used earlier in the case of a Jacobian equipped with its canonical polarization. By the reduction argument already given in the preceding section, this is enough to imply the general statement for abelian varieties.

For $\ell \neq p$, the proof is essentially the cohomological Picard–Lefschetz formula for

$$H^1(X_{\bar\eta}, \mathbb{Z}_\ell) \simeq \operatorname{Hom}(T_\ell(A_\eta), \mathbb{Z}_\ell). \tag{12.7.1}$$

This formula is proved later in the seminar in a more general situation; in the present case it is exactly the assertion above, once the definition of the monodromy pairing is unfolded. It remains to treat the case $\ell = p$. We reduce it to characteristic zero.

The reduction uses the same principle as the proof of the essential tameness theorem for monodromy on T_ℓ. We may suppose S complete. Let W be a Cohen p-ring with residue field k, and choose a homomorphism $W \to V$ inducing the identity on residue fields. Let L be the set of singular points of X_s. Consider the formal moduli space $\mathfrak{S}$ of X_s over W, with universal curve

$$\mathfrak{X} \longrightarrow \mathfrak{S}.$$

As in the local deformation theory recalled earlier, $\mathfrak{S}$ is isomorphic to the spectrum of a formal power series ring $W[[T_\lambda]]_{\lambda \in L}$, and in $\mathfrak{S}$ there is a normal-crossings divisor

$$D = \bigcup_{\lambda \in L} D_\lambda, \tag{12.8.1}$$

where the component D_λ corresponds to smoothing all singularities except the point indexed by λ. The given family over S is obtained by pullback from the universal family:

$$\begin{array}{ccc} X & \longrightarrow & \mathfrak{X} \\ \downarrow & & \downarrow \\ S & \xrightarrow{g} & \mathfrak{S}. \end{array} \tag{12.8.2}$$

Let

$$P = \operatorname{Pic}^0_{\mathfrak{X}/\mathfrak{S}}. \tag{12.8.3}$$

Since the special fibre is geometrically reduced of dimension one, $\mathfrak{X}$ is cohomologically flat in dimension zero, and P is formally smooth over $\mathfrak{S}$. We use the representability of this Picard functor by a smooth group object over $\mathfrak{S}$. Alternatively, Artin's representability theorem supplies an algebraic space, which suffices for the argument.

Let

$$U = \mathfrak{S} - \operatorname{Supp} D. \tag{12.8.4}$$

Over U, the universal curve is smooth and $P_U = \operatorname{Pic}^0_{\mathfrak{X}_U/U}$ is an abelian scheme. Pulling back along $g : S \to U$ gives the abelian variety A_η we are studying. The isomorphism (12.1.10) gives, on $\mathfrak{S}$,

$$P^0 \times_{\mathfrak{S}} S \simeq A^0. \tag{12.8.5}$$

Since the special fibre of P is ℓ-divisible for every ℓ, so is P itself. We may therefore consider $T_\ell(P)$ over $\mathfrak{S}$; its pullback to S is $T_\ell(A_\eta)$. Applying the constructions of the previous sections gives a filtration by the fixed part and the toric part

$$T_\ell(P)^f \subset T_\ell(P)^t \subset T_\ell(P). \tag{12.8.6}$$

Over U the canonical autoduality of the Jacobian gives a pairing

$$T_\ell(P_U) \times T_\ell(P_U) \to \mathbb{Z}_\ell(1). \tag{12.8.7}$$

For this pairing, the fixed part and the toric part are mutual orthogonals. Thus the quotient $E = T_\ell(P_U)^t/T_\ell(P_U)^f$ is canonically dual to $F = T_\ell(P_U)/T_\ell(P_U)^t$, and extends canonically over $\mathfrak{S}$ as

$$E = D(F), \qquad F = T_\ell(P)^t/T_\ell(P)^f. \tag{12.8.8}$$

We are therefore in the typical situation of a mixed extension. The group $T_\ell(P_U)$ is a mixed extension of E by F, where E and F arise from two dual extensions. Schematically the situation is

$$\begin{tikzcd}
0 \arrow[r] & F \arrow[r] & R \arrow[r] & M \arrow[r] & 0 \\
 & & T_\ell(P_U) \arrow[u] \arrow[d] & & \\
0 \arrow[r] & D(M) \arrow[r] & R^\vee \arrow[r] & D(F) \arrow[r] & 0.
\end{tikzcd} \tag{12.8.9}$$

The only difference from the earlier discussion is that the resulting pairing takes values not in a canonically fixed copy of $\mathbb{Z}_\ell$, but in

$$\Gamma(U, \mathcal{O}_U^*)/\Gamma(\mathfrak{S}, \mathcal{O}_{\mathfrak{S}}^*) \otimes \mathbb{Z}_\ell,$$

which is identified with $\mathbb{Z}^L \otimes \mathbb{Z}_\ell$ by the functions defining the components D_λ. This gives a canonical pairing

$$u : M \otimes M \longrightarrow \mathbb{Z}^L. \tag{12.8.11}$$

The monodromy pairing for the given curve over S is the composite of this pairing with the homomorphism $\mathbb{Z}^L \to \mathbb{Z}$ induced by $g : S \to \mathfrak{S}$. The latter homomorphism is given by an element

$$(d_\lambda)_{\lambda \in L} \in \mathbb{Z}^L, \qquad d_\lambda \geq 1, \tag{12.8.12}$$

where d_λ is the multiplicity with which the pullback of D_λ appears on S. For fixed λ, one has $d_\lambda = 1$ precisely when X is regular at the corresponding double point. Since regularity is assumed in Theorem 12.5, all d_λ's are equal to 1. It remains to prove the following local assertion.

Lemma 12.8.13. For $\lambda \in L$, the composite of (12.8.11) with the projection $\mathbb{Z}^L \to \mathbb{Z}$ onto the factor λ is the form $\varphi_{x_\lambda} \otimes \varphi_{x_\lambda}$ of (12.4.4).

Complete the strictly local ring at the generic point of D_λ and pull the family back to this trait. The new special fibre has exactly one ordinary quadratic singularity. The preceding constructions commute with this localization: the torus, its character group, the monodromy form, and the specialization morphism all localize compatibly. Thus the assertion reduces to Theorem 12.5 in the case where the base is a trait of characteristic zero and the special fibre has a single ordinary double point. This is the classical Picard–Lefschetz case. The reduction proves the theorem.

Remarks 12.9–12.10. The reduction just made also reduces to the classical situation in which the special fibre has one double point. In that case there are one or two irreducible components; these correspond respectively to rank $M = 1$ and rank $M = 0$, the latter being the case of good reduction for the Jacobian. Without assuming regularity of X, the same argument gives a more general formula in which the contribution of each double point is multiplied by the integer d_λ of (12.8.12): the monodromy pairing is obtained from

$$\sum_{\lambda \in L} d_\lambda \, \varphi_{x_\lambda} \otimes \varphi_{x_\lambda}. \tag{12.10.1}$$

13. Links with transcendental theory: the complex analytic case

For general facts on analytic spaces we refer to the standard sources. We review only the analytic dictionary needed to compare the preceding algebraic constructions with Hodge-theoretic ones.

13.1. Analytic groups over a base

Let S be a complex analytic space, and let A be a commutative analytic group over S. Assume A is smooth over S, and put

$$E = \mathrm{Lie}(A/S), \tag{13.1.1}$$

the locally free $\mathcal{O}_S$-module pulled back from the relative tangent sheaf along the zero section. We shall use the exponential homomorphism

$$\exp : E \longrightarrow A, \tag{13.1.2}$$

where E denotes the analytic vector bundle with sheaf of holomorphic sections E. It is defined locally near the zero section by the standard theory of formal groups in characteristic zero and then extended by multiplicativity. Its image is the open analytic subgroup A^0, the union of the neutral connected components of the fibres of A. If $R = \mathrm{Ker}(\exp)$, then one has an exact sequence of relative analytic groups

$$0 \longrightarrow R \longrightarrow E \longrightarrow A^0 \longrightarrow 0. \tag{13.1.3}$$

The group R is S-étale and is a closed analytic subspace of E precisely when A^0 is separated over S.

The exact sequence (13.1.3) is functorial in A and commutes with arbitrary base change. In this way one obtains a base-change-compatible equivalence between smooth relative analytic groups with connected fibres and pairs (E, R), where E is a locally free analytic vector bundle on S and $R \subset E$ is an S-étale analytic subgroup.

When A is abeloid over S, meaning smooth, proper and separated over S with connected fibres, then $A = A^0$, and R is locally constant. Its rank in a fibre is twice the complex rank of E. Conversely, an abeloid analytic group is described by such a local system R together with a complex analytic vector-bundle structure on the real analytic bundle $R \otimes_{\mathbb{Z}} \mathbb{R}$.

13.2. Formal completion along a fibre

Fix a point $s \in S$. For $n \geq 0$, let S_n be the n-th infinitesimal neighbourhood of s in S, and for an analytic space X over S, put $X_n = X \times_S S_n$. The system $\widehat{X} = (X_n)_n$ is the formal completion of X along the fibre X_s. It is functorial in X, and formation of $\widehat{X}$ commutes with finite projective limits.

Lemma 13.2.1. Let A be as in 13.1, and suppose that:

a) $A = A^0$;

b) R is constant in a neighbourhood of s;

c) R_s generates the complex vector space E_s.

If A' is another smooth analytic group over S, then the natural map from germs of homomorphisms $A \to A'$ near s to homomorphisms of formal completions

$$\varinjlim_{U \ni s} \mathrm{Hom}_U(A_U, A'_U) \longrightarrow \mathrm{Hom}_{\widehat{S}}(\widehat{A}, \widehat{A}') \tag{13.2.1.1}$$

is bijective.

Proof. Using (13.1.3), we may replace a homomorphism $A \to A'$ by a homomorphism of the corresponding vector bundles carrying R into R'. Passing to formal completions amounts to passing from an $\mathcal{O}_s$-linear homomorphism to its completion. The injectivity is clear. For surjectivity, Nakayama's lemma and the hypothesis that R_s generates E_s show that any compatible formal homomorphism of vector bundles sends the local system R into R' in a neighbourhood of s. Thus it comes from a genuine germ of homomorphisms.

Consequently, under the hypotheses of the lemma, the analytic group A is determined, up to unique isomorphism in a neighbourhood of s, by its formal completion $\widehat{A}$.

13.3. The analytic Raynaud extension

Let A be as in 13.1 with $A = A^0$. Write R_s for the lattice in the fibre. After shrinking S around s, the group of germs of sections of R gives an injective homomorphism from the constant group R_s to R. Its image is the largest constant subgroup of R; call it the fixed part:

$$R^f \subset R, \qquad R^f_s \xrightarrow{\sim} R^f_s. \tag{13.3.2}$$

The quotient

$$A^\natural = E/R^f \tag{13.3.3}$$

fits into an exact sequence

$$0 \longrightarrow R/R^f \longrightarrow A^\natural \longrightarrow A \longrightarrow 0. \tag{13.3.4}$$

The morphism $A^\natural \to A$ is ét, hence induces an isomorphism

$$\mathrm{Lie}(A^\natural/S) \xrightarrow{\sim} \mathrm{Lie}(A/S) = E. \tag{13.3.5}$$

For every $n \geq 0$, it also induces an isomorphism on the n-th infinitesimal neighbourhood of the fibre, and therefore

$$\widehat{A^\natural} \xrightarrow{\sim} \widehat{A}. \tag{13.3.6}$$

Thus $A^\natural$ is the analytic counterpart of the group considered algebraically in the Raynaud construction.

Suppose now that a complex torus

$$T_s \subset A_s \tag{13.3.7}$$

is given. Let T be the constant torus over S with fibre T_s. The induced homomorphism of formal completions comes, by Lemma 13.2.1, from a unique germ of analytic homomorphisms

$$T \longrightarrow A^\natural. \tag{13.3.9}$$

After shrinking S, this homomorphism is defined on all of S. Its differential identifies $\mathrm{Lie}(T/S)$ with a locally direct-factor submodule $E^t \subset E$:

$$\mathrm{Lie}(T/S) \simeq E^t \subset E. \tag{13.3.10}$$

Let

$$R^t \subset R_s, \qquad R^t \subset R, \tag{13.3.11}$$

be the sublattice defined by the torus T_s and the corresponding constant subgroup. The immersion condition implies, after shrinking S, that $T \to A^\natural$ is a closed immersion. Put

$$B = A^\natural/T. \tag{13.3.12}$$

Then $A^\natural$ is an extension

$$0 \longrightarrow T \longrightarrow A^\natural \longrightarrow B \longrightarrow 0, \tag{13.3.13}$$

where B is smooth and separated over S, with connected fibres, and

$$\mathrm{Lie}(T) = E^t. \tag{13.3.14}$$

The exponential sequence for B is

$$0 \longrightarrow R/R^t \longrightarrow E/E^t \longrightarrow B \longrightarrow 0. \tag{13.3.15}$$

Since

$$R^t/R^f = (R/R^f)^t \tag{13.3.16}$$

is constant, B is abeloid near s precisely when its fibre $B_s = A_s/T_s$ is abeloid. Therefore every expression of A_s as an extension of an abeloid analytic group by a complex torus gives, near s, a unique extension structure of $A^\natural$ by a torus whose fibre is the given torus.

13.4. Mixed Hodge structures

We reformulate the dictionary (13.1.4). Let $\mathcal{R}$ be the sheaf of sections of R, and put

$$\mathcal{E}(R) = \mathcal{R} \otimes_{\mathbb{Z}} \mathcal{O}_S. \tag{13.4.1}$$

When R is locally constant this is a locally free $\mathcal{O}_S$-module. The inclusion $R \subset E$ gives a canonical homomorphism

$$\mathcal{E}(R) \longrightarrow E. \tag{13.4.2}$$

Let F be its kernel, so that
$$0 \longrightarrow F \longrightarrow \mathcal{E}(R) \longrightarrow E \tag{13.4.3}$$
is exact on the left. In the most useful case the last map is surjective; then
$$0 \longrightarrow F \longrightarrow \mathcal{E}(R) \longrightarrow E \longrightarrow 0 \tag{13.4.4}$$
is exact. The pair (E, R) is then equivalent to the pair (R, F), where R is locally constant with finitely generated free $\mathbb{Z}$-modules as fibres and $F \subset \mathcal{E}(R)$ is a locally direct-factor submodule satisfying
$$F \cap \overline{F} = 0 \tag{13.4.5}$$
in the usual fibrewise sense. We regard $\mathcal{E}(R)$ as endowed with the Hodge filtration
$$\mathrm{Fil}^0 \mathcal{E}(R) = \mathcal{E}(R), \qquad \mathrm{Fil}^1 \mathcal{E}(R) = F, \qquad \mathrm{Fil}^2 \mathcal{E}(R) = 0. \tag{13.4.6}$$
The exact sequence (13.4.3) is functorial in A and commutes with base change.

Returning to 13.3, set
$$\mathcal{E}^f = R^f \otimes \mathcal{O}_S, \qquad \mathcal{E}^t = R^t \otimes \mathcal{O}_S. \tag{13.4.7}$$
These form the increasing weight filtration
$$\mathcal{E}^f \subset \mathcal{E}^t \subset \mathcal{E} = \mathcal{E}(R), \tag{13.4.8}$$
where $\mathcal{E}^f$, $\mathcal{E}^t/\mathcal{E}^f$, and $\mathcal{E}/\mathcal{E}^t$ are to be thought of as of weights -2, -1, and 0, respectively. The functoriality of (13.4.3), applied to $T \to A^\natural \to A$, gives
$$F(A^\natural) = F \cap \mathcal{E}^f, \qquad F(T) = 0. \tag{13.4.10}$$
Consequently the composite $\mathcal{E}^t \to E$ induces an isomorphism
$$\mathcal{E}^t/\mathcal{E}^f \xrightarrow{\sim} E^t = \mathrm{Lie}(T). \tag{13.4.11}$$
Assume moreover that R_s generates E_s, for instance in the favourable case where A_s is an extension of an abeloid analytic group by a torus. Then the corresponding map for $A^\natural$ is surjective, and one has exact sequences
$$0 \longrightarrow F^\natural \longrightarrow \mathcal{E}^f \longrightarrow E \longrightarrow 0, \tag{13.4.12}$$
with
$$F \cap \mathcal{E}^t = 0. \tag{13.4.13}$$
For the quotient $B = A^\natural/T$, one has
$$0 \longrightarrow F(B) \longrightarrow \mathcal{E}(B) \longrightarrow \mathrm{Lie}(B) \longrightarrow 0, \tag{13.4.14}$$
and explicitly
$$\mathcal{E}(B) = \mathcal{E}/\mathcal{E}^t, \qquad \mathrm{Lie}(B) = E/E^t. \tag{13.4.15}$$
Using (13.4.11) and (13.4.13), one obtains an isomorphism
$$F^\natural \xrightarrow{\sim} F(B). \tag{13.4.16}$$

Proposition 13.4.17. In the favourable situation just described, the Hodge filtration and the weight filtration
$$\mathcal{E} \supset F \supset 0, \qquad \mathcal{E} \supset \mathcal{E}^t \supset \mathcal{E}^f \supset 0 \tag{13.4.17.1}$$
on $\mathcal{E} = R \otimes \mathcal{O}_S$ satisfy:

a) the weight filtration comes by tensoring with $\mathcal{O}_S$ from a filtration

$$R \supset R^t \supset R^f \supset 0$$

by locally constant analytic groups whose fibres are finitely generated free $\mathbb{Z}$-modules;

b) one has

$$F \cap \mathcal{E}^t = 0, \qquad F + \mathcal{E}^f = \mathcal{E};$$

thus the filtration induced by the Hodge filtration on $\mathcal{E}^t$ is concentrated in degree -1, while that induced on $\mathcal{E}/\mathcal{E}^f$ is concentrated in degree 0;

c) the filtration induced on

$$\mathcal{E}^t/\mathcal{E}^f = (R^t/R^f) \otimes \mathcal{O}_S$$

by the Hodge filtration corresponds to the analytic abeloid group B of (13.3.13).

Remark 13.4.18. In Deligne's terminology, after restriction to an open set on which R is locally constant, the preceding proposition says that R, together with the Hodge and weight filtrations on $R \otimes \mathcal{O}_S$, is an analytic family of mixed Hodge structures. This situation was one of the models for Deligne's conjectures on Néron theory in higher dimensions. However, a feature present in the algebraic theory of Section 7 is not visible in this naive analytic description: when A is abeloid over the complement of a rare analytic subset, the local system R/R^t should be constant, and more precisely dual to the corresponding toric system for a suitable dual analytic group. This follows algebraically from the duality theorem used earlier. It is not clear from the elementary analytic methods used here whether the analogous assertion, or even unipotence of the local monodromy action, follows without additional hypotheses. Under algebraicity hypotheses the next section shows that the expected behaviour does hold and relates it to the construction of Section 7.

13.5. The relatively algebraic situation

We now suppose that the analytic group A comes from a relative group scheme over S, smooth and separated, whose fibres are extensions of abelian schemes by tori. We denote this relative scheme again by A. This abuse of notation is justified by the fact that the functor

$$G \longmapsto G^{\mathrm{an}}$$

from relative group schemes of this type to analytic groups over S is fully faithful. This is a relative GAGA-type assertion. For the notion of a relative scheme, and for the relation between relative schemes over S and analytic spaces over S, the references are Hakim's book and the corresponding earlier exposés on relative analytic geometry.

Let O be the local ring of S at s. Near s, the relative scheme A is determined by the ordinary scheme A_O over $\operatorname{Spec} O$ obtained by localizing A at s. The formal completion of A_O along its special fibre, which is isomorphic to $A_s = A_0$, is the same as the formal completion of the analytic group A along A_s, after identifying

$$A_n = A \times_S S_n$$

with algebraic schemes over S_n. In the favourable case where this formal group scheme, viewed over the completed local ring $\widehat{O}$, is algebraizable, one obtains a group scheme over $\operatorname{Spec} \widehat{O}$, extension of an abelian scheme $B_{\widehat{O}}$ by a torus. To simplify the statement, assume that $B_{\widehat{O}}$ is projective over $\widehat{O}$; this is the case, for instance, when S is normal at s.

Proposition 13.5.1. *Under the preceding hypotheses, the analytic group $A^\natural$ of (13.3.3) comes, in a neighbourhood of s, from a relative group scheme determined up to a unique isomorphism. This group scheme is an extension of a projective relative abelian scheme over S by a torus. If $A^\natural_{\widehat{O}}$ denotes the group scheme over $\widehat{O} = \widehat{O}_{S,s}$ that it defines, then $A^\natural_{\widehat{O}}$, with its extension structure, is canonically isomorphic to the Raynaud extension introduced above.*

The last assertion follows at once from the first and from 7.2.1, since $A^\natural_{\widehat{O}}$ has, along its special fibre, the formal completion prescribed by (13.3.6). To prove algebraicity of $A^\natural$ near s, one proceeds as in 7.2.1 and is reduced immediately to proving algebraicity of the abelian quotient B of $A^\natural$.

Lemma 13.5.2. *Let B be an analytic abeloid group over the analytic space S. Let $s \in S$ be a point such that the formal completion $\widehat{B}$ along the fibre B_s is algebraizable. Thus the spaces $B_n = B \times_S S_n$ are algebraizable, and $\widehat{B}$, viewed as a formal scheme over $\widehat{O}_{S,s}$, comes from an abelian scheme $B_{\widehat{O}}$ over $\widehat{O}$. Suppose moreover that $B_{\widehat{O}}$ is projective over $\widehat{O}$. Then B is projective over S after replacing S by a neighbourhood of s.*

The hypothesis means that the formal analytic space $\widehat{B}$ has a polarization, that is, a compatible system of polarizations p_n on the B_n. The conclusion means that there is a polarization p of B over a neighbourhood of s. Let P be the analytic space over S of relative polarizations of B. It represents the functor which assigns to $S' \to S$ the set of polarizations of $B_{S'}$ over S'. This is an open subfunctor of the relative Néron-Severi functor $\mathrm{NS}_{B/S}$. Standard arguments show that $\mathrm{NS}_{B/S}$, hence also P, is represented by an unramified analytic space over S. The formal polarization gives a formal section of P at s. Since P is unramified and locally quasi-finite over S, a formal section is induced by an analytic section over a neighbourhood of s. This analytic section is the desired polarization of B.

Remarks 13.5.3. (a) If the projectivity hypothesis on $B_{\widehat{O}}$ is omitted, one can still conclude that B is algebraic near s. One uses an unpublished theorem, independently due to Artin and Deligne, saying that if X is proper over S and if the formal completion of X along X_s is algebraizable, then X itself is algebraizable. This reduces matters to the case where S is reduced. When S is normal, an abelian scheme over $\widehat{O}$ is projective, and Lemma 13.5.2 applies. The general reduced case is obtained by non-flat descent from the normalization of S, using the fact that a morphism for which $\mathcal{O}_S \to f_*\mathcal{O}_{S'}$ is injective is an effective descent morphism for relative abelian schemes.

(b) The same method should also prove the analogous result for any analytic space B proper over S. The more delicate question is whether an analytic space X proper over S, whose formal completion along X_s is algebraizable, is itself algebraizable near s. Here algebraizability must be understood in a wider sense than in Hakim's book: near s, the space should come from an algebraic space over the local base, in Artin's sense.

Still under the hypotheses of Proposition 13.5.1, choose for every $n > 0$ the standard isomorphism

$$\mu_n \simeq (\mathbb{Z}/n\mathbb{Z})_S, \qquad m \longmapsto \exp(2\pi i m/n),$$

and, by passing to inverse limits over powers of a fixed prime ℓ, identify the usual Tate twist with the constant $\mathbb{Z}_\ell$-system. Applying the exact sequence of $\mathrm{Ext}^1(\mathbb{Z}/n\mathbb{Z}, -)$ to the exact sequence (13.1.3) gives the corresponding canonical projective systems. In the present algebraic situation the finite sheaves that occur are represented by relative finite group schemes, and the analytic symbols used earlier therefore acquire an algebraic meaning.

The resulting identifications are compatible with fixed parts and toric parts. If A' satisfies the same hypotheses and if W is a biextension of (A, A') by $\mathbb{G}_m$, defining a duality over the open set where A and A' are abelian schemes, then the induced pairings on Tate modules agree with the algebraic pairings. After shrinking around s, the ℓ-adic pairings glue to an integral pairing

$$R \times R' \longrightarrow \mathbb{Z}. \tag{13.5.11}$$

The algebraic orthogonality theorem can then be applied to R. The pairing (13.5.11) is zero on the fixed and toric products prescribed by the algebraic theory. If S is nonsingular of dimension one and if, over $U = S - \{s\}$, the restrictions of A and A' are dual abelian schemes, the fixed and toric subobjects are mutual annihilators near s. Since (13.5.11) is a perfect duality over U, it follows that R/R^t is locally constant and that the local monodromy action on R is unipotent of exponent at most two. One also obtains a perfect duality between R/R^f and the dual toric lattice. Under this identification the homomorphism of (13.3.4) may be written

$$i : M \longrightarrow A^{\natural}. \tag{13.5.13}$$

This is the form in which it will reappear below.

14. Links with transcendental theory: the rigid-analytic case

14.1. The canonical homomorphism $M_K \to G_K$

We assume the foundations of rigid-analytic spaces. Let S be a complete trait and let A_K be an abelian scheme over its fraction field K. Section 8.1 constructs, by descent from the semi-stable case and from the Raynaud extension, an extension

$$0 \longrightarrow T_K \longrightarrow G_K \longrightarrow B_K \longrightarrow 0 \tag{14.1.1}$$

of an abelian scheme B_K by a torus T_K. For the dual abelian scheme A'_K one has similarly

$$0 \longrightarrow T'_K \longrightarrow G'_K \longrightarrow B'_K \longrightarrow 0, \tag{14.1.2}$$

where B'_K is dual to B_K. Let

$$M_K = D(T'_K), \qquad M'_K = D(T_K) \tag{14.1.3}$$

be the character lattices. By Exposé VIII, the two extension structures are equivalently described by homomorphisms

$$M_K \xrightarrow{p_K} B'_K, \qquad M'_K \xrightarrow{p'_K} B_K. \tag{14.1.4}$$

Let W_K be the biextension of (B_K, B'_K) by $\mathbb{G}_m$ that gives the canonical duality, and let E_K be its inverse image on (M_K, M'_K). A trivialization of E_K is canonically the same as a homomorphism

$$i : M_K \longrightarrow G_K \tag{14.1.5}$$

lifting p_K, or, symmetrically, as a homomorphism

$$i' : M'_K \longrightarrow G'_K \tag{14.1.6}$$

lifting p'_K.

Writing a superscript an for the associated rigid-analytic groups, Raynaud defines a canonical exact sequence

$$0 \longrightarrow M_K^{\mathrm{an}} \xrightarrow{i^{\mathrm{an}}} G_K^{\mathrm{an}} \xrightarrow{e} A_K^{\mathrm{an}} \longrightarrow 0. \tag{14.1.7}$$

The homomorphism i^{an} can also be read as an ordinary homomorphism $i : M_K \to G_K$, lifting p_K. The sequence is the rigid-analytic analogue of (13.3.4), functorial in A_K and compatible with change of traits.

By this compatibility one reduces to the semi-stable case. There the groups $T_K, G_K, B_K, M_K, \ldots$ are generic fibres of group schemes $T, G, B, M, \ldots$ over S, and G is an extension of B by T. Let $\widehat{G}$ be the formal completion of G along the special fibre. There is a canonical isomorphism

$$\widehat{G} \xrightarrow{\sim} \widehat{A}^0, \tag{14.1.8}$$

and, on rigid generic fibres,

$$\widehat{G}^{\mathrm{an}} \xrightarrow{\sim} \widehat{A}^{0,\mathrm{an}}. \tag{14.1.9}$$

There are also canonical open immersions

$$\widehat{G}^{\mathrm{an}} \hookrightarrow G_K^{\mathrm{an}}, \qquad \widehat{A}^{0,\mathrm{an}} \hookrightarrow A_K^{\mathrm{an}}. \tag{14.1.10}$$

The map e is characterized uniquely by the commutative square

$$\begin{tikzcd} \widehat{G}^{\mathrm{an}} \arrow[r] \arrow[d] & \widehat{A}^{0,\mathrm{an}} \arrow[d] \\ G_K^{\mathrm{an}} \arrow[r, "e"'] & A_K^{\mathrm{an}}, \end{tikzcd} \tag{14.1.11}$$

where the unnamed arrows are those of (14.1.9) and (14.1.10).

Remarks 14.1.12. (a) The uniqueness of e is immediate from connectedness of $\widehat{G}^{\mathrm{an}}$; existence is the non-trivial part and is supplied by Raynaud's rigid-analytic theory. The definition of i^{an} is likewise transcendental, although once G is constructed by formal geometry the resulting i is an algebraic datum of the form (14.1.5). At this stage no purely algebraic-geometric description is known. The arithmetic characterization below is obtained by reduction to a situation of finite type over $\mathbb{Z}$.

(b) The homomorphism i is canonically associated with a determined trivialization of E_K, or equivalently with the dual homomorphism i'.

14.2. Relation with monodromy

Let $n > 0$. Applying the exact sequence of $\mathrm{Ext}^1(\mathbb{Z}/n\mathbb{Z}, -)$ to (14.1.7) on the rigid-analytic site over K, and using that M_K^{an} is torsion-free and G_K^{an} is n-divisible, gives an exact sequence of finite group schemes

$$0 \longrightarrow G_K[n] \longrightarrow A_K[n] \longrightarrow M_K/nM_K \longrightarrow 0. \tag{14.2.1}$$

Passing to inverse limits over powers of a prime ℓ yields

$$0 \longrightarrow T_\ell(G_K) \longrightarrow T_\ell(A_K) \longrightarrow M_K \otimes \mathbb{Z}_\ell \longrightarrow 0. \tag{14.2.2}$$

The purely algebraic formulas of 8.1.2 and 8.1.7 give the same exact sequence:

$$0 \longrightarrow T_\ell(G_K) \longrightarrow T_\ell(A_K) \longrightarrow T_\ell(A_K)/T_\ell(A_K)^f \simeq M_K \otimes \mathbb{Z}_\ell \longrightarrow 0. \tag{14.2.3}$$

The compatibility follows from (14.1.11). This compatibility, for a fixed ℓ, already characterizes i transcendently: the map e identifies the quotients $T_\ell(A_K)/T_\ell(A_K)^f$ with the ℓ-adic completions of a single lattice $\operatorname{Ker} e$, compatibly with the integral structure coming from the algebraic theory. Thus

$$i : M_K \xrightarrow{\sim} \operatorname{Ker} e. \tag{14.2.3 bis}$$

The same homomorphism reconstructs the monodromy pairing

$$u : M \times M' \longrightarrow \mathbb{Z}. \tag{14.2.4}$$

In the semi-stable case, assume for simplicity that M and M' are constant lattices. Pulling the Poincaré biextension on the abelian quotients back by the two homomorphisms from M and M' gives a biextension E of (M, M') by $\mathbb{G}_m$. For $(m, m') \in M \times M'$, the fibre $E_{m,m'}$ corresponds to an invertible V-module. Choosing a basis defines, independently of the choice, a valuation map

$$v : E_{m,m'}(K) \longrightarrow \mathbb{Z}.$$

A trivialization φ of E gives

$$(m, m') \longmapsto v(\varphi(m, m')), \tag{14.2.5}$$

and the monodromy homomorphism is this pairing for the splitting attached to i.

Remarks 14.2.6. Thus the rigid-analytic theory recovers integrality and independence of ℓ of the monodromy pairing. Raynaud's theory also recovers its positivity.

Remark 14.2.7. Regard $M \otimes \mathbb{Q}$ as an extension of $(M \otimes \mathbb{Q})/M$ by M. Using $i : M \to G_K$ gives

$$0 \longrightarrow G_K \longrightarrow G_K^\natural \longrightarrow (M \otimes \mathbb{Q})/M \longrightarrow 0. \tag{14.2.8}$$

Since G_K is divisible, this is equivalent to the family of extensions (14.2.2). It is therefore defined by algebraic data and is canonically the group introduced in (2.2.3), after identifying its component group with $(M \otimes \mathbb{Q})/M$.

14.3. An arithmetic characterization of i

For every prime ℓ, the extension (14.2.2) is known algebraically. Hence one knows the image of i under

$$\mathrm{Hom}(M_K, G_K) \longrightarrow \prod_\ell \mathrm{Ext}^1(M_K \otimes \mathbb{Z}_\ell, T_\ell(G_K)). \tag{14.3.1}$$

When M_K is constant with value M, this is the homomorphism

$$\mathrm{Hom}(M, G_K(K)) \longrightarrow \prod_\ell \mathrm{Hom}(M, H^1(K, T_\ell(G_K))) \tag{14.3.2}$$

obtained from the Kummer boundary maps

$$G_K(K) \longrightarrow \prod_\ell H^1(K, T_\ell(G_K)). \tag{14.3.3}$$

The kernel consists of those elements of $G_K(K)$ divisible by every integer $n > 1$.

Lemma 14.3.4. *Let V be a discrete valuation ring with residue field k of characteristic $p > 0$, of finite type over the prime field. Let G_K be a finite type group scheme over K, with zero unipotent rank if* $\mathrm{char}\, K = 0$*. Then the subgroup of infinitely divisible elements of $G_K(K)$ is zero. Equivalently, when G_K is divisible, (14.3.3) is injective.*

Proof. After a finite extension of K and normalization of V, reduce to the case where G_K has a composition series with quotients among finite constant groups, abelian varieties, $\mathbb{G}_m$, and additive groups. The finite constant and multiplicative cases are immediate. For an abelian variety, pass to the Néron model G and reduce to proving that the infinitely divisible subgroup of $G(S)$ is zero. This is true on the special fibre by the usual dévissage and the Mordell-Weil theorem over a field of finite type. The kernel of $G(S) \to G(k)$ is filtered by vector groups over k, killed by p on successive quotients; intersection over all infinitesimal levels is zero. The additive case is handled similarly.

Remark 14.3.4.1. It is essential to require divisibility by all integers, not only by powers of one prime. For a smooth group over a henselian trait, elements in the kernel of reduction are infinitely ℓ-divisible for every $\ell \neq p$. The preceding argument shows that the infinitely p-divisible subgroup is finite, and the full intersection is zero.

When k has characteristic $p > 0$ and is of finite type in the absolute sense, the homomorphism i is therefore determined by its image under (14.3.1), equivalently by the family of extensions (14.2.2). To obtain a general characterization, let $S = \mathrm{Spec}\, V$, with V a normal noetherian ring, separated and complete for an adic topology. Let A be a smooth quasi-projective group scheme over S, with connected fibres, such that the special fibre is an extension of an abelian scheme by an isotrivial torus and such that A_U is an abelian scheme over a non-empty open $U \subset S - S_0$. Section 7 constructs a group scheme G over S, an extension of an abelian scheme B by a torus T, characterized by an isomorphism of formal completions

$$\widehat{G} \xrightarrow{\sim} \widehat{A}^0. \tag{14.3.5.1}$$

For every prime ℓ one obtains on U an exact sequence

$$0 \longrightarrow T_\ell(G_U) \longrightarrow T_\ell(A_U) \longrightarrow M \otimes \mathbb{Z}_\ell \longrightarrow 0. \tag{14.3.5.3}$$

If A' satisfies the same hypotheses and W is a biextension of (A, A') by $\mathbb{G}_m$, inducing a duality over U, the resulting pairings on Tate modules give the orthogonality theorem and hence the identification

$$T_\ell(A_U)/T_\ell(A_U)^f \simeq M \otimes \mathbb{Z}_\ell. \tag{14.3.5.5}$$

Thus one obtains the extension

$$0 \longrightarrow T_\ell(G_U) \longrightarrow T_\ell(A_U) \longrightarrow M \otimes \mathbb{Z}_\ell \longrightarrow 0. \tag{14.3.5.6}$$

Conjecture 14.4. *In the situation described in 14.3.5, there exists a homomorphism*

$$i : M_U \longrightarrow G_U \tag{14.4.1}$$

such that, for every prime ℓ, the extension (14.3.5.6) is isomorphic to the extension deduced from i as in (14.2.2). More precisely, there is one and only one way to assign such an i to every situation of the type of 14.3.5, compatibly with all base changes $S' \to S$ arising from continuous homomorphisms $V \to V'$ of noetherian adic rings.

The existence of i, compatibly with base change, should follow from Raynaud's rigid-analytic theory in its general adic form. For uniqueness, one reduces to the case where S_0 is of finite type over $\mathbb{Z}$. Indeed, V is a filtered inductive limit of normal subrings of finite type over $\mathbb{Z}$, and the situation descends, for large index, to such a subring or to its separated completion for the induced adic topology. Since $S - S_0$ is Jacobson, two possible homomorphisms are equal if they agree at all closed points of U. Normalizing the closure of such a point gives a complete trait with finite residue field, and the uniqueness result of the preceding one-dimensional case applies. This proves uniqueness, and then base-change compatibility follows from uniqueness, assuming existence.

Remark 14.5. It remains to formulate and prove the expected compatibility between the rigid-analytic homomorphism i and the analogous complex-analytic homomorphism (13.5.13), or better, to find a common generalization of both constructions.

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